

21 世纪高等院校教材



# Ordinary Differential Equations and Their Applications ——*Theories and Models*

(常微分方程及其应用——理论与模型)

周宇虹 罗建书 编著



科学出版社

(O-4178.0101)

[www.sciencep.com](http://www.sciencep.com)

ISBN 978-7-03-030125-3



高等教育出版中心 数理出版分社  
联系电话: 010-64034725  
E-mail: [mph@mail.sciencep.com](mailto:mph@mail.sciencep.com)

定价: 29.00 元

21 世纪高等院校教材

# **Ordinary Differential Equations and Their Applications ——Theories and Models**

(常微分方程及其应用——理论与模型)

周宇虹 罗建书 编著

**科 学 出 版 社**

北 京

## 内 容 简 介

本书是常微分方程课程的英文教材,是作者结合多年的双语教学经验编写而成.全书共5章,包括一阶线性微分方程,高阶线性微分方程,线性微分方程组, Laplace 变换及其在微分方程求解中的应用,以及微分方程的稳定性理论.书中配有大量的应用实例和用 Matlab 软件绘制的微分方程解的相图,并介绍了绘制相图的程序.

本书可作为高等院校理工科偏理或非数学专业的本科双语教材,也可供相关专业的研究生、教师 and 广大科技人员参考.

### 图书在版编目(CIP)数据

常微分方程及其应用:理论与模型=Ordinary Differential Equations and Their Applications: Theories and Models: 英文/周宇虹,罗建书编著. —北京:科学出版社, 2011

21 世纪高等院校教材

ISBN 978-7-03-030125-3

I. ①常… II. ①周… ②罗… III. ①常微分方程-教材-英文 IV. O175.1

中国版本图书馆 CIP 数据核字(2011) 第 014597 号

**科学出版社 出版**

北京东黄城根北街 16 号

邮政编码: 100717

<http://www.sciencep.com>

**北京市安泰印刷厂印刷**

科学出版社发行 各地新华书店经销

\*

2011 年 1 月第 一 版 开本: B5 (720 × 1000)

2011 年 1 月第一次印刷 印张: 14

印数: 1—3 000 字数: 275 000

**定价: 29.00 元**

(如有印装质量问题, 我社负责调换)

# 前 言

“常微分方程”是数学专业的一门专业基础课,随着常微分方程理论与方法的应用领域不断扩大,它也被许多工科专业列为本科生和研究生的选修课.本书是作者根据多年为信息科学与计算专业本科生上的“常微分方程”双语课、公共“常微分方程”选修课,以及数学专业的“科技英语”课的讲稿整理而成的.科学技术的进步和发展,使得常微分方程的理论与方法得到了越来越广泛的应用,其应用领域涉及自然科学、工程技术、社会、经济、生物、生态与环境、医学、交通、军事等,硕果累累.因此,在内容的编排方面,作者除了积极吸纳国内外优秀教材在反映最新教学理念和知识结构方面的精彩之处外,还在每一章中精选了不同领域的应用案例,以期让学生了解常微分方程是如何用于解决实际问题的.根据学生的专业特性,书中介绍了用 Laplace 变换解微分方程和方程组的方法和技巧.书中还首次介绍了 lode 软件及其在平衡点附近解的几何特征研究和平衡点稳定性分析中的应用(这部分内容由罗建书负责编写).

本书是一本双语课教材.将常微分方程作为双语课,是因为学生不仅可以学到有关常微分方程、微积分和高等代数方面的专业术语,还可以通过不同领域的应用案例拓展阅读范围,从而提高学生的专业英语水平.为了使学生更好地理解和掌握学科知识,提高双语教学的教学质量和教学效果,书中对专业术语、比较难懂的词句作了注释.

华中师范大学的严国政教授和中北大学的靳祯教授对书稿作了认真仔细的审阅,并提出了宝贵的修改意见.在此,作者向他们表示衷心的感谢.本书的出版得到南京林业大学“精品教材”建设项目和国防科学技术大学专业建设项目的资助.对两所学校多年来对双语教学给予的鼓励和支持,作者在此表示衷心的感谢.

对于书中不可避免的各种疏漏及不妥之处,敬请各位读者和同行批评指教,作者将不胜感激.

作 者

2010 年 10 月于南京

# Contents

|                  |                                                                      |           |
|------------------|----------------------------------------------------------------------|-----------|
| <b>Chapter 1</b> | <b>First-order Differential Equations</b>                            | <b>1</b>  |
| 1.1              | Introduction                                                         | 1         |
|                  | Exercise 1.1                                                         | 7         |
| 1.2              | First-order Linear Differential Equations                            | 8         |
| 1.2.1            | First-order Homogeneous Linear Differential Equations                | 8         |
| 1.2.2            | First-order Nonhomogeneous Linear Differential Equations             | 11        |
| 1.2.3            | Bernoulli Equations                                                  | 16        |
|                  | Exercise 1.2                                                         | 18        |
| 1.3              | Separable Equations                                                  | 19        |
| 1.3.1            | Separable Equations                                                  | 19        |
| 1.3.2            | Homogeneous Equations                                                | 23        |
|                  | Exercise 1.3                                                         | 26        |
| 1.4              | Applications                                                         | 27        |
|                  | Module 1 The Spread of Technological Innovations                     | 27        |
|                  | Module 2 The Van Meegeren Art Forgeries                              | 30        |
| 1.5              | Exact Equations                                                      | 35        |
| 1.5.1            | Criterion for Exactness                                              | 35        |
| 1.5.2            | Integrating Factor                                                   | 39        |
|                  | Exercise 1.5                                                         | 42        |
| 1.6              | Existence and Uniqueness of Solutions                                | 43        |
|                  | Exercise 1.6                                                         | 50        |
| <b>Chapter 2</b> | <b>Second-order Differential Equations</b>                           | <b>51</b> |
| 2.1              | General Solutions of Homogeneous Second-order Linear Equations       | 51        |
|                  | Exercise 2.1                                                         | 59        |
| 2.2              | Homogeneous Second-order Linear Equations with Constant Coefficients | 60        |
| 2.2.1            | The Characteristic Equation Has Distinct Real Roots                  | 61        |
| 2.2.2            | The Characteristic Equation Has Repeated Roots                       | 62        |
| 2.2.3            | The Characteristic Equation Has Complex Conjugate Roots              | 63        |

|                                                                                                            |            |
|------------------------------------------------------------------------------------------------------------|------------|
| Exercise 2.2 .....                                                                                         | 65         |
| 2.3 Nonhomogeneous Second-order Linear Equations .....                                                     | 66         |
| 2.3.1 Structure of General Solutions .....                                                                 | 66         |
| 2.3.2 Method of Variation of Parameters .....                                                              | 68         |
| 2.3.3 Methods for Some Special Form of the Nonhomogeneous Term $g(t)$ ..                                   | 70         |
| Exercise 2.3 .....                                                                                         | 76         |
| 2.4 Applications .....                                                                                     | 77         |
| Module 1 An Atomic Waste Disposal Problem .....                                                            | 77         |
| Module 2 Mechanical Vibrations .....                                                                       | 82         |
| <b>Chapter 3 Linear Systems of Differential Equations .....</b>                                            | <b>90</b>  |
| 3.1 Basic Concepts and Theorems .....                                                                      | 90         |
| Exercise 3.1 .....                                                                                         | 98         |
| 3.2 The Eigenvalue-Eigenvector Method of Finding Solutions .....                                           | 99         |
| 3.2.1 The Characteristic Polynomial of $A$ Has $n$ Distinct Real<br>Eigenvalues .....                      | 100        |
| 3.2.2 The Characteristic Polynomial of $A$ Has Complex Eigenvalues .....                                   | 101        |
| 3.2.3 The Characteristic Polynomial of $A$ Has Equal Eigenvalues .....                                     | 104        |
| Exercise 3.2 .....                                                                                         | 108        |
| 3.3 Fundamental Matrix Solution; Matrix-valued Exponential<br>Function $e^{At}$ .....                      | 109        |
| Exercise 3.3 .....                                                                                         | 113        |
| 3.4 Nonhomogeneous Equations; Variation of Parameters .....                                                | 115        |
| Exercise 3.4 .....                                                                                         | 120        |
| 3.5 Applications .....                                                                                     | 121        |
| Module 1 The Principle of Competitive Exclusion in Population Biology ..                                   | 121        |
| Module 2 A Model for the Blood Glucose Regular System .....                                                | 127        |
| <b>Chapter 4 Laplace Transforms and Their Applications in Solving<br/>    Differential Equations .....</b> | <b>136</b> |
| 4.1 Laplace Transforms .....                                                                               | 136        |
| Exercise 4.1 .....                                                                                         | 138        |
| 4.2 Properties of Laplace Transforms .....                                                                 | 138        |
| Exercise 4.2 .....                                                                                         | 145        |
| 4.3 Inverse Laplace Transforms .....                                                                       | 146        |
| Exercise 4.3 .....                                                                                         | 148        |
| 4.4 Solving Differential Equations by Laplace Transforms .....                                             | 148        |

---

|                  |                                                                                                 |            |
|------------------|-------------------------------------------------------------------------------------------------|------------|
| 4.4.1            | The Right-Hand Side of the Differential Equation is Discontinuous ···                           | 152        |
| 4.4.2            | The Right-Hand Side of Differential Equation is an Impulsive<br>Function ···········            | 154        |
|                  | Exercise 4.4 ···········                                                                        | 156        |
| 4.5              | Solving Systems of Differential Equations by Laplace Transforms ·····                           | 157        |
|                  | Exercise 4.5 ···········                                                                        | 159        |
| <b>Chapter 5</b> | <b>Introduction to the Stability Theory ···········</b>                                         | <b>161</b> |
| 5.1              | Introduction ···········                                                                        | 161        |
|                  | Exercise 5.1 ···········                                                                        | 164        |
| 5.2              | Stability of the Solutions of Linear System ···········                                         | 164        |
|                  | Exercise 5.2 ···········                                                                        | 171        |
| 5.3              | Geometrical Characteristics of Solutions of the System of Differential<br>Equations ··········· | 173        |
| 5.3.1            | Phase Space and Direction Field ···········                                                     | 173        |
| 5.3.2            | Geometric Characteristics of the Orbits near a Singular Point ·····                             | 176        |
| 5.3.3            | Stability of Singular Points ···········                                                        | 180        |
|                  | Exercise 5.3 ···········                                                                        | 183        |
| 5.4              | Applications ···········                                                                        | 183        |
|                  | Module 1 Volterra's Principle ···········                                                       | 183        |
|                  | Module 2 Mathematical Theories of War ···········                                               | 188        |
|                  | <b>Answers to Selected Exercises ···········</b>                                                | <b>196</b> |
|                  | <b>References ···········</b>                                                                   | <b>209</b> |
| <b>附录</b>        | <b>软件包 Iode 简介 ···········</b>                                                                  | <b>210</b> |

# Chapter 1 First-order Differential Equations<sup>[1]</sup>

## 1.1 Introduction

### Example 1.1.1 Dating of art works<sup>[2]</sup>

On May 29, 1945, H.A. Van Meegeren, a *third rate Dutch painter*<sup>[3]</sup>, was arrested on the charge of collaborating with the enemy for<sup>[4]</sup> the sale to Goering of a painting of famed 17th century Dutch painter. Van Meegeren refused to accept the charge and announced, in his *prison cell*<sup>[5]</sup>, that he had never sold painting to Goering. He stated that all the *questioned paintings*<sup>[6]</sup> were his own works. To settle the question an international panel of distinguished chemists, physicists and art historians was appointed to investigate the matter. The panel took X-rays of the paintings to determine whether other paintings were underneath those paintings, analyzed the *pigments*<sup>[7]</sup> (coloring materials) used in the paintings, and examined the paintings for certain *signs of old age*<sup>[8]</sup>.

The panel of experts found traces of the modern pigment *cobalt blue*<sup>[9]</sup> in some paintings. In addition, they also detected *phenoformaldehyde*<sup>[10]</sup>, which was not discovered until *the turn of the 19th century*<sup>[11]</sup>, in several paintings. On the basis of these evidences Van Meegeren was convicted, of *forgery*<sup>[12]</sup>, on October 12, 1947 and sentenced to one year in prison<sup>[13]</sup>. Two months later, he died of a heart attack.

However, many people refused to believe that the famed “*Disciples at Emmaus*<sup>[14]</sup>” was a *forgery*<sup>[15]</sup>. In 1967, almost twenty years later, scientists at Carnegie Mellon University proved that the “Disciples at Emmaus” was indeed a forgery.

The key to the dating of materials lies in *the phenomenon*

[1] 一阶微分方程

[2] 艺术品的断代

[3] 一个三流的德国画家

[4] 因...被判通敌罪

[5] 单人牢房

[6] 受质疑的绘画作品

[7] 颜料

[8] 旧年代的痕迹

[9] 钴蓝 (一种现代绘画颜料)

[10] 苯酚甲醛

[11] 19 世纪末

[12] 因伪造被判有罪

[13] 被判服狱一年

[14] Emmaus 的信徒们

[15] 赝品

- [16] 放射性现象
- [17] 某些放射性元素的原子是不稳定的
- [18] 在一个给定的时间段内一定比例的原子会自然衰减并生成一种新元素的原子
- [19] 与...成正比
- [20] 其中
- [21] 物质的衰变系数
- [22] 半衰期
- [23] 读乐谱的妇人
- [24] 弹曼陀林的妇人
- [25] 糖尿病的检测
- [26] 糖尿病(一种代谢疾病)
- [27] 尿液
- [28] 葡萄糖耐量测试
- [29] 一整夜的禁食
- [30] 公认标准

of radioactivity<sup>[16]</sup> discovered at the turn of the 20th century by the physicist Rutherford and his colleagues. They showed that *the atoms of certain “radioactive” elements are unstable*<sup>[17]</sup> and that *within a given time period a fixed proportion of the atoms spontaneously disintegrates to form atoms of a new element*<sup>[18]</sup>. Rutherford also showed that the radioactivity of a substance is *directly proportional to*<sup>[19]</sup> the number of atoms of the substance present. Let  $N(t)$  denotes the number of atoms present at time  $t$ , then  $dN/dt$ , the number of atoms that disintegrate per unit time, is proportional to  $N$ , thus we have the following equation

$$\frac{dN}{dt} = -\lambda N, \quad (1.1.1)$$

where<sup>[20]</sup> constant  $\lambda$  is positive and is known as *the decay constant of the substance*<sup>[21]</sup>. Usually, we use **half-life**<sup>[22]</sup>, the time required for half of a given quantity of radioactive atoms to decay, to measure the rate of disintegration of a substance. Assume that  $N(t_0) = N_0$ , then we have the mathematical model for computing half-life

$$\frac{dN}{dt} = -\lambda N, \quad N(t_0) = N_0. \quad (1.1.2)$$

By evaluating the present disintegration rates of the radioactive pigments in Van Meegeren's questioned paintings, the experts concluded that the paintings “*Disciples at Emmaus*”, “*Woman Reading Music*<sup>[23]</sup>” and “*Woman Playing Mandolin*<sup>[24]</sup>” must be modern forgeries. ■

### Example 1.1.2 Detection of diabetes<sup>[25]</sup>

*Diabetes mellitus*<sup>[26]</sup> is a disease of metabolism which is characterized by too much sugar in the blood and *urine*<sup>[27]</sup>. *Glucose tolerance test*<sup>[28]</sup> (GTT) is a commonly used method to diagnose the disease. In this test, the patient is asked to take a large dose of glucose after *an overnight fast*<sup>[29]</sup>. During the next three to five hours, several measurements of the concentration of glucose are made in the patient's blood, and these measurements are used in the diagnosis of diabetes. Unfortunately, there is no *universally accepted criterion*<sup>[30]</sup> exist for interpreting the results

of a GTT. Different *physicians*<sup>[31]</sup> interpreting the results of a GTT may *come up with*<sup>[32]</sup> different diagnoses. Here is a case. A Rhode Island physician, after reviewing the results of a GTT, came up with a diagnosis of diabetes. But another physician declared the patient to be normal after reviewing the results of the same GTT. To settle the question, the results of the GTT were sent to a specialist in Boston. After examining these results, the specialist concluded that the patient was suffering from a *pituitary tumor*<sup>[33]</sup>.

In the mid-1960's, Drs. Rosevear and Molnar of the Mayo Clinic and Drs. Ackerman and Gatewood of the University of Minnesota discovered a fairly reliable criterion for interpreting the results of a GTT. They constructed a model which could accurately describe *the blood glucose regulatory system*<sup>[34]</sup> during a glucose tolerance test and in which one or two *parameters*<sup>[35]</sup> would *yield criteria for distinguishing normal individuals from mild diabetics and prediabetics*<sup>[36]</sup>.

The basic model is described *analytically*<sup>[37]</sup> by following *system of equations*<sup>[38]</sup>

$$\frac{dG}{dt} = F_1(G, H) + J(t), \quad (1.1.3)$$

$$\frac{dH}{dt} = F_2(G, H), \quad (1.1.4)$$

where  $G$  denotes the concentration of glucose in the blood and  $H$  denotes the concentration of the net *hormonal*<sup>[39]</sup> concentration. The function  $J(t)$  is the external rate at which the blood glucose concentration is being increased. ■

**Definition 1.1.1** An equation relating an unknown function, its *derivatives*<sup>[40]</sup> and *independent variables*<sup>[41]</sup> is called a **differential equation**<sup>[42]</sup>.

Eq.(1.1.1)-Eq.(1.1.4) are all examples of differential equation.

Differential equations are frequently used to describe the changing universe.

**Example 1.1.3** According to *Newton's law of cooling*<sup>[43]</sup>, the rate of change with respect to time  $t$  of the temperature  $T(t)$

[31] (内科) 医生

[32] 得出

[33] 脑垂体肿瘤

[34] 血糖调节系统

[35] 参数

[36] 产生能区别出正常人与中度糖尿病患者和早期糖尿病患者的标准

[37] 解析地

[38] 方程组

[39] 荷尔蒙、激素

[40] 它的各阶导数

[41] 自变量

[42] 微分方程

[43] 牛顿冷却定律

[44] 数学模型

[45] 种群

[46] 出生率和死亡率

[47] 比例系数

[48] 微分方程的解

[49] 对... 微分

[50] 通解

of a body is proportional to the difference between  $T$  and the temperature  $A$  of the surrounding medium, then the *mathematical model*<sup>[44]</sup> describing this law is

$$\frac{dT}{dt} = -k(T - A), \quad (1.1.5)$$

where  $k$  is a positive constant. ■

**Example 1.1.4** Let  $P(t)$  denote a *population*<sup>[45]</sup> with constant *birth and death rates*<sup>[46]</sup>, then the rate of change of  $P(t)$  is, in many simple cases, proportional to the size of the population, thus we have the following differential equation

$$\frac{dP}{dt} = kP(t), \quad (1.1.6)$$

where  $k$  is *the constant of proportionality*<sup>[47]</sup>. ■

If a continuous function together with its derivatives satisfy a differential equation, then we call this function a ***solution of the differential equation***<sup>[48]</sup>.

**Example 1.1.5** Is  $y(t) = c_1 \sin 2t + c_2 \cos 2t$ , where  $c_1$  and  $c_2$  are arbitrary constants, a solution of the differential equation  $y'' + 4y = 0$ ?

**Solution** *Differentiating*<sup>[49]</sup>  $y(t)$  with respect to  $t$ , we have that

$$y' = 2c_1 \cos 2t - 2c_2 \sin 2t, \quad y'' = -4c_1 \sin 2t - 4c_2 \cos 2t.$$

Hence,

$$\begin{aligned} y'' + 4y &= (-4c_1 \sin 2t - 4c_2 \cos 2t) + 4(c_1 \sin 2t + c_2 \cos 2t) \\ &= (-4c_1 + 4c_1) \sin 2t + (-4c_2 + 4c_2) \cos 2t = 0. \end{aligned}$$

$y$  and its derivative satisfy the differential equation. Moreover,  $y$  is obviously a continuous function. So  $y(t) = c_1 \sin 2t + c_2 \cos 2t$  is a solution of the differential equation. ■

From Example 1.1.5 we see that a differential equation may have infinite many solutions. The set of all solutions of a differential equation is called the ***general solution***<sup>[50]</sup> of the differential equation. For example, function  $y(t) = c_1 \sin 2t + c_2 \cos 2t$ , where  $c_1$  and  $c_2$  are arbitrary constants, is the general solution of equation  $y'' + 4y = 0$  since every solution of the differential equation is

of this form. Any one solution of a differential equation is called a **particular solution**<sup>[51]</sup> of the differential equation. It is not difficult to *verify*<sup>[52]</sup> that

- (a)  $y = \sin 2t + \cos 2t$ ,
- (b)  $y = 4\sin 2t$ ,
- (c)  $y = -3\cos 2t$ , and
- (d)  $y = 0$

are all solutions of the differential equation  $y'' + 4y = 0$ . So they are all particular solutions of the differential equation.

**Remark**<sup>[53]</sup> 1.1.1 The general solution of a differential equation cannot always be expressed by a single formula.

**Example 1.1.6** We can verify that the function  $y = cx^4$  is a family of solutions of the differential equation  $xy' - 4y = 0$  on the *interval*<sup>[54]</sup>  $(-\infty, +\infty)$ . Moreover, we can also verify that the *piecewise-defined function*<sup>[55]</sup>

$$y = \begin{cases} -x^4, & x < 0, \\ x^4, & x \geq 0 \end{cases}$$

is a particular solution of the differential equation. Obviously, this particular solution cannot be obtained from  $y = cx^4$  by a choice of the parameter  $c$ . We refer such extra solution as **singular solution**<sup>[56]</sup> of the differential equation. So the family of solutions  $y = cx^4$  is not the general solution of the equation. ■

A solution of a differential equation that *is identically zero*<sup>[57]</sup> on an interval  $I$  is called a **trivial solution**<sup>[58]</sup>.

A differential equation along with *subsidiary conditions*<sup>[59]</sup> on the unknown function and its derivatives, all given at the same value of the independent variable, constitutes an **initial-value problem**<sup>[60]</sup> (or **Cauchy problem**<sup>[61]</sup>):

$$\begin{cases} F(t; y, y', y'', \dots, y^{(n)}) = 0, \\ y(t_0) = y_0, y'(t_0) = y'_0, y''(t_0) = y''_0, \dots, y^{(n)}(t_0) = y_0^{(n)}. \end{cases} \quad (1.1.7)$$

Eq.(1.1.2) is an initial-value problem.

If the subsidiary conditions are given at more than one value of the independent variable, the problem is called a **boundary-**

- [51] 特解
- [52] 证明
- [53] 注
- [54] 区间
- [55] 分段定义函数
- [56] 奇(异)解
- [57] 恒为零
- [58] 平凡解
- [59] 附加条件
- [60] 初值问题
- [61] 柯西问题

- [62] 边值问题
- [63] 边界条件
- [64] 类型
- [65] 阶
- [66] 线性性
- [67] 常微分方程
- [68] 偏微分方程
- [69] 微分方程的  
阶
- [70] 一阶微分方  
程
- [71] 二阶微分方  
程
- [72]  $n$  阶常微分  
方程
- [73] 实值函数
- [74] 线性的

**value problem**<sup>[62]</sup> and the conditions are called **boundary conditions**<sup>[63]</sup>.

#### Example 1.1.7 Problem

$$3y''' + 5y'' - y' + 7y = 0; \quad y(1) = 0, y'(1) = 0, y''(1) = 0 \quad (1.1.8)$$

is an initial-value problem. While the problem

$$y'' + 2y' = e^t; \quad y(0) = 1, y'(1) = 1 \quad (1.1.9)$$

is a boundary-value problem. ■

Differential equations can be classified in three ways: *type*<sup>[64]</sup>, *order*<sup>[65]</sup>, and *linearity*<sup>[66]</sup>.

(i) A differential equation is an **ordinary differential equation**<sup>[67]</sup> if the unknown function depends on only one independent variable, such as Eq.(1.1.1)—Eq.(1.1.9). If the unknown function depends on two or more independent variables, then we call it a **partial differential equation**<sup>[68]</sup>. For example, equation

$$\frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2} = 1 \quad (1.1.10)$$

is a partial differential equation. We will be concerned solely with ordinary differential equations in this textbook.

(ii) The order of the highest derivative of the unknown function that appears in a differential equation is called the **order of the differential equation**<sup>[69]</sup>. Thus Eq.(1.1.1) is a *first-order differential equation*<sup>[70]</sup>, Eq.(1.1.9) and Eq.(1.1.10) are *second-order differential equations*<sup>[71]</sup>. The general form of an *n*th-order ordinary differential equation<sup>[72]</sup> is

$$F(t, y, y', \dots, y^{(n)}) = 0, \quad (1.1.11)$$

where  $F$  is a *real-valued function*<sup>[73]</sup> of  $(n+2)$  variables,  $t, y, y', \dots, y^{(n)}$ .

(iii) *n*th-order ordinary differential equation (1.1.11) is said to be **linear**<sup>[74]</sup> if  $F$  is linear about  $y, y', \dots, y^{(n)}$ . Thus the equations

$$(y - t)dt + 4tdy = 0,$$

$$y'' + y = 0,$$

$$4t^2 \frac{d^4 y}{dt^4} - 3 \frac{dy}{dt} + 5ty = 0$$

[75] 非线性的

[76] 证明

are, in turn, linear first-, second-, and fourth-order ordinary differential equations. A **nonlinear**<sup>[75]</sup> ordinary differential equation is simply one that is not linear. Therefore,

$$(y - t)^3 dy + 4t dt = 0,$$

$$y'' + y^2 = 0,$$

$$4t^2 \frac{d^3 y}{dt^3} - 3 \left( \frac{dy}{dt} \right)^3 + 5ty = 0$$

are examples of nonlinear first, second, and third-order ordinary differential equations, respectively.

### Exercise 1.1

In Problems 1-5 determine whether the given functions  $y(t)$  are solutions of the differential equations on left?

1.  $y'' + 2y' + y = 0$ ;  $y(t) = 2e^{-t} + te^{-t}$ .

2.  $y'' + 2y' + y = t$ ;  $y(t) \equiv 1$ .

3.  $(y')^4 + y^2 = -1$ ;  $y = t^2 - 1$ .

4.  $y' + y^2 = 0$ ;  $y_1 = \frac{1}{t}$ ,  $y_2 \equiv 0$ .

5.  $y'' + 4y = 0$ ;  $y(0) = 0$ ,  $y'(0) = 1$ ;

$$y_1(t) = \sin 2t, \quad y_2(t) = t, \quad y_3(t) = \frac{1}{2} \sin 2t.$$

6. Show that<sup>[76]</sup>  $y = \ln t$  is a solution of  $ty'' + y' = 0$  on interval  $(0, +\infty)$  but is not a solution on  $(-\infty, +\infty)$ .

7. Show that  $y = \frac{1}{t^2 - 1}$  is a solution of  $y' + 2ty^2 = 0$  on the interval  $I = (-1, 1)$  but not on any larger interval containing  $I$ .

8. Determine the order and linearity in each of the following differential equations.

(a)  $y''' - 5ty' = e^t + 1$ ;

(b)  $ty'' + t^2 y' - (\sin t)\sqrt{y} = t^2 - 1$ ;

(c)  $y \frac{d^2 x}{dy^2} = y^2 + 1$ ;

(d)  $y \left( \frac{dx}{dy} \right)^2 = x^2 + 1$ ;

(e)  $y' = (\sin t)y + e^t$ ;

(f)  $y' = t \sin y + e^t$ .

## 1.2 First-order Linear Differential Equations

[77] 可积函数

[78] 对 ... 求关于  $t$  的积分

[79] 积分常数

[80] 积分

[81] 齐次的

The standard form for a first-order differential equation in the unknown function  $y(t)$  is

$$\frac{dy}{dt} = f(t, y), \quad (1.2.1)$$

where  $f$  is any *integrable function*<sup>[77]</sup> of  $t$ .

Consider first the simple case where  $f(t, y)$  is only the function of one variable  $t$ ,

$$\frac{dy}{dt} = f(t). \quad (1.2.2)$$

*Integrating* both sides of Eq.(1.2.2) *with respect to*  $t$ <sup>[78]</sup> which yields

$$y(t) = \int f(t)dt + c. \quad (1.2.3)$$

Here  $c$  is an arbitrary *constant of integration*<sup>[79]</sup>. Unfortunately, we will not be able to solve most differential equations simply by *integration*<sup>[80]</sup> since, in most cases, we cannot integrate the function  $f(t)$  directly. For example, the solution of the initial-value problem

$$\frac{dy}{dt} + e^{t^2}y = 0; \quad y(1) = 2$$

is

$$y(t) = 2 \exp \left( - \int_1^t e^{s^2} ds \right).$$

We cannot integrate the function  $e^{t^2}$  directly!

We should, therefore, turn our attention to those differential equations that we can solve. The “simplest” equations that we can solve are those which are linear.

### 1.2.1 First-order Homogeneous<sup>[81]</sup> Linear Differential Equations

**Definition 1.2.1** The general form of the first-order linear differential equation is

$$\frac{dy}{dt} + a(t)y = b(t), \quad (1.2.4)$$

where  $a(t)$  and  $b(t)$  are continuous functions of  $t$ . If  $b(t) \equiv 0$ , the corresponding equation

$$\frac{dy}{dt} + a(t)y = 0 \quad (1.2.5)$$

is called a first-order **homogeneous** linear differential equation. If  $b(t) \not\equiv 0$ , then Eq.(1.2.4) is called a first-order **nonhomogeneous**<sup>[82]</sup> linear differential equation.

To solve the homogeneous equation (1.2.5), we divide both sides of the equation by  $y$  and rewrite it in the form

$$\frac{y'}{y} = -a(t). \quad (1.2.6)$$

Integrating both sides of Eq.(1.2.6) gives

$$\ln |y(t)| = - \int a(t)dt + c_1,$$

where  $c_1$  is an arbitrary constant of integration. Taking exponentials<sup>[83]</sup> of both sides of last equation<sup>[84]</sup> yields

$$|y(t)| = \exp \left( - \int a(t)dt + c_1 \right) = c \exp \left( - \int a(t)dt \right)$$

or

$$\left| y(t) \exp \left( \int a(t)dt \right) \right| = c. \quad (1.2.7)$$

If there exist two different  $t_1$  and  $t_2$  such that

$$\left( y(t) \exp \left( \int a(t)dt \right) \right) \Big|_{t=t_1} = c,$$

$$\left( y(t) \exp \left( \int a(t)dt \right) \right) \Big|_{t=t_2} = -c,$$

then, by the intermediate value theorem<sup>[85]</sup>,  $y(t) \exp \left( \int a(t)dt \right)$  must achieve all values between  $-c$  and  $+c$ , this is contrary to (1.2.7). Hence,

$$y(t) \exp \left( \int a(t)dt \right) = c.$$

Thus,

$$y(t) = c \exp \left( - \int a(t)dt \right). \quad (1.2.8)$$

[82] 非齐次的

[83] 取指数

[84] 上式

[85] 微积分的介  
值定理

[86] 变化趋势  
 [87] 趋于无穷大  
 [88] 对 ... 求从  
 $t_0$  到  $t$  的积分

Eq.(1.2.8) is the general solution of the homogeneous equation (1.2.5).

**Example 1.2.1** Solve  $y' + y\sqrt{t}\sin t = 0$ .

**Solution** Here  $a(t) = \sqrt{t}\sin t$ , so

$$y(t) = c \exp \left( - \int \sqrt{t} \sin t dt \right). \quad \blacksquare$$

**Example 1.2.2** Find the general solution of differential equation  $y'(t) + ay(t) = 0$ , and then determine the *behavior*<sup>[86]</sup> of all solutions as  $t \rightarrow \infty$ , where  $a$  is a constant.

**Solution** The general solution is

$$y(t) = c \exp \left( - \int a dt \right) = ce^{-at}.$$

If  $a < 0$ , all solutions, with the exception of  $y(t) = 0$ , *approach infinity*<sup>[87]</sup> as  $t \rightarrow \infty$ .

If  $a > 0$ , all solutions approach zero as  $t \rightarrow \infty$ . ■

The initial-value problem corresponding to Eq.(1.2.5) is

$$\frac{dy}{dt} + a(t)y = 0; \quad y(t_0) = y_0. \quad (1.2.9)$$

To find the solution of Eq.(1.2.9) we *integrate* both sides of Eq. (1.2.6) *between*  $t_0$  *and*  $t$ <sup>[88]</sup>,

$$\int_{t_0}^t \frac{d}{ds} \ln |y(s)| ds = - \int_{t_0}^t a(s) ds.$$

Thus,

$$\ln |y(t)| - \ln |y(t_0)| = - \int_{t_0}^t a(s) ds$$

or

$$\ln \left| \frac{y(t)}{y(t_0)} \right| = - \int_{t_0}^t a(s) ds.$$

Taking exponentials of both sides of last equation we obtain that

$$\left| \frac{y(t)}{y(t_0)} \right| = \exp \left( - \int_{t_0}^t a(s) ds \right)$$

or

$$\left| \frac{y(t)}{y(t_0)} \exp \left( \int_{t_0}^t a(s) ds \right) \right| = 1.$$

Since

[89] 表达式

$$\frac{y(t_0)}{y(t_0)} \exp \left( \int_{t_0}^{t_0} a(s) ds \right) = 1,$$

we have that

$$\frac{y(t)}{y(t_0)} \exp \left( \int_{t_0}^t a(s) ds \right) = 1.$$

Hence

$$y(t) = y(t_0) \exp \left( - \int_{t_0}^t a(s) ds \right) = y_0 \exp \left( - \int_{t_0}^t a(s) ds \right). \quad (1.2.10)$$

**Example 1.2.3** Find the solution of initial-value problem

$$\frac{dy}{dt} + \left( \frac{1}{t} \right) y = 0; \quad y(1) = \pi.$$

**Solution** Here  $a(t) = \frac{1}{t}$ ,  $y_0 = \pi$ , so that

$$y(t) = \pi \exp \left( - \int_1^t \frac{1}{s} ds \right) = \pi e^{-(\ln t)} = \frac{\pi}{t}. \quad \blacksquare$$

**Example 1.2.4** Solve the initial-value problem

$$y' + \sqrt{1+t^2} y = 0; \quad y(0) = 5.$$

**Solution** Here  $a(t) = \sqrt{1+t^2}$ ,  $y_0 = 5$ , so

$$\begin{aligned} y(t) &= 5 \exp \left( - \int_0^t \sqrt{1+s^2} ds \right) \\ &= 5 \exp \left[ - \frac{s}{2} \sqrt{1+s^2} - \frac{1}{2} \ln \left( s + \sqrt{1+s^2} \right) \right]_0^t \\ &= 5 \exp \left[ - \frac{t}{2} \sqrt{1+t^2} - \frac{1}{2} \ln \left( t + \sqrt{1+t^2} \right) \right]. \quad \blacksquare \end{aligned}$$

### 1.2.2 First-order Nonhomogeneous Linear Differential Equations

We now come back to the nonhomogeneous equation (1.2.4)

$$\frac{dy}{dt} + a(t)y = b(t).$$

The solutions of Eq.(1.2.4) must be those functions which satisfy the following *expression*<sup>[89]</sup>

$$\frac{d}{dt}(\text{some function}) = b(t). \quad (1.2.11)$$

[90] 等价方程

[91] 当且仅当、充要条件

[92] 积分因子

[93] 不定积分

However, in most cases, the expression  $y' + a(t)y$  does not appear to be the derivative of some function. How can we make a differential equation in Eq.(1.2.4) to be of the form (1.2.11)?

Multiplying both sides of Eq.(1.2.4) by a continuous function  $\mu(t)$  we obtain an *equivalent equation*<sup>[90]</sup>

$$\mu(t) \frac{dy}{dt} + \mu(t)a(t)y = \mu(t)b(t). \quad (1.2.12)$$

To find a  $\mu(t)$  so that  $\mu(t)y' + \mu(t)a(t)y$  is the derivative of some function, we note that

$$\frac{d}{dt}\mu(t)y = \mu(t) \frac{dy}{dt} + \frac{d\mu}{dt}y,$$

this means that  $\mu(t)y' + \mu(t)a(t)y$  equals to the derivative of  $\mu(t)y$  *if and only if*<sup>[91]</sup>

$$\frac{d\mu}{dt} = a(t)\mu(t).$$

Thus,

$$\mu(t) = \exp \left( \int a(t)dt \right). \quad (1.2.13)$$

Since we only need one such function  $\mu(t)$ , we set the integral constant  $c$  in Eq.(1.2.13) equal to one. The function  $\mu(t) = \exp \left( \int a(t)dt \right)$  is called an **integrating factor**<sup>[92]</sup> for the non-homogeneous equation (1.2.4), and for this  $\mu(t)$ , Eq.(1.2.12) can be written in the form

$$\frac{d}{dt}\mu(t)y = \mu(t)b(t). \quad (1.2.14)$$

Taking the *indefinite integral*<sup>[93]</sup> of both sides of Eq.(1.2.14) yields

$$\mu(t)y = \int \mu(t)b(t)dt + c$$

or

$$\begin{aligned} y &= \frac{1}{\mu(t)} \left( \int \mu(t)b(t)dt + c \right) \\ &= \exp \left( - \int a(t)dt \right) \left( \int \mu(t)b(t)dt + c \right). \end{aligned} \quad (1.2.15)$$

**Example 1.2.5** Find the general solution of differential equation

$$y' - \frac{2}{t}y = \frac{2}{3}t^4.$$

**Solution** Here  $a(t) = -\frac{2}{t}$ , so the integrating factor is

$$\begin{aligned}\mu(t) &= \exp\left(\int a(t)dt\right) = \exp\left(-\int \frac{2}{t}dt\right) = e^{-2\ln|t|} \\ &= e^{\ln t^{-2}} = t^{-2}.\end{aligned}$$

Multiplying both sides of the differential equation by  $\mu(t)$  gives the equivalent equation

$$t^{-2}y' - 2t^{-3}y = \frac{2}{3}t^2$$

or

$$\frac{d}{dt}(t^{-2}y) = \frac{2}{3}t^2.$$

Integrating both sides of this last equation we have

$$t^{-2}y = \frac{2}{9}t^3 + c$$

or

$$y(t) = ct^2 + \frac{2}{9}t^5. \quad \blacksquare$$

**Example 1.2.6** Find the general solution of equation

$$\frac{dy}{dt} + 2ty = t.$$

**Solution** Here  $a(t) = 2t$ , so that the integrating factor

$$\mu(t) = \exp\left(\int a(t)dt\right) = \exp\left(\int 2tdt\right) = e^{t^2}.$$

Multiplying both sides of the differential equation by  $\mu(t)$  we obtain that

$$e^{t^2}\left(\frac{dy}{dt} + 2ty\right) = te^{t^2} \quad \text{or} \quad \frac{d}{dt}e^{t^2}y = te^{t^2}.$$

Integrating both sides of the last equation with respect to  $t$  we find that

$$e^{t^2}y = \frac{1}{2}e^{t^2} + c \quad \text{or} \quad y = \frac{1}{2} + ce^{-t^2}. \quad \blacksquare$$

According to the above discussion, we can solve the initial-value problem

$$\frac{dy}{dt} + a(t)y = b(t); \quad y(t_0) = y_0 \quad (1.2.16)$$

by two ways.

[94] 定积分

**Method I** Inserting the initial condition into the general solution of the equation

$$\frac{dy}{dt} + a(t)y = b(t)$$

to determine the value of the constant  $c$ .

**Method II** Taking the *definite integral*<sup>[94]</sup> of both sides of Eq.(1.2.14) between  $t_0$  and  $t$  gives

$$\mu(t)y - \mu(t_0)y_0 = \int_{t_0}^t \mu(s)b(s)ds$$

or

$$y = \frac{1}{\mu(t)} \left( \mu(t_0)y_0 + \int_{t_0}^t \mu(s)b(s)ds \right). \quad (1.2.17)$$

**Example 1.2.7** Find the solution of initial-value problem

$$\frac{dy}{dt} + 2ty = t; \quad y(1) = 2.$$

**Solution** The general solution of the differential equation is

$$y = \frac{1}{2} + ce^{-t^2}$$

(see Example 1.2.6). Inserting the initial condition  $y(1)=2$  into the last equation we obtain that

$$2 = \frac{1}{2} + ce^{-1}, \quad \text{or} \quad c = \frac{3}{2}e.$$

So solution of the initial-value problem is

$$y = \frac{1}{2} + \frac{3e}{2}e^{-t^2} = \frac{1}{2} + \frac{3}{2}e^{1-t^2}. \quad \blacksquare$$

**Example 1.2.8** Find the solution of initial-value problem

$$\frac{dy}{dt} + y = \frac{1}{1+t^2}; \quad y(2) = 3.$$

**Solution** The integrating factor is

$$\mu(t) = \exp \left( \int a(t)dt \right) = \exp \left( \int 1dt \right) = e^t.$$

Multiplying both sides of the differential equation by  $\mu(t)$  yields

$$e^t \left( \frac{dy}{dt} + y \right) = \frac{e^t}{1+t^2} \quad \text{or} \quad \frac{d}{dt} e^t y = \frac{e^t}{1+t^2}.$$

Taking the definite integral of both sides of the last equation from 2 to  $t$  we obtain

$$e^t y - 3e^2 = \int_2^t \frac{e^s}{1+s^2} ds$$

or

$$y = e^{-t} \left( 3e^2 + \int_2^t \frac{e^s}{1+s^2} ds \right). \quad \blacksquare$$

This solution cannot be given *explicitly*<sup>[95]</sup> since we cannot integrate the function  $\frac{e^t}{1+t^2}$  directly. However, it is equally as valued and as useful as those solutions which can be given explicitly. In fact, we can evaluate the above integral to any *degree of accuracy*<sup>[96]</sup> with the aid of a computer.

**Example 1.2.9** Find a continuous solution of initial-value problem

$$y' + y = g(t); \quad y(0) = 0,$$

where

$$g(t) = \begin{cases} 2, & 0 \leq t < 1, \\ 0, & t \geq 1. \end{cases}$$

**Solution** When  $0 \leq t < 1$ , we have initial-value problem

$$y' + y = 2; \quad y(0) = 0.$$

The integrating factor for this nonhomogeneous problem is

$$\mu(t) = \exp \left( \int 1 dt \right) = e^t,$$

so solution over the interval  $[0, 1)$  is

$$y(t) = e^{-t} \left( \int_0^t 2e^s ds + 0 \right) = 2(1 - e^{-t}).$$

When  $t \geq 1$ , we have initial-value problem

$$y' + y = 0; \quad y(1) = 2(1 - e^{-1}).$$

Then the solution over the interval  $[1, +\infty)$  is

$$y(t) = 2(1 - e^{-1}) \exp \left( - \int_1^t 1 ds \right) = 2(1 - e^{-1}) e^{1-t} = 2(e - 1) e^{-t}.$$

[95] 以显式的形式

[96] 精度

[97] 所求解

[98] 因变量

[99] 代换

[100] 作 (1.2.18)

式定义的代换

[101] 原函数、不

定积分

[102] 伯努利方程

Since

$$\lim_{t \rightarrow 1^-} 2(1 - e^{-t}) = 2(1 - e^{-1}) \quad \text{and} \quad \lim_{t \rightarrow 1^+} 2(e-1)e^{-t} = 2(1 - e^{-1}),$$

the required solution<sup>[97]</sup> is

$$y(t) = \begin{cases} 2(1 - e^{-t}), & 0 \leq t < 1, \\ 2(e-1)e^{-t}, & t \geq 1. \end{cases} \quad \blacksquare$$

**Example 1.2.10** Solve the differential equation  $ty'' + 2y' = 6t$ . In this example, the dependent variable<sup>[98]</sup>  $y$  is missing. Making substitution<sup>[99]</sup>

$$p = y' = \frac{dy}{dt}, \quad y'' = \frac{dp}{dt}, \quad (1.2.18)$$

it gives the first-order differential equation

$$F(t, p, p') = 0. \quad (1.2.19)$$

**Solution** Taking substitution defined in (1.2.18)<sup>[100]</sup> gives

$$t \frac{dp}{dt} + 2p = 6t$$

or equivalently

$$\frac{dp}{dt} + \frac{2}{t}p = 6.$$

This is a linear equation and its integrating factor is

$$\mu(t) = \exp \int \frac{2}{t} dt = e^{2 \ln t} = t^2.$$

Thus,

$$p = y' = t^{-2} \int 6s^2 ds = 2t + c't^{-2}.$$

Taking anti-derivative<sup>[101]</sup> with respect to  $t$  yields the general solution

$$y(t) = t^2 + c_1 t^{-1} + c_2 \quad (c_1 = -c'). \quad \blacksquare$$

### 1.2.3 Bernoulli Equations<sup>[102]</sup>

We call equation of the form

$$y' + p(t)y = q(t)y^n \quad (1.2.20)$$

**Bernoulli equation**, where  $n$  is a real number.

Bernoulli equation is not linear. Rewrite Eq.(1.2.20) in the form

$$y^{-n} \frac{dy}{dt} + p(t)y^{1-n} = q(t)$$

or

$$\frac{1}{1-n} \frac{dy^{1-n}}{dt} + p(t)y^{1-n} = q(t).$$

[103] 变换为

[104] 原微分方程

Make the substitution

$$z = y^{1-n}, \quad (1.2.21)$$

then the Bernoulli equation (1.2.20) can be transformed into<sup>[103]</sup> a linear equation in the unknown function  $z(t)$ .

**Example 1.2.11** Solve  $y' + ty = ty^2$ .

**Solution** This is a Bernoulli equation with  $p(t) = q(t) = t$ , and  $n = 2$ . Make the substitution

$$z = y^{1-2} = y^{-1},$$

then,

$$y = \frac{1}{z} \quad \text{and} \quad y' = -\frac{z'}{z^2}.$$

Substituting these variables into the original differential equation<sup>[104]</sup>, we obtain that

$$-\frac{z'}{z^2} + \frac{t}{z} = \frac{t}{z^2} \quad \text{or} \quad z' - t = -t.$$

Thus, the solution of the original differential equation is

$$y = \frac{1}{z} = \frac{1}{ce^{(t^2/2)} + 1}. \quad \blacksquare$$

**Example 1.2.12** Solve  $y' - \frac{3}{t}y = t^4 y^{\frac{1}{3}}$ .

**Solution** This is a Bernoulli equation with

$$p(t) = -\frac{3}{t}, \quad q(t) = t^4, \quad n = \frac{1}{3}.$$

Make the substitution

$$z = y^{1-(1/3)} = y^{2/3},$$

then,

$$y = z^{3/2} \quad \text{and} \quad y' = \frac{3}{2}z^{1/2}z'.$$

Substituting these values into the original differential equation we obtain that

$$\frac{3}{2}z^{1/2}z' - \frac{3}{t}z^{3/2} = t^4 z^{1/2}$$

[105] 隱(含)式地 or

$$z' - \frac{2}{t}z = \frac{2}{3}t^4.$$

This last equation is linear, its solution is

$$z = ct^2 + \left(\frac{2}{9}\right)t^5.$$

Since  $y = z^{3/2}$ , solution of the original problem is given *implicitly*<sup>[105]</sup> by

$$y^{2/3} = ct^2 + \left(\frac{2}{9}\right)t^5. \quad \blacksquare$$

When  $n = 1$  or  $n = 0$ , a Bernoulli equation reduces to a linear equation.**Exercise 1.2**

Find the general solution of each of following differential equations.

- |                                            |                                                   |
|--------------------------------------------|---------------------------------------------------|
| 1. $y' + 5y = 0.$                          | 2. $y' - 5y = 0.$                                 |
| 3. $y' + 3t^2y = 0.$                       | 4. $y' - t^2y = 0.$                               |
| 5. $y' + \frac{2}{t}y = 0.$                | 6. $y' - \frac{1}{t^2}y = 0.$                     |
| 7. $y' + (\sin t)y = 0.$                   | 8. $y' - (\cos t)y = 0.$                          |
| 9. $y' - 3y = 6.$                          | 10. $y' + \frac{4}{t}y = t^4.$                    |
| 11. $y' + y = \sin t.$                     | 12. $(1 + t^2)y' + 3ty = 6t.$                     |
| 13. $y' + \frac{2}{10 + 2t}y = 4.$         | 14. $y' + \frac{1}{3}y = \frac{1}{3}(1 - 2t)y^4.$ |
| 15. $y' - \frac{3}{t^2}y = \frac{1}{t^2}.$ | 16. $y' + y = y^2.$                               |
| 17. $y' + 2ty + ty^4 = 0.$                 | 18. $y' + t^2y = t^2.$                            |
| 19. $y' + y = y^2(\cos t - \sin t).$       | 20. $y' + ty = 6t\sqrt{y}.$                       |
| 21. $y'' = y' + t.$                        | 22. $ty'' + y' = 0.$                              |

Solve the following initial-value problems.

23.  $y' + \frac{2}{t}y = 0; y(1) = 0.$
24.  $y' + 6ty = 0; y(\pi) = 5.$
25.  $y' + (\sin t)y = 0; y(0) = \frac{3}{2}.$
26.  $y' + \sqrt{1 + t^2}e^{-t}y = 0; y(0) = 1.$
27.  $y' + \frac{2}{10 + 2t}y = 4; y(2) = 100.$
28.  $y' + y = \sin t; y(\pi) = 1.$
29.  $y' - ty = -t; y(0) = -4.$

30.  $y' + ty = 1 + t$ ;  $y\left(\frac{3}{2}\right) = 0$ .

[106] 可分离变量  
方程

31. Show that every solution of equation  $(dy/dt) + ay = be^{-ct}$ , where  $a$  and  $c$  are positive constants and  $b$  is any real number, approaches zero as  $t$  approaches infinity.

32. Given the differential equation  $(dy/dt) + a(t)y = f(t)$  with  $a(t)$  and  $f(t)$  continuous for  $-\infty < t < \infty$ ,  $a(t) \geq c > 0$ , and  $\lim_{t \rightarrow \infty} f(t) = 0$ , show that every solution tends to zero as  $t$  approaches infinity.

Determine the behavior of all solutions of following differential equations.

33.  $\frac{dy}{dt} + \frac{1}{t}y = \frac{1}{t^2}$  (as  $t \rightarrow 0$ ).

34.  $\frac{dy}{dt} + \frac{1}{\sqrt{t}}y = e^{\sqrt{t}/2}$  (as  $t \rightarrow 0$ ).

35.  $\frac{dy}{dt} + \frac{1}{t}y = \cos t + \frac{\sin t}{t}$  (as  $t \rightarrow 0$ ).

36.  $\frac{dy}{dt} + y \tan t = \sin t \cos t$  (as  $t \rightarrow \pi/2$ ).

## 1.3 Separable Equations<sup>[106]</sup>

### 1.3.1 Separable Equations

**Definition 1.3.1** A differential equation is said to be **separable** if it can be put into the following form

$$\frac{dy}{dt} = \frac{g(t)}{f(y)}, \quad (1.3.1)$$

where  $f$  and  $g$  are continuous functions of  $y$  and  $t$  respectively.

Rewrite Eq.(1.3.1) in the following equivalent form

$$f(y)dy = g(t)dt. \quad (1.3.2)$$

Integrating both sides of Eq.(1.3.2) gives

$$F(y(t)) = \int g(t)dt + c, \quad (1.3.3)$$

where  $c$  is an arbitrary constant of integration,  $F(y)$  is the anti-derivative of  $y(t)$ . Solving for  $y(t)$  from Eq.(1.3.3) we can find the general solution of Eq.(1.3.1).

**Example 1.3.1** Find the general solution of equation

$$e^y \frac{dy}{dt} - (t + t^3) = 0.$$

[107] 取对数

**Solution** Rewrite the differential equation in the following form

$$e^y dy = (t + t^3) dt.$$

Thus,

$$e^{y(t)} = \frac{1}{2}t^2 + \frac{1}{4}t^4 + c.$$

*Taking logarithms*<sup>[107]</sup> of both sides of this equation gives the general solution

$$y(t) = \ln \left( \frac{1}{2}t^2 + \frac{1}{4}t^4 + c \right). \quad \blacksquare$$

**Example 1.3.2** Find the general solution of equation

$$\frac{dy}{dt} = (1+t)(1+y).$$

**Solution** Rewrite the differential equation in the form

$$\frac{dy}{1+y} = (1+t)dt, \quad y \neq -1.$$

Integrating both sides of the last equation we obtain that

$$\ln |1+y| = \frac{1}{2}t^2 + t + c'$$

or

$$y(t) = \pm e^{\frac{1}{2}t^2 + t + c'} - 1 = ce^{\frac{1}{2}t^2 + t} - 1, \quad c = \pm e^{c'} \neq 0.$$

When  $c = 0$ ,  $y = -1$ . It is a solution of the equation. Thus, the general solution is

$$y(t) = ce^{\frac{1}{2}t^2 + t} - 1,$$

$c$  is an arbitrary constant. ■

The initial-value problem corresponding to the differential equation (1.3.1) is

$$\frac{dy}{dt} = \frac{g(t)}{f(y)}; \quad y(t_0) = y_0. \quad (1.3.4)$$

We can solve such initial-value problem in two ways. Either we use the initial condition  $y(t_0) = y_0$  to solve for the constant  $c$  in Eq.(1.3.3), or else we integrate both sides of Eq.(1.3.2) between  $t_0$  and  $t$  to obtain that

$$F(y) - F(y_0) = \int_{t_0}^t g(s)ds. \quad (1.3.5)$$

Since

$$F(y) - F(y_0) = \int_{y_0}^y f(r)dr, \quad (1.3.6)$$

[108] ♣

we can rewrite Eq.(1.3.5) in a simpler form

$$\int_{y_0}^y f(r)dr = \int_{t_0}^t g(s)ds. \quad (1.3.7)$$

**Example 1.3.3** Find the solution of initial-value problem

$$e^y \frac{dy}{dt} - (t + t^3) = 0; \quad y(1) = 1.$$

**Solution** Method (i). From Example 1.3.1, we know that the general solution of this equation is

$$y(t) = \ln \left( \frac{t^2}{2} + \frac{t^4}{4} + c \right).$$

Setting<sup>[108]</sup>  $t = 1$  and  $y = 1$  gives  $1 = \ln(3/4 + c)$ , or  $c = e - 3/4$ .

Hence,

$$y(t) = \ln \left( \frac{t^2}{2} + \frac{t^4}{4} + e - \frac{3}{4} \right).$$

Method (ii). From Eq.(1.3.7),

$$\int_1^y e^r dr = \int_1^t (s + s^3)ds.$$

Consequently,

$$e^y - e = \frac{t^2}{2} + \frac{t^4}{4} - \frac{1}{2} - \frac{1}{4}$$

or

$$y(t) = \ln \left( \frac{t^2}{2} + \frac{t^4}{4} + e - \frac{3}{4} \right). \quad \blacksquare$$

**Example 1.3.4** Solve the initial-value problem

$$e^t dt - y dy = 0; \quad y(0) = 1.$$

**Solution** From Eq.(1.3.7),

$$\int_1^y r dr = \int_0^t e^s ds.$$

Consequently, solution of the initial-value problem is

$$y^2 = 2e^t - 1$$

[109] 平方根

[110] 由这个方程不能解出  $y$  关于  $t$  的显函数

[111] 认可

[112] 这不会给我们带来任何应用方面的困难

[113] 借助于数字计算机总可以求出  $y(t)$  的数字解

[114] 非负的

or

$$y = \sqrt{2e^t - 1}.$$

We choose the positive *square root*<sup>[109]</sup> here because  $y(0) = 1 > 0$ .

This solution is only defined when

$$2e^t - 1 \geq 0 \quad \text{or} \quad e^t \geq \frac{1}{2}.$$

which implies that  $t \geq -\ln 2$ . ■**Example 1.3.5** Solve the initial-value problem

$$(1 + e^y) \frac{dy}{dt} = \cos t; \quad y\left(\frac{\pi}{2}\right) = 3.$$

**Solution** From Eq.(1.3.7),

$$\int_3^y (1 + e^r) dr = \int_{\pi/2}^t \cos s ds,$$

so that

$$y + e^y = 2 + e^3 + \sin t.$$

*This equation cannot be solved explicitly for  $y$  as a function of  $t$ <sup>[110]</sup>, we must content with<sup>[111]</sup> the solution in its implicit form.* ■

Most separable equations cannot be solved explicitly for  $y$  as a function of  $t$ . *This does not present us with any difficulties in applications<sup>[112]</sup>, since we can always find  $y(t)$  numerically with the aid of a digital computer<sup>[113]</sup>.*

**Example 1.3.6** Solve  $yy'' = (y')^2$ .In this equation, the independent variable  $t$  is missing, then substitutions

$$p = y' = \frac{dy}{dt}, \quad y'' = \frac{dp}{dy} \frac{dy}{dt} = p \frac{dp}{dy} \quad (1.3.8)$$

will result in the first-order differential equation

$$F\left(y, p, p \frac{dp}{dy}\right) = 0.$$

**Solution** Assume that  $y$  and  $y'$  are both *nonnegative*<sup>[114]</sup>. Then the substitutions defined in Eq.(1.3.8) give the first-order equation

$$yp \frac{dp}{dy} = p^2.$$

This equation is separable. *Separating the variables*<sup>[115]</sup> we obtain that

$$\int \frac{dp}{p} = \int \frac{dy}{y}, \quad p, y \neq 0$$

or

$$\ln |p| = \ln |c'y|,$$

$$p = c_1 y, \quad c_1 = \pm c' \neq 0,$$

$$\frac{dy}{y} = c_1 dt,$$

$$\ln |y| = c_1 t + c_2,$$

$$y = ce^{c_1 t}, \quad c = \pm e^{c_2} \neq 0.$$

Since both  $y = 0$  and  $y = c$  (when  $p = 0$ ) are solutions of the equation, the general solution of the equation is

$$y = ce^{c_1 t},$$

where  $c$  and  $c_1$  are arbitrary constants. ■

### 1.3.2 Homogeneous Equations<sup>[116]</sup>

A differential equation  $y' = f(t, y)$  is **homogeneous** if

$$f(kt, ky) = f(t, y) \quad (1.3.9)$$

for every real number  $k$ .

**Remark 1.3.1** Note the difference of the meaning of the terminology “homogeneous” defined here and that defined in Section 1.2. *Only in the context of first-order differential equations does “homogeneous” have the meaning defined above*<sup>[117]</sup>.

A homogeneous differential equation can be transformed into a separable equation by making the substitution

$$y = tv \quad (1.3.10)$$

along with its corresponding derivative

$$\frac{dy}{dt} = v + t \frac{dv}{dt}. \quad (1.3.11)$$

**Example 1.3.7** Determine whether the following differential equations are homogeneous.

$$(a) \ y' = \frac{y+t}{t}; \quad (b) \ y' = \frac{y^2}{t};$$

[115] 分离变量

[116] 齐次方程

[117] 只有在一阶微分方程的范围内“齐次”才有上述定义的含义

[118] 经代数简化为

$$(c) y' = \frac{2tye^{\frac{t}{y}}}{t^2 + y^2 \sin \frac{t}{y}}; \quad (d) y' = \frac{y + t^2}{t^3}.$$

[119] 将... 代换回...

**Solution**

(a)

$$f(kt, ky) = \frac{ky + kt}{kt} = \frac{y + t}{t} = f(t, y).$$

So the equation is homogeneous.

(b)

$$f(kt, ky) = \frac{(ky)^2}{kt} = k \frac{y^2}{t} \neq f(t, y).$$

So the equation is not homogeneous.

(c)

$$f(kt, ky) = \frac{2(kt)(ky)e^{\frac{kt}{ky}}}{(kt)^2 + (ky)^2 \sin \frac{kt}{ky}} = \frac{2tye^{\frac{t}{y}}}{t^2 + y^2 \sin \frac{t}{y}} = f(t, y).$$

So the equation is homogeneous.

(d)

$$f(kt, ky) = \frac{ky + (kt)^2}{(kt)^3} = \frac{y + kt^2}{k^2 t^3} \neq f(t, y).$$

So the equation is not homogeneous. ■

**Example 1.3.8** Solve  $y' = \frac{y + t}{t}$ .

**Solution** This is a homogeneous equation. Making the substitutions in Eq.(1.3.10) and Eq.(1.3.11) we obtain that

$$v + t \frac{dv}{dt} = \frac{tv + t}{t},$$

which can be algebraically simplified to<sup>[118]</sup>

$$t \frac{dv}{dt} = 1.$$

The last equation is separable, and integrating both sides gives

$$\int dv = \int \frac{dt}{t},$$

which yields

$$v = \ln |t| + c. \quad (1.3.12)$$

Substituting  $v = y/t$  back into<sup>[119]</sup> (1.3.12) we obtain the solution to the given differential equation

$$y = t \ln |t| + ct. \quad \blacksquare$$

**Example 1.3.9** Solve  $\frac{dy}{dt} = \frac{2y^4 + t^4}{ty^3}$ .

**Solution** Since

$$\frac{2(ky)^4 + (kt)^4}{(kt)(ky)^3} = \frac{k^4(2y^4 + t^4)}{k^4ty^3} = \frac{2y^4 + t^4}{ty^3},$$

the given equation is homogeneous. Substituting Eq.(1.3.10) and Eq.(1.3.11) into the differential equation we obtain that

$$v + t \frac{dv}{dt} = \frac{2(tv)^4 + t^4}{t(tv)^3},$$

which can be algebraically simplified to

$$t \frac{dv}{dt} = \frac{v^4 + 1}{v^3}.$$

This last equation is separable. Integrating both sides we obtain that

$$\begin{aligned} \int \frac{dt}{t} &= \int \frac{v^3}{v^4 + 1} dv, \\ \ln |t| &= \frac{1}{4} \ln(v^4 + 1) + c' \end{aligned}$$

or

$$v^4 + 1 = (c''t)^4,$$

where  $c'' = e^{-c'}$ . Finally, substituting  $v = y/t$  back into the last equation, we obtain the solution of the original differential equation

$$y^4 = ct^8 - t^4, \quad c = (c'')^4. \quad \blacksquare$$

**Example 1.3.10** Solve the initial value problem

$$\frac{dy}{dt} = \frac{t^2 + y^2}{ty}; \quad y(1) = -2.$$

**Solution** This is a homogeneous equation. By the same way as in Example 1.3.8 and Example 1.3.9, we can obtain the solution of the equation

$$y^2 = t^2 \ln t^2 + ct^2.$$

Applying the initial condition we have that

$$(-2)^2 = 1^2 \ln(1)^2 + c(1)^2 \Rightarrow c = 4.$$

Thus, solution to the initial-value problem is

$$y^2 = t^2 \ln t^2 + 4t^2$$

[120]  $n$  度齐次的 or

$$y = -\sqrt{t^2 \ln t^2 + 4t^2}, \quad \text{for } y(1) < 0. \quad \blacksquare$$

**Example 1.3.11** A function  $g(t, y)$  is *homogeneous of degree  $n$* <sup>[120]</sup> if

$$g(kt, ky) = k^n g(t, y)$$

for all  $k$ . Determine whether the following functions are homogeneous, and if so, point out their degrees.

(a)  $ty + y^2$ ; (b)  $t + y \sin(y/t)^2$ ; (c)  $t^3 + ty^2 e^{t/y}$ ; (d)  $t + ty$ .

**Solution**

- (a) The function is homogeneous of degree two.
- (b) The function is homogeneous of degree one.
- (c) The function is homogeneous of degree three.
- (d) The function is not homogeneous. ■

### Exercise 1.3

Find the general solution for each of following differential equations.

1.  $\frac{dy}{dt} = t^3 y^2$ .
2.  $\frac{dy}{dt} = \frac{t^2 + 2}{y}$ .
3.  $\frac{dy}{dt} = e^{t+y+3}$ .
4.  $\cos y \sin t \frac{dy}{dt} = \sin y \cos t$ .
5.  $dy = 2t(y^2 + 9)dt$ .
6.  $\frac{dy}{dt} = t^3 - 2t + 2$ .
7.  $t dt - y^3 dy = 0$ .
8.  $(t^2 + 1)dt + (y^2 + y)dy = 0$ .
9.  $yy'' + 1 = y'^2$ .
10.  $y^3 y'' - 1 = 0$ .

Solve the following initial-value problems.

11.  $t \cos t dt + (1 - 6y^5)dy = 0$ ;  $y(\pi) = 0$ .
12.  $\frac{dy}{dt} = 1 + y^2$ ;  $y(0) = 1$ .
13.  $y \frac{dy}{dt} + (y^2 + 1) \sin t = 0$ ;  $y(0) = 1$ .
14.  $t^2(1 + y^2) + 2y \frac{dy}{dt} = 0$ ;  $y(0) = 1$ .
15.  $\frac{dy}{dt} = \frac{2t}{y + yt^2}$ ;  $y(2) = 3$ .
16.  $y^3 y'' + 1 = 0$ ;  $y(1) = 1, y'(1) = 0$ .
17. Determine whether each of following functions of  $t$  and  $y$  is homogeneous.

- (a)  $\frac{y^2 + 2ty}{y^2}$ ;
- (b)  $\frac{y^3 + t^3}{yt^2 + y^3}$ ;

(c)  $\frac{y^3 + t^3}{t^2 + y^3};$

(d)  $\ln y - \ln t + \frac{t+y}{t-y};$

(e)  $\frac{e^{t+y}}{e^{t-y}};$

(f)  $\frac{(t^2 + 7ty + 9y^2)^{1/2}}{3t + 5y}.$

Solve the following differential equations.

18.  $2ty \frac{dy}{dt} = 3y^2 - t^2.$

19.  $(t - \sqrt{ty}) \frac{dy}{dt} = y.$

20.  $t \frac{dy}{dt} = y + \sqrt{t^2 + y^2}; y(1) = 0.$

[121] 技术革新的传播

[122] 一个行业

[123] 一个企业

[124] 跟着

[125] 我们近似地假设  $p(t)$  是时间的连续函数

[126] 作整数变化

[127] 还不知道这项革新

## 1.4 Applications

### Module 1 The Spread of Technological Innovations<sup>[121]</sup>

How does a technological change or innovation spread in *an industry*<sup>[122]</sup>? Once an innovation is introduced by *one firm*<sup>[123]</sup>, how soon do others in the industry *come to*<sup>[124]</sup> adopt it? And what factors determine how rapidly they follow? The following is a model of the spread of innovations among farmers.

Assume that a new innovation is introduced into a fixed community of  $N$  farmers at time  $t = 0$ . Let  $p(t)$  denote the number of farmers who have adopted the innovation at time  $t$ . We make the approximation that  $p(t)$  is a continuous function of time<sup>[125]</sup>, even though it obviously changes by integer amounts<sup>[126]</sup>. The simplest realistic assumption that we can make concerning the spread of this innovation is that a farmer adopts the innovation only after he has been told of it by a farmer who had already adopted it. Then the number of farmers  $\Delta p$  who adopt the innovation in a small time interval  $\Delta t$  is directly proportional to the number of farmers  $p$  who have already adopted the innovation and the number of farmers  $(N - p)$  who are as yet unaware<sup>[127]</sup>. Hence

$$\Delta p = cp(N - p)\Delta t,$$

or

$$\Delta p / \Delta t = cp(N - p)$$

for some positive constant  $c$ . Letting  $\Delta t \rightarrow 0$  we obtain the differential equation

[128] 采纳过程加速上升直到一半的农场主知道了这项革新

$$\frac{dp}{dt} = cp(N - p). \quad (1.4.1)$$

Assuming that  $p(0) = 1$ , then  $p(t)$  satisfies the initial-value problem

[129] 累计数

$$\frac{dp}{dt} = cp(N - p); \quad p(0) = 1. \quad (1.4.2)$$

[130] 2,4-D 除草喷雾器

This differential equation is separable and its solution is

[131] 杂交玉米占玉米种植面积的累计百分比

$$p(t) = \frac{Ne^{cNt}}{N - 1 + e^{cNt}}. \quad (1.4.3)$$

Setting  $p''(t) = 0$  we obtain that

[132] 实际测量值

$$cN - 2cp = 0$$

[133] 逻辑曲线

or

[134] 给出了与模型非常好的一致性

$$p = \frac{N}{2}.$$

性

[135] 偏离

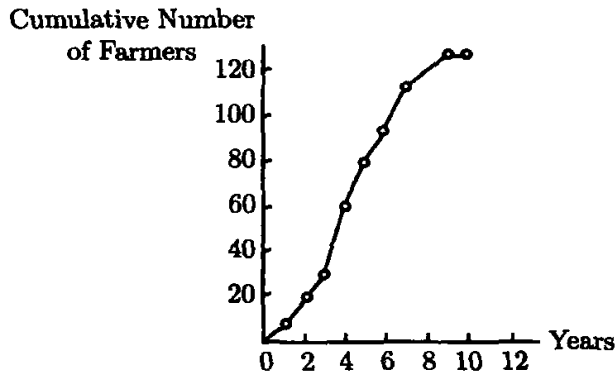
[136] 根源

[137] 大众传媒

Hence our model predicts that *the adoption process accelerates up to the point at which half the community is aware of the innovation*<sup>[128]</sup>. After that point, the adoption process begins to decelerate until it eventually reaches zero.

Figure 1.4.1 represents *the cumulative number*<sup>[129]</sup> of farmers in Iowa during 1944—1955 who adopted 2,4-D *weed spray*<sup>[130]</sup>, and Figure 1.4.2 represents *the cumulative percentage of corn acreage in hybrid corn*<sup>[131]</sup> in three American states during the years 1934—1958. The circles in the figure are *the actual measurements*<sup>[132]</sup>. As can be seen that these curves have all the properties of *logistic curves*<sup>[133]</sup> and, on the whole, *offer very good agreement with our model*<sup>[134]</sup>. However, there are two *discrepancies*<sup>[135]</sup>. First, from Figure 1.4.2 we can see that the adoption process for hybrid corn began to decelerate in Alabama only after nearly 60% of the farmers had adopted the innovation. Secondly, the agreement with our model is much better in the later stages of the adoption process than in the earlier stages.

The *source*<sup>[136]</sup> of the second discrepancy is our assumption that a farmer only learns of an innovation through contact with another farmer. This is not entirely true. In fact, *mass communication media*<sup>[137]</sup> such as radio, television, newspapers, and farmers magazines play a large role in the early stages of the adoption



[138] 把这一点  
考虑进去

Figure 1.4.1 Cumulative Number of Farmers Who Adopted 2,4-D Weed Spray in Iowa

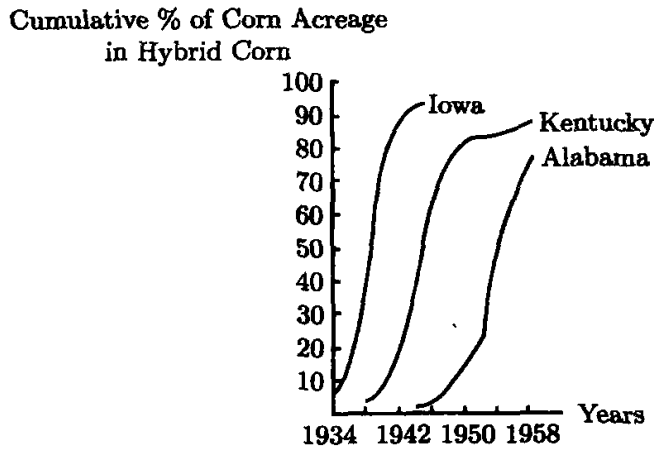


Figure 1.4.2 Cumulative Percentage of Corn Acreage in Hybrid Corn in Three American States

process. Therefore, we must add a term to the differential equation (1.4.1) *to take this into account*<sup>[138]</sup>. Assume that the number of farmers  $\Delta p$  who learn of the innovation through the mass communication media in a short period of time  $\Delta t$  is proportional to the number of farmers who do not yet know; i.e.,

$$\Delta p = c'(N - p)\Delta t$$

for some positive constant  $c'$ . Letting  $\Delta t \rightarrow 0$ , we see that  $(N - p)$  farmers per unit time learn of the innovation through the mass communication media. Thus, if  $p(0) = 0$ , then  $p(t)$  satisfies the initial value problem

$$\frac{dp}{dt} = cp(N - p) + c'(N - p); \quad p(0) = 0. \quad (1.4.4)$$

The solution of Eq.(1.4.4) is

[139] 改进后的曲线

$$p(t) = \frac{Nc' [e^{(c'+cN)t} - 1]}{cN + c'e^{(c'+cN)t}}. \quad (1.4.5)$$

[140] 一位农场主从他首次获知一项革新技术到他实际采纳该项技术所需的时间延迟区间

[141] 我们的模型必须对此加以考虑

[142] 铀

[143] 衰变

[144] 如此下去形成一个元素链, 它以铅为末端元素

[145] 碳-14

The corrected curve<sup>[139]</sup> (1.4.5) now gives remarkably good agreement with Figure 1.4.1 and Figure 1.4.2 for suitable choices of  $c$  and  $c'$ . The fact that the adoption of hybrid corn in Alabama only began to decelerate after 60% of the farmers had adopted the innovation indicates that other factors, such as *the time interval that elapses between when a farmer first learns of an innovation and when he actually adopts it*<sup>[140]</sup> may play an important role in the adoption process and *must be taken into account in our model*<sup>[141]</sup>.

## Module 2 The Van Meegeren Art Forgeries

We have known (see Example 1.1.1) that the key to the dating of materials lies in the phenomenon of radioactivity. Almost all rocks in the earth's crust contain a small quantity of *uranium*<sup>[142]</sup>. The uranium in the rock *decays to*<sup>[143]</sup> another radioactive element, and that one decays to another and another, and *so forth in a series of elements that results in lead*<sup>[144]</sup>, which is not radioactive (see Figure 1.4.3). The method, scientists used to determine the age of Van Meegeren's questioned paintings, is to evaluate disintegration rates of the radioactive elements in the pigments of the paintings. One measure of the rate of disintegration of a substance is its half-life which is defined as the time required for half of a given quantity of radioactive atoms to decay. The half-lives of many substances have been determined and recorded. For example, the half-life of *carbon-14*<sup>[145]</sup> is 5568 years and the half-life of uranium-238 is 4.5 billion years.

Let  $N(t)$  denote the number of atoms present at time  $t$ , then (see Example 1.1.1) the radioactivity of a substance satisfies the differential equation

$$\frac{dN}{dt} = -\lambda N, \quad (1.4.6)$$

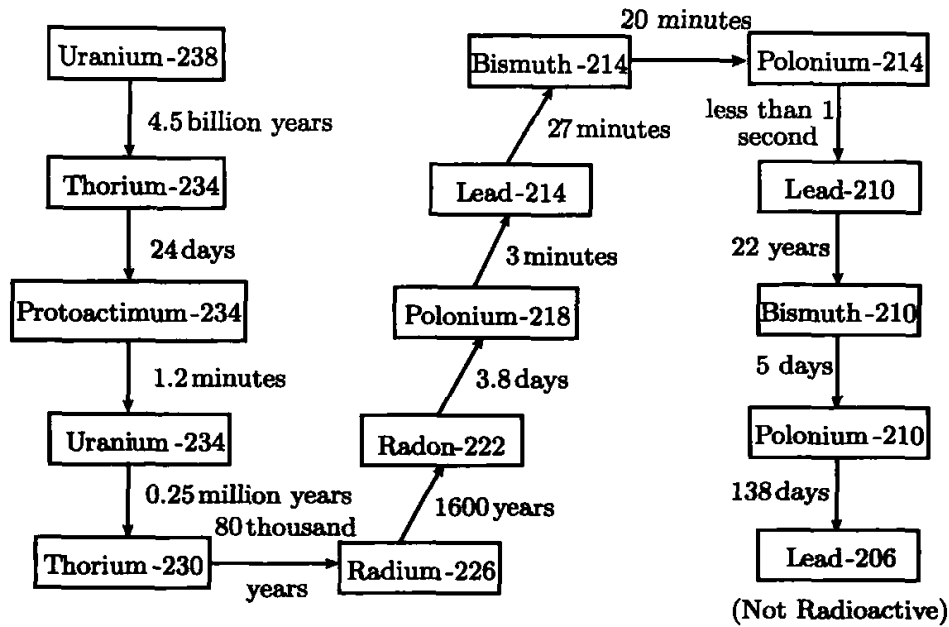


Figure 1.4.3 The Uranium series

The time shown on the arrows is the half-life of each step

where constant  $\lambda$  is positive and is known as the decay constant of the substance. Assume, moreover, that  $N(t_0) = N_0$ , then the solution of initial-value problem

$$\frac{dN}{dt} = -\lambda N, \quad N(t_0) = N_0 \quad (1.4.7)$$

is

$$N(t) = N_0 \exp \left( -\lambda \int_{t_0}^t ds \right) = N_0 e^{-\lambda(t-t_0)},$$

or

$$\frac{N(t)}{N_0} = \exp(-\lambda(t-t_0)).$$

Taking logarithms of both sides we obtain that

$$-\lambda(t-t_0) = \ln \frac{N}{N_0}. \quad (1.4.8)$$

Now, if  $N/N_0 = 1/2$ , then  $-\lambda(t-t_0) = \ln(1/2)$ , so that

$$t-t_0 = \frac{\ln 2}{\lambda} = \frac{0.6931}{\lambda}. \quad (1.4.9)$$

From Eq.(1.4.8) we can solve for

$$t-t_0 = -\frac{1}{\lambda} \ln \frac{N}{N_0} = \frac{1}{\lambda} \ln \frac{N_0}{N}. \quad (1.4.10)$$

[146] 物质最初形成或产生的时刻

[147] 年代

[148] 我们可以间接地确定  $N_0$  或确定  $N_0$  的某个合适的取值范围

[149] 在 Van Meegeren 的案例中用的就是这种方法

[150] 铅-210

[151] 镭-226

[152] 白铅

[153] 铅金属

[154] 铅矿石

[155] 熔炼

[156] 派生物

[157] 炉渣

[158] 阻断

[159] 与... 处于放射平衡

[160] 完全被镭的衰变所平衡

If  $t_0$  is the time the substance was initially formed or manufactured<sup>[146]</sup>, then the age<sup>[147]</sup> of the substance is  $(1/\lambda) \ln(N_0/N)$ . The decay constant  $\lambda$  is known or can be computed, in most instances. And we can usually evaluate  $N$  quite easily. The real difficulty is we usually do not know  $N_0$ . In some instances though, we can either determine  $N_0$  indirectly, or else determine certain suitable ranges for  $N_0$ <sup>[148]</sup>, and this is the case for the Van Meegeren's forgeries<sup>[149]</sup>.

All paintings contain a small amount of the radioactive element lead-210 ( $^{210}\text{Pb}$ )<sup>[150]</sup>, and an even smaller amount of radium-226 ( $^{226}\text{Ra}$ )<sup>[151]</sup>. These radioactive elements are contained in white lead<sup>[152]</sup> which is a pigment that artists have used for over 2000 years. White lead is made from lead metal<sup>[153]</sup>, which is extracted from lead ore<sup>[154]</sup>, in a process called smelting<sup>[155]</sup>. In this process, the lead-210 in the ore goes along with the lead metal. However, 90%—95% of the radium and its descendants<sup>[156]</sup> are removed with other waste products<sup>[157]</sup>. Thus, most of the supply of lead-210 is cut off<sup>[158]</sup> and it begins to decay very rapidly with a half-life of 22 years. This process continues until the lead-210 in the white lead is once more in radioactive equilibrium with<sup>[159]</sup> the small amount of radium present, i.e, the disintegration of lead-210 is exactly balanced by the disintegration of radium<sup>[160]</sup>.

Now we would like to use this information to compute the amount of lead-210 present in a sample. Let  $y(t)$  be the amount of lead-210 per gram of white lead at time  $t$ ,  $y_0$  be the amount of lead-210 per gram of white lead present at the time of manufacture  $t_0$ , and  $r(t)$  be the number of disintegrations of radium-226 per minute per gram of white lead at time  $t$ . If  $\lambda$  is the decay constant for lead-210, then

$$\frac{dy}{dt} = -\lambda y(t) + r(t); \quad y(t_0) = y_0. \quad (1.4.11)$$

Since we are only interested in a time period of at most 300 years and the half-life of the radium-226 is 1600 years, we assume that the radium-226 remains constant, so that  $r(t)$  is a constant  $r$ . Multiplying both sides of the differential equation by the integrating

factor  $\mu(t) = e^{\lambda t}$  we obtain that

[161] 种类

$$e^{\lambda t} \frac{dy}{dt} = r e^{\lambda t}.$$

[162] 产地

Hence,

$$e^{\lambda t} y(t) - e^{\lambda t_0} y_0 = \frac{r}{\lambda} (e^{\lambda t} - e^{\lambda t_0})$$

or

$$y(t) = \frac{r}{\lambda} (1 - e^{-\lambda(t-t_0)}) + y_0 e^{-\lambda(t-t_0)}. \quad (1.4.12)$$

Now  $y(t)$  and  $r$  can be easily measured. Thus, if we knew  $y_0$  we could use Eq.(1.4.12) to compute  $(t - t_0)$  and consequently, we could determine the age of the painting. As we pointed out, though, we cannot measure  $y_0$  directly. One possible way out of this difficulty is to use the fact that the original quantity of lead-210 was in radioactive equilibrium with a larger amount of radium-226 in the ore from which the metal was extracted. Table 1.4.1 gives the numbers of disintegrations of the radium-226 in different ores. These numbers vary from 0.18 to 1400.

**Table 1.4.1 All disintegration rates are per gram of white lead**

| <i>Description</i> <sup>[161]</sup> and <i>Source</i> <sup>[162]</sup> | Disintegrations per minute of <sup>226</sup> Ra |
|------------------------------------------------------------------------|-------------------------------------------------|
| Ore concentrate (Oklahoma-Kansas)                                      | 4.5                                             |
| Crushed raw ore (S.E.Missouri)                                         | 2.4                                             |
| Ore concentrate (S.E.Missouri)                                         | 0.7                                             |
| Ore concentrate (Idaho)                                                | 2.2                                             |
| Ore concentrate (Idaho)                                                | 0.18                                            |
| Ore concentrate (Washinton)                                            | 140.0                                           |
| Ore concentrate (British Columbia)                                     | 1.9                                             |
| Ore concentrate (British Columbia)                                     | 0.4                                             |
| Ore concentrate (Bolivia)                                              | 1.6                                             |
| Ore concentrate (Australia)                                            | 1.1                                             |

Consequently, the number of disintegrations of the lead-210 per minute per gram of white lead at the time of manufacture will vary from 0.18 to 1400. This implies that  $y_0$  will also vary over a very large interval since the number of disintegrations of lead-210 is proportional to the amount present. Thus, we cannot use Eq.(1.4.12) to obtain an accurate, or even a crude estimate, of the age of a painting.

[163] 针  
 [164] 被确认的  
 [165] 洗足

However, we will still use Eq.(1.4.12) to distinguish between a 17th century painting and a modern forgery. The basis for this statement is the simple observation that if the paint is very old compared to the 22 year half-life of lead-210, then the amount of radioactivity from the lead-210 in the paint will be nearly equal to the amount of radioactivity from the radium in the paint. On the other hand, if the painting is modern (approximately 20 years old, or so) then the amount of radioactivity from the lead-210 will be much greater than the amount of radioactivity from the radium. Now let us assume that the painting in question is either very new or about 300 years old. Set  $t - t_0 = 300$  years in Eq.(1.4.12). Then, after some simple algebra, we see that

$$\lambda y_0 = \lambda y(t)e^{300\lambda} - r(e^{300\lambda} - 1). \quad (1.4.13)$$

If the painting is indeed a modern forgery, then  $\lambda y_0$  will be absurdly large.

To evaluate  $\lambda y_0$  we must evaluate the present disintegration rate,  $\lambda y(t)$ , of the lead-210, the disintegration rate  $r$  of the radium-226, and  $e^{300\lambda}$ . Since the disintegration rate of *polonium-210*<sup>[163]</sup>( $^{210}\text{Po}$ ) equals that of lead-210 after several years, and since it is easier to measure the disintegration rate of polonium-210, we substitute these values for those of lead-210. To compute  $e^{300\lambda}$ , we observe from Eq.(1.4.9) that  $\lambda = (\ln 2/22)$ . Hence

$$e^{300\lambda} = e^{(300/22)\ln 2} = 2^{(150/11)}.$$

The disintegration rates of polonium-210 and radium-226 were measured for the "Disciples at Emmaus" and various other *alleged*<sup>[164]</sup> forgeries and are given in Table 1.4.2 below.

**Table 1.4.2 Questioned Paintings**  
**All disintegration rates are per minute per gram**  
**of white lead**

| Description                        | $^{210}\text{Po}$ disintegration | $^{226}\text{Ra}$ disintegration |
|------------------------------------|----------------------------------|----------------------------------|
| "Disciples at Emmaus"              | 8.5                              | 0.8                              |
| "Washing of Feet" <sup>[165]</sup> | 12.6                             | 0.26                             |
| "Woman Reading Music"              | 10.3                             | 0.3                              |
| "Woman Playing Mandolin"           | 8.2                              | 0.17                             |
| "Lace Maker"                       | 1.5                              | 1.4                              |
| "Laughing Girl"                    | 5.2                              | 6.0                              |

If we now evaluate  $\lambda y_0$  from Eq.(1.4.13) for the white lead in the painting “Disciples at Emmaus”, we obtain that

$$\lambda y_0 = (8.5)2^{150/11} - 0.8(2^{150/11} - 1) = 98.050,$$

which is unacceptably large. Thus, this painting must be a modern forgery. By a similar analysis, the paintings “Woman Reading Music” and “Woman Playing Mandolin” were indisputably shown to be faked Vermeers<sup>[166]</sup>. On the other hand, the paintings “Lace Maker<sup>[167]</sup>” and “Laughing Girl” cannot be recently forged Vermeers, as claimed by some experts, since for these two paintings, the polonium-210 is very nearly in radioactive equilibrium with the radium-226, and no such equilibrium has been observed in any samples from 19th or 20th century paintings<sup>[168]</sup>.

## 1.5 Exact Equations<sup>[169]</sup>

It will be convenient for our following discuss to write the differential equation in a *symmetric form*<sup>[170]</sup>

$$M(t, y)dt + N(t, y)dy = 0 \quad (1.5.1a)$$

or equivalently

$$M(t, y) + N(t, y)\frac{dy}{dt} = 0, \quad (1.5.1b)$$

where  $M(t, y)$  and  $N(t, y)$  are both *continuous and differentiable with respect to  $t$  and  $y$  in some region*<sup>[171]</sup>.

**Definition 1.5.1** If there exists a function  $\phi(t, y)$  such that<sup>[172]</sup>

$$M(t, y)dt + N(t, y)dy = d\phi(t, y), \quad (1.5.2)$$

then we call Eq.(1.5.1) an **exact differential equation**.

**Example 1.5.1** The equation  $1 + \cos(t+y) + \cos(t+y)(dy/dt) = 0$  can be written in the form  $(d/dt)[t + \sin(t+y)] = 0$ . Hence, it is exact and  $\phi(t, y) = t + \sin(t+y)$ . ■

### 1.5.1 Criterion for Exactness<sup>[173]</sup>

From the chain rule of partial differentiation<sup>[174]</sup> we obtain that

$$\frac{d}{dt}\phi(t, y(t)) = \frac{\partial \phi}{\partial t} + \frac{\partial \phi}{\partial y} \frac{dy}{dt},$$

[166] 是伪造 Vermeers 的作品

[167] 花边制作者

[168] 19 世纪或 20 世纪绘画作品

[169] 正合方程、完全方程、全微分方程

[170] 对称形式

[171] 在某个区域内关于  $t$  和  $y$  连续且可微

[172] 使得

[173] 正合性判别准则

[174] 偏导数的链式规则

[175] 可微函数

thus differential equation

[176] 洞

$$M(t, y) + N(t, y) \frac{dy}{dt} = 0$$

[177] 任何一条完

全落在  $R$  内的闭

曲线

is exact if and only if there exists a function  $\phi(t, y)$  such that

[178] 必要性

$$M(t, y) = \frac{\partial \phi}{\partial t} \quad \text{and} \quad N(t, y) = \frac{\partial \phi}{\partial y}.$$

[179] 求偏导数

Now we are facing the question: for any two *differentiable functions*<sup>[175]</sup>  $M(t, y)$  and  $N(t, y)$ , does there exist a function  $\phi(t, y)$  such that

$$M(t, y) = \frac{\partial \phi}{\partial t} \quad \text{and} \quad N(t, y) = \frac{\partial \phi}{\partial y}?$$

**Theorem 1.5.1** Eq.(1.5.1) is exact if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t} \quad (1.5.3)$$

in any region  $R$  which contain no “hole”<sup>[176]</sup>.

**Remark 1.5.1** A region  $R$  containing no “hole” means that if  $C$  is *any closed curve lying entirely in  $R$* <sup>[177]</sup>, then its interior also lies in  $R$ .

**Proof of theorem 1.5.1** (1) *Necessity*<sup>[178]</sup>. Suppose that Eq. (1.5.1) is exact, then there exists a function  $\phi(t, y)$  such that

$$M(t, y) = \frac{\partial \phi}{\partial t} \quad \text{and} \quad N(t, y) = \frac{\partial \phi}{\partial y}.$$

Taking indefinite integral of both sides of the first equation gives

$$\phi(t, y) = \int M(t, y) dt + h(y), \quad (1.5.4)$$

where  $h(y)$  is an arbitrary differentiable function of  $y$ . Then, *take partial derivatives*<sup>[179]</sup> of both sides of (1.5.4) with respect to  $y$  we obtain that

$$\frac{\partial \phi}{\partial y} = \int \frac{\partial M(t, y)}{\partial y} dt + h'(y).$$

Since  $\partial \phi / \partial y = N(t, y)$ , i.e

$$N(t, y) = \int \frac{\partial M(t, y)}{\partial y} dt + h'(y) \quad (1.5.5)$$

and take partial derivatives of both sides of (1.5.5) with respect to  $t$  we have that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

(2) *Sufficiency*<sup>[180]</sup>. If  $\partial M/\partial y = \partial N/\partial t$ , integrating both sides of the equation with respect to  $t$  gives [180] 充分性

$$N(t, y) = \int \frac{\partial M(t, y)}{\partial y} dt + g(y)$$

or

$$g(y) = N(t, y) - \int \frac{\partial M(t, y)}{\partial y} dt,$$

where  $g$  is any differentiable function. Set

$$\phi(t, y) = \int M(t, y) dt + \int g(y) dy,$$

then,

$$\begin{aligned} \frac{\partial \phi}{\partial y} &= \int \frac{\partial M(t, y)}{\partial y} dt + g(y) \\ &= \int \frac{\partial M(t, y)}{\partial y} dt + N(t, y) - \int \frac{\partial M(t, y)}{\partial y} dt \\ &= N(t, y) \end{aligned}$$

and

$$\frac{\partial \phi}{\partial t} = M(t, y) + 0. \quad \blacksquare$$

**Example 1.5.2** Find the general solution of the differential equation

$$1 + (1 + ty)e^{ty} + (1 + t^2e^{ty})\frac{dy}{dt} = 0.$$

**Solution** Here  $M(t, y) = 1 + (1 + ty)e^{ty}$  and  $N(t, y) = 1 + t^2e^{ty}$ .

This equation is exact since

$$\frac{\partial M}{\partial y} = te^{ty} + t(1 + ty)e^{ty} = 2te^{ty} + yt^2e^{ty} = \frac{\partial N}{\partial t}.$$

Hence, there exists a function  $\phi(t, y)$  such that

$$(i) \ 1 + (1 + ty)e^{ty} = \frac{\partial \phi}{\partial t} \quad \text{and} \quad (ii) \ 1 + t^2e^{ty} = \frac{\partial \phi}{\partial y}.$$

We will find  $\phi(t, y)$  by three different methods.

**Method 1** Integrating both sides of equation (i) gives

$$\phi(t, y) = t + te^{ty} + h(y).$$

Differentiating this equation with respect to  $y$  and using (ii) we obtain that

$$h'(y) + t^2e^{ty} = 1 + t^2e^{ty}.$$

Thus,  $h(y) = y + c_1$  and hence

$$\phi(t, y) = t + te^{ty} + y + c_1.$$

Setting  $\phi(t, y) = \text{constant}$ . Therefore, the general solution is of the form

$$t + te^{ty} + y = c.$$

**Method 2** Integrating both sides of equation (ii) gives

$$\phi(t, y) = y + te^{ty} + k(t).$$

Differentiating this expression with respect to  $t$ , and using (i) we obtain that

$$(1 + ty)e^{ty} + k'(t) = 1 + (1 + ty)e^{ty}.$$

Thus,

$$k(t) = t + c_2 \quad \text{and} \quad \phi(t, y) = t + te^{ty} + y + c_2.$$

**Method 3** From (i) and (ii),

$$\phi(t, y) = t + te^{ty} + h(y) \quad \text{and} \quad \phi(t, y) = y + te^{ty} + k(t).$$

Comparing these two expressions for the same function  $\phi(t, y)$ , it is obvious that  $h(y) = y$  and  $k(t) = t$ . Hence, the general solution is of the form

$$t + te^{ty} + y = c. \quad \blacksquare$$

**Example 1.5.3** Find the solution of the initial-value problem

$$2t \cos y + 3t^2 y + (t^3 - t^2 \sin y - y) \frac{dy}{dt} = 0; \quad y(0) = 2.$$

**Solution** This equation is exact since

$$\frac{\partial}{\partial y}(2t \cos y + 3t^2 y) = -2t \sin y + 3t^2 = \frac{\partial}{\partial t}(t^3 - t^2 \sin y - y).$$

Hence, there exists a function  $\phi(t, y)$  such that

$$(i) \quad 2t \cos y + 3t^2 y = \frac{\partial \phi}{\partial t} \quad \text{and} \quad (ii) \quad t^3 - t^2 \sin y - y = \frac{\partial \phi}{\partial y}.$$

From (i),

$$\phi(t, y) = t^2 \cos y + t^3 y + h(y).$$

From (ii),

$$\phi(t, y) = t^3 y + t^2 \cos y - \frac{1}{2} y^2 + k(t).$$

Thus,  $h(y) = -\frac{1}{2} y^2$ ,  $k(t) = 0$  and the general solution of the

differential equation is

[181] 积分因子

$$t^3y + t^2 \cos y - \frac{1}{2}y^2 = c.$$

From initial condition  $c = -\frac{1}{2}2^2 = -2$ . Finally, the solution of our initial-value problem is

$$t^3y + t^2 \cos y - \frac{1}{2}y^2 = -2. \quad \blacksquare$$

**Example 1.5.4** Determine whether the differential equation  $y' = y/t$  is exact.

A given differential equation may have many differential forms.

If we rewrite the differential equation in the form

$$\frac{y}{t}dt - dy = 0,$$

then  $M(t, y) = y/t$ ,  $N(t, y) = -1$ , and

$$\frac{\partial M}{\partial y} = \frac{1}{t} \neq 0 = \frac{\partial N}{\partial t}.$$

The equation is not exact. If we rewrite the differential equation in the form

$$\frac{1}{t}dt - \frac{1}{y}dy = 0,$$

then  $M(t, y) = 1/t$ ,  $N(t, y) = -1/y$ , and

$$\frac{\partial M}{\partial y} = 0 = \frac{\partial N}{\partial t}.$$

In this case, the equation is exact. ■

Exactness is only defined for equations in differential form, not standard form!

### 1.5.2 Integrating Factor<sup>[181]</sup>

If a differential equation (1.5.1)

$$M(t, y)dt + N(t, y)dy = 0$$

is not exact, can we make it exact? More precisely, can we find a function  $\mu(t, y)$  such that the equivalent differential equation

$$\mu(t, y)M(t, y)dt + \mu(t, y)N(t, y)dy = 0 \quad (1.5.6)$$

is exact? By Theorem 1.5.1, Eq.(1.5.6) is exact if and only if

$$\frac{\partial}{\partial y}\mu(t, y)M(t, y) = \frac{\partial}{\partial t}\mu(t, y)N(t, y)$$

or

$$M\frac{\partial\mu}{\partial y} + \mu\frac{\partial M}{\partial y} = N\frac{\partial\mu}{\partial t} + \mu\frac{\partial N}{\partial t}. \quad (1.5.7)$$

**Definition 1.5.2** A function  $\mu(t, y)$  satisfying Eq.(1.5.7) is called **integrating factor** for the differential equation (1.5.1).

In general, it is quite difficult to find a function  $\mu(t, y)$  from Eq.(1.5.7). In fact, there are only two special cases where we can find an explicit solution of Eq.(1.5.7): the differential equation (1.5.1) has an integrating factor which is either a function of  $t$  alone, or a function of  $y$  alone.

If  $\mu$  is a function of  $t$  alone, then Eq.(1.5.7) is reduced to

$$\frac{d\mu}{dt} = \frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial t} \right) \mu(t). \quad (1.5.8)$$

This equation is meaningless unless the expression

$$\frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial t} \right)$$

is also a function of  $t$  alone. If it is, that is

$$\frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial t} \right) = R(t),$$

then,

$$\mu(t) = \exp \left( \int R(t) dt \right). \quad (1.5.9)$$

Similarly, if  $\mu$  is a function of  $y$  alone, then Eq.(1.5.7) is reduced to

$$\frac{d\mu}{dy} = \frac{1}{-M} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial t} \right) \mu(y).$$

Moreover, if

$$\frac{1}{-M} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial t} \right) = Q(y),$$

then,

$$\mu(t) = \exp \left( \int Q(y) dy \right). \quad (1.5.10)$$

**Example 1.5.5** Use the methods of this section to find the general solution of linear differential equation  $y' + a(t)y = b(t)$ .

**Solution** Writing the equation in the form (1.5.1) with

$$M(t, y) = a(t)y - b(t) \quad \text{and} \quad N(t, y) = 1.$$

This equation is not exact since

$$\frac{\partial M}{\partial y} = a(t) \quad \text{and} \quad \frac{\partial N}{\partial t} = 0.$$

Note that

$$\frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial t} \right) = a(t),$$

so

$$\mu(t) = \exp \left( \int a(t) dt \right)$$

is an integrating factor for the given equation. Hence, there exists a function  $\phi(t, y)$  such that

$$(i) \mu(t)[a(t)y - b(t)] = \frac{\partial \phi}{\partial t} \quad \text{and} \quad (ii) \mu(t) = \frac{\partial \phi}{\partial y}.$$

From (ii) we have that

$$\phi(t, y) = \mu(t)y + k(t).$$

Differentiating this equation with respect to  $t$  and using (i) we see that

$$\mu'(t)y + k'(t) = \mu(t)a(t)y - \mu(t)b(t).$$

Thus,

$$\mu'(t) = \mu(t)a(t)$$

and

$$k'(t) = -\mu(t)b(t).$$

Hence,

$$\phi(t, y) = \mu(t)y - \int \mu(t)b(t)dt.$$

Consequently, the general solution of the first-order linear equation is

$$\phi(t) = c$$

or

$$y(t) = \frac{1}{\mu(t)} \left( \int \mu(t)b(t)dt + c \right).$$

This is the result we obtained in Section 1.2. ■

Since the expressions

$$\frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial t} \right) \quad \text{and} \quad \frac{1}{-M} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial t} \right)$$

are almost always functions of both  $t$  and  $y$ , it is for this reason that we cannot solve very many differential equations.

### Exercise 1.5

Determine whether the following differential equations are exact.

1.  $3t^2ydt + (y + t^2)dy = 0$ .

2.  $tydt + y^2dy = 0$ .

3.  $2tydt + (1 + t^2)dy = 0$ .

4.  $ydt - tdy = 0$ .

Find the general solutions of following differential equations.

5.  $3y + e^t + (3t + \cos y) \frac{dy}{dt} = 0$ .

6.  $(t + \sin y)dt + (t \cos y - 2y)dy = 0$ .

7.  $y^2dt + (2ty + 1)dy = 0$ .

8.  $\frac{dy}{dt} = \frac{2 + ye^{ty}}{2y - te^{ty}}$ .

Solve the following initial value problems.

9.  $3t^2y + 8ty^2 + (t^3 + 8t^2y + 12y^2) \frac{dy}{dt} = 0$ ;  $y(0) = 1$ .

10.  $4t^3e^{t+y} + t^4e^{t+y} + 2t + (t^4e^{t+y} + 2y) \frac{dy}{dt} = 0$ ;  $y(2) = 1$ .

11.  $2tydt + (1 + t^2)dy = 0$ ;  $y(2) = 5$ .

12.  $y^2dt + (2ty + 1)dy = 0$ ,  $y(1) = -2$ .

Convert the following differential equations into exact.

13.  $\frac{y^2}{2} + 2ye^t + (y + e^t) \frac{dy}{dt} = 0$ .

14.  $y' = 2ty - t$ .

15.  $y^2dt + tydy = 0$ .

16.  $y' = \frac{ty^2 - y}{t}$ .

In each of following problems determine the constant  $a$  so that the equation is exact, and then solve the resulting equation.

17.  $\frac{1}{t^2} + \frac{1}{y^2} + \frac{(at + 1)}{y^3} \frac{dy}{dt} = 0$ .

18.  $e^{at+y} + 3t^2y^2 + (2yt^3 + e^{at+y}) \frac{dy}{dt} = 0$ .

19. Show that a separable equation is always exact.

20. Find all functions  $f(t)$  that the differential equation  $y^2 \sin t + yf(t)(dy/dt) = 0$  is exact. Solve the differential equations for these  $f(t)$ .

[182] 解的存在性与唯一性

[183] 矩形区域

[184] Lipschitz 条件

[185] Lipschitz 常数

## 1.6 Existence and Uniqueness of Solutions<sup>[182]</sup>

As indicated in Section 1.4 that we will, almost always, be unable to solve an initial-value problem

$$\frac{dy}{dt} = f(t, y); \quad y(t_0) = y_0 \quad (1.6.1)$$

explicitly. Thus, it is quite natural to ask following questions:

(1) How do we know whether the initial-value problem (1.6.1) has a solution if we can't exhibit it?

(2) If the initial-value problem (1.6.1) has solution, then how many solutions it may have? Only one? Or perhaps it has two, three, or even infinitely many solutions?

The first question is about the existence of solutions, while the second question is about the uniqueness of solutions. An initial-value problem can fail to have a unique solution. For example,  $y_1(t) = 1$  and  $y_2(t) = \cos t$  are two distinct solutions of the initial-value problem

$$y' = -\sqrt{1 - y^2}; \quad y(0) = 1$$

on the interval  $(0, \pi)$ . In this section we will investigate the conditions on function  $f(t, y)$  that suffice to guarantee that the initial-value problem (1.6.1) has one and only one solution.

**Theorem 1.6.1 (Picard Theorem)** Let  $f(t, y)$  be continuous on the rectangle<sup>[183]</sup>  $R = \{(t, y) | |t - t_0| \leq a, |y - y_0| \leq b\}$  and satisfy **Lipschitz condition**<sup>[184]</sup> with respect to  $y$ , that is, for any  $(t, y_1), (t, y_2) \in R$ , there exists a constant  $L$ , called **Lipschitz constant**<sup>[185]</sup>, such that

$$|f(t, y_1) - f(t, y_2)| \leq L |y_1 - y_2|, \quad (1.6.2)$$

then the initial-value problem (1.6.1) has unique solution  $y(t)$  on interval  $|t - t_0| \leq h$  where

[186] 序列

[187] 一致收敛于

[188] 积分方程

[189] 逐次逼近

[190] Picard 迭代

$$h = \min \left( a, \frac{b}{M} \right), \quad M = \max_{(t,y) \in R} |f(t,y)|. \quad (1.6.3)$$

The proof of Theorem 1.6.1 will follow the procedure:

(a) Construct a *sequence*<sup>[186]</sup> of functions  $y_n(t)$ .

(b) Show that the sequence  $y_n(t)$  *converges uniformly to*<sup>[187]</sup> the solution  $y(t)$  of Eq.(1.6.4) on some suitable interval.

(c) Prove that  $y(t)$  is the unique solution of Eq.(1.6.1).

Let  $y(t)$  be a continuous solution of Eq.(1.6.1). Since

$$\int_{t_0}^t \frac{dy(s)}{ds} ds = \int_{t_0}^t f(s, y(s)) ds,$$

we have that

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds. \quad (1.6.4)$$

Conversely, if  $y(t)$  is continuous and satisfies (1.6.4), then

$$\frac{dy}{dt} = f(t, y(t)).$$

Eq.(1.6.4) is called an *integral equation*<sup>[188]</sup>. Therefore,  $y(t)$  is a solution of Eq.(1.6.1) if and only if it is a continuous solution of Eq.(1.6.4), and the existence and uniqueness of solutions of Eq.(1.6.1) is equivalent to the existence and uniqueness of solutions of Eq.(1.6.4).

Setting

$$y_0(t) \equiv y_0,$$

then construct a sequence of functions  $y_n(t)$  as follows

$$\begin{aligned} y_1(t) &= y_0 + \int_{t_0}^t f(s, y_0(s)) ds, \\ y_2(t) &= y_0 + \int_{t_0}^t f(s, y_1(s)) ds, \\ &\dots\dots\dots \\ y_n(t) &= y_0 + \int_{t_0}^t f(s, y_{n-1}(s)) ds, \end{aligned} \quad (1.6.5)$$

These functions  $y_n(t)$  are called *successive approximations*<sup>[189]</sup>, or *Picard iterates*<sup>[190]</sup> after the French mathematician *Picard* who first discovered them.

**Example 1.6.1** Compute the first two *Picard* iterates for initial value problem  $y' = t^2 + y^2; y(0) = 1$ .

**Solution** Here  $y_0(t) \equiv y(0) = 1$  and  $f(t, y) = t^2 + y^2$ . Then

$$\begin{aligned} y_1(t) &= y_0 + \int_0^t (s^2 + y_0^2(s)) ds = 1 + \int_0^t (s^2 + 1) ds = 1 + t + \frac{1}{3}t^3, \\ y_2(t) &= y_0 + \int_0^t (s^2 + y_1^2(s)) ds = 1 + \int_0^t \left( s^2 + \left( 1 + s + \frac{1}{3}s^3 \right)^2 \right) ds \\ &= \frac{1}{63}t^7 + \frac{2}{15}t^5 + \frac{1}{6}t^4 + \frac{2}{3}t^3 + t^2 + t + 1. \quad \blacksquare \end{aligned}$$

**Lemma<sup>[191]</sup> 1.6.1** The sequence  $\{y_n(t)\}$  defined in Eq.(1.6.5) is uniformly bounded<sup>[192]</sup> on interval  $t_0 \leq t \leq t_0 + h$ .

**Proof** We will prove the conclusion by *mathematic induction on  $n$* <sup>[193]</sup>. Observe first that

$$|y_n(t) - y_0| \leq Mh \quad (1.6.6)$$

is true for  $n = 0$  since  $y_0(t) = y_0$ , where  $M$  is the maximum value of  $f(t, y)$  on  $R$ .

Next, suppose that *inequality*<sup>[194]</sup> (1.6.6) is true for  $n = j$ , then for  $n = j + 1$ ,

$$\begin{aligned} |y_{j+1}(t) - y_0| &= \left| \int_{t_0}^t f(s, y_j(s)) ds \right| \leq \int_{t_0}^t |f(s, y(s))| ds \\ &\leq M(t - t_0) \leq Mh, \quad t_0 \leq t \leq t_0 + h. \end{aligned}$$

Consequently, inequality (1.6.6) is true for all  $n$ .  $\blacksquare$

**Lemma 1.6.2** *Picard* iterates (1.6.5) converges uniformly on interval  $t_0 \leq t \leq t_0 + h$ .

**Proof** Writing  $y_n(t)$  in the form

$$y_n(t) = y_0 + [y_1(t) - y_0(t)] + \cdots + [y_n(t) - y_{n-1}(t)].$$

Clearly, sequence  $\{y_n(t)\}$  converges if and only if the *infinite series*<sup>[195]</sup>

$$y_0 + \sum_{n=1}^{\infty} [y_n(t) - y_{n-1}(t)] \quad (1.6.7)$$

converges. By *Lipschitz* condition (1.6.2) and Lemma 1.6.1, we have that

$$|y_k(t) - y_{k-1}(t)| = \left| \int_{t_0}^t f(s, y_{k-1}(s)) - f(s, y_{k-2}(s)) ds \right|$$

[191] 引理

[192] 一致有界

[193] 对  $n$  实施  
数学归纳法

[194] 不等式

[195] 无穷级数

[196] 正项无穷级  
数

[197] Weierstrass  
判别准则

$$\begin{aligned} &\leq \int_{t_0}^t |f(s, y_{n-1}(s)) - f(s, y_{n-2}(s))| ds \\ &\leq \int_{t_0}^t L |y_{n-1}(s) - y_{n-2}(s)| ds. \end{aligned} \quad (1.6.8)$$

Set  $n = 2$  in (1.6.8) and use the result of Lemma 1.6.1 we have that

$$\begin{aligned} |y_2(t) - y_1(t)| &\leq L \int_{t_0}^t |y_1(s) - y_0| ds \\ &\leq L \int_{t_0}^t M(s - t_0) ds = \frac{ML}{2}(t - t_0)^2 \leq \frac{MLh^2}{2} \end{aligned}$$

for  $t_0 \leq t \leq t_0 + h$ . It follows that

$$\begin{aligned} |y_3(t) - y_2(t)| &\leq L \int_{t_0}^t |y_2(s) - y_1(s)| ds \\ &\leq L \int_{t_0}^t \frac{ML(s - t_0)^2}{2} ds = \frac{ML^2}{3}(t - t_0)^3 \leq \frac{ML^2h^3}{3!}. \end{aligned}$$

In general, we have that

$$|y_n(t) - y_{n-1}(t)| \leq \frac{ML^{n-1}h^n}{n!}. \quad (1.6.9)$$

Hence, the infinite series with positive items<sup>[196]</sup>

$$|y_0| + \sum_{n=1}^{\infty} \frac{ML^{n-1}h^n}{n!}$$

is stronger than the infinite series (1.6.7), therefore the series (1.6.7) is, by *Weierstrass criterion*<sup>[197]</sup>, uniformly convergent on interval  $[t_0, t_0 + h]$ , and so is the sequence  $\{y_n(t)\}$ . ■

**Lemma 1.6.3** Let  $g(t)$  be a nonnegative function with

$$g(t) \leq L \int_{t_0}^t g(s) ds. \quad (1.6.10)$$

Then  $g(t)$  is identically zero.

**Proof** Set

$$G(t) = \int_{t_0}^t g(s) ds.$$

Then,

$$\frac{dG}{dt} = g(t) \leq L \int_{t_0}^t g(s) ds = LG(t).$$

Integrating the inequality

[198] 连续性

$$\frac{dG}{dt} \leq LG(t)$$

[199] 一致收敛性

from  $t_0$  to  $t$  gives

$$\ln G(t) - \ln G(t_0) \leq L(t - t_0)$$

or

$$e^{-L(t-t_0)} G(t) \leq G(t_0) = 0.$$

On the other hand,

$$G(t) = \int_{t_0}^t g(s) ds \geq 0$$

since  $g(t)$  is nonnegative. Thus,  $G(t) = 0$ . Note that

$$0 \leq g(t) \leq L \int_{t_0}^t g(s) ds = LG(t) = 0,$$

so  $g(t) \equiv 0$ . ■

**Proof of Theorem 1.6.1** We give our proof on the interval  $[t_0, t_0 + h]$ . By a similar way one can proof the case on interval  $[t_0 - h, t_0]$ .

(1) Existence of solution.

Let  $\varphi(t)$  denote the limit of the *Picard* iterates  $y_n(t)$ . Then, by *continuity*<sup>[198]</sup> and *uniform convergence*<sup>[199]</sup> of the sequence  $\{y_n(t)\}$ ,

$$\begin{aligned} \varphi(t) &= \lim_{n \rightarrow +\infty} y_n(t) = y_0 + \int_{t_0}^t f(s, \lim_{n \rightarrow +\infty} y_{n-1}(s)) ds \\ &= y_0 + \int_{t_0}^t f(s, \varphi(s)) ds. \end{aligned}$$

$\varphi(t)$  is a solution of Eq.(1.6.4).

(2) Uniqueness of solution.

Suppose that  $\psi(t)$  is another solution of the integral equation (1.6.4) and  $\varphi(t) \neq \psi(t)$  on interval  $[t_0, t_0 + h]$ . Then,

$$|\psi(t) - \varphi(t)| = \left| \int_{t_0}^t [f(s, \psi(s)) - f(s, \varphi(s))] ds \right| \leq L \int_{t_0}^t |\psi(s) - \varphi(s)| ds,$$

where  $L$  is the *Lipschitz* constant. By Lemma 1.6.3  $|\psi(t) - \varphi(t)| \equiv 0$ , i.e.,  $\psi(t) \equiv \varphi(t)$  on interval  $[t_0, t_0 + h]$ . Therefore, the integral equation (1.6.4), and thus the Eq.(1.6.1), has only one solution. ■

**Example 1.6.2** Construct the *Picard* iterates for the initial-value problem

$$y' = 2t(y + 1); \quad y(0) = 0,$$

and show that  $y(t) = e^{t^2} - 1$  is the solution of the problem.

**Solution** Here  $y_0(t) = y(0) = 0$  and  $f(t, y) = 2t(y + 1)$ . Then, the integral equation corresponding to this initial-value problem is

$$y(t) = \int_0^t 2s(y + 1)ds.$$

Hence, the *Picard* iterates for the initial-value problem is

$$y_1(t) = \int_0^t 2s(0 + 1)ds = t^2 \neq y_0(t),$$

so  $y_0(t)$  is not the solution.

$$y_2(t) = \int_0^t 2s(y_1(s) + 1)ds = \frac{1}{4}t^4 + t^2 \neq y_1(t),$$

so  $y_1(t)$  is not the solution.

$$\begin{aligned} y_3(t) &= \int_0^t 2s(y_2(s) + 1)ds = \frac{1}{6}t^4 + \frac{1}{2}t^4 + t^2 \\ &= \frac{1}{3!}t^4 + \frac{1}{2!}t^4 + t^2 \neq y_2(t), \end{aligned}$$

so  $y_2(t)$  is not the solution. In general, we have that

$$\begin{aligned} y_n(t) &= \int_0^t 2s(y_{n-1}(s) + 1)ds \\ &= \int_0^t 2s \left[ \frac{s^{2(n-1)}}{(n-1)!} + \cdots + \frac{1}{2!}s^4 + s^2 + 1 \right] ds \\ &= \frac{t^{2n}}{n!} + \cdots + \frac{t^6}{3!} + \frac{1}{2!}t^4 + t^2 \\ &\rightarrow \sum_{n=1}^{+\infty} \frac{t^2}{n!} = e^{t^2} - 1 \quad (n \rightarrow +\infty). \end{aligned}$$

So  $y(t) = e^{t^2} - 1$  is the solution of the initial value problem. ■

**Example 1.6.3** Prove that  $y(t) = -1$  is the only solution of the initial value problem

$$y' = t(1 + y); \quad y(0) = -1.$$

**Proof** Construct the *Picard* iterates

$$\begin{aligned} y_0(t) &= y(0) = -1, \\ y_1(t) &= y_0(0) + \int_0^t s(1 + y_0(s))ds \\ &= -1 + \int_0^t s(1 - 1)ds = -1 = y_0(t). \end{aligned}$$

Hence,  $y(t) = -1$  is a solution of the initial value problem. It can be proved that function  $f(t, y) = t(1 + y)$  satisfies the conditions of *Picard* Theorem,  $y(t) = -1$  is the only solution of the given initial value problem. ■

**Example 1.6.4** Show that solution  $y(t)$  of the initial-value problem

$$\frac{dy}{dt} = t^2 + y^2; \quad y(0) = 0$$

exists for  $0 \leq t \leq 1/\sqrt{2}$ .

**Proof** Let  $R$  be the rectangle  $0 \leq t \leq a, |y(t)| \leq b$ . Computing

$$M = \max_{(t,y) \in R} (t^2 + y^2) = a^2 + b^2.$$

Then, by *Picard* Theorem, there exists a solution  $y(t)$  on the interval

$$0 \leq t \leq h = \min \left( a, \frac{b}{a^2 + b^2} \right).$$

Note that

$$a^2 + b^2 \geq 2ab$$

or

$$\frac{b}{a^2 + b^2} \leq \frac{b}{2ab} = \frac{1}{2a},$$

and when  $b = a$ ,

$$\frac{b}{a^2 + b^2} = \frac{b}{2ab} = \frac{1}{2a}.$$

Thus, we have

$$\max \left( \frac{b}{a^2 + b^2} \right) = \frac{1}{2a}.$$

It is not difficult to show that

$$h < \frac{1}{\sqrt{2}}, \quad a < 1/(2a) \text{ or } a > 1/(2a),$$

and

$$h = \frac{1}{\sqrt{2}}, \quad a = 1/(2a).$$

So solution of the initial value problem exists on the interval  $[0, 1/\sqrt{2}]$ . And this is the largest interval of existence that *Picard* Theorem predicts for solution of the initial value problem. ■

### Exercise 1.6

1. Construct the *Picard* iterates for the initial-value problem  $y' = 4ty; y(0) = 3$ .

2. Construct the *Picard* iterates for the initial-value problem  $y' = y; y(0) = 1$  and show that they converge to the solution  $y(t) = e^t$ .

3. Compute the first two *Picard* iterates for the initial-value problem  $y' = 1 + y^3; y(1) = 1$ .

4. Compute the first three *Picard* iterates for the initial-value problem  $y' = e^t + y^2; y(0) = 0$ .

Show that the solution  $y(t)$  of each of following initial-value problems exists on the given interval.

5.  $y' = y^2 + \cos t^2, y(0) = 0; 0 \leq t \leq 1/2$ .

6.  $y' = e^{-y^2} + t^2, y(0) = 0; 1 \leq t \leq 1/2$ .

7.  $y' = 1 + y + y^2 \cos t, y(0) = 0; 0 \leq t \leq 1/3$ .

8.  $y' = e^{-t^2} + y^3, y(0) = 1; 0 \leq t \leq 1/9$ .

9. Find a nontrivial solution of the initial-value problem  $y' = ty^a; y(0) = 0, a > 1$ . Does this violate Theorem 1.6.3? Explain.

10. Find a solution of the initial-value problem  $y' = t\sqrt{1 - y^2}; y(0) = 1$ , other than  $y(t) = 1$ . Does this violate Theorem 1.6.3? Explain.

# Chapter 2 Second-order Differential Equations

## 2.1 General Solutions of Homogeneous Second-order Linear Equations

The most famous second-order differential equation is **Newton's law of motion**<sup>[1]</sup>

$$m \frac{d^2 y}{dt^2} = F \left( t, y, \frac{dy}{dt} \right),$$

where  $m$  is the mass of a particle<sup>[2]</sup>,  $y = y(t)$  is its position at the time  $t$ ,  $dy/dt$  is its velocity<sup>[3]</sup>, and  $F$  is the total force acting on the particle. For example, if  $y(t)$  denotes the position at time  $t$  of a particle moving under the influence of gravity<sup>[4]</sup>, then,  $y(t)$  satisfies the initial value problem

$$\frac{d^2 y}{dt^2} = -g; \quad y(t_0) = y_0, y'(t_0) = y'_0,$$

where  $y_0$  is the initial position of the particle and  $y'_0$  is the initial velocity of the particle.

Though we only succeed in solving **second-order linear differential equation**

$$\frac{d^2 y}{dt^2} + p(t) \frac{dy}{dt} + q(t)y = g(t), \quad (2.1.1)$$

fortunately, many of the second-order equations that arise in applications are of the form Eq.(2.1.1). For example, both

$$y'' + 3ty' + t^2y = e^t$$

and

$$y'' + e^t y' + 2y = \sin t$$

[1] 牛顿运动定律

[2] 物质的质量

[3] 速度

[4] 重力

[5] 微分运算的线性性 are linear equations, while

$$y'' + 3y' + e^y = t^3$$

and

$$y'' + (y')^2 = 1$$

are nonlinear.

When  $g(t) = 0$ , equation

$$\frac{d^2y}{dt^2} + p(t)\frac{dy}{dt} + q(t)y = 0 \quad (2.1.2)$$

is called a homogeneous second-order linear equation.

**Theorem 2.1.1 (Existence-uniqueness Theorem)** Let functions  $p(t)$ ,  $q(t)$  and  $g(t)$  be continuous on an interval  $I$  containing point  $t_0$ . Then, for any two numbers  $y_0$  and  $y'_0$ , there exists one and only one function  $y(t)$  satisfying differential equation (2.1.1) on the entire interval  $I$  and initial conditions  $y(t_0) = y_0, y'(t_0) = y'_0$ . In particular, any solution  $y = y(t)$  of Eq.(2.1.2) which satisfies  $y(t_0) = 0, y'(t_0) = 0$  must be identically zero.

Theorem 2.1.1 is extremely important for us in solving Eq.(2.1.1).

**Theorem 2.1.2** Let  $y_1(t)$  and  $y_2(t)$  be two solutions of Eq.(2.1.2) on the interval  $I$ , then, for any constants  $c_1$  and  $c_2$ , the linear combination

$$y(t) = c_1y_1(t) + c_2y_2(t)$$

is also a solution of Eq.(2.1.2) on  $I$ .

**Proof** By the linearity of the operation of differentiation<sup>[5]</sup> we obtain immediately that

$$y'(t) = c_1y'_1(t) + c_2y'_2(t) \quad \text{and} \quad y''(t) = c_1y''_1(t) + c_2y''_2(t).$$

Then,

$$\begin{aligned} & y'' + p(t)y' + q(t)y \\ &= (c_1y_1 + c_2y_2)'' + p(c_1y_1 + c_2y_2)' + q(c_1y_1 + c_2y_2) \end{aligned}$$

$$\begin{aligned}
&= c_1(y_1'' + py_1' + qy_1) + c_2(y_2'' + py_2' + qy_2) \\
&= c_1 \cdot 0 + c_2 \cdot 0 = 0.
\end{aligned}$$

[6] 疊加原理

Thus,  $y(t) = c_1y_1(t) + c_2y_2(t)$  is a solution of Eq.(2.1.2). ■

Does any solution of Eq.(2.1.2) can be expressed in the form of a combination of two given solutions? The answer is yes if we impose some condition on the given solutions.

**Theorem 2.1.3** (*Principle of superposition*<sup>[6]</sup>) Let  $y_1(t)$  and  $y_2(t)$  be two solutions of Eq.(2.1.2) on interval  $I$  and

$$y_1(t)y_2'(t) - y_1'(t)y_2(t) \neq 0. \quad (2.1.3)$$

Then,

$$y(t) = c_1y_1(t) + c_2y_2(t) \quad (2.1.4)$$

is the general solution of Eq.(2.1.2).

**Proof** Let  $y(t)$  be a solution of Eq.(2.1.2) and  $y_0 = y(t_0)$ ,  $y'_0 = y'(t_0)$ . Solving following system of equations

$$\begin{cases} c_1y_1(t_0) + c_2y_2(t_0) = y_0, \\ c_1y_1'(t_0) + c_2y_2'(t_0) = y'_0 \end{cases}$$

for constants  $c_1$  and  $c_2$ .

Multiplying the first equation by  $y_2'(t_0)$ , the second equation by  $y_2(t_0)$  and subtracting gives

$$c_1[y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)] = y_0y_2'(t_0) - y'_0y_2(t_0).$$

Similarly, multiplying the first equation by  $y_1'(t_0)$ , the second equation by  $y_1(t_0)$  and subtracting gives

$$c_2[y_1'(t_0)y_2(t_0) - y_1(t_0)y_2'(t_0)] = y_0y_1'(t_0) - y'_0y_1(t_0).$$

Then,

$$c_1 = \frac{y_0y_2'(t_0) - y'_0y_2(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

and

$$c_2 = \frac{y'_0y_1(t_0) - y_0y_1'(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

if  $y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0) \neq 0$ . Set

$$\phi(t) = c_1y_1(t) + c_2y_2(t)$$

[7] 基本解集

[8] Wronski 行列式

[9] 线性相关

[10] 线性无关

for above constants  $c_1$  and  $c_2$ . Then  $\phi(t)$  is a solution of Eq.(2.1.2). Moreover,  $\phi(t_0) = y_0$  and  $\phi'(t_0) = y'_0$ . Thus,  $y(t)$  and  $\phi(t)$  are both solutions of the same initial-value problem. Therefore, by the uniqueness,  $y(t)$  must be identically equal to  $\phi(t)$ , that is,

$$y(t) = c_1 y_1(t) + c_2 y_2(t)$$

on the interval  $I$ . ■

Theorem 2.1.3 is extremely useful in solving Eq.(2.1.2) since it reduces the problem of finding all solutions of Eq.(2.1.2) to the much simpler problem of finding just two solutions. If solutions  $y_1(t)$  and  $y_2(t)$  satisfy condition (2.1.3), we say that  $y_1(t)$  and  $y_2(t)$  are **a fundamental set of solutions**<sup>[7]</sup> of Eq.(2.1.2). Denote

$$W[y_1, y_2](t) = y_1(t)y'_2(t) - y'_1(t)y_2(t) \quad (2.1.5)$$

and call it **the Wronskian determinant**<sup>[8]</sup> ( named after the Polish mathematician J.M.H.Wronski, 1778—1853 ) of  $y_1(t)$  and  $y_2(t)$ . Thus, condition (2.1.3) is equivalent to

$$W[y_1, y_2](t) \neq 0.$$

**Definition 2.1.1** Functions  $y_1(t)$  and  $y_2(t)$  are said to be **linearly dependent**<sup>[9]</sup> on an interval  $I$  if one of these two functions is a constant multiple of the other on  $I$ . Functions  $y_1(t)$  and  $y_2(t)$  are said to be **linearly independent**<sup>[10]</sup> on an interval  $I$  if they are not linearly dependent on  $I$ .

**Theorem 2.1.4** Let  $y_1(t)$  and  $y_2(t)$  be two solutions of Eq.(2.1.2). Then,

(a)  $y_1(t)$  and  $y_2(t)$  are linearly independent if and only if

$$W[y_1, y_2](t) \neq 0$$

on an interval  $I$ .

(b)  $y_1(t)$  and  $y_2(t)$  are linearly dependent if and only if

$$W[y_1, y_2](t) \equiv 0$$

on an interval  $I$ .

**Lemma 2.1.1** Let  $y_1(t)$  and  $y_2(t)$  be two solutions of Eq.(2.1.2). Then,

$$W'(t) + p(t)W(t) = 0,$$

where  $W(t)$  is the *Wronskian* of  $y_1(t)$  and  $y_2(t)$ .

**Proof** Note that both  $y_1(t)$  and  $y_2(t)$  are solutions of the equation  $y'' + p(t)y' + q(t)y = 0$ , then

$$y_1'' = -p(t)y_1' - q(t)y_1, \quad y_2'' = -p(t)y_2' - q(t)y_2.$$

Hence,

$$\begin{aligned} W'(t) &= \frac{d}{dt}(y_1 y_2' - y_1' y_2) = y_1 y_2'' - y_1'' y_2 \\ &= y_1 [-p(t)y_2' - q(t)y_2] - y_2 [-p(t)y_1' - q(t)y_1] \\ &= -p(t)[y_1 y_2' - y_1' y_2] \\ &= -p(t)W(t). \end{aligned} \quad \blacksquare$$

**Lemma 2.1.2** Let  $p(t)$  and  $q(t)$  be continuous on an interval  $I$ , and  $y_1(t)$  and  $y_2(t)$  be two solutions of Eq.(2.1.2). Then,  $W[y_1, y_2](t)$  is either identically zero, or never zero, on this interval.

**Proof** Chose arbitrarily some  $t_0$  in  $I$ , by Lemma 2.1.1,

$$W[y_1, y_2](t) = W[y_1, y_2](t_0) \exp \left( - \int_{t_0}^t p(s) ds \right).$$

Since  $\exp \left( - \int_{t_0}^t p(s) ds \right)$  can never equal to zero on  $I$ ,  $W[y_1, y_2](t)$  is either identically zero, or never zero.  $\blacksquare$

**Proof of Theorem 2.1.4** We first prove part (b), and part (a) can be proved immediately by contradiction.

Let  $y_1(t)$  and  $y_2(t)$  be two solutions of Eq.(2.1.2) on an interval  $I$ , and suppose that  $W[y_1, y_2](t_0) = 0$  for some  $t_0$  in this interval. Then, the system of equations

$$\begin{cases} c_1 y_1(t_0) + c_2 y_2(t_0) = 0, \\ c_1 y_1'(t_0) + c_2 y_2'(t_0) = 0 \end{cases}$$

has a nontrivial solution  $c_1, c_2$ :  $|c_1| + |c_2| \neq 0$ . Set

$$y(t) = c_1 y_1(t) + c_2 y_2(t),$$

then  $y(t)$  is a solution of Eq.(2.1.2), and moreover  $y(t_0) = 0$ ,  $y'(t_0) = 0$ . Therefore, by Theorem 2.1.1,  $y(t)$  is identically zero. Hence,

$$c_1 y_1(t) + c_2 y_2(t) = 0, \quad t \in I.$$

If  $c_1 \neq 0$  then  $y_1(t) = -(c_2/c_1)y_2(t)$ ; if  $c_2 \neq 0$  then  $y_2(t) = -(c_1/c_2)y_1(t)$ . In either case, one of these solutions is a constant multiple of the other. So  $y_1(t)$  and  $y_2(t)$  are linearly dependent.

If  $y_1(t)$  and  $y_2(t)$  are linearly dependent, say  $y_1(t) = ky_2(t)$ , then

$$W[y_1, y_2](t) = ky_2(t)y_2'(t) - ky_2'(t)y_2(t) \equiv 0. \quad \blacksquare$$

Now we discuss the method for finding a second solution  $y_2(t)$  of Eq.(2.1.2) which is independent with the first solution  $y_1(t)$ . Consider function

$$y_2(t) = v(t)y_1(t), \quad (2.1.6)$$

then

$$\frac{dy_2}{dt} = v \frac{dy_1}{dt} + y_1 \frac{dv}{dt}$$

and

$$\frac{d^2 y_2}{dt^2} = v \frac{d^2 y_1}{dt^2} + 2 \frac{dv}{dt} \frac{dy_1}{dt} + y_1 \frac{d^2 v}{dt^2}.$$

Thus,

$$\begin{aligned} & \frac{d^2 y_2}{dt^2} + p(t) \frac{dy_2}{dt} + q(t)y_2 \\ &= \left( v \frac{d^2 y_1}{dt^2} + 2 \frac{dv}{dt} \frac{dy_1}{dt} + y_1 \frac{d^2 v}{dt^2} \right) \\ & \quad + p(t) \left( v \frac{dy_1}{dt} + y_1 \frac{dv}{dt} \right) + q(t)vy_1 \\ &= y_1 \frac{d^2 v}{dt^2} + \left( 2 \frac{dy_1}{dt} + p(t)y_1 \right) \frac{dv}{dt} \\ & \quad + \left( \frac{d^2 y_1}{dt^2} + p(t) \frac{dy_1}{dt} + q(t)y_1 \right) v \\ &= y_1 \frac{d^2 v}{dt^2} + \left( 2 \frac{dy_1}{dt} + p(t)y_1 \right) \frac{dv}{dt}. \end{aligned}$$

Hence,  $y_2(t) = v(t)y_1(t)$  is a solution of Eq.(2.1.2) if  $v(t)$  satisfies

differential equation

[11] 降阶法

$$y_1 \frac{d^2 v}{dt^2} + \left( 2 \frac{dy_1}{dt} + p(t)y_1 \right) \frac{dv}{dt} = 0. \quad (2.1.7)$$

This is a first-order linear equation for  $dv/dt$ , and its solution is

$$\begin{aligned} \frac{dv}{dt} &= c \exp \left[ - \int \left( 2 \frac{y_1'(t)}{y_1(t)} + p(t) \right) dt \right] \\ &= c \exp \left( - \int p(t) dt \right) \exp \left[ -2 \int \left( \frac{y_1'(t)}{y_1(t)} \right) dt \right] \\ &= \frac{c \exp \left( - \int p(t) dt \right)}{y_1^2(t)} = \mu(t). \end{aligned} \quad (2.1.8)$$

Set  $c = 1$  in Eq.(2.1.8) because we need only one solution. Thus we can obtain  $v(t)$  by integrating Eq.(2.1.8) with respect to  $t$  and, for the same reason, set the integration constant equal to some constant (one, or zero, or some other constant). We call such method for finding a second solution of Eq.(2.1.2) **the method of reduction of order**<sup>[11]</sup>, since it reduces the problem of solving a second order differential equation to a first order differential equation.

**Example 2.1.1** Find the solution  $y(t)$  of initial-value problem

$$(1-t^2) \frac{d^2 y}{dt^2} + 2t \frac{dy}{dt} - 2y = 0; \quad y(0) = 3, y'(0) = -4$$

on the interval  $-1 < t < 1$ .

**Solution** Clearly,  $y_1(t) = t$  is one solution of the differential equation

$$(1-t^2) \frac{d^2 y}{dt^2} + 2t \frac{dy}{dt} - 2y = 0. \quad (2.1.9)$$

Dividing both sides of Eq.(2.1.9) by  $1-t^2$  to obtain the equivalent equation

$$\frac{d^2 y}{dt^2} + \frac{2t}{1-t^2} \frac{dy}{dt} - \frac{2}{1-t^2} y = 0.$$

Then, from Eq.(2.1.8)

$$\mu(t) = \frac{c \exp \left( - \int \frac{2t}{1-t^2} dt \right)}{(y_1(t))^2} = \frac{e^{\ln(1-t^2)}}{t^2} = \frac{1-t^2}{t^2} = \frac{dv}{dt},$$

thus

$$y_2(t) = v(t)y_1(t) = t \int \frac{1-t^2}{t^2} dt = -t \left( \frac{1}{t} + t \right) = -(1+t^2)$$

is a second solution of Eq.(2.1.9) which is independent with  $y_1(t)$ . Therefore, the general solution is

$$y(t) = c_1 t - c_2(1+t^2).$$

From the initial conditions

$$3 = y(0) = -c_2 \text{ and } -4 = y'(0) = c_1.$$

Hence, solution of the initial-value problem is

$$y(t) = -4t + 3(1+t^2). \quad \blacksquare$$

**Example 2.1.2** It is not difficult to verify that functions  $y_1(t) = \sqrt{t}$  and  $y_2(t) = \frac{1}{t}$  are solutions of differential equation

$$2t^2 y''(t) + 3t y'(t) - y(t) = 0 \quad (2.1.10)$$

on the interval  $0 < t < \infty$ .

(a) Compute  $W[y_1, y_2](t)$ . What happens as  $t$  approaches zero?

(b) Show that  $y_1(t)$  and  $y_2(t)$  form a fundamental set of solutions of Eq.(2.1.10) on the interval  $0 < t < \infty$ .

(c) Solve the initial-value problem  $2t^2 y''(t) + 3t y'(t) - y(t) = 0$ ;  $y(1) = 2, y'(1) = 1$ .

**Solution**

$$(a) \quad W[y_1, y_2](t) = y_1 y_2' - y_1' y_2 = -\frac{3}{2} t^{-\frac{3}{2}} \rightarrow -\infty \quad (t \rightarrow 0).$$

$$(b) \quad W[y_1, y_2](t) = y_1 y_2' - y_1' y_2 = -\frac{3}{2} t^{-\frac{3}{2}} \neq 0 \quad (0 < t < +\infty).$$

Hence,  $y_1(t)$  and  $y_2(t)$  form a fundamental set of solutions of Eq.(2.1.10) on the interval  $0 < t < \infty$ .

(c) From (b), the general solution of Eq.(2.1.10) is

$$y(t) = c_1 y_1(t) + c_2 y_2(t).$$

Substituting the initial conditions into the last equation gives

$$\begin{cases} y(1) = c_1 + c_2 = 2, \\ y'(1) = c_1 \frac{1}{2} + c_2(-1) = 1, \end{cases} \quad [12] \text{ 为零}$$

thus we obtain that  $c_1 = 2$  and  $c_2 = 0$ . So solution of the initial value problem is

$$y = 2\sqrt{t}. \quad \blacksquare$$

### Exercise 2.1

1. Compute the *Wronskian* of each of the following pairs of functions.

(a)  $e^t, e^{-t}$ ; (b)  $\sin 3t, \cos 3t$ ; (c)  $\sin^2 t, 1 - \cos 2t$ ; (d)  $t, t \ln t$ .

2. Determine whether the pair of functions  $t^3$  and  $|t^3|$  are linearly dependent on  $[-1, 1]$ .

3. Two solutions of  $y'' - 2y' + y = 0$  are  $e^t$  and  $te^t$ . Is the general solution  $y = c_1 e^t + c_2 te^t$ ?

4. Two solutions of  $y'' - 2y' + y = 0$  are  $e^{-t}$  and  $5e^{-t}$ . Is the general solution  $y = c_1 e^{-t} + c_2 5e^{-t}$ ?

5. Let  $y_1(t)$  and  $y_2(t)$  be solutions of Eq.(2.1.2) on the interval  $-\infty < t < \infty$  with  $y_1(0) = 3, y_1'(0) = 1, y_2(0) = 1$ , and  $y_2'(0) = 1/3$ . Show that  $y_1(t)$  and  $y_2(t)$  are linearly dependent on the interval  $-\infty < t < \infty$ .

6. Let  $y_1(t) = t^2$  and  $y_2(t) = t|t|$ .

(a) Show that  $y_1(t)$  and  $y_2(t)$  are linearly dependent on the interval  $0 \leq t \leq 1$ .

(b) Show that  $y_1(t)$  and  $y_2(t)$  are linearly independent on the interval  $-1 \leq t \leq 1$ .

7. Suppose that  $y_1(t)$  and  $y_2(t)$  are linearly independent on the interval  $I$ . Prove that  $z_1 = y_1(t) + y_2(t)$  and  $z_2 = y_1(t) - y_2(t)$  are also linearly independent on  $I$ .

8. Suppose that the *Wronskian* of any two solutions of Eq.(2.1.2) is constant. Prove that  $p(t) = 0$ .

9. Assume that  $p(t)$  and  $q(t)$  are continuous, and that the functions  $y_1(t)$  and  $y_2(t)$  are solutions of Eq.(2.1.2) on the interval  $\alpha < t < \beta$ . Proof that

(a) If  $y_1$  and  $y_2$  vanish<sup>[12]</sup> at same point in the interval  $\alpha < t < \beta$ , then they cannot form a fundamental set of solutions on this interval.

(b) If  $y_1$  and  $y_2$  achieve a maximum or minimum at same point in the interval  $\alpha < t < \beta$ , then they cannot form a fundamental set of

- [13] 拐点  
 [14] 二阶常系数  
 线性齐次方程  
 [15] 相互抵消  
 [16] 特征方程  
 [17] 二次公式

solutions on this interval.

(c) If  $y_1$  and  $y_2$  form a fundamental set of solutions, then they cannot have a common *point of inflexion*<sup>[13]</sup> in  $\alpha < t < \beta$  unless  $p$  and  $q$  vanish simultaneously there.

Use the method of reduction of order to find the general solution of the following differential equations.

$$10. \frac{d^2y}{dt^2} - 4t \frac{dy}{dt} + (4t^2 - 2)y = 0; y_1(t) = e^{t^2}.$$

$$11. (1 + t^2) \frac{d^2y}{dt^2} - 2t \frac{dy}{dt} + 2y = 0; y_1(t) = t.$$

$$12. (1 + 2t) \frac{d^2y}{dt^2} - 4(t + 1) \frac{dy}{dt} + 4y = 0; y_1(t) = t + 1.$$

## 2.2 Homogeneous Second-order Linear Equations with Constant Coefficients<sup>[14]</sup>

In this section, we discuss the methods for finding solutions of the following homogeneous second-order linear equation

$$ay'' + by' + cy = 0, \quad (2.2.1)$$

where  $a, b$  and  $c$  are constants with  $a \neq 0$ . According to Theorem 2.1.2, we need only find two independent solutions  $y_1(t)$  and  $y_2(t)$  of Eq.(2.2.1). Note that terms  $ay'', by'$  and  $cy$  can *cancel each other*<sup>[15]</sup> only if functions  $y''(t)$ ,  $y'(t)$  and  $y(t)$  are of the “same type”. Function  $e^{rt}$  has such property! This suggests us to make the substitution  $y(t) = e^{rt}$  in Eq.(2.2.1)

$$a(e^{rt})'' + b(e^{rt})' + c(e^{rt}) = (ar^2 + br + c)e^{rt}.$$

Thus  $y(t) = e^{rt}$  is a solution of Eq.(2.2.1) if and only if

$$ar^2 + br + c = 0. \quad (2.2.2)$$

Eq.(2.2.2) is called the **characteristic equation**<sup>[16]</sup> of Eq.(2.2.1). Its two roots are given by the following *quadratic formula*<sup>[17]</sup>

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}. \quad (2.2.3)$$

### 2.2.1 The Characteristic Equation Has Distinct Real Roots

If  $b^2 - 4ac > 0$ , then  $r_1$  and  $r_2$  are real and distinct. In this case,  $y_1(t) = e^{r_1 t}$  and  $y_2(t) = e^{r_2 t}$  are two distinct and linearly independent solutions of Eq.(2.2.1) on any interval since

$$W[e^{r_1 t}, e^{r_2 t}] = (r_1 - r_2)e^{(r_1 - r_2)t} \neq 0.$$

Hence, the general solutions of Eq.(2.2.1) is

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}. \quad (2.2.4)$$

**Example 2.2.1** Find the general solution of equation

$$3y'' + 6y' + 2y = 0.$$

**Solution** The characteristic equation

$$3r^2 + 6r + 2 = 0$$

has two distinct real roots

$$r_{1,2} = \frac{-3 \pm \sqrt{3}}{3}.$$

Thus, the general solution is

$$y(t) = c_1 e^{\frac{-3-\sqrt{3}}{3}t} + c_2 e^{\frac{-3+\sqrt{3}}{3}t}. \quad \blacksquare$$

**Example 2.2.2** Find the solution  $y(t)$  of initial-value problem

$$2y'' + y' - 10y = 0; \quad y(1) = 5, y'(1) = 2.$$

**Solution** Two roots of the characteristic equation  $2r^2 + r - 10 = 0$  are

$$r_{1,2} = \frac{-1 \pm 9}{4}.$$

Hence,  $y_1(t) = e^{r_1 t}$  and  $y_2(t) = e^{r_2 t}$  form a fundamental set of solutions of the given equation, so that the general solution is

$$y(t) = c_1 e^{2t} + c_2 e^{-\frac{5}{2}t}.$$

[18] 重根

Using the initial conditions

$$\begin{cases} c_1 e^2 + c_2 e^{-\frac{5}{2}} = 5, \\ 2c_1 e^2 - \frac{5}{2} c_2 e^{-\frac{5}{2}} = 2, \end{cases}$$

we obtain that

$$c_1 = \frac{29}{9e^2}, \quad c_2 = \frac{16e^{\frac{5}{2}}}{9}.$$

Therefore, solution of initial value problem is

$$y(t) = \frac{29}{9} e^{2(t-1)} + \frac{16}{9} e^{\frac{5}{2}(1-t)}.$$

■

### 2.2.2 The Characteristic Equation Has Repeated Roots<sup>[18]</sup>

If  $b^2 - 4ac = 0$ , then the characteristic equation (2.2.2) has equal real roots  $r_1 = r_2 = -b/(2a)$ . In this case, we can obtain only one solution

$$y_1(t) = e^{-bt/(2a)}.$$

Suppose that

$$y_2(t) = y_1(t)v(t) \tag{2.2.5}$$

is a second solution which is linearly independent with  $y_1(t)$  and  $v(t)$  is undetermined. Then

$$\begin{aligned} y_2' &= v y_1' + y_1 v', \\ y_2'' &= v y_1'' + 2v' y_1' + y_1 v''. \end{aligned}$$

Consequently,

$$\begin{aligned} a y_2'' + b y_2' + c y_2 &= a(v y_1'' + 2v' y_1' + y_1 v'') + b(v y_1' + y_1 v') + c y_1 v \\ &= (a y_1'' + b y_1' + c y_1) v + (2a y_1' + b y_1) v' + a y_1 v'' \\ &= a y_1 v'' \end{aligned}$$

since  $y_1(t)$  is a solution of Eq.(2.2.1) and  $-b/(2a)$  is a root of Eq.(2.2.2). Hence,  $y_2(t) = y_1(t)v(t)$  is a solution of Eq.(2.2.1) if  $v(t)$  satisfies the differential equation

$$v''(t) = 0.$$

The simplest choice of  $v$  is  $v(t) = t$ . Thus the general solutions of Eq.(2.2.1) is

$$y(t) = (c_1 + c_2 t)e^{-\frac{b}{2a}t}. \quad (2.2.6) \quad [19] \text{ 共轭复根}$$

[20] 复值解

**Example 2.2.3** Find the solution  $y(t)$  of initial-value problem

$$\frac{d^2 y}{dt^2} + 4\frac{dy}{dt} + 4y = 0; \quad y(0) = 1, y'(0) = 3.$$

**Solution** The characteristic equation  $r^2 + 4r + 4 = (r + 2)^2 = 0$  has two equal roots  $r_1 = r_2 = -2$ . Hence, the general solution is

$$y(t) = c_1 e^{-2t} + c_2 t e^{-2t}.$$

From initial conditions we obtain that

$$1 = y(0) = c_1 \quad \text{and} \quad 3 = y'(0) = -2c_1 + c_2.$$

These imply that  $c_1 = 1$  and  $c_2 = 5$ , so solution of the initial-value problem is

$$y(t) = e^{-2t} + 5te^{-2t}. \quad \blacksquare$$

### 2.2.3 The Characteristic Equation Has Complex Conjugate Roots<sup>[19]</sup>

If  $b^2 - 4ac < 0$ , then the characteristic equation  $r^2 + 4r - 2 = 0$  has complex roots

$$r_1 = \frac{-b + i\sqrt{b^2 - 4ac}}{2a}, \quad r_2 = \frac{-b - i\sqrt{b^2 - 4ac}}{2a}. \quad (2.2.7)$$

Thus  $y_1(t) = e^{r_1 t}$  and  $y_2(t) = e^{r_2 t}$  are two *complex-valued solutions*<sup>[20]</sup> of Eq.(2.2.1). Now the problem before us is how to find two real-valued solutions of Eq.(2.2.1).

Assume that  $y(t) = u(t) + iv(t)$  is a complex-valued solution of Eq.(2.2.1). Then,

$$a[u''(t) + iv''(t)] + b[u'(t) + iv'(t)] + c[u(t) + iv(t)] = 0. \quad (2.2.8)$$

**Lemma 2.2.1** Let  $y(t) = u(t) + iv(t)$  be a complex-valued solution of Eq.(2.2.1) with  $a, b$  and  $c$  are real. Then,  $y_1(t) = u(t)$  and  $y_2(t) = v(t)$  are two real-valued solutions of Eq.(2.2.1).

**Proof** From Eq.(2.2.8),

$$[au''(t) + bu'(t) + cu(t)] + i[av''(t) + bv'(t) + cv(t)] = 0.$$

[21] 欧拉公式

Consequently,

$$au''(t) + bu'(t) + cu(t) = 0 \text{ and } av''(t) + bv'(t) + cv(t) = 0,$$

this proves Lemma 2.2.1. ■From *Euler's formula*<sup>[21]</sup>

$$e^{(\alpha+i\beta)t} = e^{\alpha t} e^{i\beta t} = e^{\alpha t} (\cos \beta t + i \sin \beta t), \quad (2.2.9)$$

we have that

$$\begin{aligned} y(t) &= e^{[-b+i\sqrt{4ac-b^2}]t/(2a)} \\ &= e^{-bt/(2a)} [\cos(\sqrt{4ac-b^2}t/2a) + i \sin(\sqrt{4ac-b^2}t/2a)]. \end{aligned}$$

Therefore, by Lemma 2.2.1,

$$\begin{aligned} y_1(t) &= e^{-bt/(2a)} \cos \beta t \quad \text{and} \quad y_2(t) = e^{-bt/(2a)} \sin \beta t, \\ \beta &= \frac{\sqrt{4ac-b^2}}{2a} \end{aligned} \quad (2.2.10)$$

are two real-valued solutions of Eq.(2.2.1). Moreover, these two functions are linearly independent on any interval  $I$  since their *Wronskian* is never zero. Consequently, the general solution of Eq.(2.2.1), for  $b^2 - 4ac < 0$ , is

$$y(t) = e^{-bt/(2a)} [c_1 \cos \beta t + c_2 \sin \beta t]. \quad (2.2.11)$$

**Example 2.2.4** Find two linearly independent real-valued solutions of differential equation

$$4 \frac{d^2 y}{dt^2} - \frac{dy}{dt} + y = 0. \quad (2.2.12)$$

**Solution** The characteristic equation  $4r^2 - r + 1 = 0$  has complex roots

$$r_{1,2} = \frac{1 \pm i\sqrt{15}}{8}.$$

Thus,

$$e^{r_1 t} = e^{\left(\frac{1+i\sqrt{15}}{8}\right)t} = e^{\frac{1}{8}t} \left( \cos \frac{\sqrt{15}}{8}t + i \sin \frac{\sqrt{15}}{8}t \right)$$

is a complex-valued solution of Eq.(2.2.12). Therefore, by Lemma 2.2.1,

$$\operatorname{Re}\{e^{r_1 t}\} = e^{\frac{1}{8}t} \cos \frac{\sqrt{15}}{8}t \quad \text{and} \quad \operatorname{Im}\{e^{r_1 t}\} = e^{\frac{1}{8}t} \sin \frac{\sqrt{15}}{8}t$$

are two linearly independent real-valued solutions of Eq.(2.2.12). ■

**Example 2.2.5** Find the solution  $y(t)$  of initial-value problem

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = 0; \quad y(0) = 0, y'(0) = 2.$$

**Solution** The characteristic equation  $r^2 + 2r + 5 = 0$  has complex roots

$$r_{1,2} = -1 \pm i2.$$

Hence,

$$e^{r_1 t} = e^{(-1+i2)t} = e^{-t} \cos 2t + ie^{-t} \sin 2t$$

is a complex-valued solution of  $y'' + 2y' + 5 = 0$ . Therefore, both

$$\operatorname{Re}\{e^{r_1 t}\} = e^{-t} \cos 2t \quad \text{and} \quad \operatorname{Im}\{e^{r_1 t}\} = e^{-t} \sin 2t$$

are real-valued solutions. Hence, the general solution is

$$y(t) = e^{-t}(c_1 \cos 2t + c_2 \sin 2t).$$

Using the initial value conditions we have that

$$1 = y(0) = c_1, \quad 2 = y'(0) = -c_1 + 2c_2,$$

these imply that  $c_1 = 1, c_2 = 3/2$ . Consequently, solution of the initial value problem is

$$y(t) = e^{-t} \left( \cos 2t + \frac{3}{2} \sin 2t \right). \quad \blacksquare$$

### Exercise 2.2

Find the general solution of following equations

1.  $y'' + 5y' + 4y = 0.$
2.  $y'' - y' - 2y = 0.$
3.  $y'' - 7y = 0.$
4.  $y'' + 10y' + 21y = 0.$
5.  $y'' - 8y' + 16y = 0.$
6.  $y'' = 0.$
7.  $100y'' - 20y' + y = 0.$
8.  $y'' + 4y' + 5y = 0.$
9.  $4y'' + 4y' + 5y = 0.$
10.  $y'' - 3y' + 4y = 0.$
11.  $y'' + 4y = 0.$
12.  $y'' - 6y' + 25y = 0.$

Solve following initial-value problems.

13.  $y'' + 4y' - 2y = 0; y(0) = 1, y'(0) = 2.$
14.  $y'' - 4y' + 3y = 0; y(0) = 7, y'(0) = 11.$
15.  $9y'' + 6y' + 4y = 0; y(0) = 3, y'(0) = 4.$
16.  $y'' + 4y' + 4y = 0; y(0) = 1, y'(0) = 3.$

17.  $y'' + 2y' + 4y = 0$ ;  $y(0) = 1, y'(0) = 1$ .

18.  $y'' - 6y' + 25y = 0$ ;  $y(0) = 3, y'(0) = 1$ .

19. Let  $y(t)$  be the solution of the initial-value problem

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 0; \quad y(0) = 1, y'(0) = V.$$

For what values of  $V$  does  $y(t)$  remain nonnegative for all  $t \geq 0$ ?

20. (a) Assume that  $b^2 - 4ac < 0$ . Show that

$$y_1(t) = e^{(-bt/(2a))(t-t_0)} \cos \beta(t-t_0)$$

and

$$y_2(t) = e^{(-bt/(2a))(t-t_0)} \sin \beta(t-t_0), \quad \beta = \frac{\sqrt{4ac - b^2}}{2a}$$

are solutions of Eq.(2.2.1) for any number  $t_0$ .

(b) Suppose that  $b^2 - 4ac = 0$ . Show that

$$y_1(t) = e^{-b(t-t_0)/(2a)}$$

and

$$y_2(t) = (t-t_0)e^{-b(t-t_0)/(2a)}$$

are solutions of Eq.(2.2.1) for every choice of  $t_0$ .

## 2.3 Nonhomogeneous Second-order Linear Equations

We turn our attention now back to the nonhomogeneous equation

$$y'' + p(t)y' + q(t)y = g(t) \quad (2.3.1)$$

where functions  $p(t)$ ,  $q(t)$  and  $g(t)$  are continuous on an interval  $I$ .

### 2.3.1 Structure of General Solutions

**Theorem 2.3.1** Let  $y_1(t)$  and  $y_2(t)$  be two linearly independent solutions of the homogeneous equation

$$y'' + p(t)y' + q(t)y = 0 \quad (2.3.2)$$

and let  $\psi(t)$  be any particular solution of the nonhomogeneous equation (2.3.1). Then, every solution  $y(t)$  of Eq.(2.3.1) must be of the form

$$y(t) = c_1y_1(t) + c_2y_2(t) + \psi(t) \quad (2.3.3)$$

for some choice of constants  $c_1, c_2$ .

**Lemma 2.3.1** If  $y_1(t)$  and  $y_2(t)$  are two solutions of the nonhomogeneous equation (2.3.1), then the difference of  $y_1(t)$  and  $y_2(t)$  is a solution of the corresponding homogeneous equation (2.3.2).

**Proof**

$$\begin{aligned} & [y_1 - y_2]'' + p(t)[y_1 - y_2]' + q(t)[y_1 - y_2] \\ &= [y_1'' + p(t)y_1' + q(t)y_1] - [y_2'' + p(t)y_2' + q(t)y_2] \\ &= g(t) - g(t) = 0. \end{aligned}$$

Hence,  $y_1(t) - y_2(t)$  is a solution of Eq.(2.3.2). ■

**Proof of Theorem 2.3.1** Let  $y(t)$  be any solution of Eq.(2.3.1). By Lemma 2.3.1, function  $\phi(t) = y(t) - \psi(t)$  is a solution of the homogeneous equation (2.3.2). On the other hand, every solution  $\phi(t)$  of the homogeneous equation (2.3.2) is of the form

$$\phi(t) = c_1 y_1(t) + c_2 y_2(t).$$

Therefore,

$$y(t) = \phi(t) + \psi(t) = c_1 y_1(t) + c_2 y_2(t) + \psi(t)$$

for some choice of the constants  $c_1, c_2$ . ■

The importance of Theorem 2.3.1 is that it reduces the problem of finding all solutions of the nonhomogeneous equation (2.3.1) to the much simpler problem of finding just two linearly independent solutions of the corresponding homogeneous equation (2.3.2), and one particular solution of Eq.(2.3.1).

**Example 2.3.1** Find the general solution of equation

$$\frac{d^2 y}{dt^2} + y = t. \quad (2.3.4)$$

**Solution** Since functions  $y_1(t) = \cos t$  and  $y_2(t) = \sin t$  are two linearly independent solutions of the homogeneous equation  $y'' + y = 0$ , and  $\psi(t) = t$  is obviously a particular solution of Eq.(2.3.4). By Theorem 2.3.1, every solution  $y(t)$  of Eq.(2.3.4) must be of the form

$$y(t) = c_1 \cos t + c_2 \sin t + t. \quad \blacksquare$$

**Example 2.3.2** We have known that

[22] 参数变易法

$$\psi_1(t) = t^2, \quad \psi_2(t) = t^2 + e^{2t} \quad \text{and} \quad \psi_3(t) = 1 + t^2 + 2e^{2t}$$

are three solutions of a certain second-order nonhomogeneous linear equation. Find the general solution of this equation.

**Solution** By Lemma 2.3.1, functions

$$\psi_2(t) - \psi_1(t) = e^{2t} \quad \text{and} \quad \psi_3(t) - \psi_2(t) = 1$$

are solutions of the corresponding homogeneous equation. And it is not difficult to verify that these two functions are linearly independent. Therefore, by Theorem 2.3.1, the general solution  $y(t)$  of this equation must be of the form

$$y(t) = c_1 e^{2t} + c_2 + t^2. \quad \blacksquare$$

### 2.3.2 Method of Variation of Parameters<sup>[22]</sup>

We now introduce a very general method for finding a particular solution  $\psi(t)$  of the nonhomogeneous equation (2.3.1)

$$y'' + p(t)y' + q(t)y = g(t)$$

once the solutions of homogeneous equation (2.3.2)

$$y'' + p(t)y' + q(t)y = 0$$

are known.

Let  $y_1(t)$  and  $y_2(t)$  be two linearly independent solutions of the homogeneous equation (2.3.2). The structure of general solution (2.1.4) suggests us to try finding a particular  $\psi(t)$  of nonhomogeneous equation (2.3.1) of the form

$$\psi(t) = u_1(t)y_1(t) + u_2(t)y_2(t). \quad (2.3.5)$$

This method is referred to as **the method of variation of parameters**.

Computing

$$\begin{aligned} \frac{d}{dt}\psi(t) &= \frac{d}{dt}[u_1y_1 + u_2y_2] \\ &= [u_1y_1' + u_2y_2'] + [u_1'y_1 + u_2'y_2]. \end{aligned}$$

If we wish that  $d^2\psi/dt^2$ , and consequently the left-hand side of Eq.(2.3.1) will contain no second-order derivatives of  $u_1(t)$  and  $u_2(t)$ , then

$$u_1' y_1 + u_2' y_2 = 0 \quad (2.3.6)$$

should hold. This suggests us to impose condition (2.3.6) on functions  $u_1(t)$  and  $u_2(t)$ . In this case

$$\begin{aligned} & [u_1 y_1 + u_2 y_2]'' + p(t)[u_1 y_1 + u_2 y_2]' + q(t)[u_1 y_1 + u_2 y_2] \\ &= u_1' y_1' + u_2' y_2' + u_1[y_1'' + p(t)y_1' + q(t)y_1] + u_2[y_2'' + p(t)y_2' + q(t)y_2] \\ &= u_1' y_1' + u_2' y_2' \end{aligned}$$

since both  $y_1(t)$  and  $y_2(t)$  are solutions of the homogeneous equation (2.3.2). Consequently,

$$\psi(t) = u_1 y_1 + u_2 y_2$$

is a solution of the nonhomogeneous equation (2.3.1) if  $u_1(t)$  and  $u_2(t)$  satisfy following two equations:

$$\begin{aligned} y_1(t)u_1'(t) + y_2(t)u_2'(t) &= 0, \\ y_1'(t)u_1'(t) + y_2'(t)u_2'(t) &= g(t). \end{aligned}$$

Multiplying the first equation by  $y_2'(t)$ , the second equation by  $y_2(t)$ , and subtracting gives

$$[y_1(t)y_2'(t) - y_1'(t)y_2(t)]u_1'(t) = -g(t)y_2(t).$$

Multiplying the first equation by  $y_1'(t)$ , the second equation by  $y_1(t)$ , and subtracting gives

$$[y_1(t)y_2'(t) - y_1'(t)y_2(t)]u_2'(t) = g(t)y_1(t).$$

Hence,

$$\begin{aligned} u_1'(t) &= -\frac{g(t)y_2(t)}{y_1(t)y_2'(t) - y_1'(t)y_2(t)} = -\frac{g(t)y_2(t)}{W[y_1, y_2](t)}, \\ u_2'(t) &= \frac{g(t)y_1(t)}{y_1(t)y_2'(t) - y_1'(t)y_2(t)} = \frac{g(t)y_1(t)}{W[y_1, y_2](t)}. \end{aligned} \quad (2.3.7)$$

Integrating both sides of Eq.(2.3.7) we finally obtain  $u_1(t)$  and  $u_2(t)$ . Note that integration will produce two constants. We usually take these constants to be zero since we only need one solution  $\psi(t)$  here.

[23] 合理推断法

**Example 2.3.3** Find the general solution of differential equation

$$y'' - 4y' + 4y = te^{2t}.$$

**Solution** The characteristic equation  $r^2 - 4r + 4 = 0$  has two equal roots  $r_{1,2} = 2$ , so functions

$$y_1(t) = e^{2t}, \quad y_2(t) = te^{2t}$$

are two linearly independent solutions of the corresponding homogeneous equation

$$y'' - 4y' + 4y = 0.$$

Computing

$$\begin{aligned} u_1'(t) &= \frac{-g(t)y_2(t)}{W[y_1, y_2](t)} = \frac{-te^{2t}te^{2t}}{e^{4t}} = -t^2, \\ u_2'(t) &= \frac{g(t)y_1(t)}{W[y_1, y_2](t)} = \frac{te^{2t}e^{2t}}{e^{4t}} = t. \end{aligned}$$

Integrating both sides of the last two equations and setting the arbitrary constants zero give functions

$$u_1(t) = \int -t^2 dt = -\frac{1}{3}t^3, \quad u_2(t) = \int t dt = \frac{1}{2}t^2.$$

Thus, the given differential equation has a particular solution

$$\begin{aligned} \varphi(t) &= u_1(t)y_1(t) + u_2(t)y_2(t) \\ &= -\frac{1}{3}t^3e^{2t} + \frac{1}{2}t^2te^{2t} = \frac{1}{6}t^3e^{2t}. \end{aligned}$$

Finally, we obtain the general solution of the original equation

$$y(t) = c_1e^{2t} + c_2te^{2t} + \frac{1}{6}t^3e^{2t}. \quad \blacksquare$$

### 2.3.3 Methods for Some Special Form of the Nonhomogeneous Term $g(t)$

A serious disadvantage of the method of variation of parameters is that the integrations required for solving the system (2.3.7) are often quite difficult. It is usually much simpler, for some special form of the nonhomogeneous term  $g(t)$  in Eq.(2.3.1), to guess the form of a particular solution. And it is called *the method of judicious guessing*<sup>[23]</sup>.

$$1. \quad g(t) = (a_0 + a_1t + \cdots + a_nt^n)e^{\lambda t}$$

[24] 加起来成为

Denote

[25] 多项式

$$P_n(t)e^{\lambda t} = (a_0 + a_1t + \cdots + a_nt^n)e^{\lambda t}. \quad (2.3.8)$$

[26] 次数待定

[27]  $n$  次多项式

Now, we want a solution  $\psi(t)$  which will make functions  $\psi''(t)$ ,  $p\psi'(t)$ , and  $q\psi(t)$  add up to<sup>[24]</sup>  $P_n(t)e^{\lambda t}$ . Thus, a reasonable choice of  $\psi(t)$  is

[28] 待定系数

[29] 二重根

$$\psi(t) = Q(t)e^{\lambda t}, \quad (2.3.9)$$

where  $Q(t)$  is a *polynomial*<sup>[25]</sup> with *undetermined degree*<sup>[26]</sup>. Computing

$$\psi''(t) + p\psi'(t) + q\psi(t)$$

and reducing  $e^{\lambda t}$  from the resulting equation we obtain that

$$Q''(t) + (2\lambda + p)Q'(t) + (\lambda^2 + p\lambda + q)Q(t) = P_n(t).$$

Now we determine the degree of the polynomial  $Q(t)$ . Note that the left side of the last equation is a *polynomial of degree*  $n$ <sup>[27]</sup>.

(i)  $\lambda$  is not a root of the characteristic equation  $\lambda^2 + p\lambda + q = 0$ .

In this case,  $\lambda^2 + p\lambda + q \neq 0$ , so  $Q(t)$  is a polynomial of degree  $n$ . Thus, we can assume that

$$Q(t) = Q_n(t) = A_0t^n + A_1t^{n-1} + \cdots + A_{n-1}t + A_n, \quad (2.3.10)$$

where  $A_i (i = 0, 1, 2, \dots, n)$  are *undetermined constants*<sup>[28]</sup>.

(ii)  $\lambda$  is one of the two distinct roots of the characteristic equation  $\lambda^2 + p\lambda + q = 0$ .

Then  $\lambda^2 + p\lambda + q = 0$ , but  $2\lambda + p \neq 0$ . Thus  $Q'(t)$  is a polynomial of degree  $n$ , so  $Q(t)$ 's degree is  $(n+1)$ . Therefore, we can set

$$Q(t) = tQ_n(t) = t(A_0t^n + A_1t^{n-1} + \cdots + A_{n-1}t + A_n). \quad (2.3.11)$$

(iii)  $\lambda$  is a root of the characteristic equation *with multiplicity two*<sup>[29]</sup>.

Thus  $\lambda^2 + p\lambda + q = 0$  and  $2\lambda + p = 0$ . For the similar reason,  $Q(t)$ 's degree is  $(n+2)$ . Hence, we set

[30] 待定系数法  $Q(t) = t^2 Q_n(t) = t^2(A_0 t^n + A_1 t^{n-1} + \cdots + A_{n-1}t + A_n). \quad (2.3.12)$

[31] 幂

**Example 2.3.4** Find a particular solution of differential equation

$$y'' + y' + y = 1 + t + t^2.$$

**Solution** Since  $\lambda = 0$  is not a root of the characteristic equation  $r^2 + r + 1 = 0$ , the given differential equation has a solution of the form

$$Q_2(t)e^{0t} = A_0 + A_1 t + A_2 t^2.$$

By the method of undetermined coefficients<sup>[30]</sup>, we obtain that

$$(A_0 + A_1 + 2A_2) + (A_1 + 2A_2)t + A_2 t^2 = 1 + t + t^2.$$

Comparing the coefficients of  $t$  with same power<sup>[31]</sup> gives

$$\begin{cases} A_0 + A_1 + 2A_2 = 1, \\ A_1 + 2A_2 = 1, \\ A_2 = 1. \end{cases}$$

These imply that  $A_2 = 1, A_1 = -1, A_0 = 0$ . So the original differential equation has a particular solution

$$\psi(t) = t^2 - t. \quad \blacksquare$$

**Example 2.3.5** Find a particular solution of differential equation

$$y'' - y' = t^2 e^t.$$

**Solution** Since  $\lambda = 1$  is one of the two distinct roots of the characteristic equation  $r^2 - r = 0$ , the given differential equation has a particular solution of the form

$$tQ_2(t)e^t = t(A_0 + A_1 t + A_2 t^2)e^t.$$

By the same method as in Example 2.3.4, we obtain that

$$A_0 = \frac{1}{4}, \quad A_1 = -\frac{1}{4}, \quad A_2 = \frac{1}{6}.$$

Hence,

$$\psi(t) = t \left( \frac{1}{4} - \frac{1}{4}t + \frac{1}{6}t^2 \right) e^t. \quad \blacksquare$$

**Example 2.3.6** Find a particular solution of differential equation [32] 虚部

$$y'' + 2y' + y = e^{-t}.$$

**Solution** Since  $\lambda = -1$  is a root with multiple 2 of the characteristic equation  $r^2 + 2r + 1 = 0$ , we set

$$Q_2(t) = A_0 t^2.$$

Thus, we obtain that  $A_0 = \frac{1}{2}$ . Hence, a particular solution of the given equation is

$$\psi(t) = \frac{1}{2} t^2 e^{-t}. \quad \blacksquare$$

## 2. $g(t) = P_n(t) \cos \omega t$ or $P_n(t) \sin \omega t$

**Lemma 2.3.2** Let  $y(t) = u(t) + iv(t)$  be a complex-valued solution of equation

$$y'' + py' + qy = g(t) = g_1(t) + ig_2(t), \quad (2.3.13)$$

where  $p$  and  $q$  are real. Then

$$u'' + pu' + qu = g_1(t), \quad v'' + pv' + qv = g_2(t).$$

**Proof** Inserting  $y(t) = u(t) + iv(t)$  into Eq.(2.3.13) and equating real and *imaginary parts*<sup>[32]</sup> give

$$u''(t) + pu'(t) + qu(t) = g_1(t),$$

$$v''(t) + pv'(t) + qv(t) = g_2(t). \quad \blacksquare$$

Now, we back to the discussion about Eq.(2.3.13). Let

$$\varphi(t) = u(t) + iv(t)$$

be a particular solution of equation

$$y'' + py' + qy = P_n(t)e^{i\omega t}. \quad (2.3.14)$$

Since

$$\operatorname{Re}\{P_n(t)e^{i\omega t}\} = P_n(t) \cos \omega t, \quad \operatorname{Im}\{P_n(t)e^{i\omega t}\} = P_n(t) \sin \omega t,$$

by Lemma 2.3.2,

$$u(t) = \operatorname{Re}\{\varphi(t)\}$$

is a solution of

$$y'' + py' + qy = P_n(t) \cos \omega t,$$

while

$$v(t) = \operatorname{Im}\{\varphi(t)\}$$

is a solution of

$$y'' + py' + qy = P_n(t) \sin \omega t.$$

**Example 2.3.7** Find a particular solution of the differential equation

$$\frac{d^2 y}{dt^2} + 4y = t \sin 2t.$$

**Solution** Consider following differential equation

$$\frac{d^2 y}{dt^2} + 4y = te^{i2t}. \quad (2.3.15)$$

Observe that the characteristic equation  $r^2 + 4 = 0$  has complex roots  $r_{1,2} = \pm 2i$ . Therefore, Eq.(2.3.15) has a particular solution of the form

$$\phi(t) = t(A_0 + A_1 t)e^{i2t}.$$

Since

$$\phi'(t) = [A_0 + 2(A_1 + iA_0)t + i2A_1 t^2]e^{i2t},$$

$$\phi''(t) = [(2A_1 + i4A_0) + (i8A_1 - 4A_0)t - 4A_1 t^2]e^{i2t}.$$

thus,

$$\phi''(t) + 4\phi(t) = [(2A_1 + i4A_0) + i8A_1 t]e^{i2t}. \quad (2.3.16)$$

Comparing the right-hand sides of both Eq.(2.3.15) and Eq.(2.3.16) we obtain that

$$\begin{cases} 2A_1 + 4iA_0 = 0, \\ i8A_1 = 1. \end{cases}$$

It gives  $A_0 = \frac{1}{16}$ ,  $A_1 = \frac{1}{8i} = -\frac{1}{8}i$ . Hence,

$$\phi(t) = t \left( \frac{1}{16} - i\frac{1}{8}t \right) e^{i2t} = \left( \frac{t}{16} - i\frac{t^2}{8} \right) (\cos 2t + i \sin 2t)$$

$$= \left( \frac{t}{16} \cos 2t + \frac{t^2}{8} \sin 2t \right) + i \left( \frac{t}{16} \sin 2t - \frac{t^2}{8} \cos 2t \right).$$

Therefore,

$$\psi(t) = \operatorname{Im}\{\phi(t)\} = \frac{t}{16} \sin 2t - \frac{t^2}{8} \cos 2t$$

is a particular solution of the given equation.  $\blacksquare$

**3.  $g(t) = \sum_{j=1}^n g_j(t)e^{\alpha_j t}$ , where  $g_j(t), j = 1, \dots, n$  are polynomials in  $t$**

Let  $\psi_j(t)$  be a particular solution of equation

$$y'' + py' + qy = g_j(t)e^{\alpha_j t}, \quad j = 1, \dots, n.$$

Then,

$$\psi(t) = \sum_{j=1}^n \psi_j(t)$$

is a solution of Eq.(2.3.1) since, by the linearity of differentiation,

$$\begin{aligned} \psi'' + p\psi' + q\psi &= \left[ \sum_{j=1}^n \psi_j \right]'' + p \left[ \sum_{j=1}^n \psi_j \right]' + q \left[ \sum_{j=1}^n \psi_j \right] \\ &= \sum_{j=1}^n [\psi_j'' + p\psi_j' + q\psi_j] = \sum_{j=1}^n g_j(t)e^{\alpha_j t}. \end{aligned}$$

Thus, to find a particular solution of equation

$$y'' + y' + y = e^t + t \sin t,$$

we can find particular solutions of equations

$$y'' + y' + y = e^t \text{ and } y'' + y' + y = t \sin t$$

respectively, and then the sum of these two solutions is a solution of the original equation.

**Example 2.3.8** Find a particular solution of equation

$$y'' + y' - 6y = \cos t + te^{2t}.$$

**Solution** Let  $\psi_1(t)$  and  $\psi_2(t)$  be the particular solutions of equations

$$y'' + y' - 6y = \cos t \tag{2.3.17}$$

and

$$y'' + y' - 6y = te^{2t} \quad (2.3.18)$$

respectively. Consider equation

$$y'' + y' - 6y = e^{it}.$$

Since  $\lambda = i$  is not a root of the characteristic equation, Eq.(2.3.17) has a particular solution of the form

$$\varphi_1(t) = A_0 e^{it}$$

Then we can find  $A_0 = -\frac{1}{50}(i + 7)$ . Thus,

$$\varphi_1(t) = -\frac{1}{50}(i + 7)e^{it} = -\frac{1}{50}[(7 \cos t - \sin t) + i(7 \sin t + \cos t)].$$

Therefore,

$$\psi_1(t) = \operatorname{Re} \left\{ -\frac{1}{50}(i + 7)e^{it} \right\} = -\frac{1}{50}(7 \cos t - \sin t).$$

Similarly,

$$\psi_2(t) = t \left( \frac{1}{2}t - \frac{1}{5} \right) e^{2t}.$$

Hence,

$$\begin{aligned} \psi(t) &= \psi_1(t) + \psi_2(t) \\ &= -\frac{1}{50}(7 \sin t + \cos t) + t \left( \frac{1}{10}t - \frac{1}{25} \right) e^{2t} \end{aligned}$$

is a particular solution of the original equation. ■

### Exercise 2.3

1. Three solutions of a certain second-order nonhomogeneous linear equation are

$$\psi_1(t) = t, \quad \psi_2(t) = t + e^t \quad \text{and} \quad \psi_3(t) = 1 + t + e^t.$$

Find the general solution of this equation.

2. Three solutions of a second-order linear equation

$$L[y] = y'' + p(t)y' + q(t)y = g(t)$$

are

$$\psi_1(t) = 3e^t + e^{t^2}, \quad \psi_2(t) = 7e^t + e^{t^2} \quad \text{and} \quad \psi_3(t) = 5e^t + e^{-t^2} + e^{t^2}.$$

Find the solution of initial-value problem

$$L[y] = g(t); \quad y(0) = 1, y'(0) = 2.$$

Find a particular solution of following equations by the method of variation of parameters

3.  $y'' - 3y' + 2y = (1+t)e^{3t}.$

4.  $y'' - 2y' + y = \frac{e^t}{t}.$

5.  $y'' - y' - 2y = e^{3t}.$

6.  $y'' + 4y = \sin^2 2t.$

7.  $y'' + y = \tan t; y(0) = 1, y'(0) = 1.$

8.  $y'' - 3y' + 2y = \sqrt{t+1}; y(0) = y'(0) = 0.$

9.  $3y'' + 4y' + y = (\sin)e^{-1}; y(0) = 1, y'(0) = 0.$

Find a particular solution of the following equations by judicious guessing.

10.  $y'' + y' + y = t^2.$

11.  $y'' - y' - 2y = 4t^2.$

12.  $y'' - 4y' + 4y = (1+t+\cdots+t^{27})e^{2t}.$

13.  $y'' + 4y = \sin 2t.$

14.  $y'' + 4y = \cos 2t.$

15.  $y'' - 6y' + 25y = 2 \sin \frac{t}{2} - \cos \frac{t}{2}.$

16.  $y'' + 2y' + y = te^t \cos t.$

17.  $y'' - 3y' + 2y = e^t + e^{2t}.$

18.  $y'' + y' - 6y = \sin t + te^{2t}.$

19. Let  $a, b$  and  $c$  be positive constants. Show that the difference of any two solutions of equation  $ay'' + by' + cy = g(t)$  approaches zero as  $t$  approaches infinity.

20. Let  $\psi(t)$  be a solution of nonhomogeneous equation (2.3.1), and let  $\phi(t)$  be a solution of homogeneous equation (2.3.2). Show that  $\psi(t) + \phi(t)$  is again a solution of Eq.(2.3.1).

21. (a) Let  $L[y] = y'' - 2tr_1y' + r_1^2y$ . Show that  $L[e^{r_1t}v(t)] = e^{r_1t}v''(t).$

(b) Find the general solution of equation  $y'' - 6y' + 9y = t^{\frac{3}{2}}e^{3t}.$

[33] 一个原子垃圾处理问题

[34] 原子能委员会

## 2.4 Applications

### Module 1 An Atomic Waste Disposal Problem<sup>[33]</sup>

Ecologists and scientists had questioned the *Atomic Energy Commission*<sup>[34]</sup> (A.E.C) (now is known as the *Nuclear Regulatory Com-*

- [35] 核调节委员会
- [36] 浓缩的放射性废弃物
- [37] 密封的
- [38] 被丢进 300 深的海水中
- [39] 承诺
- [40] 发生泄漏
- [41] 破裂
- [42] 碰撞
- [43] 桶与海底碰撞时的速度
- [44] 合力
- [45] 重力加速度
- [46] 以秒计
- [47] 斯勒格

*mission*<sup>[35]</sup>) for the way of disposing *concentrated radioactive waste material*<sup>[36]</sup> by placing it in *tightly sealed*<sup>[37]</sup> drums which were then *dumped at sea in 300 feet of water*<sup>[38]</sup>. They were *assumed*<sup>[39]</sup> by the A.E.C. that the drums would never *develop leaks*<sup>[40]</sup>. Exhaustive tests on the drums proved the A.E.C. was right. However, several engineers then raised the question of whether the drums could *crack*<sup>[41]</sup> from the *impact*<sup>[42]</sup> of hitting the ocean floor. "Never," said the A.E.C. "We'll see about that." said the engineers. After performing numerous experiments, the engineers found that the drums could crack on impact if their velocity exceeded forty feet per second. The problem before us, therefore, is to compute *the velocity of the drums upon impact with the ocean floor*<sup>[43]</sup>.

Let  $y(t)$  denote the position of the center of gravity of the object (the drum). (We assume that the object move in only one direction.) Those forces acting on the object tending to increase  $y$  are considered positive, while those forces tending to decrease  $y$  are considered negative. The *resultant forces*<sup>[44]</sup>  $F$  acting on an object is defined to be the sum of all positive forces minus the sum of all negative forces. By *Newton's second law of motion*, the acceleration  $d^2y/dt^2$  of an object is propotional to the resultant force  $F$  acting on it; i.e.,

$$\frac{d^2y}{dt^2} = \frac{1}{m}F. \quad (2.4.1)$$

The constant  $m$  is the mass of the object. It is related to the weight  $W$  of the object by the relation  $W = mg$ , where  $g$  is *the acceleration of gravity*<sup>[45]</sup>. Unless otherwise stated, we assume that the weight of an object and the acceleration of gravity are constant. We will adopt the English system of units in the following computation, so that  $t$  is *measured in seconds*<sup>[46]</sup>,  $y$  is measured in feet, and  $F$  is measured in pounds. The units of  $m$  are then *slugs*<sup>[47]</sup>, and the graviational acceleration  $g$  equals  $32.2 \text{ ft/s}^2$ .

**Remark 2.4.1** If we use the mks system of units, then  $y$  is measured in meters and  $F$  is measured in newtons. The units of  $m$  are then kilograms, and the gravitational acceleration equals

9.8 m/s<sup>2</sup>.

We now return to our atomic disposal problem. As a drum descends through the water, it is acted upon by three forces  $W$ ,  $B$  and  $D$ . The force  $W$  is the weight of the drum pulling it down, and in magnitude  $W = 527.436$  lb. The force  $B$  is the *buoyancy force*<sup>[48]</sup> of the water acting on the drum. This force pushes the drum up, and its magnitude is the weight of the water displaced by the drum. Now, the A.E.C. used 55 gallon drums, whose volume is 7.35 ft<sup>3</sup>. The weight of one cubic foot of salt water is 63.99 lb. Hence  $B = (63.99)(7.35) = 470.327$  lb.

The force  $D$  is the *drag force*<sup>[49]</sup> of the water acting on the drum. It resists the motion of the drum through the water. Experiments have shown that any medium such as water, oil, and air resists the motion of an object through it. This resisting force acts in the direction opposite the motion, and is usually directly proportional to the velocity  $V$  of the object. Thus,  $D = cV$  for some positive constant  $c$ . Notice that the drag force increases as  $V$  increases, and decreases as  $V$  decreases. To calculate  $D$ , the engineers conducted numerous *towing experiments*<sup>[50]</sup>. They concluded that the orientation of the drum had little effect on the drag force, and that

$$D = 0.08V \frac{(\text{lb})(\text{s})}{\text{ft}}.$$

Now, set  $y = 0$  at *sea level*<sup>[51]</sup>, and let the direction of increasing  $y$  be downwards. Then,  $W$  is a positive force, while  $B$  and  $D$  are negative forces. Consequently, from Eq.(2.4.1),

$$\frac{d^2y}{dt^2} = \frac{1}{m}(W - B - cV) = \frac{g}{W}(W - B - cV).$$

We can rewrite this equation as a first-order linear differential equation for  $V = dy/dt$ ; i.e.,

$$\frac{dV}{dt} + \frac{cg}{W}V = \frac{g}{W}(W - B). \quad (2.4.2)$$

Initially, when the drum is released in the ocean, its velocity is zero. Thus,  $V(t)$ , the velocity of drum, satisfies the initial-value

[48] 浮力

[49] 拖拽力

[50] 拖拽试验

[51] 海平面

[52] 单调递增函数 problem

数

[53] 终端速度

$$\frac{dV}{dt} + \frac{cg}{W}V = \frac{g}{W}(W - B), \quad V(0) = 0, \quad (2.4.3)$$

and this implies that

$$V(t) = \frac{W - B}{c} \left[ 1 - e^{(-cg/W)t} \right]. \quad (2.4.4)$$

In order to determine the impact velocity of the drum, we must compute the time  $t$  at which the drum hits the ocean floor. Unfortunately, though, it is impossible to find  $t$  as an explicit function of  $y$ . Therefore, we can not use Eq.(2.4.4) to find the velocity of the drum when it hits the ocean floor. However, the A.E.C. can use this equation to try and prove that the drum does not crack on impact. To wit, observe from Eq.(2.4.4) that  $V(t)$  is a *monotonic increasing function*<sup>[52]</sup> of time which approaches the limiting value

$$V_T = \frac{W - B}{c}$$

as  $t$  approaches infinity. The quantity  $V_T$  is called *the terminal velocity*<sup>[53]</sup> of the drum. Clearly,  $V(t) \leq V_T$ , so that the velocity of the drum when it hits the ocean floor is certainly less than  $(W - B)/c$ . Now, if this terminal velocity is less than 40ft/s, then the drums could not possibly break on impact. However,

$$\frac{W - B}{c} = \frac{527.436 - 470.327}{0.08} = 713.86\text{ft/s},$$

and this is too large.

It should be clear now that the only way we can resolve the dispute between the A.E.C. and the engineers is to find  $v(y)$ , the velocity of the drum as a function of position. The function  $v(y)$  is very different from the function  $V(t)$ , which is the velocity of the drum as a function of time. However, these two functions are related through the equation

$$V(t) = v(y(t)),$$

if we express  $y$  as a function of  $t$ . By the chain rule of differentiation

$$\frac{dV}{dt} = \frac{dv}{dy} \frac{dy}{dt},$$

hence

$$\frac{W}{g} \frac{dv}{dy} \frac{dy}{dt} = W - B - cV.$$

[54] 忽略

But  $dy/dt = V(t) = v(y(t))$ . Thus, *suppressing*<sup>[54]</sup> the dependence of  $y$  on  $t$ , we see that  $v(y)$  satisfies the first-order differential equation

$$\frac{W}{g} v \frac{dv}{dy} = W - B - cv, \text{ or } \frac{v}{W - B - cv} \frac{dv}{dy} = \frac{g}{W}.$$

Moreover,

$$v(0) = v(y(0)) = V(0) = 0.$$

Hence,

$$\int_0^v \frac{r dr}{W - B - cr} = \int_0^y \frac{g}{W} ds = \frac{gy}{W}.$$

Now,

$$\begin{aligned} \int_0^v \frac{r dr}{W - B - cr} &= \int_0^v \frac{r - (W - B)/c}{W - B - cr} dr + \frac{W - B}{c} \int_0^v \frac{dr}{W - B - cr} \\ &= -\frac{1}{c} \int_0^v dr + \frac{W - B}{c} \int_0^v \frac{dr}{W - B - cr} \\ &= -\frac{v}{c} - \frac{(W - B)}{c^2} \ln \frac{|W - B - cv|}{W - B}. \end{aligned}$$

We know already that  $v < (W - B)/c$ . Consequently,  $W - B - cv$  is always positive, and

$$\frac{gy}{W} = -\frac{v}{c} - \frac{(W - B)}{c^2} \ln \frac{|W - B - cv|}{W - B}. \quad (2.4.5)$$

Unfortunately, we cannot find  $v$  as an explicit function of  $y$  from Eq.(2.4.5). This is not an insurmountable difficulty, though. It is quite simple, with the aid of a digit computer, to find  $v(300)$  from Eq.(2.4.5). We need only supply the computer with a good approximation of  $v(300)$  and this is obtained in the following manner. The velocity  $v(y)$  of the drum satisfies the initial-value problem

$$\frac{W}{g} v \frac{dv}{dy} = W - B - cv; \quad v(0) = 0. \quad (2.4.6)$$

Setting  $c = 0$  in Eq.(2.4.6), we obtain a new initial-value problem

$$\frac{W}{g} u \frac{du}{dy} = W - B, \quad u(0) = 0. \quad (2.4.6')$$

[55] 机械振动

[56] 弹簧

[57] 刚性水平支撑

[58] 回复力

[59] 弹性系数

[60] 硬度

[61] 弹簧-物质-  
阻尼器系统

[62] 地震仪

[63] 地震仪

(We have replaced  $v$  by  $u$  to avoid confusion later.) Integrating Eq.(2.4.6') gives

$$\frac{W}{g} \frac{u^2}{2} = (W - B)y \quad \text{or} \quad u(y) = \left[ \frac{2g}{W} (W - B)y \right]^{1/2}.$$

In particular,

$$\begin{aligned} u(300) &= \left[ \frac{2g}{W} (W - B)300 \right]^{1/2} = \left[ \frac{2(32.2)(57.109)(300)}{527.436} \right]^{1/2} \\ &= \sqrt{2092} \approx 45.7 \text{ ft/s.} \end{aligned}$$

We can prove that  $u(300)$  is a very good approximation of  $v(300)$ . In fact, observe that the velocity of drum is always greater if there is no drag force opposing the motion. Hence,

$$v(300) < u(300).$$

Secondly, the velocity  $v$  increases as  $y$  increases, so that  $v(y) \leq v(300)$  for  $y \leq 300$ . Consequently, the drag force  $D$  of the water acting on the drum is always less than  $0.08 \times u(300) \approx 3.7$  lb. Now, the resultant force  $W - B$  pulling the drum down is approximately 57.1lb, which is very large compared to  $D$ . It stands to reason, therefore, that  $u(t)$  should be a very good approximation of  $v(y)$ . And indeed, this is the case, since we find numerically that  $v(300) = 45.1$  ft/s. Thus, the drums can break upon impact, and the engineers were right.

## Module 2 Mechanical Vibrations<sup>[55]</sup>

Consider the case where a small object of mass  $m$  attached to an *elastic spring*<sup>[56]</sup> of length  $l$  which is suspended from a *rigid horizontal support*<sup>[57]</sup> (see Figure 2.4.1). The elastic spring has the property that if it stretched or compressed a distance  $\Delta l$  which is small compared to its natural length  $l$ , then it will exert a *restoring force*<sup>[58]</sup> of magnitude  $k\Delta l$ . The constant  $k$  is called the *spring-constant*<sup>[59]</sup>, and is a measure of the *stiffness*<sup>[60]</sup> of the spring. Engineers usually refer to such systems as *spring-mass-dashpot system*<sup>[61]</sup>, or as *seismic instruments*<sup>[62]</sup> since they are similar, in principle, to a *seismograph*<sup>[63]</sup> which is used to detect motions of earth's surface.

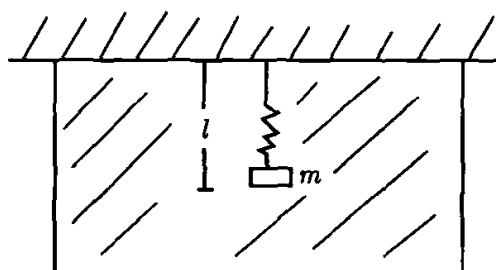


Figure 2.4.1

- [64] 减震器
- [65] 重型武器装备
- [66] 后坐力
- [67] 外力
- [68] 伸长
- [69] 压缩

Spring-mass-dashpot systems have many diverse applications. For example, the *shock absorbers*<sup>[64]</sup> in our automobiles are simple spring-mass-dashpot systems. Also, most *heavy gun emplacements*<sup>[65]</sup> are attached to such systems so as to minimize the “*recoil*”<sup>[66]</sup> effect of the gun. The usefulness of these devices will become apparent after we set up and solve the differential equation of motion of mass  $m$ .

In calculating the motion of the mass  $m$ , it will be convenient for us to measure distances from the equilibrium position of the mass, rather than the horizontal support. The equilibrium position of the mass is that point where the mass will hang at rest if no *external forces*<sup>[67]</sup> act upon it. In equilibrium, the weight  $mg$  of the mass exactly balanced by the restoring force of the spring. Thus, in its equilibrium position, the spring has been stretched a distance  $\Delta l$ , where  $k\Delta l = mg$ . Let  $y = 0$  denote this equilibrium position,  $y(t)$  denote the position of the mass at time  $t$ , and we take the downward direction as positive. To find  $y(t)$ , we must compute the total force acting on mass  $m$ . This force is the sum of four separate forces  $W$ ,  $R$ ,  $D$  and  $F$ .

(i) The force  $W = mg$  is the weight of mass pulling it downward. This force is positive.

(ii) The force  $R$  is the restoring force of the spring, and it is proportional to the *elongation*<sup>[68]</sup>, or *compression*<sup>[69]</sup>,  $\Delta l + y$  of the spring. It always acts to restore the spring to its natural length. If  $\Delta l + y > 0$ , then  $R$  is negative, so that  $R = -k(\Delta l + y)$ ; and if  $\Delta l + y < 0$ , then  $R$  is positive, so that  $R = -k(\Delta l + y)$ . In either case,

$$R = -k(\Delta l + y).$$

- [70] 阻力  
 [71] 自由振动  
 [72] 无阻尼运动

(iii) The force  $D$  is the *damping*<sup>[70]</sup>, or drag force, which the medium exerts on the mass  $m$ . Most medium, such as oil and air, tend to resist the motion of an object through it. This force always acts in the direction opposite the direction of motion, and is usually directly proportional to the magnitude of the velocity  $dy/dt$ . If the velocity is positive, that is, the mass is moving in the downward direction, then  $D = -c dy/dt$ ; and if the velocity is negative, then  $D = -c dy/dt$ . In either case,

$$D = -c \frac{dy}{dt}.$$

(iv) The force  $F$  is the external force applied to the mass. This force is directed upward or downward, depending on whether  $F$  is positive or negative. In general, this external force will depend explicitly on time.

From *Newton's* second law of motion

$$\begin{aligned} m \frac{d^2 y}{dt^2} &= W + R + D + F \\ &= mg - k(\Delta l + y) - c \frac{dy}{dt} + F(t) \\ &= -ky - c \frac{dy}{dt} + F(t) \end{aligned}$$

since  $mg = k\Delta l$ . Hence, the position  $y(t)$  of the mass satisfies the second-order linear differential equation

$$m \frac{d^2 y}{dt^2} + c \frac{dy}{dt} + ky = F(t), \quad (2.4.7)$$

where  $m$ ,  $c$  and  $k$  are nonnegative constants. We adopt the mks system of units so that  $F$  is measured in newtons,  $y$  is measured in meters, and  $t$  is measured in seconds. In this case, the units of  $k$  are N/m, the units of  $c$  are N·s/m, and the units of  $m$  are kilograms (N·s<sup>2</sup>/m).

(a) **Free vibrations**<sup>[71]</sup>:

We consider first the simplest case of free *undamped motion*<sup>[72]</sup>. In this case, Eq.(2.4.7) reduces to

$$m \frac{d^2 y}{dt^2} + ky = 0 \quad \text{or} \quad m \frac{d^2 y}{dt^2} + \omega^2 y = 0, \quad (2.4.8)$$

where  $\omega_0^2 = k/m$ . The general solution of Eq.(2.4.8) is

$$y(t) = a \cos \omega_0 t + b \sin \omega_0 t. \quad (2.4.9)$$

In order to analyze the solution (2.4.9), it is convenient to rewrite it as a single cosin function

$$y(t) = R \cos(\omega_0 t - \delta), \quad (2.4.10)$$

where  $R = \sqrt{a^2 + b^2}$  and  $\delta = \arctan(b/a)$ .

From Figure 2.4.2 we see that  $y(t)$  always lies between  $-R$  and  $+R$ , and that the motion of the mass is periodic. This type of motion is called *simple harmonic motion*<sup>[73]</sup>.  $R$  is called the amplitude of the motion,  $\delta$  the *phase angle*<sup>[74]</sup> of the motion.  $T_0 = 2\pi/\omega_0$  the *natural period*<sup>[75]</sup> of the motion, and  $\omega_0 = \sqrt{k/m}$  the *natural frequency*<sup>[76]</sup> of the system.

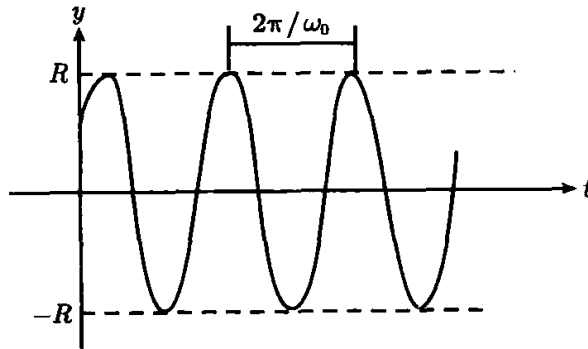


Figure 2.4.2 Graph of  $y(t) = R \cos(\omega_0 t - \delta)$

(b) **Damped free vibrations:**

If we now include the effect of damping, then the differential equation governing the motion of the mass is

$$m \frac{d^2 y}{dt^2} + c \frac{dy}{dt} + ky = 0. \quad (2.4.11)$$

The roots of the characteristic equation  $mr^2 + cr + k = 0$  are

$$r_1 = \frac{-c + \sqrt{c^2 - 4km}}{2m} \quad \text{and} \quad r_2 = \frac{-c - \sqrt{c^2 - 4km}}{2m}.$$

(i)  $c^2 - 4km > 0$ . In this case both  $r_1$  and  $r_2$  are negative, and every solution  $y(t)$  of Eq.(2.4.11) has the form

$$y(t) = ae^{r_1 t} + be^{r_2 t}.$$

[73] 简谐运动

[74] 相角

[75] 自然周期

[76] 自然频率

[77] 过阻尼

(ii)  $c^2 - 4km = 0$ . In this case, every solution  $y(t)$  of

[78] 临界阻尼

Eq.(2.4.11) is of the form

[79] 缓缓地返回  
到

$$y(t) = (a + bt)e^{-ct/(2m)}.$$

[80] 越过

(iii)  $c^2 - 4km < 0$ . In this case, every solution  $y(t)$  of

[81] 受阻尼运动

Eq.(2.4.11) is of the form

[82] 震荡

$$y(t) = (a \cos \mu t + b \sin \mu t)e^{-ct/(2m)}, \quad \mu = \frac{\sqrt{4km - c^2}}{2m}.$$

[83] 消失

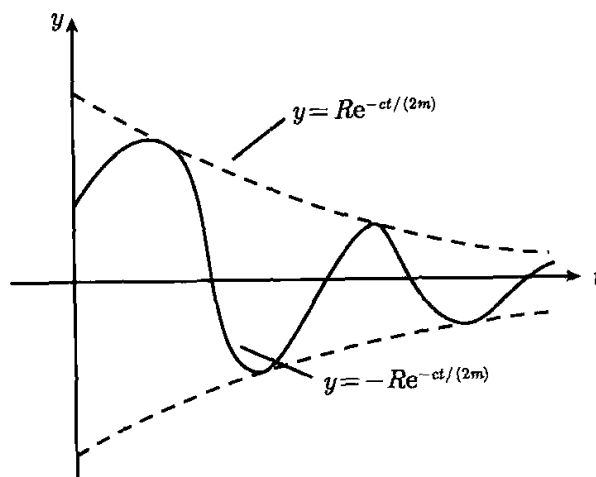
The first two cases are referred to as *overdamped*<sup>[77]</sup> and *critically damped*<sup>[78]</sup>, respectively. They represent motions in which the originally displaced mass *creeps back to*<sup>[79]</sup> its equilibrium position. Depending on the initial conditions, it may be possible to *overshoot*<sup>[80]</sup> the equilibrium position once, but no more than once. The third case, which is referred to as an *underdamped motion*<sup>[81]</sup>, occurs quite often in mechanical systems and represents a damped vibration. To see this, we rewrite the function

$$y(t) = (a \cos \mu t + b \sin \mu t)e^{-ct/(2m)}$$

in the form

$$y(t) = R \cos(\mu t - \delta)e^{-ct/(2m)}.$$

Thus the displacement  $y$  *oscillates*<sup>[82]</sup> between the curves  $y(t) = \pm Re^{-ct/(2m)}$  with decreasing amplitude, as shown in Figure 2.4.3.

Figure 2.4.3 Graph of  $R \cos(\mu t - \delta)e^{-ct/(2m)}$ 

Now, observe that the motion of the mass always *dies out*<sup>[83]</sup> eventually if there is damping in the system. In other words, any

initial disturbance of the system *is dissipated by*<sup>[84]</sup> the damping present in the system. This is one reason why spring-dashpot systems are so useful in mechanical systems: they can be used to damp out any undesirable disturbances. For example, the shock transmitted to an automobile by a bump in the road is dissipated by the shock absorbers in car, and the *momentum*<sup>[85]</sup> from the recoil of a *gun barrel*<sup>[86]</sup> is dissipated by a spring-dashpot system attached to the gun.

[84] 被...削减掉

[85] 冲撞

[86] 枪管

[87] 有阻尼强迫  
振动

[88] 稳定态部分

(c) **Damped forced vibrations**<sup>[87]</sup>:

If we now introduce an external force  $F(t) = F_0 \cos \omega t$ , then the differential equation governing the motion of the mass is

$$m \frac{d^2 y}{dt^2} + c \frac{dy}{dt} + ky = F_0 \cos \omega t. \quad (2.4.12)$$

Using the method of judicious guessing, we can find a particular solution  $\psi(t)$  of Eq.(2.4.12) of the form

$$\begin{aligned} \psi(t) &= \frac{F_0}{(k - m\omega^2)^2 + c^2\omega^2} [(k - m\omega^2) \cos \omega t + c\omega \sin \omega t] \\ &= \frac{F_0}{(k - m\omega^2)^2 + c^2\omega^2} [(k - m\omega^2) + c^2\omega^2]^{1/2} \cos(\omega t - \delta) \\ &= \frac{F_0 \cos(\omega t - \delta)}{[(k - m\omega^2)^2 + c^2\omega^2]^{1/2}}, \end{aligned} \quad (2.4.13)$$

where  $\tan \delta = c\omega / (k - m\omega^2)$ . Hence, every solution  $y(t)$  of Eq.(2.4.12) must be of the form

$$y(t) = \phi(t) + \psi(t) = \phi(t) + \frac{F_0 \cos(\omega t - \delta)}{[(k - m\omega)^2 + c^2\omega^2]^{1/2}}, \quad (2.4.14)$$

where  $\phi(t)$  is a solution of the homogeneous equation

$$m \frac{d^2 y}{dt^2} + c \frac{dy}{dt} + ky = 0. \quad (2.4.15)$$

We have already seen, though, that every solution  $y = \phi(t)$  of Eq.(2.4.15) approaches zero as  $t$  approaches infinity. Thus, for large  $t$ , the equation  $y(t) = \psi(t)$  describes very accurately the position of the mass  $m$ , regardless of its initial position and velocity. For this reason,  $\psi(t)$  is called *the steady state part*<sup>[88]</sup> of

[89] 瞬变部分

[90] 强迫自由振

动

[91] 共振

the solution (2.4.14), while  $\phi(t)$  is called *the transient part*<sup>[89]</sup> of the solution.

(d) **Forced free vibrations**<sup>[90]</sup>:

We now remove the damping from our system and consider the case of forced free vibrations where the forcing term is periodic and has the form  $F(t) = F_0 \cos \omega t$ . In this case, the differential equation governing the motion of the mass  $m$  is

$$\frac{d^2 y}{dt^2} + \omega_0^2 y = \frac{F_0}{m} \cos \omega t, \quad \omega_0^2 = \frac{k}{m}. \quad (2.4.16)$$

The case  $\omega \neq \omega_0$  is uninteresting. Every solution  $y(t)$  of Eq.(2.4.16) has the form

$$y(t) = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos \omega t,$$

and this is the sum of two periodic functions of different periods. The interesting case is when  $\omega = \omega_0$ , that is, when the frequency  $\omega$  of the external force equals the natural frequency of the system. This case is called the *resonance*<sup>[91]</sup> case, and the differential equation of motion for the mass  $m$  is

$$\frac{d^2 y}{dt^2} + \omega_0^2 y = \frac{F_0}{m} \cos \omega t. \quad (2.4.17)$$

A particular solution of following equation

$$\frac{d^2 y}{dt^2} + \omega_0^2 y = \frac{F_0}{m} e^{i\omega_0 t} \quad (2.4.18)$$

is

$$\begin{aligned} \phi(t) &= \frac{-iF_0 t}{2m\omega_0} (\cos \omega_0 t + i \sin \omega_0 t) \\ &= \frac{F_0 t}{2m\omega_0} \sin \omega_0 t - i \frac{F_0 t}{2m\omega_0} \cos \omega_0 t. \end{aligned}$$

Hence,

$$\psi(t) = \operatorname{Re}\{\phi(t)\} = \frac{F_0 t}{2m\omega_0} \sin \omega_0 t$$

is a particular solution of Eq.(2.4.17). Consequently, every solution  $y(t)$  of Eq.(2.4.17) is of the form

$$y(t) = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t + \frac{F_0 t}{2m\omega_0} \sin \omega_0 t \quad (2.4.19)$$

for some choice of constants  $c_1, c_2$ .

Now, the sum of the first two terms in Eq.(2.4.19) is a periodic function of time. The third term, though, represents an oscillation with increasing amplitude, as shown in Figure 2.4.4. Thus, the forcing term  $F_0 \cos \omega t$ , if it is in resonance with the natural frequency of the system, will always cause *unbounded oscillations*<sup>[92]</sup>. Such a phenomenon was responsible for the *collapse*<sup>[93]</sup> of the Tacoma Bridge, and many other *mechanical catastrophes*<sup>[94]</sup>.

[92] 无界震荡

[93] 坍塌、倒塌

[94] 机械性灾难

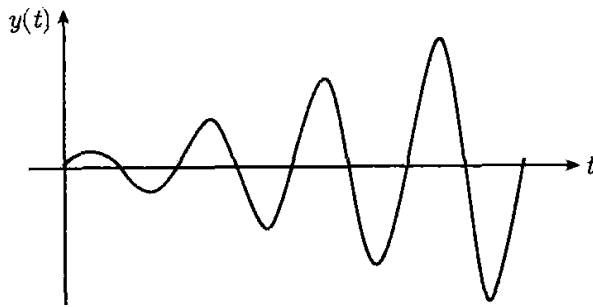


Figure 2.4.4 Graph of  $f(t) = At \sin \omega_0 t$

# Chapter 3 Linear Systems of Differential Equations <sup>[1]</sup>

## 3.1 Basic Concepts and Theorems

[1] 线性微分方程组

[2] 联立的

As we will see in Section 3.5, many applications require us to solve a system of *simultaneous*<sup>[2]</sup> differential equations. Furthermore, a higher-order differential equation for a single variable can be converted into a **system of simultaneous first-order differential equations** (or simply referred to as system of first-order differential equations). It provides us a way to solve higher-order differential equations. For example, set  $x_1(t) = y$ ,  $x_2(t) = y'$ ,  $x_3(t) = y''$ , then 3th-order differential equation

$$y''' + \cos y = e^t$$

can be converted into the following system of first-order differential equations

$$\frac{dx_1}{dt} = x_2, \quad \frac{dx_2}{dt} = x_3, \quad \frac{dx_3}{dt} = e^t - \cos x_1.$$

In general, every  $n$ th-order differential equation for the single variable  $y$

$$a_n(t) \frac{d^n y}{dt^n} + a_{n-1}(t) \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1(t) \frac{dy}{dt} + a_0 y = 0$$

can be converted into a system of  $n$  first-order differential equations

$$\left\{ \begin{array}{l} \frac{dx_1}{dt} = x_2, \\ \frac{dx_2}{dt} = x_3, \\ \dots\dots\dots \\ \frac{dx_{n-1}}{dt} = x_n, \\ \frac{dx_n}{dt} = -\frac{a_{n-1}(t)x_n + a_{n-2}(t)x_{n-1} + \cdots + a_0(t)x_1}{a_n(t)} \end{array} \right. \quad (3.1.1a)$$

by setting

$$x_1 = y, \quad x_2 = \frac{dy}{dt}, \quad \dots, \quad x_n = \frac{d^{n-1}y}{dt^{n-1}}. \quad (3.1.1b)$$

Consider the system of first-order differential equations in several variables

$$\begin{cases} \frac{dx_1}{dt} = f_1(t, x_1, \dots, x_n), \\ \frac{dx_2}{dt} = f_2(t, x_1, \dots, x_n), \\ \dots\dots\dots \\ \frac{dx_n}{dt} = f_n(t, x_1, \dots, x_n). \end{cases} \quad (3.1.2)$$

A **solution** of system (3.1.2) is consisted of  $n$  functions  $x_1(t), \dots, x_n(t)$  which satisfy

$$\frac{dx_j}{dt} = f_j(t, x_1, \dots, x_n), \quad j = 1, 2, \dots, n.$$

For example, functions  $x_1(t) = e^{2t}$ ,  $x_2(t) = 2e^{2t}$ , and  $x_3(t) = 4e^{2t}$  consist a solution of system of first-order differential equations

$$\begin{cases} \frac{dx_1}{dt} = x_2, \\ \frac{dx_2}{dt} = x_3, \\ \frac{dx_3}{dt} = 2x_1 - x_2 + 2x_3, \end{cases}$$

since

$$x_1'(t) = 2e^{2t} = x_2(t),$$

$$x_2'(t) = 4e^{2t} = x_3(t),$$

$$x_3'(t) = 8e^{2t} = 2x_1(t) - x_2(t) + 2x_3(t).$$

System (3.1.2), together with following initial conditions

$$x_1(t_0) = x_1^0, \quad x_2(t_0) = x_2^0, \quad \dots, \quad x_n(t_0) = x_n^0, \quad (3.1.3)$$

is referred to as an initial-value problem. That is,

$$\begin{cases} \frac{dx_j}{dt} = f_j(t, x_1, \dots, x_n), \quad j = 1, 2, \dots, n, \\ x_1(t_0) = x_1^0, \quad \dots, \quad x_n(t_0) = x_n^0. \end{cases} \quad (3.1.4)$$

A solution of initial-value problem (3.1.4) is  $n$  functions  $x_1(t), \dots, x_n(t)$  which satisfy the equations in system (3.1.2) and the initial conditions in (3.1.3). For example,  $x_1(t) = e^{2-t}$  and  $x_2(t) = -e^{2-t}$  form a solution of the initial-value problem

$$\begin{cases} x_1' = x_2, \\ x_2' = -x_1 - 2x_2, \\ x_1(2) = 1, \quad x_2(2) = -1, \end{cases}$$

since

$$\begin{aligned} x_1'(t) &= -e^{2-t}, & x_2'(t) &= e^{2-t}, & x_1(2) &= e^{2-2} = 1, \\ x_2(2) &= -e^{2-2} = -1. \end{aligned}$$

If each of the functions  $f_1, f_2, \dots, f_n$  in system (3.1.2) is a linear function in variables  $x_1, x_2, \dots, x_n$ , then we call the system (3.1.2) a **linear system of differential equations**. The general form of a linear system is

$$\begin{cases} x_1' = a_{11}(t)x_1 + \dots + a_{1n}(t)x_n + g_1(t), \\ \dots\dots\dots \\ x_n' = a_{n1}(t)x_1 + \dots + a_{nn}(t)x_n + g_n(t). \end{cases} \quad (3.1.5)$$

If each of the functions  $g_1, g_2, \dots, g_n$  is identically zero, that is

$$\begin{cases} x_1' = a_{11}(t)x_1 + \dots + a_{1n}(t)x_n, \\ \dots\dots\dots \\ x_n' = a_{n1}(t)x_1 + \dots + a_{nn}(t)x_n. \end{cases} \quad (3.1.6)$$

then we call the system (3.1.6) a **homogeneous system**; otherwise it is **nonhomogeneous**. In this chapter, we only consider the case where the coefficients  $a_{ij}$  are constants,

$$\begin{cases} x_1' = a_{11}x_1 + \dots + a_{1n}x_n + g_1(t), \\ \dots\dots\dots \\ x_n' = a_{n1}x_1 + \dots + a_{nn}x_n + g_n(t). \end{cases} \quad (3.1.7)$$

**Function**

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$$

is referred to as **vector-valued function**<sup>[3]</sup>. Its derivative is [3] 向量(值)函数  
defined as the vector-valued function 数

$$\frac{d\mathbf{x}}{dt} = \begin{pmatrix} dx_1/dt \\ dx_2/dt \\ \vdots \\ dx_n/dt \end{pmatrix}. \quad (3.1.8)$$

Thus, we can write the homogeneous linear system (3.1.6) in a concise form

$$\dot{\mathbf{x}} = \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}, \quad (3.1.9a)$$

where

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}. \quad (3.1.9b)$$

The initial-value problem corresponding to system (3.1.9) is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}; \quad \mathbf{x}(t_0) = \mathbf{x}^0, \quad (3.1.10a)$$

where

$$\mathbf{x}^0 = \begin{pmatrix} x_1^0 \\ x_2^0 \\ \vdots \\ x_n^0 \end{pmatrix} = \begin{pmatrix} x_1(t_0) \\ x_2(t_0) \\ \vdots \\ x_n(t_0) \end{pmatrix}. \quad (3.1.10b)$$

**Theorem 3.1.1** Let  $\mathbf{x}(t)$  and  $\mathbf{y}(t)$  be two solutions of homogeneous equation (3.1.9). Then,

- (a)  $c\mathbf{x}(t)$  is a solution of Eq.(3.1.9) for any constant  $c$ ;
- (b)  $\mathbf{x}(t) + \mathbf{y}(t)$  is also a solution of Eq.(3.1.9).

**Proof** (a) If  $\mathbf{x}(t)$  is a solution of Eq.(3.1.9), then

$$\frac{d}{dt}(c\mathbf{x}) = c \frac{d\mathbf{x}}{dt} = c\mathbf{A}\mathbf{x} = \mathbf{A}(c\mathbf{x}).$$

Hence,  $c\mathbf{x}(t)$  is a solution of Eq.(3.1.9).

(b) If  $\mathbf{x}(t)$  and  $\mathbf{y}(t)$  are solutions of Eq.(3.1.9), then

[4] 维数

$$\begin{aligned}\frac{d}{dt}(\mathbf{x} + \mathbf{y}) &= \frac{d\mathbf{x}}{dt} + \frac{d\mathbf{y}}{dt} = \mathbf{A}\mathbf{x} + \mathbf{A}\mathbf{y} \\ &= \mathbf{A}(\mathbf{x} + \mathbf{y}).\end{aligned}$$

Hence,  $\mathbf{x}(t) + \mathbf{y}(t)$  is again a solution of Eq.(3.1.9). ■

**Corollary 3.1.1** Any linear combination of solutions of Eq. (3.1.9) is also a solution of Eq.(3.1.9).

**Theorem 3.1.2 (Existence-uniqueness Theorem)** There exists one, and only one, solution of initial-value problem

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}; \quad \mathbf{x}(t_0) = \mathbf{x}^0 \quad (3.1.11a)$$

on interval  $-\infty < t < \infty$ , where

$$\mathbf{x}^0 = \begin{pmatrix} x_1^0 \\ x_2^0 \\ \vdots \\ x_n^0 \end{pmatrix}. \quad (3.1.11b)$$

From Theorem 3.1.2 we can obtain, immediately, that if  $\mathbf{x}(t)$  is a nontrivial solution, then  $\mathbf{x}(t) \neq \mathbf{0}$  for any  $t$ . In fact, if  $\mathbf{x}(t^*) = \mathbf{0}$  for some  $t^*$ , then, by the uniqueness,  $\mathbf{x}(t)$  must be identically zero since this solution and the trivial solution  $\mathbf{x}(0) \equiv \mathbf{0}$  satisfy the same differential equation and have the same value at  $t = t^*$ .

**Theorem 3.1.3** The *dimension*<sup>[4]</sup> of the space  $V$  of all solutions of homogeneous linear system (3.1.9) is  $n$ .

**Proof** Let  $\phi^j(t)$  ( $j = 1, 2, \dots, n$ ) be the solutions of initial-value problem

$$\frac{d\mathbf{x}^j}{dt} = \mathbf{A}\mathbf{x}^j; \quad \mathbf{x}^j(0) = \mathbf{e}^j = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \leftarrow j\text{th row}. \quad (3.1.12)$$

That is,

$$\begin{cases} \frac{d\phi^j}{dt} = A\phi^j, & j = 1, 2, \dots, n, \\ \phi^1(0) = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, & \phi^2(0) = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \phi^n(0) = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}. \end{cases} \quad [5] \text{ 生成}$$

Constructing equation

$$c_1\phi^1(t) + c_2\phi^2(t) + \dots + c_n\phi^n(t) = \mathbf{0}. \quad (3.1.13)$$

Evaluating both sides of Eq.(3.1.13) at  $t = 0$  gives

$$c_1\phi^1(0) + c_2\phi^2(0) + \dots + c_n\phi^n(0) = \mathbf{0}$$

or

$$c_1\mathbf{e}^1 + c_2\mathbf{e}^2 + \dots + c_n\mathbf{e}^n = \mathbf{0}.$$

Since  $\mathbf{e}^1, \mathbf{e}^2, \dots, \mathbf{e}^n$  are linearly independent in  $\mathbf{R}^n$ , thus, we have  $c_1 = c_2 = \dots = c_n = 0$ . Therefore,  $\phi^1(t), \phi^2(t), \dots, \phi^n(t)$  are linearly independent vectors in  $V$ .

Pick arbitrarily an  $\mathbf{x}(t)$  in  $V$  and choose  $\mathbf{c} = \mathbf{x}(0) = (c_1, c_2, \dots, c_n)^T$ . With these constants construct the vector-value function

$$\phi(t) = c_1\phi^1(t) + c_2\phi^2(t) + \dots + c_n\phi^n(t).$$

Then,  $\phi(t)$  is a solution of the homogeneous linear system, and satisfies the initial condition

$$\begin{aligned} \phi(0) &= c_1\phi^1(0) + c_2\phi^2(0) + \dots + c_n\phi^n(0) \\ &= c_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \dots + c_n \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} = \mathbf{x}(0). \end{aligned}$$

Now, both  $\mathbf{x}(t)$  and  $\phi(t)$  are the solutions of the same initial-value problem. Consequently, by the uniqueness part of Theorem 3.1.2,  $\mathbf{x}(t)$  and  $\phi(t)$  must be identical. Thus,  $\phi^1(t), \phi^2(t), \dots, \phi^n(t)$   $\text{span}^{[5]} V$ , so, by the theory of linear algebra,  $\dim V = n$ . ■

From Theorem 3.1.3, we know that we need only find  $n$  independent solutions of Eq.(3.1.12).

**Theorem 3.1.4** Let  $\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^k(t)$  be  $k$  solutions of system (3.1.9). Then,  $\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^k(t)$  are linearly dependent solutions if and only if  $\mathbf{x}^1(t_0), \mathbf{x}^2(t_0), \dots, \mathbf{x}^k(t_0)$  are linearly dependent vectors in  $\mathbf{R}^n$  for some  $t_0$ .

**Proof** Suppose that  $\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^k(t)$  are linearly dependent solutions of system (3.1.9). Then, there exist constants  $c_1, c_2, \dots, c_k$ , not all zero, such that

$$c_1 \mathbf{x}^1(t) + c_2 \mathbf{x}^2(t) + \dots + c_k \mathbf{x}^k(t) = \mathbf{0}.$$

Thus,

$$c_1 \mathbf{x}^1(t_0) + c_2 \mathbf{x}^2(t_0) + \dots + c_k \mathbf{x}^k(t_0) = \mathbf{0}.$$

Hence,  $\mathbf{x}^1(t_0), \mathbf{x}^2(t_0), \dots, \mathbf{x}^k(t_0)$  are linearly dependent.

Conversely, suppose that  $\mathbf{x}^1(t_0), \mathbf{x}^2(t_0), \dots, \mathbf{x}^k(t_0)$  are linearly dependent vectors in  $\mathbf{R}^n$ . Then, there exist constants  $c_1, c_2, \dots, c_k$ , not all zero, such that

$$c_1 \mathbf{x}^1(t_0) + c_2 \mathbf{x}^2(t_0) + \dots + c_k \mathbf{x}^k(t_0) = \mathbf{0}.$$

With these constants  $c_1, c_2, \dots, c_k$  construct the vector-valued function

$$\phi(t) = c_1 \mathbf{x}^1(t) + c_2 \mathbf{x}^2(t) + \dots + c_k \mathbf{x}^k(t).$$

This function satisfies Eq.(3.1.9) since it is a linear combination of solutions of Eq.(3.1.9). Moreover,  $\phi(t_0) = \mathbf{0}$ . Hence, by Theorem 3.1.2,  $\phi(t) \equiv \mathbf{0}$  for all  $t$ . This implies that  $\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^k(t)$  are linearly dependent. ■

**Corollary 3.1.2** Let  $\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^k(t)$  be  $k$  solutions of system (3.1.9). Then,  $\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^k(t)$  are linearly independent if and only if  $\mathbf{x}^1(t_0), \mathbf{x}^2(t_0), \dots, \mathbf{x}^k(t_0)$  are linearly independent for some  $t_0$ .

**Example 3.1.1** Functions

$$\mathbf{x}^1(t) = \begin{pmatrix} e^{-t} \\ -e^{-t} \end{pmatrix} \quad \text{and} \quad \mathbf{x}^2(t) = \begin{pmatrix} te^{-t} \\ (1-t)e^{-t} \end{pmatrix}$$

are two solutions of equation

$$\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}. \quad (3.1.14)$$

Determine whether  $\mathbf{x}^1(t)$  and  $\mathbf{x}^2(t)$  are linearly dependent or linearly independent.

**Solution** Consider vectors

$$\mathbf{x}^1(0) = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{and} \quad \mathbf{x}^2(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Setting

$$c_1 \mathbf{x}^1(0) + c_2 \mathbf{x}^2(0) = \begin{pmatrix} c_1 \\ -c_1 + c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This equation implies that both  $c_1$  and  $c_2$  are zero. Hence,  $\mathbf{x}^1(0)$  and  $\mathbf{x}^2(0)$  are linearly independent vectors in  $\mathbf{R}^2$ . Consequently, by Corollary 3.1.2,  $\mathbf{x}^1(t)$  and  $\mathbf{x}^2(t)$  are linearly independent solutions of differential equation (3.1.14). ■

**Example 3.1.2** Solving the initial-value problem

$$\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix} \mathbf{x}; \quad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

**Solution** From Example 3.1.1, every solution  $\mathbf{x}(t)$  must be of the form

$$\mathbf{x}(t) = c_1 \begin{pmatrix} e^{-t} \\ -e^{-t} \end{pmatrix} + c_2 \begin{pmatrix} te^{-t} \\ (1-t)e^{-t} \end{pmatrix}.$$

Using the initial condition we have that

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \mathbf{x}(0) = \begin{pmatrix} c_1 \\ c_2 - c_1 \end{pmatrix}.$$

Therefore,  $c_1 = 1$  and  $c_2 = 1 + c_1 = 2$ . Hence,

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \begin{pmatrix} (1+2t)e^{-t} \\ (1-2t)e^{-t} \end{pmatrix}. \quad \blacksquare$$

**Exercise 3.1**

1. Convert each of following differential equations into a system of first-order equations.

(a)  $y''' + 4y'' + (y')^3 - 2y = 0$ .

(b)  $y^{(5)} + y' + \cos y = te^t$ .

(c)  $y'' - y^2 = 0$ ;  $y(0) = 1$ ;  $y'(0) = 3/2$ .

(d)  $y''' + (y')^2 + 3y = e^t$ ;  $y(0) = 1$ ,  $y'(0) = 0$ ,  $y''(0) = 0$ .

2. Write each of the given systems of differential equations in the form of a vector equation.

(a)  $\dot{x}_1 = x_2 + x_3$ ,  $\dot{x}_2 = x_1 + x_2 - x_3$ ,  $\dot{x}_3 = x_2 + x_3$ .

(b)  $\dot{x}_1 = 3x_1 - x_2 + x_3$ ,  $\dot{x}_2 = 2x_1 + x_3$ ,  $\dot{x}_3 = x_1 - x_2 + 2x_3$ .

(c)  $\dot{x}_1 = 5x_1 + 5x_2$ ,  $\dot{x}_2 = -x_1 + 7x_2$ ;  $x_1(3) = 0$ ,  $x_2(3) = 6$ .

(d)  $\dot{x}_1 = -x_3$ ,  $\dot{x}_2 = x_1$ ,  $\dot{x}_3 = -x_2$ ;

$$x_1(-1) = 2, \quad x_2(-1) = 3, \quad x_3(-1) = 4.$$

3. (a) Let  $y(t)$  be a solution of the equation  $y'' + y' + y = 0$ . Show that  $\mathbf{x}(t) = \begin{pmatrix} y(t) \\ y'(t) \end{pmatrix}$  is a solution of the system of equations

$$\dot{\mathbf{x}} = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} \mathbf{x}.$$

(b) Let  $\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$  be a solution of the system of equations

$$\dot{\mathbf{x}}(t) = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} \mathbf{x}.$$

Show that  $y = x_1(t)$  is a solution of the equation  $y'' + y' + y = 0$ .

Determine whether the given solutions form a basis for the set of all solutions.

4.  $\dot{\mathbf{x}} = \begin{pmatrix} 4 & -2 & 2 \\ -1 & 3 & 1 \\ 1 & -1 & 5 \end{pmatrix} \mathbf{x}$ ;

$$\mathbf{x}^1(t) = \begin{pmatrix} e^{2t} \\ e^{2t} \\ 0 \end{pmatrix}, \quad \mathbf{x}^2(t) = \begin{pmatrix} 0 \\ e^{4t} \\ e^{4t} \end{pmatrix}, \quad \mathbf{x}^3(t) = \begin{pmatrix} e^{6t} \\ 0 \\ e^{6t} \end{pmatrix}.$$

$$5. \dot{\mathbf{x}} = \begin{pmatrix} -5 & 2 & -2 \\ 1 & -4 & -1 \\ -1 & 1 & -6 \end{pmatrix} \mathbf{x};$$

[6] 求解的特征  
值-特征向  
量法

[7] 标量

$$\mathbf{x}^1(t) = \begin{pmatrix} e^{-3t} \\ e^{-3t} \\ 0 \end{pmatrix}, \quad \mathbf{x}^2(t) = \begin{pmatrix} 0 \\ e^{-5t} \\ e^{-5t} \end{pmatrix},$$

$$\mathbf{x}^3(t) = \begin{pmatrix} e^{-3t} + e^{-7t} \\ e^{-3t} \\ e^{-7t} \end{pmatrix}, \quad \mathbf{x}^4 = \begin{pmatrix} 2e^{-7t} \\ e^{-5t} \\ e^{-5t} + 2e^{-7t} \end{pmatrix}.$$

### 3.2 The Eigenvalue-Eigenvector Method of Finding Solutions [6]

Consider the first-order linear homogeneous differential equation

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}, \quad \mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}. \quad (3.2.1)$$

As what we have done for the first-order and second-order linear homogeneous *scalar*<sup>[7]</sup> equations, we try

$$\mathbf{x}(t) = e^{\lambda t} \mathbf{v}, \quad (3.2.2)$$

where  $\mathbf{v}$  is a constant vector, as a solution of Eq.(3.2.1).

Observe that

$$\frac{d}{dt} e^{\lambda t} \mathbf{v} = \lambda e^{\lambda t} \mathbf{v} = e^{\lambda t} \lambda \mathbf{v}$$

and

$$\mathbf{A} e^{\lambda t} \mathbf{v} = e^{\lambda t} \mathbf{A} \mathbf{v}.$$

Hence,  $\mathbf{x}(t) = e^{\lambda t} \mathbf{v}$  is a solution of Eq.(3.2.1) if and only if

$$e^{\lambda t} \lambda \mathbf{v} = e^{\lambda t} \mathbf{A} \mathbf{v}$$

or

$$\mathbf{A} \mathbf{v} = \lambda \mathbf{v}. \quad (3.2.3)$$

- [8]  $A$  关于特征值  $\lambda$  的一个特征向量  
 [9] 特征多项式

Thus,  $\mathbf{x}(t) = e^{\lambda t} \mathbf{v}$  is a solution of Eq.(3.2.1) if and only if  $\mathbf{v}$  is an eigenvector of  $A$  with eigenvalue  $\lambda$  [8].

### 3.2.1 The Characteristic Polynomial<sup>[9]</sup> of $A$ Has $n$ Distinct Real Eigenvalues

If  $A$  has  $n$  distinct real eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_n, \mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^n$  are eigenvectors of  $A$  with eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_n$  respectively, then, from the theory of linear algebra,  $\mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^n$  are linearly independent. Thus,

$$\mathbf{x}^j(t) = e^{\lambda_j t} \mathbf{v}^j, \quad j = 1, 2, \dots, n \quad (3.2.4)$$

are  $n$  linearly independent solutions of Eq.(3.2.1). In this case, every solution  $\mathbf{x}(t)$  of Eq.(3.2.1) is of the form

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}^1 + c_2 e^{\lambda_2 t} \mathbf{v}^2 + \dots + c_n e^{\lambda_n t} \mathbf{v}^n. \quad (3.2.5)$$

Expression (3.2.5) is the “general solution” of Eq.(3.2.1).

**Example 3.2.1** Solve the initial-value problem

$$\dot{\mathbf{x}} = \begin{pmatrix} 1 & -3 \\ -2 & 2 \end{pmatrix} \mathbf{x}; \quad \mathbf{x}(0) = \begin{pmatrix} 0 \\ 5 \end{pmatrix}.$$

**Solution** The characteristic polynomial is

$$p(\lambda) = \det \begin{pmatrix} 1 - \lambda & -3 \\ -2 & 2 - \lambda \end{pmatrix} = (\lambda - 4)(\lambda + 1).$$

Thus, the eigenvalues of  $A$  are  $\lambda_1 = -1$  and  $\lambda_2 = 4$ .

(i)  $\lambda_1 = -1$ . From equation

$$(A + I)\mathbf{v} = \begin{pmatrix} 2 & -3 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

we obtain that  $v_1 = (3/2)v_2$ . Thus,

$$\mathbf{v}^1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

is an eigenvector of  $A$  with eigenvalue  $-1$ . Therefore,

$$\mathbf{x}^1(t) = e^{-t} \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

is a solution of the given differential equation.

(ii)  $\lambda_2 = 4$ . From equation

$$(\mathbf{A} - 4\mathbf{I})\mathbf{v} = \begin{pmatrix} -3 & -3 \\ -2 & -2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

we have  $v_1 = -v_2$ . Therefore,

$$\mathbf{v}^2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

is an eigenvector of  $\mathbf{A}$  with eigenvalue 4, and

$$\mathbf{x}^2(t) = e^{4t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

is a second solution of the differential equation.

Since  $\lambda_1$  and  $\lambda_2$  are distinct,  $\mathbf{x}^1(t)$  and  $\mathbf{x}^2(t)$  are linearly independent. Hence, the general solution of the original differential equation is

$$\mathbf{x}(t) = c_1 \mathbf{x}^1(t) + c_2 \mathbf{x}^2(t).$$

Using the initial condition we have that

$$\begin{pmatrix} 0 \\ 5 \end{pmatrix} = c_1 \mathbf{x}^1(0) + c_2 \mathbf{x}^2(0) = \begin{pmatrix} 3c_1 \\ 2c_1 \end{pmatrix} + \begin{pmatrix} c_2 \\ -c_2 \end{pmatrix}.$$

Thus,  $c_1 = 1$  and  $c_2 = -3$ . Consequently, the required solution is

$$\mathbf{x}(t) = e^{-t} \begin{pmatrix} 3 \\ 2 \end{pmatrix} - 3e^{4t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3e^{-t} - 3e^{4t} \\ 2e^{-t} + 3e^{4t} \end{pmatrix}. \quad \blacksquare$$

### 3.2.2 The Characteristic Polynomial of $\mathbf{A}$ Has Complex Eigenvalues

If  $\mathbf{A}$  has a complex eigenvalue  $\lambda = \alpha + i\beta$ ,  $\mathbf{v} = \mathbf{v}^1 + i\mathbf{v}^2$  is an eigenvector of  $\mathbf{A}$  with eigenvalue  $\lambda$ , then,  $\mathbf{x}(t) = e^{\lambda t} \mathbf{v}$  is a complex-valued solution of system (3.2.1).

**Lemma 3.2.1** Let  $\mathbf{v}(t) = \mathbf{y}(t) + i\mathbf{z}(t)$  be a complex-valued solution of Eq.(3.2.1). Then, both  $\mathbf{y}(t)$  and  $\mathbf{z}(t)$  are real-valued solutions of Eq.(3.2.1).

**Proof**

$$\dot{\mathbf{y}} + i\dot{\mathbf{z}} = \mathbf{A}(\mathbf{y} + i\mathbf{z}) = \mathbf{A}\mathbf{y} + i\mathbf{A}\mathbf{z}. \quad (3.2.6)$$

Equating real and imaginary parts of Eq.(3.2.6) gives

$$\dot{\mathbf{y}} = \mathbf{A}\mathbf{y}, \quad \dot{\mathbf{z}}(t) = \mathbf{A}\mathbf{z}.$$

Consequently, both

$$\mathbf{y}(t) = \operatorname{Re}\{\mathbf{v}(t)\} \quad \text{and} \quad \mathbf{z}(t) = \operatorname{Im}\{\mathbf{v}(t)\}$$

are real-valued solutions of Eq.(3.2.1). ■By *Euler's* formula, a complex-valued function

$$\mathbf{x}(t) = e^{(\alpha+i\beta)t}(\mathbf{v}^1 + i\mathbf{v}^2)$$

can be written in the form

$$\begin{aligned} \mathbf{x}(t) &= e^{\alpha t}(\cos \beta t + i \sin \beta t)(\mathbf{v}^1 + i\mathbf{v}^2) \\ &= e^{\alpha t}[(\mathbf{v}^1 \cos \beta t - \mathbf{v}^2 \sin \beta t) \\ &\quad + i(\mathbf{v}^1 \sin \beta t + \mathbf{v}^2 \cos \beta t)]. \end{aligned}$$

Hence, if  $\lambda = \alpha + i\beta$  is a complex eigenvalue of  $\mathbf{A}$  corresponding to the eigenvector  $\mathbf{v} = \mathbf{v}^1 + i\mathbf{v}^2$ , then

$$\mathbf{y}(t) = e^{\alpha t}(\mathbf{v}^1 \cos \beta t - \mathbf{v}^2 \sin \beta t) \quad (3.2.7)$$

and

$$\mathbf{z}(t) = e^{\alpha t}(\mathbf{v}^1 \sin \beta t + \mathbf{v}^2 \cos \beta t) \quad (3.2.8)$$

are two real-valued solutions of Eq.(3.2.1). Moreover, these two solutions are obviously linearly independent.

**Example 3.2.2** Solve the initial-value problem

$$\dot{\mathbf{x}} = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix} \mathbf{x}; \quad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

**Solution** The characteristic equation is

$$p(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 3 - \lambda & -2 \\ 4 & -1 - \lambda \end{vmatrix} = \lambda^2 - 2\lambda + 5 = 0.$$

Hence, the eigenvalues of  $A$  are  $\lambda_{1,2} = 1 \pm 2i$ .

For  $\lambda = 1 + 2i$ , from following equation

$$[A - (1 + 2i)I]v = \begin{pmatrix} 3 - (1 + 2i) & -2 \\ 4 & -1 - (1 + 2i) \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

we obtain that  $v_1 = 1 + i$ ,  $v_2 = 2$ . Thus,

$$x(t) = e^{(1+2i)t} \begin{pmatrix} 1 + i \\ 2 \end{pmatrix}$$

is a complex-valued solution of differential equation  $\dot{x} = Ax$ .

Since,

$$\begin{aligned} e^{(1+2i)t} \begin{pmatrix} 1 + i \\ 2 \end{pmatrix} &= e^t (\cos 2t + i \sin 2t) \left[ \begin{pmatrix} 1 \\ 2 \end{pmatrix} + i \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right] \\ &= e^t \begin{pmatrix} \cos 2t - \sin 2t \\ 2 \cos 2t \end{pmatrix} + i e^t \begin{pmatrix} \sin 2t + \cos 2t \\ 2 \sin 2t \end{pmatrix}. \end{aligned}$$

Consequently, by Lemma 3.2.1,

$$x^1(t) = e^t \begin{pmatrix} \cos 2t - \sin 2t \\ 2 \cos 2t \end{pmatrix}, \quad x^2(t) = e^t \begin{pmatrix} \sin 2t + \cos 2t \\ 2 \sin 2t \end{pmatrix}$$

are real-valued solutions. These two solutions  $x^1(t)$ ,  $x^2(t)$  are linearly independent since

$$x^1(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad x^2(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

are linearly independent vectors in  $\mathbf{R}^2$ . Therefore, the general solution of equation  $\dot{x} = Ax$  is

$$x(t) = c_1 e^t \begin{pmatrix} \cos 2t - \sin 2t \\ 2 \cos 2t \end{pmatrix} + c_2 e^t \begin{pmatrix} \sin 2t + \cos 2t \\ 2 \sin 2t \end{pmatrix}.$$

Set  $t = 0$ , we obtain that

$$\begin{pmatrix} 1 \\ 5 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} c_1 + c_2 \\ 2c_1 \end{pmatrix}.$$

Consequently,  $c_1 = 5/2$ ,  $c_2 = -3/2$  and solution of the initial-value problem is

$$x(t) = \frac{5}{2} e^t \begin{pmatrix} \cos 2t - \sin 2t \\ 2 \cos 2t \end{pmatrix} - \frac{3}{2} e^t \begin{pmatrix} \sin 2t + \cos 2t \\ 2 \sin 2t \end{pmatrix}. \quad \blacksquare$$

[10] 矩阵指数函数 3.2.3 The Characteristic Polynomial of  $A$  Has Equal  
 [11] 逐项微分 Eigenvalues

If the characteristic polynomial of  $A$  does not have  $n$  distinct roots, then we would not be able to find  $n$  linearly independent solutions directly from the eigenvalues of  $A$ . Recall that  $e^{at}$  is a solution of the scalar differential equation  $x' = ax$ , this suggests us consider vector function

$$\mathbf{x}(t) = e^{At} \mathbf{v} \quad (3.2.9)$$

as a solution of vector differential equation

$$\dot{\mathbf{x}} = A\mathbf{x}, \quad (3.2.10)$$

where  $e^{At}$  is a *matrix-valued exponential function*<sup>[10]</sup> and  $\mathbf{v}$  is a constant vector.

It can be shown (omitted here) that infinite series

$$I + At + \frac{A^2 t^2}{2!} + \cdots + \frac{A^n t^n}{n!} + \cdots$$

converges to matrix-valued exponential function  $e^{At}$ . That is,

$$e^{At} \equiv I + At + \frac{A^2 t^2}{2!} + \cdots + \frac{A^n t^n}{n!} + \cdots \quad (3.2.11)$$

Moreover, this infinite series can be *differentiated by term*<sup>[11]</sup>. Thus,

$$\begin{aligned} \frac{d}{dt} e^{At} &\equiv A + A^2 t + \cdots + \frac{A^{n+1} t^n}{n!} + \cdots \\ &= A \left[ I + At + \frac{A^2 t^2}{2!} + \cdots + \frac{A^n t^n}{n!} + \cdots \right] = A e^{At}. \end{aligned} \quad (3.2.12)$$

This implies that  $e^{At} \mathbf{v}$  is a solution of Eq.(3.2.9) for every constant vector  $\mathbf{v}$  since

$$\frac{d}{dt} e^{At} \mathbf{v} = A e^{At} \mathbf{v} = A(e^{At} \mathbf{v}).$$

**Remark 3.2.1** The matrix-valued exponential  $e^{At}$  and the scalar exponential  $e^{at}$  satisfy many similar properties.

- (1)  $(e^{At})^{-1} = e^{-At}$ .
- (2)  $e^{A(t+s)} = e^{At}e^{As}$ .
- (3)  $e^{At+Bt} = e^{At}e^{Bt}$  only if  $AB = BA$ .

From these properties we have that

$$\begin{aligned} e^{At}\mathbf{v} &= e^{\lambda t}e^{(A-\lambda I)t}\mathbf{v} \\ &= e^{\lambda t}\left[I + t(A-\lambda I)I + \cdots + \frac{t^{m-1}}{(m-1)!}(A-\lambda I)^{m-1}I + \cdots\right]\mathbf{v} \\ &= e^{\lambda t}\left[\mathbf{v} + t(A-\lambda I)\mathbf{v} + \cdots \right. \\ &\quad \left. + \frac{t^{m-1}}{(m-1)!}(A-\lambda I)^{m-1}\mathbf{v} + \cdots\right] \end{aligned}$$

for any constant  $\lambda$ . Moreover,

(i) If  $(A - \lambda I)^m \mathbf{v} = \mathbf{0}$ , then  $(A - \lambda I)^{m+l} \mathbf{v} = \mathbf{0}$  for every positive integer  $l$  since

$$(A - \lambda I)^{m+l} \mathbf{v} = (A - \lambda I)^l [(A - \lambda I)^m \mathbf{v}].$$

(ii) If  $\mathbf{v}$  satisfies  $(A - \lambda I)^m \mathbf{v} = \mathbf{0}$  for some integer  $m$ , then the infinite series  $e^{(A-\lambda I)t}\mathbf{v}$  terminate after  $m$  terms.

Consequently,

$$\begin{aligned} e^{At}\mathbf{v} &= e^{\lambda t}e^{(A-\lambda I)t}\mathbf{v} \\ &= e^{\lambda t}\left[\mathbf{v} + t(A-\lambda I)\mathbf{v} + \cdots + \frac{t^{m-1}}{(m-1)!}(A-\lambda I)^{m-1}\mathbf{v}\right]. \end{aligned}$$

These suggest us an algorithm for finding  $n$  linearly independent solutions of Eq.(3.2.1) when  $A$  has only  $k < n$  linearly independent eigenvectors.

### Algorithm

(a) Pick an eigenvalue  $\lambda$  of  $A$  and find vector  $\mathbf{v}$  for which

$$(A - \lambda I)^2 \mathbf{v} = \mathbf{0}, \quad (A - \lambda I)\mathbf{v} \neq \mathbf{0}.$$

Then,

$$e^{At}\mathbf{v} = e^{\lambda t}e^{(A-\lambda I)t}\mathbf{v} = e^{\lambda t}[\mathbf{v} + t(A - \lambda I)\mathbf{v}]$$

is an additional solution of  $\dot{\mathbf{x}} = A\mathbf{x}$ . Do this for all eigenvalues  $\lambda$  of  $A$  with multiplicity  $m \geq 2$ .

(b) If we still do not have enough solutions, find vector  $\mathbf{v}$  for which

$$(\mathbf{A} - \lambda \mathbf{I})^3 \mathbf{v} = \mathbf{0}, \quad (\mathbf{A} - \lambda \mathbf{I})^2 \mathbf{v} \neq \mathbf{0}.$$

Then,

$$e^{\mathbf{A}t} \mathbf{v} = e^{\lambda t} \left[ \mathbf{v} + t(\mathbf{A} - \lambda \mathbf{I})\mathbf{v} + \frac{t^2}{2!}(\mathbf{A} - \lambda \mathbf{I})^2 \mathbf{v} \right]$$

is an additional solution of  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ . Do this for all eigenvalues  $\lambda$  of  $\mathbf{A}$  with multiplicity  $m \geq 3$ .

(c) Keeping this proceeding until we obtain  $n$  linearly independent solutions.

**Example 3.2.3** Find three linearly independent solutions of differential equation

$$\dot{\mathbf{x}} = \begin{pmatrix} 0 & -1 & 1 \\ 2 & -3 & 1 \\ 1 & -1 & -1 \end{pmatrix} \mathbf{x}.$$

**Solution** The characteristic equation

$$|\mathbf{A} - \lambda \mathbf{I}| = \begin{vmatrix} -\lambda & -1 & 1 \\ 2 & -3 - \lambda & 1 \\ 1 & -1 & -1 - \lambda \end{vmatrix} = (\lambda + 1)^2(\lambda + 2) = 0$$

has roots  $\lambda_{1,2} = -1, \lambda_3 = -2$ .

(1)  $\lambda = -1$ . Solving equation

$$(\mathbf{A} + \mathbf{I})\mathbf{v} = \begin{pmatrix} 1 & -1 & 1 \\ 2 & -2 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

gives  $v_1 = v_2 = 1, v_3 = 0$ . Thus,

$$\mathbf{x}^1(t) = e^{-t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

is one solution of  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ .

(2)  $\lambda = -2$ . Solving equation

$$(\mathbf{A} + 2\mathbf{I})\mathbf{v} = \begin{pmatrix} 2 & -1 & 1 \\ 2 & -1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

gives  $v_1 = 0$ ,  $v_2 = v_3 = 1$ . Hence,

$$\mathbf{x}^2(t) = e^{-2t} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

is a second linearly independent solution.

(3) For  $\lambda = -1$ , we look for all solutions of equation

$$(\mathbf{A} + \mathbf{I})^2\mathbf{v} = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

From this equation we obtain that  $v_1 = v_2$ ,  $v_2$  and  $v_3$  are arbitrary.

Since the vector

$$\mathbf{v} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

satisfies  $(\mathbf{A} + \mathbf{I})^2\mathbf{v} = \mathbf{0}$  but

$$(\mathbf{A} + \mathbf{I})\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \neq \mathbf{0}.$$

Hence,

$$\mathbf{x}^3(t) = e^{\mathbf{A}t} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = e^{-t}[\mathbf{I} + t(\mathbf{A} + \mathbf{I})] \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = e^{-t} \begin{pmatrix} t \\ t \\ 1 \end{pmatrix}$$

is a third linearly independent solution. ■

**Exercise 3.2**

Find the general solution of each of following differential equations.

$$1. \dot{\mathbf{x}} = \begin{pmatrix} 4 & 2 \\ 3 & -1 \end{pmatrix} \mathbf{x}.$$

$$2. \dot{\mathbf{x}} = \begin{pmatrix} 6 & -3 \\ 2 & 1 \end{pmatrix} \mathbf{x}.$$

$$3. \dot{\mathbf{x}} = \begin{pmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{pmatrix} \mathbf{x}.$$

$$4. \dot{\mathbf{x}} = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix} \mathbf{x}.$$

$$5. \dot{\mathbf{x}} = \begin{pmatrix} 4 & -3 \\ 3 & 4 \end{pmatrix} \mathbf{x}.$$

$$6. \dot{\mathbf{x}} = \begin{pmatrix} -3 & 2 \\ -1 & -1 \end{pmatrix} \mathbf{x}.$$

$$7. \dot{\mathbf{x}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix} \mathbf{x}.$$

$$8. \dot{\mathbf{x}} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & -2 \\ 2 & 2 & 1 \end{pmatrix} \mathbf{x}.$$

$$9. \dot{\mathbf{x}} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \mathbf{x}.$$

$$10. \dot{\mathbf{x}} = \begin{pmatrix} -1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \mathbf{x}.$$

Solve each of following initial-value problems.

$$11. \dot{\mathbf{x}} = \begin{pmatrix} 1 & 12 \\ 3 & 1 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

$$12. \dot{\mathbf{x}} = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

$$13. \dot{\mathbf{x}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

$$14. \dot{\mathbf{x}} = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 3 & 3 & -1 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}.$$

$$15. \dot{\mathbf{x}} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

$$16. \dot{\mathbf{x}} = \begin{pmatrix} -3 & 0 & 2 \\ 1 & -1 & 0 \\ -2 & -1 & 0 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 0 \\ -1 \\ -2 \end{pmatrix}.$$

$$17. \dot{\mathbf{x}} = \begin{pmatrix} -1 & 1 & 2 \\ -1 & 1 & 1 \\ -2 & 1 & 3 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

$$18. \dot{\mathbf{x}} = \begin{pmatrix} 1 & 2 & -3 \\ 1 & 1 & 2 \\ 1 & -1 & 4 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

### 3.3 Fundamental Matrix Solution<sup>[12]</sup>; [12] 基本矩阵解

#### Matrix-valued Exponential Function $e^{At}$

Let  $\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^n(t)$  be  $n$  linearly independent solutions of differential equation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \quad (3.3.1)$$

then every solution  $\mathbf{x}(t)$  of Eq.(3.3.1) can be written in the form

$$\mathbf{x}(t) = c_1 \mathbf{x}^1(t) + c_2 \mathbf{x}^2(t) + \dots + c_n \mathbf{x}^n(t), \quad (3.3.2)$$

or in a concise form

$$\mathbf{x}(t) = \mathbf{X}(t)\mathbf{c}, \quad (3.3.3)$$

where

$$\mathbf{X}(t) = (\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^n(t)), \quad \mathbf{c} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}.$$

Matrix  $\mathbf{X}(t)$  is referred to as a **fundamental matrix solution** of Eq.(3.3.1).

**Example 3.3.1** The system of differential equations

$$\dot{\mathbf{x}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -1 & 2 \end{pmatrix} \mathbf{x}$$

has three linearly independent solutions

$$e^{2t} \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}, \quad \begin{pmatrix} \cos t \\ -\sin t \\ -\cos t \end{pmatrix}, \quad \begin{pmatrix} \sin t \\ \cos t \\ -\sin t \end{pmatrix}.$$

Hence,

$$\mathbf{X}(t) = \begin{pmatrix} e^{2t} & \cos t & \sin t \\ 2e^{2t} & -\sin t & \cos t \\ 4e^{2t} & -\cos t & -\sin t \end{pmatrix}$$

is a fundamental matrix solution of the equation. ■

**Theorem 3.3.1** Let  $\mathbf{X}(t)$  be a fundamental matrix solution of the differential equation  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ . Then,

$$\mathbf{e}^{\mathbf{A}t} = \mathbf{X}(t)\mathbf{X}^{-1}(0). \quad (3.3.4)$$

**Lemma 3.3.1** A matrix  $\mathbf{X}(t)$  is a fundamental matrix solution of Eq.(3.3.1) if and only if

$$\dot{\mathbf{X}}(t) = \mathbf{A}\mathbf{X}(t) \quad \text{and} \quad \det \mathbf{X}(t) \neq 0.$$

**Proof** Let  $\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^n(t)$  denote the  $n$  columns of  $\mathbf{X}(t)$ , that is,

$$\mathbf{X}(t) = (\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^n(t)).$$

Observe that

$$\dot{\mathbf{X}}(t) = (\dot{\mathbf{x}}^1(t), \dot{\mathbf{x}}^2(t), \dots, \dot{\mathbf{x}}^n(t))$$

and

$$\mathbf{A}\mathbf{X}(t) = (\mathbf{A}\mathbf{x}^1(t), \mathbf{A}\mathbf{x}^2(t), \dots, \mathbf{A}\mathbf{x}^n(t)).$$

Hence,

$$\dot{\mathbf{X}}(t) = \mathbf{A}\mathbf{X}(t) \text{ if and only if } \dot{\mathbf{x}}^j(t) = \mathbf{A}\mathbf{x}^j(t), \quad j = 1, 2, \dots, n.$$

Moreover,  $n$  solutions  $\mathbf{x}^1(t), \mathbf{x}^2(t), \dots, \mathbf{x}^n(t)$  of Eq.(3.3.1) are linearly independent if and only if  $\mathbf{x}^1(0), \mathbf{x}^2(0), \dots, \mathbf{x}^n(0)$  are linearly independent vectors in  $\mathbf{R}^n$ , and therefore, if and only if  $\det \mathbf{X}(0) \neq 0$ . Consequently,  $\mathbf{X}(t)$  is a fundamental matrix solution of Eq.(3.3.1) if and only if  $\dot{\mathbf{X}}(t) = \mathbf{A}\mathbf{X}(t)$  and  $\det \mathbf{X}(t) \neq 0$ . ■

**Lemma 3.3.2** The matrix-valued exponential function

$$\mathbf{e}^{\mathbf{A}t} \equiv \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2 t^2}{2!} + \dots + \frac{\mathbf{A}^n t^n}{n!} + \dots$$

is a fundamental matrix solution of Eq.(3.3.1).

**Proof** From (3.2.12)

$$\frac{d}{dt}\mathbf{e}^{\mathbf{A}t} = \mathbf{A}\mathbf{e}^{\mathbf{A}t}$$

we know that  $e^{At}$  is a solution of matrix differential equation

$$\dot{X}(t) = AX(t).$$

Moreover,  $\det(e^{A0}) = 1 \neq 0$  since  $e^{A0} = I$ . Therefore, by Lemma 3.3.1,  $e^{At}$  is a fundamental matrix solution of Eq.(3.3.1). ■

**Lemma 3.3.3** Let  $X(t)$  and  $Y(t)$  be two fundamental matrix solutions of Eq.(3.3.1). Then, there exists a constant matrix  $C$  such that  $Y(t) = X(t)C$ .

**Proof** By definition, the columns  $x^1(t), \dots, x^n(t)$  of  $X(t)$  and  $y^1(t), \dots, y^n(t)$  of  $Y(t)$  are linearly independent solutions of Eq. (3.3.1). Therefore, there exist constants  $c_1^j, \dots, c_n^j$  such that

$$y^j(t) = c_1^j x^1(t) + \dots + c_n^j x^n(t), \quad j = 1, 2, \dots, n. \quad (3.3.5)$$

Set  $C = (c^1, \dots, c^n)$  where  $c^j = (c_1^j, \dots, c_n^j)$ . Then, Eq.(3.3.5) can be expressed in the form  $Y(t) = X(t)C$ . ■

**Proof of Theorem 3.3.1** Let  $X(t)$  be a fundamental matrix solution of Eq.(3.3.1). Then, by Lemma 3.3.3, there exists a constant matrix  $C$  such that

$$e^{At} = X(t)C. \quad (3.3.6)$$

Setting  $t = 0$  in Eq.(3.3.6) gives  $I = X(0)C$ , which implies that  $C = X^{-1}(0)$ . Hence,

$$e^{At} = X(t)X^{-1}(0). \quad \blacksquare$$

**Example 3.3.2** Compute  $e^{At}$  if

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & -2 & 0 \end{pmatrix}.$$

**Solution** Consider differential equation

$$\dot{x} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & -2 & 1 \end{pmatrix} x. \quad (3.3.7)$$

From

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{pmatrix} 2 - \lambda & 0 & 1 \\ 1 & -\lambda & 1 \\ 1 & -2 & -\lambda \end{pmatrix} = (2 - \lambda)(\lambda^2 + 1)$$

we know that  $\mathbf{A}$  has three distinct eigenvalues  $2, \pm i3$ .

For  $\lambda = i$ , from

$$(\mathbf{A} - i\mathbf{I})\mathbf{v} = \begin{pmatrix} 2 - i & 0 & 1 \\ 1 & -i & 1 \\ 1 & -2 & -i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \mathbf{0}$$

we obtain that

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 + i \\ -2 + i \end{pmatrix}.$$

So Eq.(3.3.7) has a complex-valued solution

$$\mathbf{x}(t) = e^{it} \begin{pmatrix} 1 \\ 1 + i \\ -2 + i \end{pmatrix}.$$

By *Euler's* formula,

$$\begin{aligned} \mathbf{x}(t) &= e^{it} \begin{pmatrix} 1 \\ 1 + i \\ -2 + i \end{pmatrix} = (\cos t + i \sin t) \left( \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right) \\ &= \begin{pmatrix} \cos t \\ \cos t - \sin t \\ -2 \cos t - \sin t \end{pmatrix} + i \begin{pmatrix} \sin t \\ \sin t + \cos t \\ -2 \sin t + \cos t \end{pmatrix}. \end{aligned}$$

Thus, we obtain two linearly independent solutions of Eq. (3.3.7)

$$\mathbf{x}^1(t) = \begin{pmatrix} \cos t \\ \cos t - \sin t \\ -2 \cos t - \sin t \end{pmatrix}, \quad \mathbf{x}^2(t) = \begin{pmatrix} \sin t \\ \sin t + \cos t \\ -2 \sin t + \cos t \end{pmatrix}.$$

Similarly, the eigenvector of  $A$  with eigenvalue 2 is

$$v = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}.$$

Hence, the third required solution is

$$x^3(t) = e^{2t} \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}.$$

Therefore, a fundamental matrix solution is

$$X(t) = \begin{pmatrix} \cos t & \sin t & 2e^{2t} \\ \cos t - \sin t & \sin t + \cos t & e^{2t} \\ -2\cos t - \sin t & -2\sin t + \cos t & 0 \end{pmatrix}.$$

Computing

$$X^{-1}(0) = \begin{pmatrix} -1/5 & 2/5 & -2/5 \\ -2/5 & 4/5 & 1/5 \\ 3/5 & -1/5 & 1/5 \end{pmatrix}.$$

Consequently,

$$\begin{aligned} e^{At} &= X(t)X^{-1}(0) \\ &= \begin{pmatrix} \cos t & \sin t & 2e^{2t} \\ \cos t - \sin t & \sin t + \cos t & e^{2t} \\ -2\cos t - \sin t & -2\sin t + \cos t & 0 \end{pmatrix} \cdot \begin{pmatrix} -1/5 & 2/5 & -2/5 \\ -2/5 & 4/5 & 1/5 \\ 3/5 & -1/5 & 1/5 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{5}(6e^{2t} - \cos t - 2\sin t) & \frac{1}{5}(-2e^{2t} + 2\cos t + 4\sin t) & \frac{1}{5}(2e^{2t} - 2\cos t + \sin t) \\ \frac{1}{5}(3e^{2t} - 3\cos t - \sin t) & \frac{1}{5}(-e^{2t} + 6\cos t + 2\sin t) & \frac{1}{5}(e^{2t} - \cos t + 3\sin t) \\ \sin t & -\frac{8}{5}\sin t & \cos t \end{pmatrix}. \quad \blacksquare \end{aligned}$$

### Exercise 3.3

Find a fundamental matrix solution for each of following systems of differential equations.

$$1. \dot{x} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} x.$$

$$2. \dot{\mathbf{x}} = \begin{pmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{pmatrix} \mathbf{x}.$$

$$3. \dot{\mathbf{x}} = \begin{pmatrix} -3 & -2 & 3 \\ 1 & 0 & -3 \\ 1 & -2 & -1 \end{pmatrix} \mathbf{x}.$$

Compute  $e^{\mathbf{A}t}$  for each of following matrices  $\mathbf{A}$ .

$$4. \mathbf{A} = \begin{pmatrix} 3 & 4 & 5 \\ 0 & 5 & 4 \\ 0 & 0 & 3 \end{pmatrix}.$$

$$5. \mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 5 \end{pmatrix}.$$

$$6. \mathbf{A} = \begin{pmatrix} 0 & 2 & 1 \\ -1 & -3 & -1 \\ 1 & 1 & -1 \end{pmatrix}.$$

In each of Problems 7-9, determine whether the given matrix is a fundamental matrix solution of  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$  for some  $\mathbf{A}$ ; if yes, find  $\mathbf{A}$ .

$$7. \begin{pmatrix} e^t + e^{2t} & e^t + e^{3t} & e^t - e^{3t} \\ e^{2t} & e^{3t} & -e^{3t} \\ 0 & e^{3t} & -e^{3t} \end{pmatrix}.$$

$$8. \begin{pmatrix} e^t & e^{2t} & e^{3t} \\ e^t & 2e^{2t} & 3e^{3t} \\ e^t & 4e^{2t} & 9e^{3t} \end{pmatrix}.$$

$$9. \begin{pmatrix} e^{2t} & 2e^{-t} & e^{3t} \\ 2e^{2t} & 2e^{-t} & e^{3t} \\ 3e^{2t} & e^{-t} & 2e^{3t} \end{pmatrix}.$$

10. Let  $\phi^j(t)$  be the solution of initial-value problem  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ ;  $\mathbf{x}(0) = \mathbf{e}^j$ . Show that  $e^{\mathbf{A}t} = (\phi^1, \phi^2, \dots, \phi^n)$ .

11. Suppose that  $\mathbf{Y}(t) = \mathbf{X}(t)\mathbf{C}$ , where  $\mathbf{X}(t)$  and  $\mathbf{Y}(t)$  are fundamental matrix solutions of  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ , and  $\mathbf{C}$  is a constant matrix. Prove that  $\det \mathbf{C} \neq 0$ .

12. Let  $\mathbf{X}(t)$  be a fundamental matrix solution of Eq.(3.3.1), and  $\mathbf{C}$  a constant matrix with  $\det \mathbf{C} \neq 0$ . Show that  $\mathbf{Y}(t) = \mathbf{X}(t)\mathbf{C}$  is again a fundamental matrix solution of Eq.(3.3.1).

13. Let  $\mathbf{X}(t)$  be a fundamental matrix solution of Eq.(3.3.1). Prove that the solution  $\mathbf{x}(t)$  of initial-value problem  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ ;  $\mathbf{x}(t_0) =$

$\mathbf{x}^0$  is

$$\mathbf{x}(t) = \mathbf{X}(t)\mathbf{X}^{-1}(t_0)\mathbf{x}^0.$$

14. Let  $\mathbf{X}(t)$  be a fundamental matrix solution of Eq.(3.3.1). Prove that

$$\mathbf{X}(t)\mathbf{X}^{-1}(t_0) = e^{\mathbf{A}(t-t_0)}.$$

### 3.4 Nonhomogeneous Equations; Variation of Parameters

Consider the initial-value problem

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{f}(t); \quad \mathbf{x}(t_0) = \mathbf{x}^0. \quad (3.4.1)$$

Let  $\mathbf{x}^1(t), \dots, \mathbf{x}^n(t)$  be  $n$  linearly independent solutions of homogeneous equation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \quad (3.4.2)$$

then the general solution of Eq.(3.4.2) is

$$c_1\mathbf{x}^1(t) + \dots + c_n\mathbf{x}^n(t).$$

**The method of variation of parameters** for finding a solution of initial-value problem (3.4.1) is to set a solution of Eq.(3.4.1) of the form

$$\mathbf{x}(t) = u_1(t)\mathbf{x}^1(t) + u_2(t)\mathbf{x}^2(t) + \dots + u_n(t)\mathbf{x}^n(t), \quad (3.4.3)$$

where  $u_1(t), \dots, u_n(t)$  are undetermined functions. Eq.(3.4.3) can be written concisely in the form

$$\mathbf{x}(t) = \mathbf{X}(t)\mathbf{u}(t), \quad (3.4.4)$$

where

$$\mathbf{X}(t) = (\mathbf{x}^1(t), \dots, \mathbf{x}^n(t)), \quad \mathbf{u}(t) = \begin{pmatrix} u_1(t) \\ \vdots \\ u_n(t) \end{pmatrix}.$$

Plugging expression (3.4.4) into Eq.(3.4.1) gives

$$\dot{\mathbf{X}}(t)\mathbf{u}(t) + \mathbf{X}(t)\dot{\mathbf{u}}(t) = \mathbf{A}\mathbf{X}(t)\mathbf{u}(t) + \mathbf{f}(t). \quad (3.4.5)$$

Note that matrix  $\mathbf{X}(t)$  is a fundamental matrix solution of Eq. (3.4.2), by Lemma 3.3.1,

$$\dot{\mathbf{X}}(t) = \mathbf{A}\mathbf{X}(t).$$

Thus, Eq.(3.4.5) can be reduced to

$$\mathbf{X}(t)\dot{\mathbf{u}}(t) = \mathbf{f}(t). \quad (3.4.6)$$

Since the columns of  $\mathbf{X}(t)$  are linearly independent vectors in  $\mathbf{R}^n$  at every  $t$ ,  $\mathbf{X}^{-1}(t)$  exists, therefore,

$$\dot{\mathbf{u}}(t) = \mathbf{X}^{-1}(t)\mathbf{f}(t). \quad (3.4.7)$$

Integrating last expression between  $t_0$  to  $t$  gives

$$\begin{aligned} \mathbf{u}(t) &= \mathbf{u}(t_0) + \int_{t_0}^t \mathbf{X}^{-1}(s)\mathbf{f}(s)ds \\ &= \mathbf{X}^{-1}(t_0)\mathbf{x}^0 + \int_{t_0}^t \mathbf{X}^{-1}(s)\mathbf{f}(s)ds. \end{aligned}$$

Consequently,

$$\mathbf{x}(t) = \mathbf{X}(t)\mathbf{X}^{-1}(t_0)\mathbf{x}^0 + \mathbf{X}(t) \int_{t_0}^t \mathbf{X}^{-1}(s)\mathbf{f}(s)ds. \quad (3.4.8)$$

Moreover, if  $\mathbf{X}(t) = e^{\mathbf{A}t}$ , then

$$\begin{aligned} \mathbf{x}(t) &= e^{\mathbf{A}t}e^{-\mathbf{A}t_0}\mathbf{x}^0 + e^{\mathbf{A}t} \int_{t_0}^t e^{-\mathbf{A}s}\mathbf{f}(s)ds \\ &= e^{\mathbf{A}(t-t_0)}\mathbf{x}^0 + \int_{t_0}^t e^{\mathbf{A}(t-s)}\mathbf{f}(s)ds. \end{aligned} \quad (3.4.9)$$

**Example 3.4.1** Solve the initial-value problem

$$\dot{\mathbf{x}}(t) = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & -2 \\ 3 & 2 & 1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 0 \\ 0 \\ e^t \cos 2t \end{pmatrix}; \quad \mathbf{x}(0) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

**Solution** Computing

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{pmatrix} 1 - \lambda & 0 & 0 \\ 2 & 1 - \lambda & -2 \\ 3 & 2 & 1 - \lambda \end{pmatrix} = (1 - \lambda)(\lambda^2 - 2\lambda + 5),$$

the eigenvalues of  $\mathbf{A}$  are

$$\lambda_1 = 1, \quad \lambda_{2,3} = \frac{2 \pm \sqrt{4 - 20}}{2} = 1 \pm 2i.$$

(i)  $\lambda = 1$ . Solve equation

$$(\mathbf{A} - \mathbf{I})\mathbf{v} = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & -2 \\ 3 & 2 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

we obtain that  $v_1 = v_3$  and  $v_2 = -3v_1/2$ . Hence,

$$\mathbf{v} = \begin{pmatrix} 2 \\ -3 \\ 2 \end{pmatrix}$$

is an eigenvector of  $\mathbf{A}$  with eigenvalue 1, and

$$\mathbf{x}^1(t) = e^t \begin{pmatrix} 2 \\ -3 \\ 2 \end{pmatrix}$$

is a solution of the homogeneous equation  $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$ .

(ii)  $\lambda = 1 + 2i$ . From

$$[\mathbf{A} - (1 + 2i)\mathbf{I}]\mathbf{v} = \begin{pmatrix} -2i & 0 & 0 \\ 2 & -2i & -2 \\ 3 & 2 & -2i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

we obtain that  $v_1 = 0$  and  $v_3 = -iv_2$ . Hence,

$$\mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ -i \end{pmatrix}$$

is an eigenvector of  $\mathbf{A}$  with eigenvalue  $1 + 2i$ , and

$$\mathbf{x}(t) = e^{(1+2i)t} \begin{pmatrix} 0 \\ 1 \\ -i \end{pmatrix}$$

is a complex-valued solution of  $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$ . By *Euler's* formula,

$$\begin{aligned} e^{(1+2i)t} \begin{pmatrix} 0 \\ 1 \\ -i \end{pmatrix} &= e^t (\cos 2t + i \sin 2t) \left[ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - i \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right] \\ &= e^t \left[ \cos 2t \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \sin 2t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right] \\ &\quad + i e^t \left[ \sin 2t \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \cos 2t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right], \end{aligned}$$

hence

$$\mathbf{x}^2(t) = e^t \begin{pmatrix} 0 \\ \cos 2t \\ \sin 2t \end{pmatrix}, \quad \mathbf{x}^3(t) = e^t \begin{pmatrix} 0 \\ \sin 2t \\ -\cos 2t \end{pmatrix}$$

are two real-valued solutions of  $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$ , and  $\mathbf{x}^1, \mathbf{x}^2, \mathbf{x}^3$  are linearly independent. Therefore,

$$\mathbf{X}(t) = \begin{pmatrix} 2e^t & 0 & 0 \\ -3e^t & e^t \cos 2t & e^t \sin 2t \\ 2e^t & e^t \sin 2t & -e^t \cos 2t \end{pmatrix}$$

is a fundamental matrix solution of  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ . Computing

$$\mathbf{X}^{-1}(0) = \begin{pmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 0 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} 1/2 & 0 & 0 \\ 3/2 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix},$$

then

$$\begin{aligned} e^{\mathbf{A}t} &= \begin{pmatrix} 2e^t & 0 & 0 \\ -3e^t & e^t \cos 2t & e^t \sin 2t \\ 2e^t & e^t \sin 2t & -e^t \cos 2t \end{pmatrix} \begin{pmatrix} 1/2 & 0 & 0 \\ 3/2 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix} \\ &= e^t \begin{pmatrix} 1 & 0 & 0 \\ -\frac{3}{2} + \frac{3}{2} \cos 2t + \sin 2t & \cos 2t & -\sin 2t \\ 1 + \frac{3}{2} \sin 2t - \cos 2t & \sin 2t & \cos 2t \end{pmatrix}. \end{aligned}$$

Consequently,

$$\begin{aligned} \mathbf{x}(t) &= e^{\mathbf{A}t} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \\ &+ e^{\mathbf{A}t} \int_0^t e^{-s} \begin{pmatrix} 1 & 0 & 0 \\ -\frac{3}{2} + \frac{3}{2} \cos 2s + \sin 2s & \cos 2s & -\sin 2s \\ 1 + \frac{3}{2} \sin 2s - \cos 2s & \sin 2s & \cos 2s \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ e^s \cos 2s \end{pmatrix} ds \\ &= e^t \begin{pmatrix} 0 \\ \cos 2t - \sin 2t \\ \cos 2t + \sin 2t \end{pmatrix} + e^{\mathbf{A}t} \int_0^t \begin{pmatrix} 0 \\ \sin 2s \cos 2s \\ \cos^2 s \end{pmatrix} ds \\ &= e^t \begin{pmatrix} 0 \\ \cos 2t - \sin 2t \\ \cos 2t + \sin 2t \end{pmatrix} + e^{\mathbf{A}t} \begin{pmatrix} 0 \\ (1 - \cos 4t)/8 \\ t/2 + (\sin 4t)/8 \end{pmatrix} \\ &= e^t \begin{pmatrix} 0 \\ \cos 2t - \sin 2t \\ \cos 2t + \sin 2t \end{pmatrix} \\ &+ e^t \begin{pmatrix} 0 \\ -\frac{t \sin 2t}{2} + \frac{\cos 2t - \cos 4t \cos 2t - \sin 4t \sin 2t}{8} \\ \frac{t \cos 2t}{2} + \frac{\sin 4t \cos 2t - \sin 2t \cos 4t + \sin 2t}{8} \end{pmatrix} \end{aligned}$$

$$= e^t \begin{pmatrix} 0 \\ \cos 2t - \left(1 + \frac{1}{2}t\right) \sin 2t \\ \left(1 + \frac{1}{2}t\right) \cos 2t + \frac{5}{4} \sin 2t \end{pmatrix}.$$

**Exercise 3.4** ■

Solve following initial-value problems by the method of variation of parameters.

1.  $\dot{x} = \begin{pmatrix} 0 & 1 \\ 8 & -2 \end{pmatrix} x + \begin{pmatrix} 0 \\ e^t \end{pmatrix}; \quad x(0) = \begin{pmatrix} 1 \\ -4 \end{pmatrix}.$

2.  $\dot{x} = \begin{pmatrix} 4 & 5 \\ -2 & -2 \end{pmatrix} x + \begin{pmatrix} 4e^t \cos t \\ 0 \end{pmatrix}; \quad x(0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$

3.  $\dot{x} = \begin{pmatrix} 0 & 1 \\ -9 & 6 \end{pmatrix} x + \begin{pmatrix} 0 \\ t \end{pmatrix}; \quad x(0) = \begin{pmatrix} 0 \\ t \end{pmatrix}.$

4.  $\dot{x} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix} x + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{2t}; \quad x(0) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$

5.  $\dot{x} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & -2 & -5 \\ 0 & 1 & 2 \end{pmatrix} x + \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} e^t; \quad x(0) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$

6. Find a particular solution of the differential equation

$$\dot{x}(t) = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & -2 \\ 3 & 2 & 1 \end{pmatrix} x + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^{ct}, \quad c \neq 1$$

by “guessing” a particular solution  $\varphi(t)$  of the form  $\varphi(t) = be^{ct}$ .

7. Consider the differential equation

$$\dot{x}(t) = Ax(t) + ve^{\lambda t}, \quad (3.4.10)$$

where  $v$  is an eigenvector of  $A$  with eigenvalue  $\lambda$ . Suppose, moreover, that  $A$  has  $n$  linearly independent eigenvectors  $v^1, v^2, \dots, v^n$  with distinct eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_n$  respectively.

(a) Show that (3.4.10) has no solution  $\psi(t)$  of the form  $\psi(t) = ae^{\lambda t}$ .

*Hint:* Write  $a = a_1 v^1 + \dots + a_n v^n$ .

(b) Show that (3.4.10) has a solution  $\psi(t)$  of the form

$$\psi(t) = ae^{\lambda t} + bte^{\lambda t}.$$

*Hint:* Show that  $b$  is an eigenvector of  $A$  with eigenvalue  $\lambda$  and choose it so that we can solve the equation

$$(A - \lambda I)a = b - v.$$

## 3.5 Applications

### Module 1 The Principle of Competitive Exclusion in Population Biology<sup>[13]</sup>

It is often observed in nature that the struggle for existence between two similar *species*<sup>[14]</sup> competing for the same limited food supply and living space nearly always ends in the complete *extinction*<sup>[15]</sup> of one of the species. This phenomenon is known as the “principle of competitive exclusion”, and was first enunciated, in a slightly different form, by Darwin in 1859. In his paper, “*The origin of species by natural selection*<sup>[16]</sup>”, Darwin writes:

“As the species of the same genus usually have, though by no means invariably, much similarity in habits and constitutions<sup>[17]</sup> and always in structure, the struggle will generally be more severe between them, if they come into competition with each other, than between the species of distinct genera<sup>[18]</sup>.”

There is a very interesting biological explanation of the principle of competitive exclusion. The *cornerstone*<sup>[19]</sup> of this theory is the idea of “*niche*<sup>[20]</sup>.” A niche indicates what place a given species occupies in a community; i.e., what are its habits, food, and mode of life. It has been observed that as a result of competition two similar species rarely occupy the same niche. Rather, each species *takes possession of*<sup>[21]</sup> those kinds of food and modes of life in which it *has an advantage over its competitor*<sup>[22]</sup>. If the two species tend to occupy the same niche, then the struggle for existence between them will be *intense*<sup>[23]</sup> and result in the extinction of the weaker species.

[13] 种群生态学中的竞争排斥原理

[14] 物种

[15] 消失、灭亡

[16] 基于自然选择的物种起源

[17] 身体结构

[18] 基因

[19] 基石

[20] 龕、生态龕

[21] 拥有、占有

[22] 具有超越它的竞争者的优势

[23] 激烈的

- [24] 领地
- [25] 燕鸥
- [26] 白嘴端燕鸥
- [27] 普通燕鸥
- [28] 黑啄燕鸥
- [29] 联合起来把捕食者赶出它们的领地
- [30] 肩并肩地
- [31] 刻画
- [32] 种群增长的逻辑率
- [33] 小环境

An excellent illustration of this theory is the colony<sup>[24]</sup> of terns<sup>[25]</sup> inhabiting the island of Jorilgatch in the Black Sea. This colony consists of four different species of terns: *sandwich tern*<sup>[26]</sup>, *common tern*<sup>[27]</sup>, *blackbeak tern*<sup>[28]</sup>, and little tern. These four species *band together to chase away predators from the colony*<sup>[29]</sup>. However, sharp differences exist among them as regards the procuring of food. The sandwich tern flies far out into the open sea to hunt certain species, while the blackbeak tern feeds exclusively on land. The common tern and little tern catch fish close to the shore. They sight the fish while flying and dive into the water after them. The little tern seizes its fish in shallow swampy places, whereas the common tern hunts somewhat further from shore. In this manner, these four similar species of tern living *side by side*<sup>[30]</sup> upon a single small island differ sharply in all their modes of feeding and procuring food. Each has a niche in which it has a distinct advantage over its competitors.

Now we derive a system of differential equations which *govern*<sup>[31]</sup> the interaction between two similar species and then show that every solution of this system approaches an equilibrium state in which one of the species is extinct.

Look again at *the logistic law of population growth*<sup>[32]</sup>

$$\frac{dN}{dt} = aN - bN^2. \quad (3.5.1)$$

This equation governs the growth of the population  $N(t)$  of a single species whose members compete among themselves for a limited amount of food and living space. Note that  $N(t)$  approaches the limiting population  $K = a/b$  as  $t$  approaches infinity. This limiting population can be thought of as the maximum population of the species which *the microcosm*<sup>[33]</sup> can support. In terms of  $K$ , the logistic law (3.5.1) can be rewritten in the form

$$\frac{dN}{dt} = aN \left( 1 - \frac{b}{a} N \right) = aN \left( 1 - \frac{N}{K} \right) = aN \left( \frac{K - N}{K} \right). \quad (3.5.2)$$

The biological interpretation about Eq.(3.5.2) is that when the population  $N$  is very low, it grows according to the *Malthusian law*<sup>[34]</sup>  $dN/dt = aN$ . The term  $aN$  is called the “*biotic potential*”<sup>[35]</sup> of the species. It is the *potential rate of increase*<sup>[36]</sup> of the species under ideal conditions, and it is realized that if no restrictions on food and living space exist and if the individual members of the species *do not excrete any toxic waste products*<sup>[37]</sup>. As the population increases though, the biotic potential is reduced by the factor  $(K - N)/K$ , which is the relative number of still vacant places in the microcosm. Ecologists call this factor the *environmental resistance to growth*<sup>[38]</sup>.

Now, let  $N_1(t)$  and  $N_2(t)$  be the populations at time  $t$  of species 1 and 2, respectively. Further, let  $K_1$  and  $K_2$  be the maximum populations of species 1 and 2 which the microcosm can support, and  $a_1N_1$  and  $a_2N_2$  be the biotic potentials of species 1 and 2. Then,  $N_1(t)$  and  $N_2(t)$  satisfy the system of differential equations

$$\begin{cases} \frac{dN_1}{dt} = a_1N_1 \left( \frac{K_1 - N_1 - m_2}{K_1} \right), \\ \frac{dN_2}{dt} = a_2N_2 \left( \frac{K_2 - N_2 - m_1}{K_2} \right), \end{cases} \quad (3.5.3)$$

where  $m_2$  is the total number of places of the species 1 which are taken up by members of the species 2, and  $m_1$  is the total number of places of the species 2 which are taken up by members of the species 1. Since equal numbers of individuals of species 1 and 2 do not, on the average, consume equal quantities of food, take up equal amounts of living space, and excrete equal amounts of *waste products of the same chemical composition*<sup>[39]</sup>, we, in general, set  $m_2 = \alpha N_2$  and  $m_1 = \beta N_1$  for some constants  $\alpha$  and  $\beta$ . The constants  $\alpha$  and  $\beta$  indicate the degree of influence of one species upon the other. If the *interests*<sup>[40]</sup> of the two species do not *clash*<sup>[41]</sup>, and they *occupy separate niches*<sup>[42]</sup>, then both  $\alpha$  and  $\beta$  are zero. If the two species *lay claim to the same niche*<sup>[43]</sup> and are very similar, then  $\alpha$  and  $\beta$  are very close to one. On the other hand, if one of the species, say species 2, *utilizes*

[34] 马尔萨斯率

[35] 繁殖能力

[36] 潜在的增长率

[37] 不排泄任何有毒废弃物

[38] 环境对增长的阻力

[39] 化学成分相同的废弃物

[40] 利益

[41] 冲突

[42] 拥有相互独立的生态龕

[43] 对同一个生态龕都声称拥有主权

[44] 以有害的方式利用环境

[45] 物种 2 的一个个体所占有的空间相当于物种 1 多个个体的占有量

[46] 平衡解

[47]  $N_1$ - $N_2$  坐标面

[48] 线段

[49] 坐标轴

[50] 四段不同轨迹的并

[51] 第一象限

*the environment unproductively*<sup>[44]</sup>, i.e., consumes a great deal of food or excretes poisonous waste products, then *one individual of species 2 takes up the place of many individuals of species 1*<sup>[45]</sup>. In this case, then, the coefficient  $\alpha$  will be very large.

We restrict ourselves now to the case where the two species are nearly identical and lay claim to the same niche. Then,  $\alpha = \beta = 1$  and  $N_1(t)$  and  $N_2(t)$  satisfy the system of differential equations.

$$\frac{dN_1}{dt} = a_1 N_1 \left( \frac{K_1 - N_1 - N_2}{K_1} \right), \quad \frac{dN_2}{dt} = a_2 N_2 \left( \frac{K_2 - N_2 - N_1}{K_2} \right). \quad (3.5.4)$$

**Theorem 3.5.1 (Principle of Competitive Exclusion)** Suppose that  $K_1$  is greater than  $K_2$ . Then, every solution  $N_1(t)$ ,  $N_2(t)$  of system (3.5.4) approaches the constant solution  $N_1 = K_1$ ,  $N_2 = 0$  as  $t$  approaches infinity.

**Remark 3.5.1** A constant solution  $N_1(t) \equiv c_1$ ,  $N_2(t) \equiv c_2$  of system (3.5.4) is called an **equilibrium solution**<sup>[46]</sup>, and the point  $(c_1, c_2)$  is called an **equilibrium point** (see Chapter 5 for details).

Theorem 3.5.1 shows that if species 1 and 2 are nearly identical and the microcosm can support more members of species 1 than that of species 2, then species 2 will ultimately become extinct. Observe that

$$N_1(t) = \frac{K_1 N_1(0)}{N_1(0) + (K_1 - N_1(0))e^{-a_1 t}}, \quad N_2(t) = 0$$

is a solution of system (3.5.4) for any choice of  $N_1(0)$ . The orbit of this solution in the  $N_1$ - $N_2$  plane<sup>[47]</sup> is the point  $(0, 0)$  for  $N_1(0) = 0$ ; the line<sup>[48]</sup>  $0 < N_1 < K_1$ ,  $N_2 = 0$  for  $0 < N_1(0) < K_1$ ; the point  $(K_1, 0)$  for  $N_1(0) = K_1$ ; and the line  $K_1 < N_1 < \infty$ ,  $N_2 = 0$  for  $N_1(0) > K_1$ . Thus the  $N_1$  axis<sup>[49]</sup>, for  $N_1 \geq 0$ , is the union of four distinct orbits<sup>[50]</sup> of system (3.5.4). Similarly, the  $N_2$  axis, for  $N_2 \geq 0$ , is the union of four distinct orbits of system (3.5.4). This implies that all solutions  $N_1(t)$ ,  $N_2(t)$  of system (3.5.4) which start in the first quadrant<sup>[51]</sup> ( $N_1 > 0$ ,  $N_2 > 0$ ) of the  $N_1$ - $N_2$  plane must remain there for all future time.

Now we split the first quadrant into regions in which both  $dN_1/dt$  and  $dN_2/dt$  have fixed signs. Let  $l_1$  and  $l_2$  be the lines  $K_1 - N_1 - N_2 = 0$  and  $K_2 - N_1 - N_2 = 0$ , respectively. Observe that  $dN_1/dt$  is negative if  $(N_1, N_2)$  lies above  $l_1$  and positive if  $(N_1, N_2)$  lies below  $l_1$ . Similarly,  $dN_2/dt$  is negative if  $(N_1, N_2)$  lies above  $l_2$  and positive if  $(N_1, N_2)$  lies below  $l_2$ . Thus the two parallel<sup>[52]</sup> lines  $l_1$  and  $l_2$  split the first quadrant of the  $N_1$ - $N_2$  plane into three regions (see Figure 3.5.1) in which both  $dN_1/dt$  and  $dN_2/dt$  have fixed signs. Both  $N_1(t)$  and  $N_2(t)$  increase with time (along any solution of the system (3.5.4)) in region I;  $N_1(t)$  increases and  $N_2(t)$  decreases with time in region II; and both  $N_1(t)$  and  $N_2(t)$  decrease with time in region III.

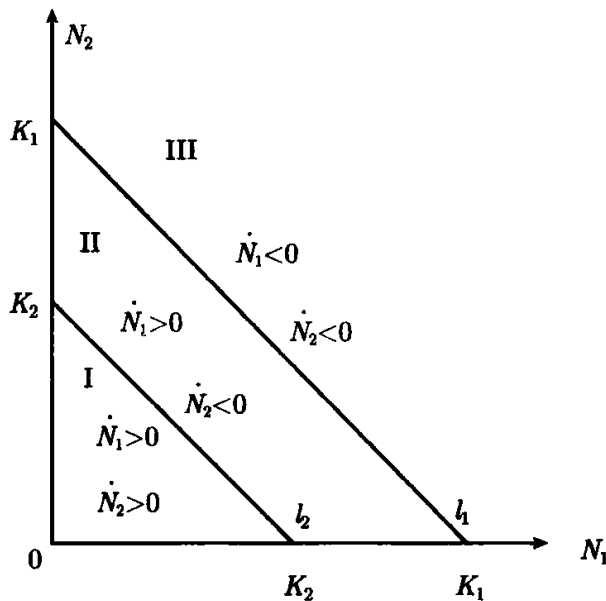


Figure 3.5.1

**Lemma 3.5.1** Any solution  $N_1(t)$ ,  $N_2(t)$  of system (3.5.4) which starts in region I at  $t = t_0$  must leave this region at some later time.

**Proof** Suppose that a solution  $N_1(t)$ ,  $N_2(t)$  of system (3.5.4) remains in region I for all time  $t \geq t_0$ . This implies that both  $N_1(t)$  and  $N_2(t)$  are monotonic increasing functions of time for  $t \geq t_0$  with  $N_1(t)$  and  $N_2(t)$  less than  $K_2$ . Consequently, both  $N_1(t)$  and  $N_2(t)$  have limits  $\xi$ ,  $\eta$  respectively, as  $t$  approaches infinity. This, in turn, implies that  $(\xi, \eta)$  is an equilibrium point of system

[53] 推导

(3.5.4). Now, the only equilibrium points of system (3.5.4) are  $(0, 0)$ ,  $(K_1, 0)$ , and  $(0, K_2)$ ,  $(\xi, \eta)$  obviously cannot equal any of these three points. We conclude, therefore, that any solution  $N_1(t)$ ,  $N_2(t)$  of system (3.5.4) which starts in region I must leave this region at a later time. ■

**Lemma 3.5.2** Any solution  $N_1(t)$ ,  $N_2(t)$  of system (3.5.4) which starts in region II at time  $t = t_0$  will remain in this region for all future time  $t \geq t_0$ , and ultimately approach the equilibrium solution  $N_1 = K_1$ ,  $N_2 = 0$ .

**Proof** Suppose that a solution  $N_1(t)$ ,  $N_2(t)$  of system (3.5.4) leaves region II at time  $t = t^*$ . Then either  $N_1(t^*)$  or  $N_2(t^*)$  is zero since the only way a solution of system (3.5.4) can leave region II is by crossing  $l_1$  or  $l_2$ . Assume that  $\dot{N}_1(t^*) = 0$ . Differentiating both sides of the first equation of system (3.5.4) with respect to  $t$  and setting  $t = t^*$  gives

$$\frac{d^2 N_1(t^*)}{dt^2} = \frac{-a_1 N_1(t^*)}{K_1} \frac{dN_2(t^*)}{dt}.$$

This quantity is positive. Hence  $N_1(t)$  has a minimum at  $t = t^*$ . However, this is impossible since  $N_1(t)$  is increasing whenever a solution  $N_1(t)$ ,  $N_2(t)$  of system (3.5.4) is in region II. Similarly, if  $\dot{N}_2(t^*) = 0$ , then

$$\frac{d^2 N_2(t^*)}{dt^2} = \frac{-a_2 N_2(t^*)}{K_2} \frac{dN_1(t^*)}{dt}.$$

This quantity is negative, and implying that  $N_2(t)$  has a maximum at  $t = t^*$ , but this is impossible since  $N_2(t)$  is decreasing whenever a solution  $N_1(t)$ ,  $N_2(t)$  of system (3.5.4) is in region II.

The previous *argument*<sup>[53]</sup> shows that any solution  $N_1(t)$ ,  $N_2(t)$  of system (3.5.4) which starts in region II at time  $t = t_0$  will remain in region II for all future time  $t \geq t_0$ . This implies that  $N_1(t)$  is monotonic increasing and  $N_2(t)$  is monotonic decreasing for  $t \geq t_0$  with  $N_1(t) < K_1$  and  $N_2(t) > K_2$ . Consequently, both  $N_1(t)$  and  $N_2(t)$  have limits  $\xi$ ,  $\eta$ , respectively, as  $t$  approaches infinity. This, in turn, implies that  $(\xi, \eta)$  is an equilibrium point of system (3.5.4). Now,  $(\xi, \eta)$  obviously cannot equal

$(0, 0)$  or  $(0, K_2)$ . Consequently,  $(\xi, \eta) = (K_1, 0)$ , and this proves Lemma 3.5.2. ■

**Lemma 3.5.3** Any solution  $N_1(t), N_2(t)$  of system (3.5.4) which starts in region III at time  $t = t_0$  and remains there for all future time must approach the equilibrium solution  $N_1(t) = K_1, N_2(t) = 0$  as  $t$  approaches infinity.

**Proof** If a solution  $N_1(t), N_2(t)$  of system (3.5.4) remains in region III for  $t \geq t_0$ , then both  $N_1(t)$  and  $N_2(t)$  are monotonic decreasing functions of time for  $t \geq t_0$  with  $N_1(t) > 0$  and  $N_2(t) > 0$ . Consequently, both  $N_1(t)$  and  $N_2(t)$  have limits  $\xi, \eta$ , respectively, as  $t$  approaches infinity. This, in turn, implies that  $(\xi, \eta)$  is an equilibrium point of system (3.5.14). Now,  $(\xi, \eta)$  obviously cannot equal  $(0, 0)$  or  $(0, K_2)$ . Consequently,  $(\xi, \eta) = (K_1, 0)$ . ■

**Proof of Theorem 3.5.1** Lemmas 3.5.1 and 3.5.2 state that every solution  $N_1(t), N_2(t)$  of system (3.5.4) which starts in regions I or II at time  $t = t_0$  must approach the equilibrium solution  $N_1 = K_1, N_2 = 0$  as  $t$  approaches infinity. Similarly, Lemma 3.5.3 shows that every solution  $N_1(t), N_2(t)$  of system (3.5.4) which starts in region III at time  $t = t_0$  and remains there for all future time must also approach the equilibrium solution  $N_1 = K_1, N_2 = 0$ . Next, observe that any solution  $N_1(t), N_2(t)$  of system (3.5.4) which starts on  $l_1$  or  $l_2$  must *immediately afterwards enter region II*<sup>[54]</sup>. Finally, if a solution  $N_1(t), N_2(t)$  of system (3.5.4) leaves region III, then it must cross line  $l_1$  and immediately afterwards enter region II. Lemma 3.5.2 then forces this solution to approach the equilibrium solution  $N_1 = K_1, N_2 = 0$ . ■

This theorem deals with the case of identical species, i.e.,  $\alpha = \beta = 1$ . By a similar analysis we can predict the outcome of the struggle for existence for all values of  $\alpha$  and  $\beta$ .

## Module 2 A Model for the Blood Glucose Regular System<sup>[55]</sup>

In the mid-1960's Drs. Rosevear and Molnar of the Mayo Clinic and Drs. Ackerman and Gatewood of the University of Minnesota

[54] 随后立即进入区域 II

[55] 血糖调节系统

[56] 葡萄糖耐量测试  
 [57] 新陈代谢  
 [58] 脊椎动物  
 [59] 最佳血糖浓度  
 [60] 过度偏离...  
 [61] 严重的病理学症状  
 [62] 倾向于自动调节  
 [63] 胰岛素(一种由胰腺的 $\beta$ 细胞分泌出的激素)  
 [64] 胰腺  
 [65] 消化系统  
 [66] 人们普遍认为胰岛素通过将其自身附着于不可渗透的膜壁上使葡萄糖可以通过隔膜进入到细胞的中心  
 [67] 生物和化学反应  
 [68] 胰高血糖素  
 [69] 多余的  
 [70] 肝脏  
 [71] 肝糖  
 [72] 分解  
 [73] 低血糖  
 [74] 禁食  
 [75] 而增加的血糖又会抑制它的分泌  
 [76] 肾上腺素  
 [77] 肾上腺髓质  
 [78] 一种紧急机制

discovered a fairly reliable criterion for interpreting the results of a GTT (*Glucose tolerance test*<sup>[56]</sup>) from a very simple model they developed for the blood glucose regulatory system. Their model is based on the following facts of elementary biology.

(1) Glucose plays an important role in the *metabolism*<sup>[57]</sup> of any *vertebrate*<sup>[58]</sup> since it is a source of energy for all tissues and organs. For each individual there is an *optimal blood glucose concentration*<sup>[59]</sup>, and *excessive deviation from*<sup>[60]</sup> this optimal concentration will lead to *severe pathological conditions*<sup>[61]</sup> and potentially death.

(2) While blood glucose levels *tend to be autoregulatory*<sup>[62]</sup>, they are also influenced and controlled by a wide variety of hormones and other metabolites. Among these are the following.

(i) *Insulin*<sup>[63]</sup>, a hormone secreted by the  $\beta$  cells of the *pancreas*<sup>[64]</sup>. After we eat any carbohydrates, our *gastrointestinal tract*<sup>[65]</sup> sends a signal to the pancreas to secrete more insulin. In addition, the glucose in our blood directly stimulates the  $\beta$  cells of the pancreas to secrete insulin. *It is generally believed that insulin facilitates tissue uptake of glucose by attaching itself to the impermeable membrane walls, thus allowing glucose to pass through the membranes to the center of the cells*<sup>[66]</sup>, where most of the *biological and chemical activity*<sup>[67]</sup> takes place. Without sufficient insulin, the body cannot avail itself of all the energy it needs.

(ii) *Glucagon*<sup>[68]</sup>, a hormone secreted by the  $\alpha$  cells of the pancreas. Any *excess*<sup>[69]</sup> glucose is stored in the *liver*<sup>[70]</sup> in the form of *glycogen*<sup>[71]</sup>. In times of need this glycogen is converted back into glucose. The hormone glucagon increases the rate of *breakdown*<sup>[72]</sup> of glycogen into glucose. Evidence collected thus far clearly indicates that *hypoglycemia*<sup>[73]</sup> (low blood sugar) and *fasting*<sup>[74]</sup> promote the secretion of glucagon *while increased blood glucose levels suppress its secretion*<sup>[75]</sup>.

(iii) *Epinephrine*<sup>[76]</sup>, a hormone secreted by the *adrenal medulla*<sup>[77]</sup>. Epinephrine is part of an *emergency mechanism*<sup>[78]</sup> to quickly increase the concentration of glucose in the blood in

times of extreme hypoglycemia. Like glucagon, breakdown of epinephrine increases the rate of glycogen into glucose. In addition, it directly inhibits glucose uptake by *muscle tissue*<sup>[79]</sup>; it acts directly on the pancreas to inhibit insulin secretion; and it aids in the conversion of *lactate*<sup>[80]</sup> to glucose in the liver.

(iv) *Glucocorticoids*<sup>[81]</sup> hormones such as *cortisol*<sup>[82]</sup> which are secreted by *the adrenal cortex*<sup>[83]</sup>. Glucocorticoids play an important role in the metabolism of carbohydrates.

(v) *Thyroxin*<sup>[84]</sup>, a hormone secreted by *the thyroid gland*<sup>[85]</sup>. This hormone aids the liver in forming glucose from noncarbohydrate sources such as *glycerol*<sup>[86]</sup>, lactate, and *amino acids*<sup>[87]</sup>.

(vi) Growth hormone, a hormone secreted by *the anterior pituitary gland*<sup>[88]</sup>. Not only does growth hormone affect glucose levels in a direct manner, but it also tends to “*block*<sup>[89]</sup>” insulin. It is believed that growth hormone decreases the sensitivity of muscle and *adipose membrane*<sup>[90]</sup> to insulin, thereby reducing the effectiveness of insulin in promoting glucose uptake.

The aim of Ackerman et al was to construct a model which would accurately describe the blood glucose regulatory system during a GTT and in which one or two parameters would yield criteria for distinguishing normal individuals from mild diabetics and prediabetics. Their model (requiring only a limited number of blood samples during a GTT) centers attention on two concentrations, that of glucose in the blood (labeled *G*), and that of the net hormonal concentration (labeled *H*). The latter is interpreted to represent the cumulative effect of all *the pertinent hormones*<sup>[91]</sup>. Those hormones such as insulin which decrease blood glucose concentrations are considered to increase *H*, while those hormones such as cortisol which increase blood glucose concentrations are considered to decrease *H*.

There are two reasons why such a simplified model can still provide an accurate description of the blood glucose regulatory system. First, studies have shown that under normal—or close to normal—conditions, the interaction of one hormone, namely insulin, with blood glucose so predominates that a simple “*lumped*

- [79] 肌肉组织
- [80] 乳酸盐
- [81] 糖皮质激素
- [82] 皮质醇
- [83] 肾上腺皮质
- [84] 甲状腺素
- [85] 甲状腺
- [86] 丙三醇
- [87] 氨基酸
- [88] 脑垂体前叶腺
- [89] 锁住
- [90] 脂肪膜
- [91] 相关激素

[92] 集总参数模型

[93] 血糖量

[94] 正常状态

[95] 动力学机制

[96] 血糖调节系统整体的运行状态

[97] 导致血糖浓度增加的外部变化速率

*parameter model*<sup>[92]</sup> is quite adequate. Second, evidence indicates that *normoglycemia*<sup>[93]</sup> does not necessarily depend on the *normalcy*<sup>[94]</sup> of each *kinetic mechanism*<sup>[95]</sup> of the blood glucose regulatory system. Rather, it depends on *the overall performance of the blood glucose regulatory system*<sup>[96]</sup>, and this system is dominated by insulin-glucose interactions.

The basic model is described analytically by the equations

$$\frac{dG}{dt} = F_1(G, H) + J(t), \quad (3.5.5)$$

$$\frac{dH}{dt} = F_2(G, H). \quad (3.5.6)$$

The function  $J(t)$  is *the external rate at which the blood glucose concentration is being increased*<sup>[97]</sup>. We assume that  $G$  and  $H$  have achieved optimal values  $G_0$  and  $H_0$  by the time the fasting patient has arrived at the hospital. Since we are interested here in the deviations of  $G$  and  $H$  from their optimal values, we make the substitutions

$$g = G - G_0, \quad h = H - H_0.$$

Then

$$\frac{dg}{dt} = F_1(G_0 + g, H_0 + h) + J(t),$$

$$\frac{dh}{dt} = F_2(G_0 + g, H_0 + h).$$

Now, observe that

$$\begin{aligned} & F_1(G_0 + g, H_0 + h) \\ &= F_1(G_0, H_0) + \frac{\partial F_1(G_0, H_0)}{\partial G} g + \frac{\partial F_1(G_0, H_0)}{\partial H} h + e_1 \end{aligned}$$

and

$$\begin{aligned} & F_2(G_0 + g, H_0 + h) \\ &= F_2(G_0, H_0) + \frac{\partial F_2(G_0, H_0)}{\partial G} g + \frac{\partial F_2(G_0, H_0)}{\partial H} h + e_2, \end{aligned}$$

where  $e_1$  and  $e_2$  are very small compared to  $g$  and  $h$ . Hence, [98] 内分泌腺 assuming that  $G$  and  $H$  deviate only slightly from  $G_0$  and  $H_0$ , and therefore neglecting the terms  $e_1$  and  $e_2$ , we see that

$$\frac{dg}{dt} = \frac{\partial F_1(G_0, H_0)}{\partial G} g + \frac{\partial F_1(G_0, H_0)}{\partial H} h + J(t), \quad (3.5.7)$$

$$\frac{dh}{dt} = \frac{\partial F_2(G_0, H_0)}{\partial G} g + \frac{\partial F_2(G_0, H_0)}{\partial H} h. \quad (3.5.8)$$

We can determine the signs of  $\partial F_1(G_0, H_0)/\partial G$ ,  $\partial F_1(G_0, H_0)/\partial H$ ,  $\partial F_2(G_0, H_0)/\partial G$ , and  $\partial F_2(G_0, H_0)/\partial H$  though there exists no way to determine their values. Moreover,  $dg/dt$  is negative for  $g > 0$  and  $h = 0$  since the blood glucose concentration will be decreasing through tissue uptake of glucose and the storing of excess glucose in the liver in the form of glycogen. Consequently,  $\partial F_1(G_0, H_0)/\partial G$  must be negative. Similarly,  $\partial F_1(G_0, H_0)/\partial H$  is negative since a positive value of  $h$  tends to decrease blood glucose levels by facilitating tissue uptake of glucose and by increasing the rate at which glucose is converted to glycogen. The number  $\partial F_2(G_0, H_0)/\partial G$  must be positive since a positive value of  $g$  causes the endocrine glands<sup>[98]</sup> to secrete those hormones which tend to increase  $H$ . Finally,  $\partial F_2(G_0, H_0)/\partial H$  must be negative since the concentration of hormones in the blood decreases through hormone metabolism.

Thus we can write Eq.(3.5.7) and Eq.(3.5.8) in the form

$$\frac{dg}{dt} = -m_1 g - m_2 h + J(t), \quad (3.5.9)$$

$$\frac{dh}{dt} = -m_3 h - m_4 g, \quad (3.5.10)$$

where  $m_1, m_2, m_3$  and  $m_4$  are positive constants. Since we only measure the concentration of glucose in the blood, we would like to remove the variable  $h$ . Differentiating Eq.(3.5.9) with respect to  $t$  gives

$$\frac{d^2 g}{dt^2} = -m_1 \frac{dg}{dt} - m_2 \frac{dh}{dt} + \frac{dJ}{dt}.$$

Substituting for  $dh/dt$  from Eq.(3.5.10) we obtain that

$$\frac{d^2 g}{dt^2} = -m_1 \frac{dg}{dt} - m_2 m_3 h - m_2 m_4 g + \frac{dJ}{dt}. \quad (3.5.11)$$

[99] 大剂量的葡萄糖

[100]  $\delta$  函数

Next, observe from Eq.(3.5.9) that

$$m_2 h = -(dg/dt) - m_1 g + J(t).$$

Consequently,  $g(t)$  satisfies the second-order linear differential equation

$$\frac{d^2 g}{dt^2} + (m_1 + m_3) \frac{dg}{dt} + (m_1 m_3 + m_2 m_4) g = m_3 J + \frac{dJ}{dt},$$

we rewrite this equation in the form

$$\frac{d^2 g}{dt^2} + 2\alpha \frac{dg}{dt} + \omega_0^2 g = S(t), \quad (3.5.12)$$

where

$$\alpha = \frac{m_1 + m_3}{2}, \quad \omega_0^2 = m_1 m_3 + m_2 m_4$$

and

$$S(t) = m_3 J + \frac{dJ}{dt}.$$

Notice that the right side of Eq.(3.5.12) is identically zero except for the very short time interval in which *the glucose load*<sup>[99]</sup> is being ingested. Such functions can be dealt with very effectively by introducing *the delta function*<sup>[100]</sup>. For our purposes here, let  $t = 0$  be the time at which the glucose load has been completely ingested. Then, for  $t \geq 0$ ,  $g(t)$  satisfies the second-order linear homogeneous equation

$$\frac{d^2 g}{dt^2} + 2\alpha \frac{dg}{dt} + \omega_0^2 g = 0. \quad (3.5.13)$$

This equation has positive coefficients. Hence  $g(t)$  approaches zero as  $t$  approaches infinity. Thus our model certainly conforms to reality in predicting that the blood glucose concentration tends to return eventually to its optimal concentration.

The solutions  $g(t)$  of Eq.(3.5.13) are of three different types, depending on whether  $\alpha^2 - \omega_0^2$  is positive, negative, or zero. We will assume that  $\alpha^2 - \omega_0^2$  is negative; the other two cases can be treated in a similar way. If  $\alpha^2 - \omega_0^2 < 0$ , then the characteristic equation of Eq.(3.5.13) has complex roots. Then, in this case, every solution  $g(t)$  of Eq.(3.5.13) is of the form

$$g(t) = Ae^{-\alpha t} \cos(\omega t - \delta), \quad \omega^2 = \omega_0^2 - \alpha^2. \quad (3.5.14)$$

Consequently,

$$G(t) = G_0 + Ae^{-\alpha t} \cos(\omega t - \delta), \quad (3.5.15)$$

where  $G_0$ ,  $A$ ,  $\alpha$ ,  $\omega_0$  and  $\delta$  are unknowns. Since  $G_0$  denotes the patient's blood glucose concentration before the glucose load is ingested, we can determine  $G_0$  by measuring the patient's blood glucose concentration immediately upon his arrival at the hospital. Next, if we take four additional measurements  $G_1$ ,  $G_2$ ,  $G_3$  and  $G_4$  of the patient's blood glucose concentration at times  $t_1$ ,  $t_2$ ,  $t_3$  and  $t_4$ , then  $A$ ,  $\alpha$ ,  $\omega_0$  and  $\delta$  can be determined from following four equations

$$G_j = G_0 + Ae^{-\alpha t_j} \cos(\omega t_j - \delta), \quad j = 1, 2, 3, 4.$$

A better method of determining  $G_0$ ,  $A$ ,  $\alpha$ ,  $\omega_0$  and  $\delta$  is to take  $n$  measurements  $G_1, G_2, \dots, G_n$  of the patient's blood glucose concentration at times  $t_1, t_2, \dots, t_n$  respectively. Typically,  $n$  is 6 or 7. We then find optimal values for  $G_0$ ,  $A$ ,  $\alpha$ ,  $\omega_0$  and  $\delta$  such that the *least-square error*<sup>[101]</sup>

$$E = \sum_{j=1}^n [G_j - G_0 - Ae^{-\alpha t_j} \cos(\omega t_j - \delta)]^2 \quad (3.5.16)$$

is minimized. This method is preferable to the first method since it is possible to find values  $G_0$ ,  $A$ ,  $\alpha$ ,  $\omega_0$  and  $\delta$  so that Eq.(3.5.15) is satisfied exactly at four points  $t_1, t_2, t_3$  and  $t_4$  but yields a poor fit to the data at other times, while the second method usually offers a better fit to the data on the entire time interval since it involves more measurements.

In numerous experiments, Ackerman et al, observed that a *slight error in measuring  $G$* <sup>[102]</sup> could produce a very large error in the value of  $\alpha$ . Hence any criterion for diagnosing diabetes that involves the parameter  $\alpha$  is unreliable. However, the parameter  $\omega_0$ , the natural frequency of the system, was relatively insensitive to experimental error in measuring  $G$ . Thus we may regard a value of  $\omega_0$  as *the basic descriptor*<sup>[103]</sup> of the response to a glucose tolerance test. For discussion purposes, using the corresponding

[101] 最小平方误差

[102] 测量  $G$  时产生的微小误差

[103] 基本反应值

- [104] 尿崩症  
 [105] 脑垂体腺后  
 叶失调  
 [106] 独立变量  
 [107] 可想而知  
 [108] 对...感到  
 困惑

natural period  $T_0 = 2\pi/\omega_0$  is more convenient. The remarkable fact is that data from a variety of sources indicated that a value of less than four hours for  $T_0$  indicated normalcy, while appreciably more than four hours implied mild diabetes.

**Remark 3.5.2** (1) The usual period between meals in our culture is about four hours. This suggests the interesting possibility that sociological factors may also play a role in the blood glucose regulatory system.

(2) We wish to emphasize that the model described above can only be used to diagnose mild diabetes or prediabetes since we have assumed throughout that the deviation of  $G$  from its optimal value  $G_0$  is small. Very large deviation of  $G$  from  $G_0$  usually indicate severe diabetes or *diabetes insipidus*<sup>[104]</sup>, which is a disorder of the posterior lobe of the pituitary gland<sup>[105]</sup>.

A serious shortcoming of this simplified model is that it sometimes yields a poor fit to the data in the time period three to five hours after ingestion of the glucose load. This indicates, of course, that variables such as epinephrine and glucagon play an important role in this time period. Thus these variables should be included as *separate variables*<sup>[106]</sup> in our model, rather than being lumped together with insulin. In fact, evidence indicates that levels of epinephrine may rise dramatically during the recovery phase of the GTT response when glucose levels have been lowered below fasting levels. This can also be seen directly from Eq.(3.5.13). If  $\alpha^2 - \omega_0^2 > 0$ , then  $g(t)$  may have the form described in Figure 3.5.2. Note that  $g(t)$  drops very rapidly from a fairly high value to a negative one. *Quite conceivably*<sup>[107]</sup>, therefore, the body will interpret this as an extreme emergency and thereby secrete a large amount of epinephrine.

Medical researchers have long recognized the need of including epinephrine as a separate variable in any model of the blood glucose regulatory system. However, they *were stymied by*<sup>[108]</sup> the fact that the concentration of epinephrine in the blood could not be reliably measured. Thus they had to assume, for all prac-

tical purposes, that the level of epinephrine remained constant during the course of a glucose tolerance test.

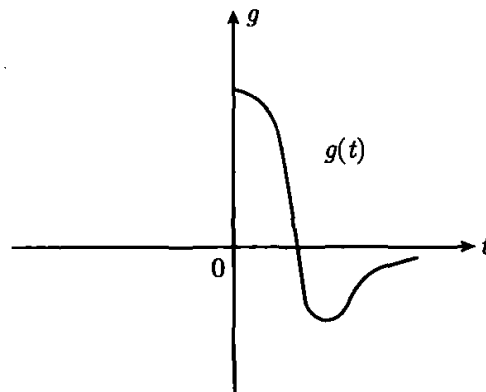


Figure 3.5.2 Graph of  $g(t)$  if  $\alpha^2 - \omega^2 > 0$

# Chapter 4 Laplace Transforms<sup>[1]</sup> and Their Applications in Solving Differential Equations

## 4.1 Laplace Transforms

[1] Laplace 变换

[2] 单位阶跃函数

In this section, we introduce a powerful tool for solving differential equations and systems of differential equations.

**Definition 4.1.1** Let  $f(t)$  be defined for  $0 \leq t < \infty$ . The **Laplace transform** of  $f(t)$ , denoted by  $F(s)$ , or  $\mathcal{L}\{f(t)\}$ , is given by formula

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt, \quad (4.1.1)$$

where

$$\int_0^{\infty} e^{-st} f(t) dt = \lim_{A \rightarrow \infty} \int_0^A e^{-st} f(t) dt. \quad (4.1.2)$$

If the limit (4.1.2) does not exist, we say that  $f(t)$  has no Laplace transform.

**Example 4.1.1** Compute the Laplace transform of **unit step function**<sup>[2]</sup>

$$u_0(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases}$$

**Solution**

$$\begin{aligned} \mathcal{L}\{u_0(t)\} &= \lim_{A \rightarrow \infty} \int_0^A e^{-st} u_0(t) dt \\ &= \lim_{A \rightarrow \infty} \frac{1 - e^{-sA}}{s} = \begin{cases} 1/s, & s > 0, \\ \infty, & s \leq 0. \end{cases} \end{aligned} \quad (4.1.3)$$

■

The general form of unit step function is

$$u_a(t) = \begin{cases} 0, & t < a, \\ 1, & t \geq a. \end{cases} \quad (4.1.4)$$

**Example 4.1.2** Compute the Laplace transform of unit step function  $u_a(t)$  if  $a > 0$ . [3] 分段连续的  
[4] 具有指数阶

**Solution**

$$\begin{aligned}\mathcal{L}\{u_a(t)\} &= \lim_{A \rightarrow \infty} \int_a^A e^{-st} u_a(t) dt = \lim_{A \rightarrow \infty} \left[ -\frac{e^{-st}}{s} \right]_a^A \\ &= \begin{cases} e^{-as}/s, & s > 0, \\ \infty, & s \leq 0. \end{cases}\end{aligned}\quad (4.1.5)$$

**Example 4.1.3** Compute the Laplace transform of function  $f(t) = e^{at}$ .

**Solution**

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \lim_{A \rightarrow \infty} \int_0^A e^{-st} e^{at} dt = \lim_{A \rightarrow \infty} \frac{1 - e^{(a-s)A}}{a - s} \\ &= \begin{cases} \frac{1}{s - a}, & s > a, \\ \infty, & s \leq a. \end{cases}\end{aligned}\quad (4.1.6)$$

**Example 4.1.4** Compute the Laplace transform of function

$$f(t) = \begin{cases} e^t, & t \leq 2, \\ 3, & t > 2. \end{cases}$$

**Solution**

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \int_0^\infty e^{-st} f(t) dt = \int_0^2 e^{-st} e^t dt + \int_2^\infty e^{-st} 3 dt \\ &= \frac{e^{(1-s)t}}{1-s} \Big|_{t=0}^{t=2} - \frac{3}{s} \lim_{A \rightarrow \infty} e^{-st} \Big|_{t=2}^{t=A} \\ &= \frac{1 - e^{-(s-1)}}{s-1} + \frac{3}{s} e^{-2s}, \quad s > 0.\end{aligned}$$

**Theorem 4.1.1 (Existence of Laplace Transform)** If function  $f(t)$  satisfies following conditions

- (1)  $f(t)$  is piecewise continuous<sup>[3]</sup> for  $t \geq 0$ ,
- (2)  $f(t)$  is of exponential order<sup>[4]</sup>, that is, there exist constants  $M$  and  $c$  such that

$$|f(t)| \leq M e^{ct}, \quad 0 \leq t < \infty, \quad (4.1.7)$$

then, the Laplace transform of  $f(t)$  exists for  $s > c$ .

**Example 4.1.5** Compute the Laplace transform of sine function  $f(t) = \sin \omega t$ .

**Solution**

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \int_0^{\infty} e^{-st}(\sin \omega t)dt \\ &= \frac{e^{-st}}{s^2 + \omega^2}(-s \sin \omega t - \omega \cos \omega t) \Big|_0^{+\infty} \\ &= \frac{\omega}{s^2 + \omega^2}, \quad s > 0.\end{aligned}\tag{4.1.8}$$

Similarly, the Laplace transform of cosine function  $f(t) = \cos \omega t$  is

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \int_0^{\infty} e^{-st}(\cos \omega t)dt \\ &= \frac{s}{s^2 + \omega^2}, \quad s > 0.\end{aligned}\tag{4.1.9}$$

■

### Exercise 4.1

Compute the Laplace transform of following functions.

1.  $f(t) = c$ , where  $c$  is a constant.
2.  $f(t) = \sqrt{3}$ .
3.  $f(t) = t$ .
4.  $f(t) = -3t$ .
5.  $f(t) = \sin 5t$ .
6.  $f(t) = \cos(-2t)$ .
7.  $f(t) = t^2$ .
8.  $f(t) = e^{-2t}$ .
9.  $f(t) = \begin{cases} -1, & t \leq 4, \\ 1, & t > 4. \end{cases}$
10.  $f(t) = \begin{cases} 1, & 0 \leq t \leq 1, \\ e^t, & 1 < t \leq 4, \\ 0, & t > 4. \end{cases}$

Show that each of the following functions is of exponential order.

11.  $t^n$
12.  $\sin at$
13.  $e^{\sqrt{t}}$ .

14. Show that  $e^{t^2}$  does not possess a Laplace transform.

*Hint:* Show that  $e^{t^2-st} > e^t$  for  $t > s + 1$ .

15. Suppose that  $f(t)$  is of exponential order. Show that  $F(s) = \mathcal{L}\{f(t)\}$  approaches 0 as  $s \rightarrow \infty$ .

## 4.2 Properties of Laplace Transforms

In this section we introduce some basic and useful properties of Laplace transforms. Using these properties we will be able to

compute the Laplace transforms of most functions without performing tedious integrations. [5] 线性算子

**Property 4.2.1 (Linearity)** Let  $\mathcal{L}\{f(t)\} = F(s)$  and  $\mathcal{L}\{g(t)\} = G(s)$ , then for any two constants  $c_1$  and  $c_2$ ,

$$\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 \mathcal{L}\{f(t)\} + c_2 \mathcal{L}\{g(t)\} = c_1 F(s) + c_2 G(s). \quad (4.2.1)$$

**Proof** From formula (4.1.1)

$$\begin{aligned} \mathcal{L}\{c_1 f(t) + c_2 g(t)\} &= \int_0^{\infty} e^{-st} [c_1 f(t) + c_2 g(t)] dt \\ &= c_1 \int_0^{\infty} e^{-st} f(t) dt + c_2 \int_0^{\infty} e^{-st} g(t) dt \\ &= c_1 F(s) + c_2 G(s). \quad \blacksquare \end{aligned}$$

According to Property 4.2.1, the Laplace transform is a *linear operator*.<sup>[5]</sup>

**Example 4.2.1** Now we redo the computation, using Property 4.2.1, for computing the Laplace transforms of functions  $\cos \omega t$  and  $\sin \omega t$ .

**Solution** From definition

$$\begin{aligned} \mathcal{L}\{\cos \omega t\} &= \int_0^{\infty} e^{-st} \cos \omega t dt, \\ \mathcal{L}\{\sin \omega t\} &= \int_0^{\infty} e^{-st} \sin \omega t dt. \end{aligned}$$

Observing that

$$\begin{aligned} \mathcal{L}\{\cos \omega t\} + i\mathcal{L}\{\sin \omega t\} &= \mathcal{L}\{\cos \omega t + i \sin \omega t\} = \mathcal{L}\{e^{i\omega t}\} \\ &= \int_0^{\infty} e^{-st} e^{i\omega t} dt = \lim_{A \rightarrow \infty} \int_0^A e^{(i\omega - s)t} dt \\ &= \lim_{A \rightarrow \infty} \frac{e^{(i\omega - s)A} - 1}{i\omega - s} \\ &= \begin{cases} \frac{1}{s - i\omega} = \frac{s + i\omega}{s^2 + \omega^2}, & s > 0, \\ \text{undefined}, & s \leq 0. \end{cases} \end{aligned}$$

Equating real and imaginary parts, respectively, gives

$$\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}, \quad (4.2.2)$$

$$\mathcal{L}\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}. \quad (4.2.3)$$

**Example 4.2.2** Compute the Laplace transform of function  $f(t) = \sin^2 at$ .

**Solution**

$$\sin^2 at = \frac{1 - \cos 2(at)}{2},$$

so

$$\begin{aligned} \mathcal{L}\{\sin^2 at\} &= \mathcal{L}\left\{\frac{1 - \cos 2(at)}{2}\right\} \\ &= \frac{1}{2}\mathcal{L}\{1\} - \frac{1}{2}\mathcal{L}\{\cos 2(at)\} \\ &= \frac{1}{2s} - \frac{1}{2} \frac{s}{s^2 + 4a^2} \\ &= \frac{2a^2}{s(s^2 + 4a^2)}. \end{aligned}$$

**Property 4.2.2** If  $\mathcal{L}\{f(t)\} = F(s)$ , then

$$\mathcal{L}\{-tf(t)\} = \frac{d}{ds}F(s). \quad (4.2.4)$$

**Proof** By the definition of  $F(s)$  (Formula (4.1.1)),

$$\begin{aligned} \frac{d}{ds}F(s) &= \frac{d}{ds} \int_0^\infty e^{-st} f(t) dt \\ &= \int_0^\infty \frac{\partial}{\partial s} (e^{-st}) f(t) dt \\ &= \int_0^\infty -t f(t) e^{-st} dt = \mathcal{L}\{-tf(t)\}. \end{aligned}$$

In general, if  $\mathcal{L}\{f(t)\} = F(s)$ , then for any positive integer  $n$

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s). \quad (4.2.5)$$

**Example 4.2.3** Compute the Laplace transform of function  $f(t) = t^n$ .

**Solution** Using Formula (4.2.5) with the result of Example 4.1.1 we have that

$$\mathcal{L}\{t^n\} = (-1)^n \frac{d^n}{ds^n} \mathcal{L}\{1\} = (-1)^n \frac{d^n}{ds^n} \frac{1}{s-1} = \frac{n!}{s^{n+1}}.$$

**Example 4.2.4** Compute the Laplace transform of function  $f(t) = 3 + 2t^2$ .

**Solution** Using Property 4.2.1 with the results of Example 4.1.1 and Example 4.2.3 we have that

$$\mathcal{L}\{3 + 2t^2\} = 3\mathcal{L}\{1\} + 2\mathcal{L}\{t^2\} = 3\frac{1}{s} + 2\frac{2!}{s^3} = \frac{3}{s} + \frac{4}{s^3}. \quad \blacksquare$$

**Example 4.2.5** Compute the Laplace transform of function  $f(t) = t^2 \cos at$ .

**Solution**

$$\begin{aligned} \mathcal{L}\{t^2 \cos at\} &= (-1)^2 \frac{d^2}{ds^2} \mathcal{L}\{\cos at\} \\ &= \frac{d^2}{ds^2} \frac{s}{s^2 + a^2} \\ &= \frac{2s^3}{(s^2 + a^2)^3}. \quad \blacksquare \end{aligned}$$

**Property 4.2.3** If  $F(s) = \mathcal{L}\{f(t)\}$ , then

$$\mathcal{L}\{e^{at} f(t)\} = F(s - a). \quad (4.2.6)$$

**Proof** By the definition,

$$\mathcal{L}\{e^{at} f(t)\} = \int_0^\infty e^{-st} e^{at} f(t) dt = \int_0^\infty e^{-(s-a)t} f(t) dt = F(s-a). \quad \blacksquare$$

**Example 4.2.6** Compute the Laplace transform of function  $f(t) = t^n e^{at}$ .

**Solution**

**Method 1** By the result of Example 4.1.3,

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad s > a.$$

Hence, using Property 4.2.2,

$$\begin{aligned} \mathcal{L}\{t^n e^{at}\} &= (-1)^n \frac{d^n}{ds^n} \mathcal{L}\{e^{at}\} \\ &= (-1)^n \frac{d^n}{ds^n} \frac{1}{s-a} \\ &= \frac{n!}{(s-a)^{n+1}}, \quad s > a. \end{aligned}$$

**Method 2** By the result of Example 4.2.3,

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}.$$

Hence, using Property 4.2.3,

$$\mathcal{L}\{t^n e^{at}\} = \frac{n!}{(s-a)^{n+1}}, \quad s > a. \quad \blacksquare$$

**Example 4.2.7** Compute the Laplace transform of function

$$f(t) = 5 \sin 3t - 17e^{-2t}.$$

**Solution** Using Property 4.2.1 together with the results of Example 4.1.2 ( $a = -2$ ) and Example 4.2.1 ( $\omega = 3$ ), we have that

$$\begin{aligned} \mathcal{L}\{5 \sin 3t - 17e^{-2t}\} &= 5\mathcal{L}\{\sin 3t\} - 17\mathcal{L}\{e^{-2t}\} \\ &= 5 \left( \frac{3}{s^2 + 3^2} \right) - 17 \left( \frac{1}{s - (-2)} \right) \\ &= \frac{15}{s^2 + 9} - \frac{17}{s + 2}. \quad \blacksquare \end{aligned}$$

**Property 4.2.4** Let  $F(s) = \mathcal{L}\{f(t)\}$ . Then,

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0) = sF(s) - f(0). \quad (4.2.7)$$

**Proof**

$$\begin{aligned} \mathcal{L}\{f'(t)\} &= \lim_{A \rightarrow \infty} \int_0^A e^{-st} f'(t) dt \\ &= \lim_{A \rightarrow \infty} e^{-st} f(t) \Big|_0^A + \lim_{A \rightarrow \infty} s \int_0^A e^{-st} f(t) dt \\ &= -f(0) + sF(s). \quad \blacksquare \end{aligned}$$

**Property 4.2.5** If  $F(s) = \mathcal{L}\{f(t)\}$ , then

$$\mathcal{L}\left\{\int_0^t f(r) dr\right\} = \frac{1}{s} F(s). \quad (4.2.8)$$

**Proof** Set

$$y(t) = \int_0^t f(s) ds,$$

then  $y'(t) = f(t)$ , and  $y(0) = 0$ . By Property 4.2.4,

$$\mathcal{L}\{y'(t)\} = s\mathcal{L}\{y(t)\} - y(0) = s\mathcal{L}\{y(t)\}.$$

So

$$\mathcal{L} \left\{ \int_0^t f(r) dr \right\} = \frac{1}{s} \mathcal{L} \{y'(t)\} = \frac{1}{s} \mathcal{L} \{f(t)\} = \frac{1}{s} F(s). \quad \blacksquare$$

In general,

$$\underbrace{\mathcal{L} \left\{ \int_0^t dr \int_0^t dr \cdots \int_0^t f(r) dr \right\}}_{n \text{ times}} = \frac{1}{s^n} F(s). \quad (4.2.9)$$

**Example 4.2.8** Compute  $\mathcal{L} \left\{ \int_0^t \sinh 2r dr \right\}$ .

**Solution** Since

$$\sinh at = \frac{e^{at} - e^{-at}}{2},$$

$$\begin{aligned} \mathcal{L} \{ \sinh at \} &= \frac{1}{2} \mathcal{L} \{ e^{at} \} - \frac{1}{2} \mathcal{L} \{ e^{-at} \} \\ &= \frac{1}{2} \left( \frac{1}{s-a} - \frac{1}{s+a} \right) \\ &= \frac{a}{s^2 - a^2}. \end{aligned} \quad (4.2.10)$$

Similarly,

$$\mathcal{L} \{ \cosh at \} = \frac{s}{s^2 - a^2}. \quad (4.2.11)$$

Hence, by Property 4.2.5,

$$\mathcal{L} \left\{ \int_0^t \sinh 2r dr \right\} = \frac{1}{s} \left( \frac{2}{s^2 - 2^2} \right) = \frac{2}{s(s^2 - 4)}. \quad \blacksquare$$

**Property 4.2.6** If  $F(s) = \mathcal{L} \{f(t)\}$  and if  $\lim_{\substack{t \rightarrow 0 \\ t > 0}} \frac{f(t)}{t}$  exists, then

$$\mathcal{L} \left\{ \frac{1}{t} f(t) \right\} = \int_s^\infty F(r) dr. \quad (4.2.12)$$

**Example 4.2.9** Compute  $\mathcal{L} \left\{ \frac{\cos at - 1}{t} \right\}$ .

**Solution** Since

$$\lim_{t \rightarrow 0^+} \frac{\cos at - 1}{t} \stackrel{L'Hospital \text{ law}}{=} \lim_{t \rightarrow 0^+} (-a \sin at) = 0,$$

$$\mathcal{L}\{\cos at - 1\} = \mathcal{L}\{\cos at\} - \mathcal{L}\{1\} = \frac{s}{s^2 + a^2} - \frac{1}{s}, \quad s > 0.$$

Hence,

$$\begin{aligned} \mathcal{L}\left\{\frac{\cos at - 1}{t}\right\} &= \int_s^\infty \left[\frac{r}{r^2 + a^2} - \frac{1}{r}\right] dr \\ &= \ln(r^2 + a^2)_s^\infty - \ln r|_s^\infty \\ &= \ln \frac{s}{\sqrt{s^2 + a^2}}. \end{aligned}$$

■

**Property 4.2.7** If  $f(t)$  is periodic with period  $\omega$ , then

$$\mathcal{L}\{f(t)\} = \frac{\int_0^\omega e^{-st} f(t) dt}{1 - e^{-\omega s}}. \quad (4.2.13)$$

**Example 4.2.10** Find the Laplace transform of function graphed in Figure 4.2.1.

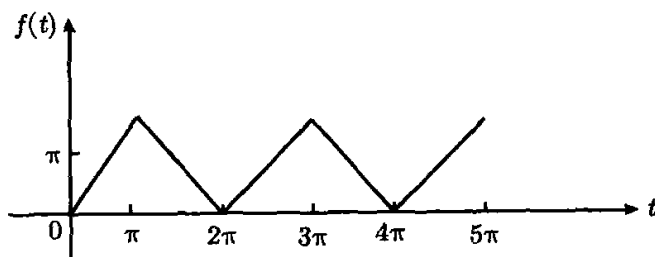


Figure 4.2.1

**Solution** Note that  $f(t)$  is periodic with period  $\omega = 2\pi$ , it can be defined analytically by

$$f(t) = \begin{cases} t, & 0 \leq t < \pi, \\ 2\pi - t, & \pi \leq t < 2\pi, \end{cases} \quad 0 \leq t < 2\pi.$$

By Property 4.2.7,

$$\mathcal{L}\{f(t)\} = \frac{\int_0^{2\pi} e^{-st} f(t) dt}{1 - e^{-2\pi s}}.$$

Since,

$$\begin{aligned} \int_0^{2\pi} e^{-st} f(t) dt &= \int_0^\pi e^{-st} t dt + \int_\pi^{2\pi} e^{-st} (2\pi - t) dt \\ &= \frac{1}{s^2} (e^{-2\pi s} - 2e^{-\pi s} + 1) \\ &= \frac{1}{s^2} (e^{-\pi s} - 1)^2, \end{aligned}$$

it follows that

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \frac{(1/s^2)(e^{-\pi s} - 1)^2}{1 - e^{-2\pi s}} = \frac{(1/s^2)(e^{-\pi s} - 1)^2}{(1 - e^{-\pi s})(1 + e^{-\pi s})} \\ &= \frac{1}{s^2} \left( \frac{1 - e^{-\pi s}}{1 + e^{-\pi s}} \right) = \frac{1}{s^2} \tanh \frac{\pi s}{2}.\end{aligned}$$

### Exercise 4.2

Compute the Laplace transforms of following functions.

1.  $f(t) = \cos^2 at.$
2.  $f(t) = \sin at \cos bt.$
3.  $f(t) = t^2 \cos at.$
4.  $f(t) = t \sin at.$
5.  $f(t) = 2t^2 - 3t + 4.$
6.  $f(t) = t^5 + 3t^2 + \sin 2t + 5.$
7.  $f(t) = te^{4t}.$
8.  $f(t) = 1 + t^n e^{at}.$
9.  $f(t) = e^{-2t} \sin 5t.$
10.  $f(t) = t \cos \sqrt{7}t.$
11.  $f(t) = e^{-t} t \cos 2t.$
12.  $f(t) = \int_0^t s \sinh s ds.$
13.  $f(t) = t \int_0^t e^{-3s} \sin 2s ds.$
14.  $f(t) = \frac{\sin 3t}{t}.$
15.  $f(t) = \frac{\sinh t}{t}.$
16.  $f(t) = \frac{e^{-3t} \sin 2t}{t}.$
17.  $f(t) = \int_0^t \frac{e^{-3s} \sin 2s}{s} ds.$
18.  $f(t) = te^{4t} \int_0^t \frac{e^{-4s} \sin 3s}{s} ds.$
- 19.

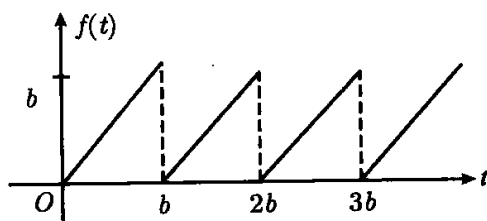


Figure 4.2.2

20.

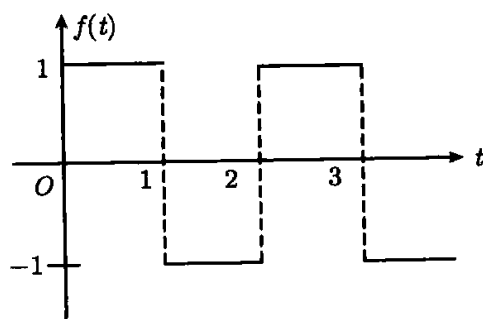


Figure 4.2.3

[6] Laplace 逆变  
换、求 Laplace  
变换的逆

### 4.3 Inverse Laplace Transforms<sup>[6]</sup>

**Definition 4.3.1** The **inverse Laplace transform** of function  $F(s)$ , designated by  $\mathcal{L}^{-1}\{F(s)\}$ , is another function  $f(t)$  having the property that  $F(s) = \mathcal{L}\{f(t)\}$ .

**Property 4.3.1 (Linearity)** If the inverse Laplace transforms of two functions  $F(s)$  and  $G(s)$  exist, then for any two constants  $c_1$  and  $c_2$ ,

$$\mathcal{L}^{-1}\{c_1 F(s) + c_2 G(s)\} = c_1 \mathcal{L}^{-1}\{F(s)\} + c_2 \mathcal{L}^{-1}\{G(s)\}.$$

The main usefulness of the properties introduced in Section 4.2 is in inverting Laplace transforms.

**Example 4.3.1** Find  $\mathcal{L}^{-1}\left\{\frac{1}{s}\right\}$ .

**Solution** Here  $F(s) = \frac{1}{s}$ . Since  $\mathcal{L}\{1\} = \frac{1}{s}$ , we have that

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1. \quad \blacksquare$$

**Example 4.3.2** What function has Laplace transform  $-\frac{1}{(s-2)^2}$ ?

**Solution**

$$-\frac{1}{(s-2)^2} = \frac{d}{ds} \frac{1}{s-2}$$

and

$$\frac{1}{s-2} = \mathcal{L}\{e^{2t}\}.$$

Hence,

$$\mathcal{L}^{-1}\left\{-\frac{1}{(s-2)^2}\right\} = -te^{2t}. \quad \blacksquare$$

**Example 4.3.3** What function has Laplace transform  $\frac{s}{s^2+6}$ ?

**Solution**

$$\mathcal{L}\{\cos \sqrt{6}t\} = \frac{s}{s^2 + (\sqrt{6})^2} = \frac{s}{s^2 + 6}.$$

Therefore,

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 6}\right\} = \cos \sqrt{6}t.$$

Hence, the required function is  $\cos \sqrt{6}t$ . ■

**Example 4.3.4** Find  $\mathcal{L}^{-1} \left\{ \frac{1}{s^2 - 4s + 9} \right\}$ .

**Solution**

$$\frac{1}{s^2 - 4s + 9} = \frac{1}{(s - 2)^2 + 5}$$

and

$$\frac{1}{s^2 + 5} = \mathcal{L} \left\{ \frac{1}{\sqrt{5}} \sin \sqrt{5}t \right\}.$$

Hence,

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 - 4s + 9} \right\} = \frac{1}{\sqrt{5}} e^{2t} \sin \sqrt{5}t. \quad \blacksquare$$

**Example 4.3.5** Find  $\mathcal{L}^{-1} \left\{ \frac{1}{s(s^2 + 4)} \right\}$ .

**Solution**

$$\frac{1}{s(s^2 + 4)} = \frac{1}{4} \frac{1}{s} - \frac{1}{4} \frac{s}{s^2 + 4}.$$

Therefore,

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{1}{s(s^2 + 4)} \right\} &= \frac{1}{4} \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} - \frac{1}{4} \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 4} \right\} \\ &= \frac{1}{4} - \frac{1}{4} \cos 2t. \end{aligned} \quad \blacksquare$$

**Example 4.3.6** Find  $\mathcal{L}^{-1} \left\{ \frac{8}{s^3(s^2 - s - 2)} \right\}$ .

**Solution** Since

$$\frac{8}{s^3(s^2 - s - 2)} = \frac{8}{s^3(s - 2)(s + 1)},$$

the fraction<sup>[7]</sup> can be decomposed into<sup>[8]</sup> the following form

$$\frac{8}{s^3(s^2 - s - 2)} = \frac{A_1}{s} + \frac{A_2}{s^2} + \frac{A_3}{s^3} + \frac{B}{s - 2} + \frac{C}{s + 1}.$$

Clearing the fractions<sup>[9]</sup>,

$$\begin{aligned} 8 &= A_1 s^2(s - 2)(s + 1) + A_2 s(s - 2)(s + 1) + A_3(s - 2)(s + 1) \\ &\quad + Bs^3(s + 1) + Cs^3(s - 2). \end{aligned}$$

Setting  $s = -1, 2, 0, 1$  and  $-2$  consecutively<sup>[10]</sup> we obtain, respectively, that

$$C = \frac{8}{3}, \quad B = \frac{1}{3}, \quad A_3 = -4, \quad A_1 = -3, \quad A_2 = 2.$$

[7] 分式、分数

[8] 分解成

[9] 去掉分式

[10] 依次地

[11] 部分分式分解式 Thus, we have the *partial-fractions decomposition*<sup>[11]</sup>

$$\frac{8}{s^3(s^2 - s - 2)} = -\frac{3}{s} + \frac{2}{s^2} - \frac{4}{s^3} + \frac{1}{3} \frac{1}{s-2} + \frac{8}{3} \frac{1}{s+1}.$$

Therefore,

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{8}{s^3(s^2 - s - 2)} \right\} &= \mathcal{L}^{-1} \left\{ -\frac{3}{s} + \frac{2}{s^2} - \frac{4}{s^3} + \frac{1}{3} \frac{1}{s-2} + \frac{8}{3} \frac{1}{s+1} \right\} \\ &= \mathcal{L}^{-1} \left\{ -\frac{3}{s} \right\} + \mathcal{L}^{-1} \left\{ \frac{2}{s^2} \right\} + \mathcal{L}^{-1} \left\{ -\frac{4}{s^3} \right\} \\ &\quad + \mathcal{L}^{-1} \left\{ \frac{1}{3} \frac{1}{s-2} \right\} + \mathcal{L}^{-1} \left\{ \frac{8}{3} \frac{1}{s+1} \right\} \\ &= -3 + 2t - 2t^2 + \frac{1}{3}e^{2t} + \frac{8}{3}e^{-t}. \quad \blacksquare \end{aligned}$$

### Exercise 4.3

Find the inverse Laplace transform of each of following functions.

1.  $F(s) = \frac{1}{s^4}.$

2.  $F(s) = \frac{1}{s-8}.$

3.  $F(s) = \frac{s}{s^2 - 2s + 5}.$

4.  $F(s) = \frac{1}{s^2 - 2s + 9}.$

5.  $F(s) = \frac{s}{s^2 - 3s - 12}.$

6.  $F(s) = \frac{s}{s^2 - 4s + 13}.$

7.  $F(s) = \frac{5s}{(s^2 + 1)^2}.$

8.  $F(s) = \frac{s+1}{s^2 - 9}.$

9.  $F(s) = \frac{s+4}{s^2 + 4s + 8}.$

10.  $F(s) = \frac{s+2}{s^2 - 3s + 4}.$

11.  $F(s) = \frac{s+3}{(s+1)(s-2)}.$

12.  $F(s) = \frac{1}{(s+1)(s^2+1)}.$

## 4.4 Solving Differential Equations by Laplace Transforms

Let  $F(s) = \mathcal{L}\{f(t)\}$ . Then, by Property 4.2.4,

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0) = sF(s) - f(0). \quad (4.4.1)$$

**Corollary 4.4.1** Let  $F(s) = \mathcal{L}\{f(t)\}$ . Then,

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0). \quad (4.4.2)$$

**Proof** Using Property 4.2.4 twice,

$$\mathcal{L}\{f''(t)\} = s\mathcal{L}\{f'(t)\} - f'(0)$$

$$\begin{aligned}
 &= s[sF(s) - f(0)] - f'(0) \\
 &= s^2 F(s) - sf(0) - f'(0).
 \end{aligned}$$

[12] 在非常宽松的  
条件下

[13] 极其巧妙的

[14] 代数方程

In general, if  $F(s) = \mathcal{L}\{f(t)\}$ , then under very broad conditions<sup>[12]</sup>, the Laplace transform of the  $n$ th-derivative ( $n = 1, 2, \dots$ ) of  $f(t)$  is

$$\begin{aligned}
 \mathcal{L}\left\{\frac{d^n f}{dt^n}\right\} &= s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - s^{n-3}f''(0) - \dots \\
 &\quad - sf^{(n-2)}(0) - f^{(n-1)}(0).
 \end{aligned} \tag{4.4.3}$$

If the initial conditions on  $f(t)$  at  $t = 0$  are given by

$$f(0) = c_0, \quad f'(0) = c_1, \quad \dots, \quad f^{(n-1)}(0) = c_{n-1},$$

then formula (4.4.3) can be rewritten as

$$\mathcal{L}\left\{\frac{d^n f}{dt^n}\right\} = s^n F(s) - c_0 s^{n-1} - c_1 s^{n-2} - c_2 s^{n-3} - \dots - c_{n-2} s - c_{n-1}. \tag{4.4.4}$$

Laplace transform is a very different and *extremely clever*<sup>[13]</sup> way in solving differential equations. In fact, from Property 4.2.4 and Corollary 4.4.1, we can reduce the problem of solving the initial-value problem

$$ay'' + by' + cy = f(t); \quad y(0) = y_0, y'(0) = y'_0(0) \tag{4.4.5}$$

to that of solving an *algebraic equation*<sup>[14]</sup>.

Let  $Y(s)$  and  $F(s)$  be the Laplace transforms of  $y(t)$  and  $f(t)$  respectively. Taking Laplace transforms of both sides of differential equation (4.4.5) gives

$$\mathcal{L}\{ay''(t) + by'(t) + cy(t)\} = F(s).$$

By the linearity of Laplace transform,

$$\begin{aligned}
 &\mathcal{L}\{ay''(t) + by'(t) + cy(t)\} \\
 &= a\mathcal{L}\{y''(t)\} + b\mathcal{L}\{y'(t)\} + c\mathcal{L}\{y(t)\}.
 \end{aligned}$$

From Property 4.2.4 and Corollary 4.4.1,

$$F(s) = a[s^2 Y(s) - sy_0 - y'_0] + b[sY(s) - sy_0] + cY(s)$$

$$=(as^2 + bs + c)Y(s) - ay'_0 - (as + b)y_0.$$

Hence,

$$Y(s) = \frac{(as + b)y_0}{as^2 + bs + c} + \frac{ay'_0}{as^2 + bs + c} + \frac{F(s)}{as^2 + bs + c}. \quad (4.4.6)$$

By inverse Laplace transform we can find the solution

$$y(t) = \mathcal{L}^{-1}\{Y(s)\}.$$

The method of Laplace transform enables us to find the solution of an initial-value problem directly without first finding all solutions of a differential equation.

**Example 4.4.1** Solve the initial-value problem  $y' - 5y = 0$ ;  $y(0) = 2$  by Laplace transform.

**Solution** Let  $Y(s) = \mathcal{L}\{y\}$ . Taking Laplace transform of both sides of the differential equation gives

$$[sY(s) - 2] - 5Y(s) = 0,$$

for which

$$Y(s) = \frac{2}{s - 5}.$$

Therefore,

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = 2e^{5t}. \quad \blacksquare$$

**Example 4.4.2** Solve the initial-value problem

$$y'' - 3y' + 2y = e^{3t}; \quad y(0) = 1, y'(0) = 0$$

by Laplace transform.

**Solution** Let  $Y(s) = \mathcal{L}\{y(t)\}$ . Taking Laplace transforms of both sides of the differential equation gives

$$s^2Y(s) - s - 3[sY(s) - 1] + 2Y(s) = \frac{1}{s - 3},$$

thus,

$$\begin{aligned} Y(s) &= \frac{1}{(s - 3)(s^2 - 3s + 2)} + \frac{s - 3}{s^2 - 3s + 2} \\ &= \frac{1}{(s - 3)(s - 2)(s - 1)} + \frac{s - 3}{(s - 2)(s - 1)}. \end{aligned}$$

*Expanding* each term on the right-hand side of last equation [15] 将 ... 展开  
*in partial fractions*<sup>[15]</sup> gives 成部分分式

$$\frac{1}{(s-3)(s-2)(s-1)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s-3}$$

or

$$A(s-2)(s-3) + B(s-1)(s-3) + C(s-1)(s-2) = 1.$$

Setting  $s = 1$  in the last equation gives  $A = 1/2$ ; setting  $s = 2$  gives  $B = -1$ ; and setting  $s = 3$  gives  $C = 1/2$ . Hence, we obtain the partial-fraction decomposition

$$\frac{1}{(s-3)(s-2)(s-1)} = \frac{1}{2} \frac{1}{s-1} - \frac{1}{s-2} + \frac{1}{2} \frac{1}{s-3}.$$

Similarly, we have that

$$\frac{1}{(s-2)(s-1)} = \frac{D}{s-1} + \frac{E}{s-2}.$$

Clearing the fractions,

$$D(s-2) + E(s-1) = s-3.$$

Setting  $s = 1$  in the last equation gives  $D = 2$ , while setting  $s = 2$  gives  $E = -1$ . Hence,

$$\begin{aligned} Y(s) &= \frac{1}{2} \frac{1}{s-1} - \frac{1}{s-2} + \frac{1}{2} \frac{1}{s-3} + \frac{2}{s-1} - \frac{1}{s-2} \\ &= \frac{5}{2} \frac{1}{s-1} - \frac{2}{s-2} + \frac{1}{2} \frac{1}{s-3} \\ &= \frac{5}{2} \mathcal{L}\{e^t\} - 2 \mathcal{L}\{e^{2t}\} + \frac{1}{2} \mathcal{L}\{e^{3t}\} \\ &= \mathcal{L}\left\{\frac{5}{2}e^t - 2e^{2t} + \frac{1}{2}e^{3t}\right\}, \end{aligned}$$

so that

$$y(t) = \frac{5}{2}e^t - 2e^{2t} + \frac{1}{2}e^{3t}. \quad \blacksquare$$

**Remark 4.4.1** Different functions may have same Laplace transform.

For example, function

[16] 跳跃式间断

$$z(t) = \begin{cases} \frac{5}{2}e^t - 2e^{2t} + \frac{1}{2}e^{3t}, & t \neq 1, 2, 3, \\ 0, & t = 1, 2, 3 \end{cases}$$

has an exactly same Laplace transform as function  $y(t)$  in Example 4.4.2, but  $z(t)$  differs from  $y(t)$  at three points.

**Theorem 4.4.1 (Uniqueness of Inverse Laplace Transforms)**

Suppose that functions  $f(t)$  and  $g(t)$  are both piecewise continuous for  $t \geq 0$  and are both of exponential orders as  $t$  approaches infinity. If their Laplace transforms  $F(s)$  and  $G(s)$  are equal for all  $s > c$  for some  $c$ , then  $f(t) = g(t)$  whenever  $f(t)$  and  $g(t)$  are continuous on interval  $[0, +\infty)$ .

**4.4.1 The Right-Hand Side of the Differential Equation is Discontinuous**

In many applications, the right-hand of differential equation

$$ay'' + by' + cy = f(t)$$

has a *jump discontinuity*<sup>[16]</sup> at one or more points. For example, a particle may be moving under the influence of a force  $f_1(t)$ , and suddenly, at time  $t_1$ , an additional force  $f_2(t)$  is applied to the particle. In this part, we will show how to handle such problems by the method of Laplace transform. The simplest example of function with a single jump discontinuity is the unit step function

$$u_a(t) = \begin{cases} 0, & 0 \leq t < a, \\ 1, & t \geq a \end{cases} \quad (4.4.7)$$

Its Laplace transform, derived in Section 4.1 (see Figure 4.4.1), is

$$\mathcal{L}\{u_a(t)\} = \frac{e^{-as}}{s}, \quad s > 0. \quad (4.4.8)$$

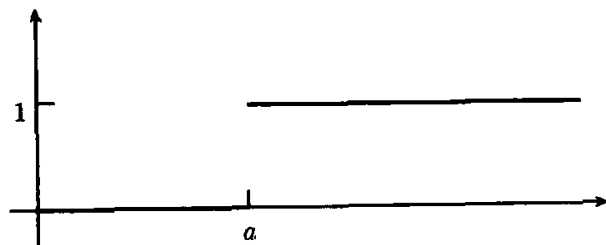


Figure 4.4.1 Graph of  $u_a(t)$

**Property 4.4.1** Let  $F(s) = \mathcal{L}\{f(t)\}$ . Then,

$$\mathcal{L}\{u_a(t)f(t-a)\} = e^{-as}F(s). \quad (4.4.9)$$

**Proof** By the definition

$$\mathcal{L}\{u_a(t)f(t-a)\} = \int_0^\infty e^{-st}u_a(t)f(t-a)dt = \int_a^\infty e^{-st}f(t-a)dt.$$

Making substitution  $\xi = t - a$  gives

$$\begin{aligned} \int_a^\infty e^{-st}f(t-a)dt &= \int_0^\infty e^{-s(\xi+a)}f(\xi)d\xi \\ &= e^{-as} \int_0^\infty e^{-s\xi}f(\xi)d\xi \\ &= e^{-as}F(s). \end{aligned}$$

Hence,  $\mathcal{L}\{u_a(t)f(t-a)\} = e^{-as}\mathcal{L}\{f(t)\}$ . ■

**Example 4.4.3** Solve the initial-value problem

$$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = f(t) = \begin{cases} 1, & 0 \leq t < 2, \\ 0, & 2 \leq t < 4, \\ 1, & 4 \leq t; \end{cases} \quad y(0) = 0, y'(0) = 0.$$

**Solution** Let  $Y(s) = \mathcal{L}\{y(t)\}$  and  $F(s) = \mathcal{L}\{f(t)\}$ . Then

$$(s^2 - 3s + 2)Y(s) = F(s),$$

so that

$$Y(s) = \frac{F(s)}{s^2 - 3s + 2} = \frac{F(s)}{(s-1)(s-2)}.$$

Write  $f(t)$  in the form

$$f(t) = [u_0(t) - u_2(t)] + u_4(t).$$

Then,

$$F(s) = \frac{1}{s} - \frac{e^{-2s}}{s} + \frac{e^{-4s}}{s}.$$

Therefore,

$$Y(s) = \frac{1 - e^{-2s} + e^{-4s}}{s(s-1)(s-2)}.$$

Expanding  $1/[s(s-1)(s-2)]$  in partial fractions

$$\frac{1}{s(s-1)(s-2)} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s-2},$$

[17] 冲量函数

then,

[18] 冲量、脉冲

$$A(s-1)(s-2) + Bs(s-2) + Cs(s-1) = 1. \quad (4.4.10)$$

Setting  $s = 0$  in Eq. (4.4.10) gives  $A=1/2$ ; setting  $s=1$  gives  $B = -1$ ; and setting  $s = 2$  gives  $C = 1/2$ . Thus,

$$\begin{aligned} \frac{1}{s(s-1)(s-2)} &= \frac{1}{2} \frac{1}{s} - \frac{1}{s-1} + \frac{1}{2} \frac{1}{s-2} \\ &= \mathcal{L} \left\{ \frac{1}{2} - e^t + \frac{1}{2} e^{2t} \right\}. \end{aligned}$$

Consequently,

$$\begin{aligned} y(t) &= \left[ \frac{1}{2} - e^t + \frac{1}{2} e^{2t} \right] - u_2(t) \left[ \frac{1}{2} - e^{t-2} + \frac{1}{2} e^{2(t-2)} \right] \\ &\quad + u_4(t) \left[ \frac{1}{2} - e^{t-4} + \frac{1}{2} e^{2(t-4)} \right]. \end{aligned} \quad \blacksquare$$

#### 4.4.2 The Right-Hand Side of Differential Equation is an Impulsive Function<sup>[17]</sup>

In many physical and biological applications we are often confronted with such initial-value problem

$$ay'' + by' + cy = f(t); \quad y(0) = y_0, \quad y'(0) = y'_0, \quad (4.4.11)$$

where  $f(t)$  is identically zero except for a very short time interval  $[t_0, t_1]$ , and its integral over this interval is a given number  $I_0 \neq 0$ . Such function  $f(t)$  is called **impulsive function**. Figure 4.4.2 gives the graph of a typical impulsive function  $f(t)$ .

Physicist P.A.M.Dirac (1902—1984, a Nobel Prize owner) developed a “strange” function (in his study on calculating *the momentum*<sup>[18]</sup> of a moving particle from time  $t_0$  to  $t_1$ ) in the early 1930’s, called **delta function**, for dealing with such impulsive functions as follows:

$$\delta(t-t_0) \begin{cases} \neq 0, & t \neq t_0, \\ = 0, & t = t_0 \end{cases} \quad (4.4.12)$$

and

$$\int_{-\infty}^{+\infty} \delta(t-t_0) dt = 1. \quad (4.4.13)$$

The function  $\delta(t - t_0)$  is, obviously, not an ordinary function<sup>[19]</sup>. However, we can operate it as if it really were an ordinary function. In fact, if we set  $f(t) = I_0\delta(t - t_0)$  in Eq.(4.4.11) and impose the condition

$$\int_a^b g(t)\delta(t - t_0)dt = \begin{cases} g(t_0), & a \leq t_0 \leq b, \\ 0, & \text{otherwise,} \end{cases} \quad (4.4.14)$$

where  $g(t)$  is any continuous function, then we will always obtain a correct solution  $y(t)$  of Eq.(4.4.11).

From the definition of Laplace transform and condition (4.4.14), we can obtain the Laplace transform of  $\delta$ -function  $\delta(t - t_0)$  ( $g(t) = e^{-st}$ )

$$\mathcal{L}\{\delta(t - t_0)\} = \int_0^{+\infty} e^{-st}\delta(t - t_0)dt = e^{-st_0} \quad (4.4.15)$$

for  $t_0 \geq 0$ . Now, we can solve the differential equation (4.4.11), with the right-hand side is an impulsive function, by the method of Laplace transform.

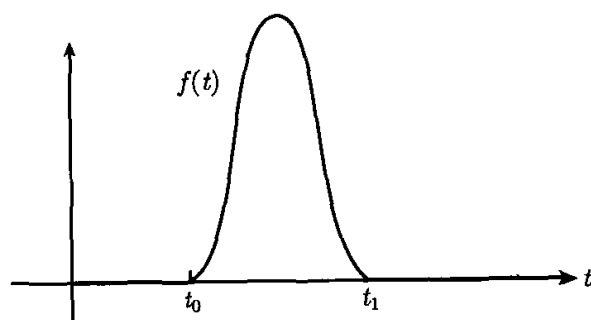


Figure 4.4.2 The graph of impulsive function  $f(t)$

**Example 4.4.4** A particle of mass 1 is attached to a spring-dashpot mechanism. The stiffness constant of the spring is 1N/ft and the drag force exerted by the dashpot mechanism on the particle is twice its velocity. At time  $t = 0$ , when the particle is at rest, an external force  $e^{-t}$  is applied to the system. At time  $t = 1$ , an additional force  $f(t)$  of very short duration<sup>[20]</sup> is applied to the particle. This force imparts an impulse of 3 N·s to the particle<sup>[21]</sup>. Find the position of the particle at any time  $t$  great than 1.

**Solution** Let  $y(t)$  denote the distance of the particle from the

[19] 一个通常意义下的函数

[20] 作用时间、持续时间

[21] 给物体一个 3N·s 的冲量

equilibrium position. Then,  $y(t)$  satisfies the initial-value problem

$$y'' + 2y' + y = e^{-t} + 3\delta(t-1); \quad y(0) = 0, y'(0) = 0.$$

Taking Laplace transform of the last differential equation and let  $Y(s) = \mathcal{L}\{y(t)\}$ , we have that

$$(s^2 + 2s + 1)Y(s) = \frac{1}{s+1} + 3e^{-s}$$

or

$$Y(s) = \frac{1}{(s+1)^3} + \frac{3e^{-s}}{(s+1)^2}.$$

Since

$$\frac{1}{(s+1)^3} = \mathcal{L}\left\{\frac{t^2 e^{-t}}{2}\right\}$$

and  $\frac{3e^{-s}}{(s+1)^2} = 3\mathcal{L}\{u_1(t)(t-1)e^{-(t-1)}\}$ ,

we have that

$$y(t) = \frac{t^2 e^{-t}}{2} + 3u_1(t)(t-1)e^{-(t-1)}.$$

Consequently,

$$y(t) = \frac{t^2 e^{-t}}{2} + 3(t-1)e^{-(t-1)}$$

for  $t > 1$ . ■

#### Exercise 4.4

Solve following initial-value problem by the method of Laplace transform.

1.  $y' + 2y = 0; y(0) = 1$ .
2.  $y' - 5y = e^{5t}; y(0) = 0$ .
3.  $y' + y = \sin t; y(0) = 1$ .
4.  $y'' + 4y = 0; y(0) = 2, y'(0) = 2$ .
5.  $y'' - 3y' + 4y = 0; y(0) = 1, y'(0) = 5$ .
6.  $y'' - y' - 2y = 4t^2; y(0) = 1, y'(0) = 4$ .
7.  $y'' - 2y' + y = te^t; y(0) = 0, y'(0) = 0$ .
8.  $y'' + 4y' + 8y = \sin t; y(0) = 1, y'(0) = 0$ .
9.  $y'' + y = \begin{cases} 2, & 0 \leq t \leq 3, \\ 3t - 7, & 3 < t < \infty; \end{cases} \quad y(0) = 0, y'(0) = 0.$

10.  $y'' + y = \begin{cases} 1, & 0 \leq t < 4, \\ 0, & t > 4; \end{cases} \quad y(0) = 3, y'(0) = -2.$
11.  $y'' + y = \begin{cases} \cos t, & 0 \leq t < \pi/2, \\ 0, & \pi/2 \leq t < \infty; \end{cases} \quad y(0) = 3, y'(0) = -1.$
12.  $y'' + y' + 7y = \begin{cases} t, & 0 \leq t < 2, \\ 0, & 2 \leq t < \infty; \end{cases} \quad y(0) = 0, y'(0) = 0.$
13.  $y'' - 2ty' + y = \begin{cases} 0, & 0 \leq t < 1, \\ t, & 1 \leq t < 2, \\ 0, & 2 \leq t < \infty; \end{cases} \quad y(0) = 0, y'(0) = 1.$
14.  $y'' + 2y' + y = e^{-t} + 3\delta(t - 3); \quad y(0) = 0, y'(0) = 3.$
15.  $y'' + y = \sin t + \delta(t - \pi); \quad y(0) = 0, y'(0) = 0.$
16.  $y'' - 4y' + 4y = 3\delta(t - 1) + \delta(t - 2); \quad y(0) = 0, y'(0) = 1.$
17. Let  $F(s) = \mathcal{L}\{f(t)\}$ . Prove that

$$\mathcal{L}\left\{\frac{d^n f(t)}{dt^n}\right\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0).$$

*Hint:* Try induction.

## 4.5 Solving Systems of Differential Equations by Laplace Transforms

The method of Laplace transforms can also be used to solve the systems of differential equations. Consider the initial-value problem

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{f}(t); \quad \mathbf{x}(0) = \mathbf{x}^0. \quad (4.5.1)$$

Let

$$\mathbf{X}(s) = \begin{pmatrix} X_1(s) \\ \vdots \\ X_n(s) \end{pmatrix} = \mathcal{L}\{\mathbf{x}(t)\} = \begin{pmatrix} \int_0^\infty e^{-st} x_1(t) dt \\ \vdots \\ \int_0^\infty e^{-st} x_n(t) dt \end{pmatrix}$$

and

$$\mathbf{F}(s) = \begin{pmatrix} F_1(s) \\ \vdots \\ F_n(s) \end{pmatrix} = \mathcal{L}\{\mathbf{f}(t)\} = \begin{pmatrix} \int_0^\infty e^{-st} f_1(t) dt \\ \vdots \\ \int_0^\infty e^{-st} f_n(t) dt \end{pmatrix}.$$

Taking Laplace transforms of both sides of Eq.(4.5.1) gives

$$\begin{aligned}\mathcal{L}\{\dot{\mathbf{x}}(t)\} &= \mathcal{L}\{\mathbf{A}\mathbf{x}(t) + \mathbf{f}(t)\} = \mathbf{A}\mathcal{L}\{\mathbf{x}(t)\} + \mathcal{L}\{\mathbf{f}(t)\} \\ &= \mathbf{A}\mathbf{X}(s) + \mathbf{F}(s),\end{aligned}$$

and from Property 4.2.4,

$$\begin{aligned}\mathcal{L}\{\dot{\mathbf{x}}_1(t)\} &= \begin{pmatrix} \mathcal{L}\{\dot{x}_1(t)\} \\ \vdots \\ \mathcal{L}\{\dot{x}_n(t)\} \end{pmatrix} = \begin{pmatrix} sX_1(s) - x_1(0) \\ \vdots \\ sX_n(s) - x_n(0) \end{pmatrix} \\ &= s\mathbf{X}(s) - \mathbf{x}^0.\end{aligned}$$

Hence,

$$s\mathbf{X}(s) - \mathbf{x}^0 = \mathbf{A}\mathbf{X}(s) + \mathbf{F}(s)$$

or

$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{x}^0 + \mathbf{F}(s). \quad (4.5.2)$$

This is a system of  $n$  simultaneous equations for  $X_1(s), \dots, X_n(s)$ . If we know  $X_1(s), \dots, X_n(s)$ , then we can find  $x_1(t), \dots, x_n(t)$  by inverting the above Laplace transforms.

**Example 4.5.1** Solve the initial-value problem

$$\dot{\mathbf{x}} = \begin{pmatrix} 1 & -3 \\ -2 & 2 \end{pmatrix} \mathbf{x}; \quad \mathbf{x}(0) = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

by Laplace transform.

**Solution** Taking Laplace transforms of both sides of the differential equation gives

$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{x}(0) + \mathbf{F}(s),$$

where  $\mathbf{F}(s) = \mathbf{0}$ . Thus,

$$\begin{pmatrix} s-1 & 3 \\ 2 & s-2 \end{pmatrix} \begin{pmatrix} X_1(s) \\ X_2(s) \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

or

$$\begin{cases} (s-1)X_1 + 3X_2 = 0, \\ 2X_1 + (s-2)X_2 = 5. \end{cases}$$

From these equations we obtain that

$$X_1(s) = -\frac{15}{(s-4)(s+1)} = \frac{3}{s+1} - \frac{3}{s-4},$$

$$X_2(s) = \frac{5(s-1)}{(s-4)(s+1)} = \frac{2}{s+1} + \frac{3}{s-4}.$$

Hence, the solution of initial value problem is

$$\begin{aligned} x_1(t) &= \mathcal{L}^{-1}\{X_1(s)\} = \mathcal{L}^{-1}\left\{\frac{3}{s+1}\right\} - \mathcal{L}^{-1}\left\{\frac{3}{s-4}\right\} \\ &= 3e^{-t} - 3e^{4t}, \\ x_2(t) &= \mathcal{L}^{-1}\{X_2(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{s+1}\right\} + \mathcal{L}^{-1}\left\{\frac{3}{s-4}\right\} \\ &= 2e^{-t} + 3e^{4t}. \end{aligned}$$

**Example 4.5.2** Solve the initial-value problem

$$\dot{\mathbf{x}} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 0 \\ 2 \end{pmatrix}; \quad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

by Laplace transform.

**Solution** Taking Laplace transforms of both sides of the differential equation gives

$$\begin{cases} [sX_1(s) - 1] + X_1(s) - X_2(s) = 0, \\ [sX_2(s) - 2] - X_1(s) + X_2(s) = \frac{2}{s}. \end{cases}$$

Thus,

$$X_1(s) = \frac{s+1}{s^2}, \quad X_2(s) = \frac{2s+1}{s^2}.$$

Taking inverse transforms, we obtain that

$$\begin{aligned} x_1(t) &= \mathcal{L}^{-1}\{X_1(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = 1 + t, \\ x_2(t) &= \mathcal{L}^{-1}\{X_2(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = 2 + t. \end{aligned}$$

### Exercise 4.5

Find the solution of each of following initial-value problems by Laplace transform.

$$1. \dot{\mathbf{x}} = \begin{pmatrix} 1 & -3 \\ -2 & 2 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 0 \\ 5 \end{pmatrix}.$$

$$2. \dot{\mathbf{x}} = \begin{pmatrix} -4 & 6 \\ -3 & 5 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

$$3. \dot{\mathbf{x}} = \begin{pmatrix} -5 & 12 \\ -2 & 5 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 8 \\ 3 \end{pmatrix}.$$

$$4. \dot{\mathbf{x}} = \begin{pmatrix} 0 & -1 \\ -4 & 0 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} t; \mathbf{x}(0) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$5. \dot{\mathbf{x}} = \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t; \mathbf{x}(0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

$$6. \dot{\mathbf{x}} = \begin{pmatrix} 3 & -2 \\ 2 & -2 \end{pmatrix} \mathbf{x} + \begin{pmatrix} t \\ 3e^t \end{pmatrix}; \mathbf{x}(0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

$$7. \dot{\mathbf{x}} = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t; \mathbf{x}(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

$$8. \dot{\mathbf{x}} = \begin{pmatrix} 4 & 5 \\ -2 & -2 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 4e^t \cos t \\ 0 \end{pmatrix}; \mathbf{x}(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

$$9. \dot{\mathbf{x}} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} \mathbf{x} + \begin{pmatrix} \sin t \\ e^t \\ 1 \end{pmatrix}; \mathbf{x}(0) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

$$10. \dot{\mathbf{x}} = \begin{pmatrix} 1 & 2 & -3 \\ 1 & 1 & 2 \\ 1 & -1 & 4 \end{pmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

$$11. \dot{\mathbf{x}} = \begin{pmatrix} -1 & -1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 3 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^t; \mathbf{x}(0) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

# Chapter 5 Introduction to the Stability Theory<sup>[1]</sup>

## 5.1 Introduction

Suppose that  $p_1(t)$  and  $p_2(t)$  denote the populations of two species, species 1 and species 2, respectively at time  $t$ . These two species live together in a *microcosm*<sup>[2]</sup> and will compete for limited food and living space. Suppose, moreover, that the rates of growth of  $p_1(t)$  and  $p_2(t)$  are governed by following system of differential equations

$$\dot{\mathbf{p}} = \mathbf{f}(t, \mathbf{p}), \quad (5.1.1)$$

where  $\mathbf{p} = (p_1(t), p_2(t))^T$  and  $\mathbf{f}(t, \mathbf{p}) = (f_1(t, \mathbf{p}), f_2(t, \mathbf{p}))^T$ .

As we have seen in Chapter 3 that what we are really interested in, in practice, is the answers of following questions rather than the values of  $p_1(t)$  and  $p_2(t)$  at every time  $t$ .

1. Does there exist a solution  $p_1(t) \equiv \xi_1$ ,  $p_2(t) \equiv \xi_2$  of Eq.(5.1.1) at which the two species coexist together in a steady state?

2. If a few members of one species, say species 1, are suddenly added to the microcosm in which two species are coexisting in equilibrium, what will happen? Will  $p_1(t)$  and  $p_2(t)$  approach to their equilibrium values as time  $t$  approaches infinitely? Or the extra few members will give species 1 a large advantage and it will *annihilate*<sup>[3]</sup> species 2?

3. Will one species ultimately become the victor, or will the struggle for existence end in a *draw*<sup>[4]</sup> as  $t$  approaches infinity?

Mathematically, these questions concern the equilibrium solutions and stability of the solutions of Eq.(5.1.1).

Consider differential equation of the form

[1] 稳定性理论初步

[2] 一个小环境

[3] 消灭

[4] 平局

[5] 一阶自治微分  
方程

$$\dot{x}(t) = f(x), \quad (5.1.2)$$

[6] 自治系统

[7] 是李雅普诺夫  
意义下稳定的

where  $f$  is a function of  $x$  alone, we call differential equation (5.1.2) an *autonomous first-order differential equation*<sup>[5]</sup>.

**Definition 5.1.1** If the solution  $x(t)$  of Eq.(5.1.2) is identically constant, that is  $x(t) \equiv c$ , then we call such solution an **equilibrium solution** of Eq.(5.1.2).

**Example 5.1.1** Find all equilibrium solutions of following *autonomous system*<sup>[6]</sup>

$$\frac{dx}{dt} = x - x^2 - 2xy, \quad \frac{dy}{dt} = 2y - 2y^2 - 3xy.$$

**Solution** Set

$$x - x^2 - 2xy = 0, \quad 2y - 2y^2 - 3xy = 0$$

or

$$x(1 - x - 2y) = 0, \quad y(2 - 2y - 3x) = 0. \quad (5.1.3)$$

If  $x = 0$  in the first equation in (5.1.3), then, from the second equation in (5.1.3), we obtain that  $y(2 - 2y) = 0$ . This implies that  $y = 0$  or  $y = 1$ . Thus we obtain two equilibrium solutions  $(0, 0)$ ,  $(0, 1)$ .

If  $1 - x - 2y = 0$  in the first equation in (5.1.3), or equivalently  $1 - x = 2y$ , then we see, from this equation, that  $x = 1$  for  $y = 0$ ; or  $2 - 2y - 3x = 0$  from the second equation in (5.1.3) for  $y \neq 0$ . Substituting  $2y = 1 - x$  into the last equation gives

$$2 - (1 - x) - 3x = 0 \Rightarrow x = \frac{1}{2},$$

for this  $x$ ,  $y = 1/4$ . Thus, we obtain another two equilibrium solutions  $(1, 0)$ ,  $(\frac{1}{2}, \frac{1}{4})$ . ■

**Definition 5.1.2** Let  $x(t) \equiv c$  be an equilibrium solution of Eq.(5.1.2).  $x(t) \equiv c$  is said to be **stable in the sense of Lyapunov**<sup>[7]</sup> if any solution  $x(t)$  of Eq.(5.1.2) with initial value  $x_0 = x(t_0)$  has the property that for every  $\varepsilon > 0$ , there exists a small number  $\delta > 0$  such that

$$|x(t) - c| < \varepsilon \quad (5.1.4)$$

for all  $t > t_0$ , whenever  $|x_0 - c| < \delta$ . The equilibrium solution is **unstable**<sup>[8]</sup> if it is not stable.

**Example 5.1.2** Consider the **Logistic differential equation**

$$\frac{dx}{dt} = Kx(M - x), \quad K, M > 0. \quad (5.1.5)$$

It is not difficult to verify that  $x(t) \equiv 0$  and  $x(t) \equiv M$  are two equilibrium solutions, and solution of Eq.(5.1.3) satisfying the initial condition  $x_0 = x(t_0)$  is

$$x(t) = \frac{Mx_0}{x_0 + (M - x_0)e^{KMt}}.$$

Since  $x(t) \rightarrow 0$  as  $t \rightarrow \infty$ , the equilibrium solution  $x \equiv 0$  is stable.

If  $x_0 > M$ , then  $x(t) \rightarrow \infty$  as  $t \rightarrow t^*$  where  $t^* = \frac{1}{KM} \ln \frac{x_0}{x_0 - M}$

$> 0$ . Therefore, the equilibrium solution  $x(t) \equiv M$  is unstable. ■

**Definition 5.1.3** Let  $x(t) \equiv c$  be an equilibrium solution of Eq.(5.1.2).  $x(t) \equiv c$  is called **asymptotic stable**<sup>[9]</sup> if it is stable in the interval  $[t_0, +\infty)$  and for every solution  $x(t)$  of Eq.(5.1.2) there exists a small number  $\delta > 0$  such that

$$\lim_{t \rightarrow \infty} x(t) = c, \quad (5.1.6)$$

whenever  $|x_0 - c| < \delta$ , where  $x(t)$  satisfies the initial condition  $x_0 = x(t_0)$ . If for any choice of number  $\delta$ , Eq.(5.1.6) always holds, then we call  $x(t) \equiv c$  is **globally asymptotic stable**<sup>[10]</sup>.

**Example 5.1.3** Now we reconsider the two coexisting populations whose growth rates are governed by differential equation (5.1.1).

If  $\xi = (\xi_1, \xi_2)$  is a stable equilibrium solution of Eq.(5.1.1), then the populations  $p_1(t)$  and  $p_2(t)$  will keep close to equilibrium point  $\xi = (\xi_1, \xi_2)$  as  $t$  approaches infinity whenever the initial populations  $p_1(t_0)$  and  $p_2(t_0)$  are sufficiently close to  $\xi_1$  and  $\xi_2$  respectively.

If  $\xi = (\xi_1, \xi_2)$  is an asymptotic stable equilibrium solution of Eq.(5.1.1), then  $p_1(t)$  and  $p_2(t)$  will approach the equilibrium

[8] 不稳定的

[9] 渐近稳定的

[10] 全局渐近稳定的

population  $\xi_1$  and  $\xi_2$ , respectively, as  $t \rightarrow \infty$  whenever the initial populations  $p_1(t_0)$  and  $p_2(t_0)$  are sufficiently close to  $\xi_1$  and  $\xi_2$  respectively, rather than merely remaining close to equilibrium solution  $\xi = (\xi_1, \xi_2)$ . ■

### Exercise 5.1

Find all equilibrium solutions of each of following system of differential equations.

1.  $x' = x - y, \quad y' = 1 - x^2.$
2.  $x' = 1 - y, \quad y' = x^3 + y.$
3.  $x' = xy^2 - x, \quad y' = x \sin \pi y.$
4.  $x' = (x - 1)(y - 1), \quad y' = (x + 1)(y + 1).$
5.  $x' = 14x - 2x^2 - xy, \quad y' = 16y - 2y^2 - xy.$
6.  $x' = ax + bxy, \quad y' = -cy + dxy, \quad z' = z + x^2 + y^2.$
7.  $x' = -1 - y - e^x, \quad y' = x^2 + y(e^x - 1), \quad z' = x + \sin z.$

## 5.2 Stability of the Solutions of Linear System

In this section, we discuss the stability of solutions of linear system

$$\dot{\mathbf{x}} = \mathbf{A}(t)\mathbf{x} + \mathbf{b}(t), \quad (5.2.1)$$

where  $\mathbf{A}(t)$  and  $\mathbf{b}(t)$  are continuous on interval  $[0, \infty)$ .

**Theorem 5.2.1** Consider the homogeneous equation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}. \quad (5.2.2)$$

(a) Every solution  $\varphi(t)$  of Eq.(5.2.2) is stable if and only if the zero solution  $\mathbf{x}(t) \equiv \mathbf{0}$  is stable.

(b) Every solution  $\varphi(t)$  of Eq.(5.2.2) is asymptotic stable if and only if the zero solution  $\mathbf{x}(t) \equiv \mathbf{0}$  is asymptotic stable.

**Proof** (a) If solution  $\mathbf{x}(t) \equiv \mathbf{0}$  is stable and  $\psi(t)$  is any solution (other than  $\varphi(t)$ ) of Eq.(5.2.2), then  $\mathbf{y}(t) = \varphi(t) - \psi(t)$  is also a solution of Eq.(5.2.2). Therefore,  $\mathbf{y}(t) = \varphi(t) - \psi(t)$  will remain close to zero point for all future  $t$  whenever  $\mathbf{y}(t_0) = \varphi(t_0) - \psi(t_0)$  is sufficiently close to zero point. That is  $\psi(t)$  will remain close to

$\varphi(t)$  as  $t$  approaches infinite whenever  $\psi(t_0)$  is sufficiently close to  $\varphi(t_0)$ . Hence,  $\varphi(t)$  is stable.

If solution  $x(t) \equiv 0$  is unstable, then, there exists a solution  $y(t)$  of Eq.(5.2.2) which is near the zero solution  $x(t) \equiv 0$  initially but diverges from it as  $t$  approaches infinite. Suppose that  $\varphi(t)$  is any solution of Eq.(5.2.2), then  $\psi(t) = y(t) + \varphi(t)$  is also a solution of Eq.(5.2.2) and  $\psi(t)$  is near  $\varphi(t)$  initially (since  $y(t)$  is near the zero solution initially) but diverges from  $\varphi(t)$  for future time  $t$ . Consequently,  $\varphi(t)$  is unstable.

Part (b) can be proved by a similar way. ■

**Theorem 5.2.2** If the zero solution of homogeneous equation (5.2.2) is stable, then every solution of the nonhomogeneous equation (5.2.1) is also stable. *Similar theorems hold for asymptotic stability and instability*<sup>[11]</sup>.

**Proof** Let  $\phi(t)$  be a solution of Eq.(5.2.1), then, for any other solution  $\varphi(t)$  of Eq.(5.2.1), the difference  $\phi(t) - \varphi(t)$  is a solution of the corresponding homogeneous equation (5.2.2). If the zero solution of homogeneous equation (5.2.2) is stable, then, for every  $\varepsilon > 0$ , there exists a number  $\delta > 0$  such that

$$|\phi(t) - \varphi(t)| < \varepsilon, \quad t > 0$$

whenever  $|\phi(0) - \varphi(0)| < \delta$ . That is,  $\varphi(t)$  will remain close to  $\phi(t)$  for all future time whenever it is close to  $\phi(t)$  at the initial time. So  $\phi(t)$  is stable. ■

Thus, we need only to discuss the stability of zero solution of homogeneous equation (5.2.2). Clearly, the zero solution  $x \equiv 0$  is an equilibrium solution of Eq.(5.2.2). We will use eigenvalue-eigenvector method.

### Theorem 5.2.3

(a) The zero solution  $x \equiv 0$  of Eq.(5.2.2) is stable if all the eigenvalues of  $A$  have negative real part.

(b) The zero solution  $x \equiv 0$  of Eq.(5.2.2) is unstable if at least one eigenvalue of  $A$  has positive real part.

(c) If the characteristic polynomial of coefficient matrix<sup>[12]</sup>  $A$

[11] 对于渐近稳定性和不稳定性有类似的定理成立

[12] 系数矩阵

[13] 分解因式成 can be factored into<sup>[13]</sup> the form

[14] 长度

[15] 分量

$$p(\lambda) = (\lambda - i\sigma_1)^{k_1} \cdots (\lambda - i\sigma_j)^{k_j} q(\lambda),$$

where all the roots of  $q(\lambda)$  have negative real part, then the zero solution  $\mathbf{x} \equiv \mathbf{0}$  of Eq.(5.2.2) is stable if and only if  $\mathbf{A}$  has  $k_j$  linearly independent eigenvectors for each eigenvalue  $\lambda_j = i\sigma_j$ .

Before give the proof of Theorem 5.2.3 we introduce the concept of *length*<sup>[14]</sup> of a vector.

**Definition 5.2.1** Let  $\mathbf{v} = (v_1, \cdots, v_n)$  be a vector with real or complex components<sup>[15]</sup>. The **length** of  $\mathbf{v}$  is defined as

$$\|\mathbf{v}\| = \max\{|v_1|, \cdots, |v_n|\}. \quad (5.2.3)$$

For example,

If  $\mathbf{v} = (1, -6, 2, 0, 3)$ , then  $\|\mathbf{v}\| = \max\{|1|, |-6|, |2|, |0|, |3|\} = 6$ ;

If  $\mathbf{v} = (1-2i, 0, 4+3i)$ , then  $\|\mathbf{v}\| = \max\{|1-2i|, |0|, |4+3i|\} = \sqrt{25} = 5$ .

It is not difficult to prove that the concept of length of a vector satisfies following properties:

(1) For any vector  $\mathbf{v}$ ,  $\|\mathbf{v}\| = 0$  if and only if  $\mathbf{v} = \mathbf{0}$ .

(2)  $\|\lambda\mathbf{v}\| = |\lambda| \|\mathbf{v}\|$ .

(3)  $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$ .

**Remark 5.2.1** We may define the length of a vector by other ways (see Exercise 11).

**Proof of Theorem 5.2.3** (a) If  $\varphi(t) = (\varphi_1(t), \cdots, \varphi_n(t))$  is a solution of  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ , then  $\varphi(t) = e^{\mathbf{A}t}\varphi(0)$ . Let  $e^{\mathbf{A}t} = (\phi_{ij}(t))$  and  $\varphi(0) = (\varphi_1^0, \cdots, \varphi_n^0)$ , thus, we have that

$$\varphi_i(t) = \sum_{j=1}^n \phi_{ij}(t) \varphi_j^0.$$

If all eigenvalues of  $\mathbf{A}$  have negative real part and suppose that  $-\lambda$  is the largest real part of the eigenvalues of  $\mathbf{A}$ , then (see Exercise 12) for any negative number  $-\alpha$  satisfying  $-\lambda < -\alpha < 0$ , there exists a positive number  $K$  such that

$$|\varphi_{ij}(t)| \leq K e^{-\alpha t}, \quad t \geq 0.$$

So, according to the definition of length of a vector,

$$|\varphi_i(t)| \leq \sum_{j=1}^n K e^{-\alpha t} |\varphi_j^0| = K e^{-\alpha t} \sum_{j=1}^n |\varphi_j^0| \leq n K e^{-\alpha t} \|\varphi(0)\|.$$

Hence,

$$\|\varphi(t)\| = \max \{ |\varphi_i(t)| \mid i = 1, \dots, n \} \leq n K e^{-\alpha t} \|\varphi(0)\|.$$

Let  $\varepsilon$  be any small positive number, choose  $\delta = \varepsilon/nK$ , then for all  $t > 0$ ,

$$\|\varphi(t)\| \leq n K e^{-\alpha t} \|\varphi(0)\| \leq n K \delta = \varepsilon$$

whenever  $\|\varphi(0)\| < \delta$ . Consequently, the zero solution  $\mathbf{x}(t) \equiv \mathbf{0}$  is stable.

(b) Let  $\lambda$  be an eigenvalue of  $\mathbf{A}$  with positive real part and  $\mathbf{v}$  be an eigenvector of  $\mathbf{A}$  with the eigenvalue  $\lambda$ , then

$$\varphi(t) = c e^{\lambda t} \mathbf{v}$$

is a solution of Eq.(5.2.2), where  $c$  is an arbitrary constant.

If  $\lambda$  is real, then  $\lambda > 0$  and  $\mathbf{v}$  is also real, thus

$$\|\varphi(t)\| = |c| e^{\lambda t} \|\mathbf{v}\| \rightarrow \infty \quad (t \rightarrow \infty)$$

whenever  $c \neq 0$ . Therefore, the zero solution  $\mathbf{x}(t) \equiv \mathbf{0}$  is unstable.

If  $\lambda = \alpha + \beta i$  is complex, then  $\alpha > 0$  and  $\mathbf{v}$  is also complex-valued. Suppose that  $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2 i$ , we have that

$$\begin{aligned} e^{(\alpha+\beta i)t} (\mathbf{v}_1 + \mathbf{v}_2 i) &= e^{\alpha t} (\cos \beta t + i \sin \beta t) (\mathbf{v}_1 + \mathbf{v}_2 i) \\ &= e^{\alpha t} [(\mathbf{v}_1 \cos \beta t - \mathbf{v}_2 \sin \beta t) + i(\mathbf{v}_1 \sin \beta t + \mathbf{v}_2 \cos \beta t)]. \end{aligned}$$

Therefore,

$$\varphi_1(t) = c e^{\alpha t} (\mathbf{v}_1 \cos \beta t - \mathbf{v}_2 \sin \beta t)$$

is a real-valued solution of Eq.(5.2.2) for any choice of constant  $c$ , and clearly,  $\|\varphi_1(t)\|$  is unbounded as  $t$  approaches infinity whenever  $c$  and either  $\mathbf{v}_1$  or  $\mathbf{v}_2$  are not zero. So the zero solution  $\mathbf{x} \equiv \mathbf{0}$ , in this case, is unstable.

(c) If  $A$  has  $k_j$  linearly independent eigenvectors for each eigenvalue  $\lambda_j = i\sigma_j$  of multiplicity  $k_j$ , then (see Exercise 13) there exists a constant  $K$  such that

$$|(e^{At})_{ij}| \leq K.$$

Therefore, for every solution of Eq.(5.2.2) following inequality holds

$$\|\varphi(t)\| \leq nK \|\varphi(0)\|.$$

Thus, we conclude immediately, from the proof of part (a), that the zero solution  $x(t) \equiv 0$  is stable.

Conversely, if  $A$  has fewer than  $k_j$  linearly independent eigenvectors with eigenvalue  $\lambda_j = i\sigma_j$ , then, according to the algorithm in Section 3.3, Eq.(5.2.2) has solutions of the form

$$\varphi(t) = ce^{i\sigma_j t} [v + t(A - i\sigma_j I)v]$$

with  $(A - i\sigma_j I)v \neq 0$ . Thus, no matter how small the  $\|\varphi(0)\| \neq 0$  is and for any choice of  $c \neq 0$ , if  $\sigma_j = 0$ , then

$$\varphi(t) = c(v + tAv)$$

is real and  $\|\varphi(t)\|$  will be unbounded as  $t$  approaches infinity; if  $\sigma_j \neq 0$ , then both real and imaginary parts of

$$\varphi(t) = ce^{i\sigma_j t} [v + t(A - i\sigma_j I)v]$$

are all unbounded as  $t$  approaches infinity. Therefore, solution  $x(t) \equiv 0$  is unstable. ■

#### Theorem 5.2.4

(a) Every solution  $x(t)$  of Eq.(5.2.2) is stable if all eigenvalues of  $A$  have negative real part.

(b) Every solution  $x(t)$  of Eq.(5.2.2) is unstable if at least one eigenvalue of  $A$  has positive real part.

(c) If the characteristic polynomial of coefficient matrix  $A$  can be factored into the form

$$p(\lambda) = (\lambda - i\sigma_1)^{k_1} \cdots (\lambda - i\sigma_j)^{k_j} Q(\lambda)$$

where all the roots of polynomial  $Q(\lambda)$  have negative real part, then every solution  $\mathbf{x}(t)$  of Eq.(5.2.2) is stable if and only if  $\mathbf{A}$  has  $k_j$  linearly independent eigenvectors for each eigenvalue  $\lambda_j = i\sigma_j$ .

This theorem can be derived directly from Theorem 5.2.2 and Theorem 5.2.3.

**Example 5.2.1** Determine whether each solution  $\mathbf{x}(t)$  of differential equation

$$\dot{\mathbf{x}} = \begin{pmatrix} -5 & 3 \\ -1 & 1 \end{pmatrix} \mathbf{x}$$

is stable or unstable.

**Solution** The characteristic polynomial of coefficient matrix is

$$p(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{pmatrix} -5 - \lambda & 3 \\ -1 & 1 - \lambda \end{pmatrix} = \lambda^2 + 4\lambda - 2.$$

Setting  $p(\lambda) = 0$  we obtain two eigenvalues  $\lambda_{1,2} = -2 \pm \sqrt{6}$ . Since the eigenvalue  $-2 + \sqrt{6} > 0$ , we conclude, by Theorem 5.2.4(b), that every solution of differential equation  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$  is unstable. ■

**Example 5.2.2** Determine the stability or instability of all solutions of equation

$$\dot{\mathbf{x}} = \begin{pmatrix} 0 & 2 & 1 \\ -1 & -3 & -1 \\ 1 & 1 & -1 \end{pmatrix} \mathbf{x}.$$

**Solution** Since the characteristic equation

$$p(\lambda) = \begin{vmatrix} -\lambda & 2 & 1 \\ -1 & -3 - \lambda & -1 \\ 1 & 1 & -1 - \lambda \end{vmatrix} = (\lambda + 2)(1 + \lambda)^2 = 0$$

has three negative eigenvalues

$$\lambda_1 = -2, \quad \lambda_{1,2} = -1,$$

all solutions of the given system are all stable. ■

**Example 5.2.3** Show that every solution of differential equation

$$\dot{\mathbf{x}} = \begin{pmatrix} 2 & -3 & 0 \\ 0 & -6 & -2 \\ -6 & 0 & -3 \end{pmatrix} \mathbf{x}$$

is unstable.

**Solution** The characteristic polynomial of coefficient matrix is

$$p(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{pmatrix} 2 - \lambda & -3 & 0 \\ 0 & -6 - \lambda & -2 \\ -6 & 0 & -3 - \lambda \end{pmatrix} = -\lambda^2(7 + \lambda).$$

Hence, the eigenvalues of  $\mathbf{A}$  are  $\lambda = -7$  and  $\lambda = 0$  (with multiplicity two). Solving equation

$$\mathbf{A}\mathbf{v} = \begin{pmatrix} 2 & -3 & 0 \\ 0 & -6 & -2 \\ -6 & 0 & -3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

we obtain that every eigenvector  $\mathbf{v}$  of  $\mathbf{A}$  with eigenvalue 0 is of the form

$$\mathbf{v} = c \begin{pmatrix} 3 \\ 2 \\ -6 \end{pmatrix}.$$

Consequently, every solution  $\mathbf{x} = \phi(t)$  of  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$  is unstable since  $\mathbf{A}$  has only one linearly independent eigenvector with eigenvalue 0. ■

Typically, for a two-dimensional linear system

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \quad (5.2.4)$$

we have following more precise conclusions.

**Theorem 5.2.5** Let  $\lambda_1$  and  $\lambda_2$  be the eigenvalues of coefficient matrix  $\mathbf{A}$  of Eq.(5.2.4) with  $|\mathbf{A}| \neq 0$ . Then the zero solution  $(0, 0)$  is

- (a) asymptotically stable if the real parts of  $\lambda_1$  and  $\lambda_2$  are both negative;
- (b) stable but not asymptotically stable if the real parts of  $\lambda_1$  and  $\lambda_2$  are both zero;
- (c) unstable if either  $\lambda_1$  or  $\lambda_2$  has a positive real part.

**Example 5.2.4** Show that the zero solution of differential equation

$$\dot{x} = \begin{pmatrix} 0 & -3 \\ 2 & 0 \end{pmatrix} x \quad [16] \text{ 可逆的}$$

is stable, but not asymptotically stable.

**Proof** It is not difficult to verify that the coefficient matrix

$$A = \begin{pmatrix} 0 & -3 \\ 2 & 0 \end{pmatrix}$$

is *invertible*<sup>[16]</sup>. The characteristic polynomial of the matrix is

$$p(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} -\lambda & -3 \\ 2 & -\lambda \end{pmatrix} = \lambda^2 + 6.$$

Thus, the eigenvalues of  $A$  are  $\lambda = \pm\sqrt{6}i$ , the real parts of both of them are zero, therefore, by Theorem 5.2.5(b), the zero solution  $x \equiv 0$  is stable but not asymptotically stable as  $t$  approaches infinity. ■

**Example 5.2.5** Consider the linear system

$$\dot{x} = \frac{1}{4} \begin{pmatrix} 5 & -3 \\ 3 & -5 \end{pmatrix} x.$$

The eigenvalues of coefficient matrix are  $\lambda_1 = 1$  and  $\lambda_2 = -1$ . According to Theorem 5.2.5(c), zero solution of the system is unstable. ■

### Exercise 5.2

Determine the stability of the zero solution of each of following systems of differential equations.

$$1. \dot{x} = \begin{pmatrix} 5 & -3 \\ 3 & -5 \end{pmatrix} x. \quad 2. \dot{x} = \begin{pmatrix} 1 & 5 \\ 5 & 1 \end{pmatrix} x.$$

$$3. \dot{x} = \frac{1}{8} \begin{pmatrix} 7 & 3 \\ -3 & 17 \end{pmatrix} x. \quad 4. \dot{x} = \frac{1}{8} \begin{pmatrix} -7 & -3 \\ 3 & -17 \end{pmatrix} x.$$

$$5. \dot{x} = \frac{1}{4} \begin{pmatrix} -9 & 15 \\ -15 & 9 \end{pmatrix} x. \quad 6. \dot{x} = \frac{1}{4} \begin{pmatrix} -10 & 15 \\ -15 & 8 \end{pmatrix} x.$$

Determine the stability of all solutions of each of following systems of differential equations.

$$7. \dot{\mathbf{x}} = \begin{pmatrix} -1 & 0 & 0 \\ -2 & -1 & 2 \\ -3 & -2 & -1 \end{pmatrix} \mathbf{x}.$$

$$8. \dot{\mathbf{x}} = \begin{pmatrix} -7 & 1 & -6 \\ 10 & -4 & 12 \\ 2 & -1 & 1 \end{pmatrix} \mathbf{x}.$$

$$9. \dot{\mathbf{x}} = \begin{pmatrix} 0 & 2 & 1 \\ -1 & -3 & -1 \\ 1 & 1 & -1 \end{pmatrix} \mathbf{x}.$$

10. Determine the stability of all solutions of differential equation

$$\dot{\mathbf{x}} = \begin{pmatrix} -1 & -1 \\ 2 & -1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

11. Let  $\mathbf{x} = (x_1, \dots, x_n)$  and define the length of  $\mathbf{x}$  as

$$\|\mathbf{x}\| = |x_1| + \dots + |x_n|.$$

Show that

- (a)  $\|\mathbf{x}\| \geq 0$  and  $\|\mathbf{x}\| = 0$  only if  $\mathbf{x} = \mathbf{0}$ ;
- (b)  $\|\lambda \mathbf{x}\| = |\lambda| \|\mathbf{x}\|$ ;
- (c)  $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ .

12. (a) Let  $f(t) = t^a e^{-bt}$  for some positive constants  $a$  and  $b$ , and let  $c$  be a positive number smaller than  $b$ . Show that we can find a positive constant  $K$  such that  $|f(t)| \leq K e^{-ct}$ ,  $0 \leq t < \infty$ .

*Hint:* Show that  $f(t)/e^{-ct}$  approaches zero as  $t$  approaches infinity.

(b) Suppose that all the eigenvalues of  $\mathbf{A}$  have real part. Show that we can find positive constants  $K$  and  $\alpha$  such that  $|(e^{\mathbf{A}t})_{ij}| \leq K e^{-\alpha t}$  for  $1 \leq i, j \leq n$ .

*Hint:* Each component of  $e^{\mathbf{A}t}$  is a finite linear combination of functions of the form  $q(t)e^{\lambda t}$ , where  $q(t)$  is a polynomial in  $t$  (of degree  $\leq n-1$ ) and  $\lambda$  is an eigenvalue of  $\mathbf{A}$ .

13. (a) Let  $\mathbf{x}(t) = e^{i\sigma t} \mathbf{v}$  be a complex-valued solution of  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$  where  $\sigma$  is real. Show that both the real and imaginary parts of  $\mathbf{x}(t)$  are bounded solutions of  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ .

(b) Suppose that all the eigenvalues of  $\mathbf{A}$  have real part  $\leq 0$  and  $\lambda_1 = i\sigma_1, \dots, \lambda_l = i\sigma_l$  have zero real part with multiplicity  $k_1, \dots, k_l$  respectively. Suppose, moreover, that  $\mathbf{A}$  has  $k_j$  linearly independent eigenvectors for each eigenvalue  $\lambda_j$ ,  $j = 1, \dots, l$ . Prove that we can find a constant  $K$  such that  $|(e^{\mathbf{A}t})_{ij}| \leq K$ .

## 5.3 Geometrical Characteristics of Solutions of the System of Differential Equations<sup>[17]</sup>

In this section, we introduce a graphical method for discussing the *behavior*<sup>[18]</sup> of solutions of the system of differential equations, it will provide us an intuitional view of the characteristics of solutions.

### 5.3.1 Phase Space<sup>[19]</sup> and Direction Field<sup>[20]</sup>

Every solution of an  $n$ -dimensional<sup>[21]</sup> system of differential equations defines a curve in an  $n$ -dimensional space. We call this  $n$ -dimensional space the **phase space** (or **phase plane** when  $n = 2$ ) of the solutions of the system and call this curve the **orbit** or **trajectory**<sup>[22]</sup> of the solution of the system, or simply called the orbit of the system. Figure 5.3.1 shows the orbit of solution  $x = \cos t$ ,  $y = \sin t$  of the 2-dimensional system of differential equations

$$\frac{dx}{dt} = -y, \quad \frac{dy}{dt} = x.$$

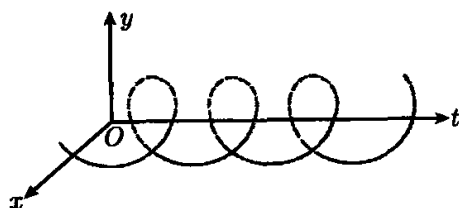


Figure 5.3.1 The orbit of the system

$$x' = -y, y' = x$$

If  $x(t) \equiv x^*$ ,  $y(t) \equiv y^*$  is an equilibrium solution of a system of differential equations, then its orbit consists of only single point  $(x^*, y^*)$ . This type of orbit is a **degenerate**<sup>[23]</sup> curve.

**Theorem 5.3.1 (Existence and Uniqueness of Orbits)** Co-

[17] 微分方程组  
解的几何特性

[18] 动态特性

[19] 相空间

[20] 方向场

[21]  $n$  维的

[22] 轨线

[23] 退化的

[24] 有一个公共点      nsider the autonomous system of differential equations

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}). \quad (5.3.1)$$

If each of the functions  $f_1(x_1, \dots, x_n), \dots, f_n(x_1, \dots, x_n)$  has continuous partial derivatives with respect to  $x_1, \dots, x_n$ , then there exists one and only one orbit through every point  $\mathbf{x}^0$  in  $\mathbf{R}^n$ . That is, if the orbits of two solutions  $\mathbf{x} = \phi(t)$  and  $\mathbf{x} = \psi(t)$  of the system (5.3.1) have one point in common<sup>[24]</sup>, then they must be identical.

**Lemma 5.3.1** If  $\mathbf{x} = \phi(t)$  is a solution of Eq.(5.3.1), then  $\mathbf{x} = \phi(t + c)$  is again a solution of Eq.(5.3.1).

**Proof** From the assumption  $\mathbf{x} = \phi(t)$  is a solution we have that

$$\dot{\phi}(t) = \mathbf{f}(\phi),$$

thus,

$$\dot{\phi}(t + c) = \mathbf{f}(\phi(t + c)).$$

This means that  $\mathbf{x} = \phi(t + c)$  is a solution of Eq.(5.3.1). ■

**Proof of Theorem 5.3.1** Let  $\mathbf{x}^0$  be any point in the  $n$ -dimensional space  $\{x_1, \dots, x_n\}$ , and  $\mathbf{x} = \phi(t)$  be a solution of initial-value problem  $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \mathbf{x}(0) = \mathbf{x}^0$ . The orbit of this solution obviously passes through the point  $\mathbf{x}^0$ . Hence, there exists at least one orbit through every point  $\mathbf{x}^0$ .

Suppose that the orbit of another solution  $\mathbf{x} = \psi(t)$  also passes through  $\mathbf{x}^0$ . This means that there exists  $t_0 (\neq 0)$  such that  $\psi(t_0) = \mathbf{x}^0$ . By Lemma 5.3.1,

$$\mathbf{x} = \psi(t + t_0)$$

is also a solution of system (5.3.1). Note that  $\psi(t + t_0)$  and  $\phi(t)$  have same value at  $t = 0$ . Hence, by the uniqueness of solution of an initial value problem,  $\psi(t + t_0)$  equals  $\phi(t)$  for all time  $t$ . Thus, if  $\xi$  is a point on the orbit of  $\phi(t)$ , that is  $\xi = \phi(t_1)$  for some time  $t_1$ , then  $\xi = \phi(t_1) = \psi(t_1 + t_0)$ , so  $\xi$  is also on the orbit of  $\psi(t)$ . Conversely, if  $\xi$  is a point on the orbit of  $\psi(t)$ , that is there exists a time  $t_2$  such that  $\psi(t_2) = \xi$ , then  $\xi = \psi(t_2) = \phi(t_2 - t_0)$ , so  $\xi$

is also on the orbit of  $\phi(t)$ . Therefore, the orbits of  $\phi(t)$  and  $\psi(t)$  are identical. ■

Because a wide variety of natural phenomena are modeled by a 2-dimensional system of differential equations, we will restrict our attention, for most part of following discussion, to the 2-dimensional autonomous system of differential equations

$$\frac{dx}{dt} = f(x, y), \quad \frac{dy}{dt} = g(x, y), \quad (5.3.2)$$

where the functions  $f$  and  $g$  are continuously differentiable in some region  $D$  in the  $xy$ -plane<sup>[25]</sup>. Here  $xy$ -plane is the phase plane of system (5.3.2).

If the point  $(x^*, y^*)$  satisfies

$$f(x^*, y^*) = 0, \quad g(x^*, y^*) = 0, \quad (5.3.3)$$

then we call point  $(x^*, y^*)$  a **singular point**<sup>[26]</sup> of the system of differential equations (5.3.2). Thus, if the point  $(x^*, y^*)$  is a singular point of the system (5.3.2), then the constant-valued function

$$x(t) \equiv x^*, \quad y(t) \equiv y^* \quad (5.3.4)$$

is an equilibrium solution of the system (5.3.2). Usually, we exhibit *qualitatively*<sup>[27]</sup> the behavior of the orbits near a singular point of the system (5.3.2) in a **direction field** which can be constructed by drawing a vector having the direction  $(f(x, y), g(x, y))$  at each point  $(x, y)$  (see Figure 5.3.2), or in a **slope field**<sup>[28]</sup> which can be constructed by drawing a *line segment*<sup>[29]</sup> having

$$\frac{dy}{dx} = \frac{g(x, y)}{f(x, y)} \quad (5.3.5)$$

at each point  $(x, y)$  (see Figure 5.3.3). A phase plane picture consisting of singular points and some typical *nondegenerate*<sup>[30]</sup> orbits of solutions is called the **phase portrait**<sup>[31]</sup> of solutions. Such phase portraits can be obtained by a MATLAB-based or Maple-based ODE *package with graphical capabilities*<sup>[32]</sup>.

[25]  $xy$  坐标面

[26] 奇点

[27] 定性地

[28] 斜率场

[29] 线段

[30] 非退化的

[31] 相图

[32] 具有绘图功能的软件包

[33] 结点

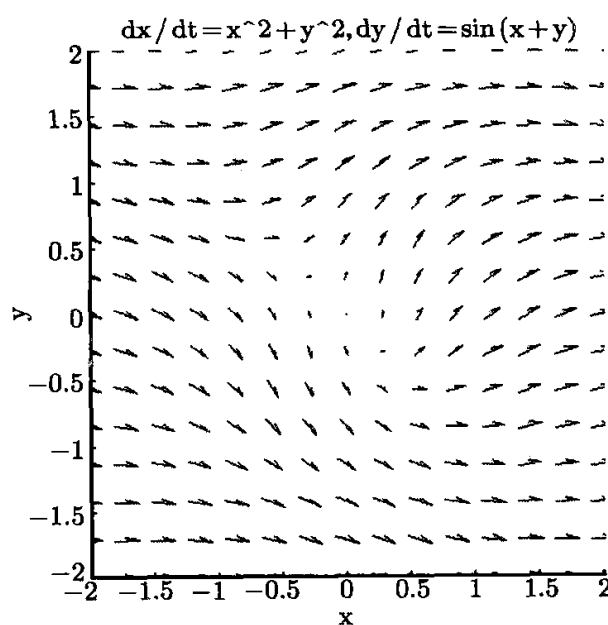


Figure 5.3.2 Direction field for the system

$$x' = x^2 + y^2, y' = \sin(x+y)$$

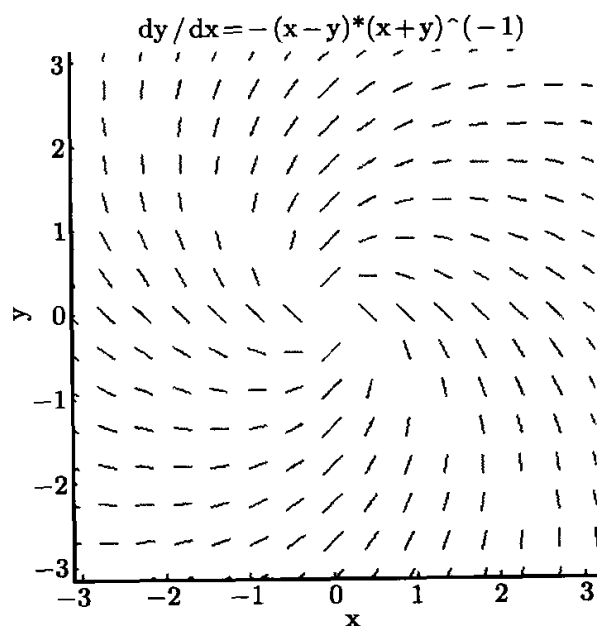


Figure 5.3.3 Slope field for the system

$$x' = -x - y, y' = x - y$$

### 5.3.2 Geometric Characteristics of the Orbits near a Singular Point

We now illustrate some of the most common types of orbits near the singular points of an autonomous system. A singular point  $(x^*, y^*)$  is called a **node**<sup>[33]</sup> if

(1) Either every orbit of solution approaches  $(x^*, y^*)$  as  $t$  approaches infinity or every orbits of the solution *recedes*<sup>[34]</sup>  $(x^*, y^*)$  as  $t$  approaches infinity, and

(2) Every orbit of solution is *tangent at  $(x^*, y^*)$  to some straight line through the singular point  $(x^*, y^*)$* <sup>[35]</sup>.

Moreover, a node is called **proper**<sup>[36]</sup> if no two different pairs of “opposite” orbits are tangent to the same straight line through the singular point  $(x^*, y^*)$ . A node is called **improper**<sup>[37]</sup> if all orbits except few are tangent to a single straight line through the singular point  $(x^*, y^*)$ .

A singular point  $(x^*, y^*)$  is called a **saddle point**<sup>[38]</sup> if all orbits but few are unbounded as  $t$  approaches infinity.

**Example 5.3.1** Consider following autonomous linear system

$$\frac{dx}{dt} = -x, \quad \frac{dy}{dt} = ky, \quad (5.3.6)$$

where  $k$  is a nonzero constant. The system has only one singular point  $(0,0)$ . Figure 5.3.4-Figure 5.3.7 give the phase portraits for different values of  $k$ . ■

If the nearby orbit of a singular point approaches to or recedes from it spirally, then we call it a **spiral point**<sup>[39]</sup>, see Figure 5.3.8-Figure 5.3.10.

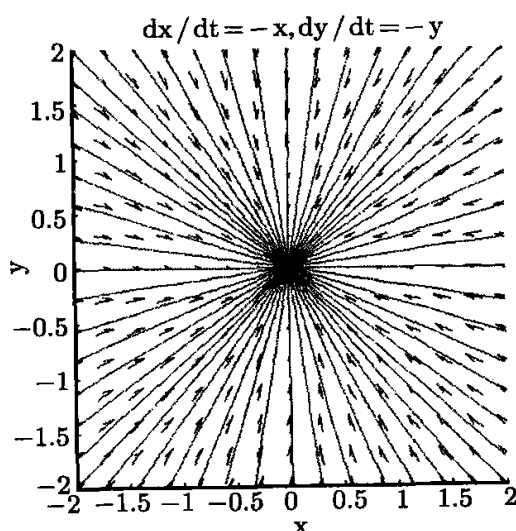


Figure 5.3.4  $k=1$ ,  $(0,0)$  is a proper node

[34] 离开

[35] 在  $(x^*, y^*)$  处与某条过奇点  $(x^*, y^*)$  的直线相切

[36] 正常的

[37] 非正常的

[38] 鞍点

[39] 螺旋点

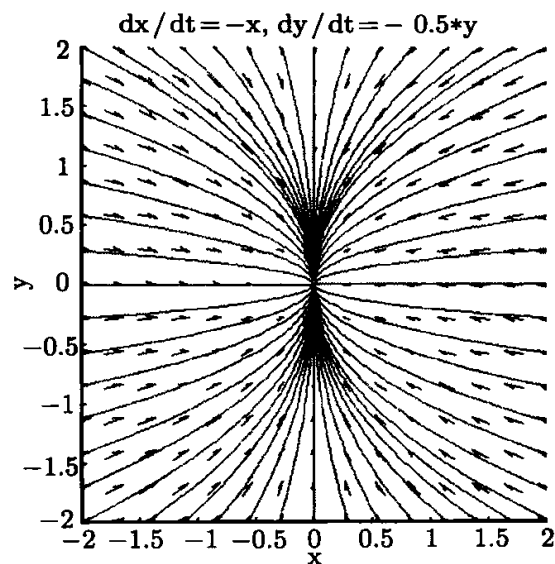


Figure 5.3.5  $k=1/2$ ,  $(0, 0)$  is an improper node

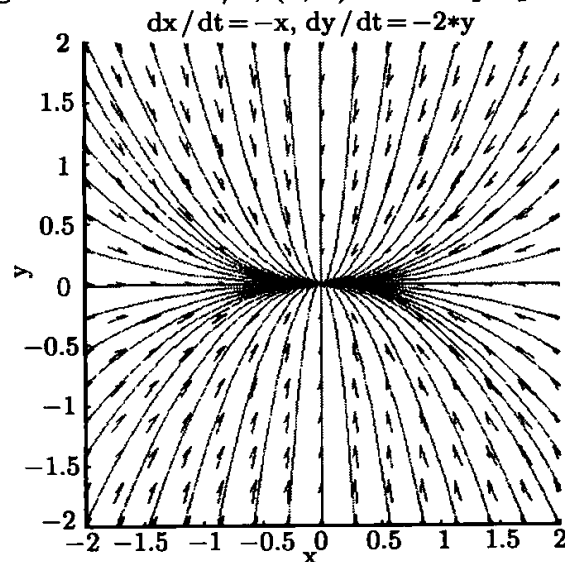


Figure 5.3.6  $k=2$ ,  $(0, 0)$  is an improper node

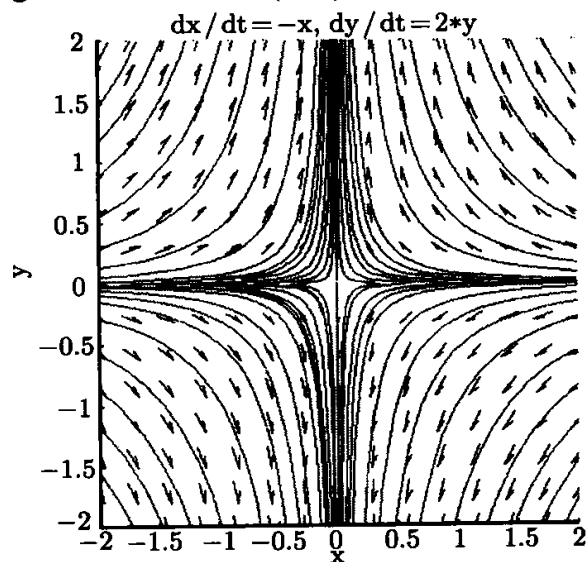


Figure 5.3.7  $k=-2$ ,  $(0, 0)$  is a saddle point

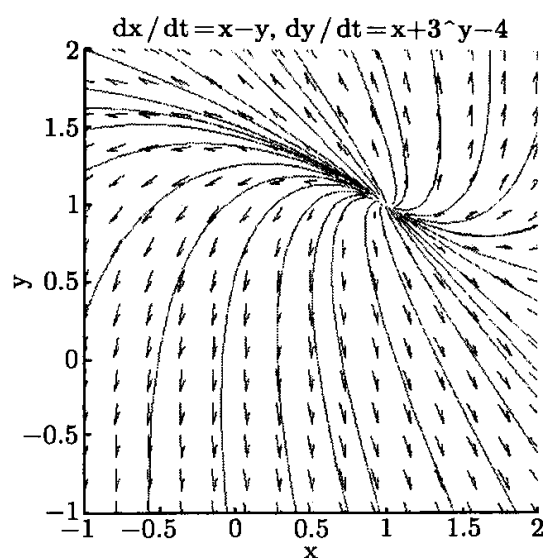


Figure 5.3.8 Spiral point (1, 1)

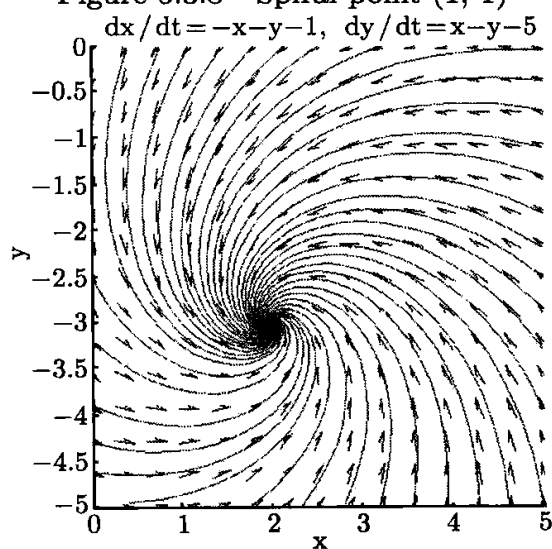


Figure 5.3.9 Spiral point (2, -3)

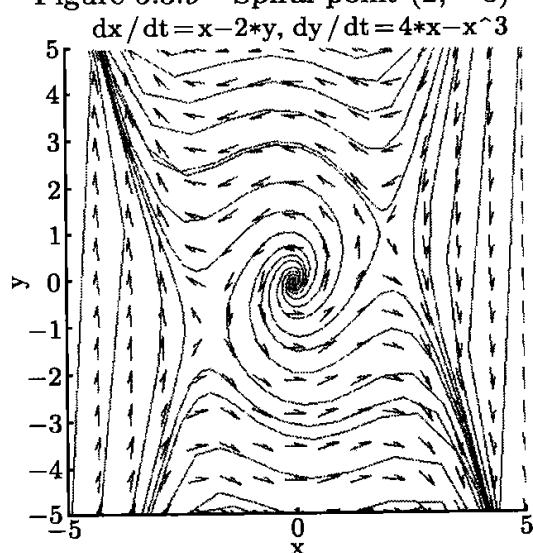


Figure 5.3.10 Spiral point (0, 0)

[40] 闭轨线

### 5.3.3 Stability of Singular Points

A singular point  $(x^*, y^*)$  of the system (5.3.1) is **stable** if every orbit remain close to  $(x^*, y^*)$  for all  $t > t_0$  whenever the initial point  $(x_0, y_0)$  is sufficiently close to  $(x^*, y^*)$ . Thus, the origin  $(0, 0)$  in Figure 5.3.4 Figure 5.3.6 and point  $(2, -3)$  in Figure 5.3.9 are all stable.

The origin  $(0, 0)$  in Figure 5.3.11 is called a **center** since it is surrounded by *closed orbits*<sup>[40]</sup> representing periodic solutions. It is stable.

A singular point  $(x^*, y^*)$  is **unstable** if it is not stable. The singular point  $(0, 0)$  in Figure 5.3.7, the singular point  $(1, 1)$  in Figure 5.3.8, and the origin  $(0, 0)$  in Figure 5.3.10 are not stable, so they are unstable.

A singular point  $(x^*, y^*)$  is **asymptotical stable** if it is stable, and every orbit will approach the singular point  $(x^*, y^*)$  as  $t$  approaches infinity whenever the initial point  $(x_0, y_0)$  is sufficiently close to  $(x^*, y^*)$ . Thus, the origin  $(0, 0)$  in Figure 5.3.4–5.3.6, and the point  $(2, -3)$  in Figure 5.3.9 are asymptotical stable singular points. The singular points in Figure 5.3.11 and Figure 5.3.12 are not asymptotical stable.

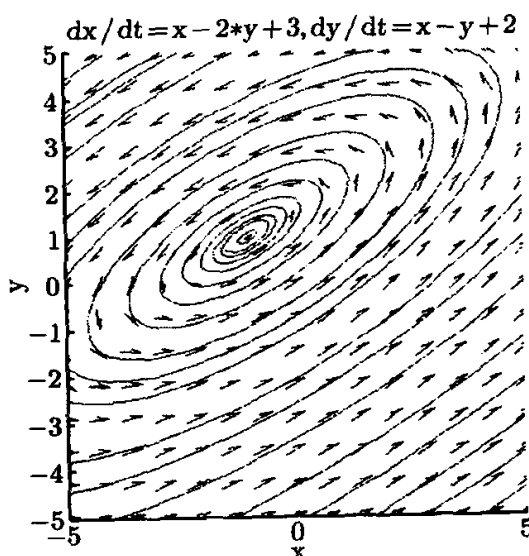


Figure 5.3.11  $(-1, 1)$  is a center

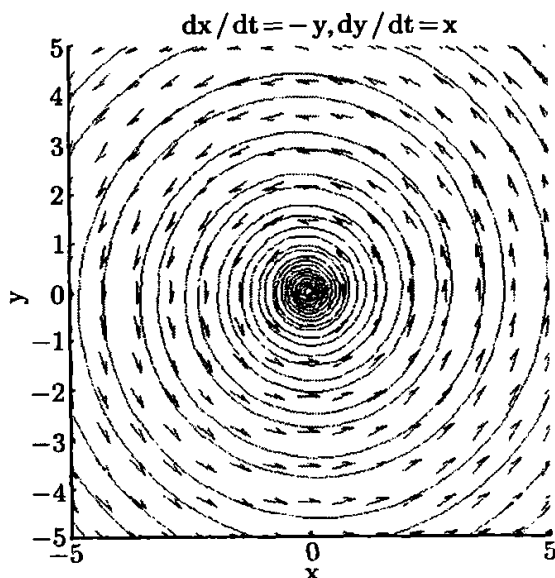
Figure 5.3.12  $(0, 0)$  is stable

Table 5.3.1 lists the relations between the eigenvalues of matrix  $A$  of the 2-dimensional system  $\dot{\mathbf{x}} = A\mathbf{x}$  and the types of singular point at  $(0, 0)$ .

Table 5.3.1

| Eigenvalue of $A$            | Type of singular point $(0, 0)$         |
|------------------------------|-----------------------------------------|
| Real, unequal, same sign     | Improper node                           |
| Real, unequal, opposite sign | Saddle node                             |
| Real, equal                  | Proper or improper node or spiral point |
| Complex conjugate            | Spiral point                            |
| Pure imaginary               | Center or spiral point                  |

**Example 5.3.2** Determine the nature of singular point  $(0, 0)$  of the system of differential equations

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -4 & 1 \\ -3 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

**Solution** Since the characteristic equation

$$\begin{vmatrix} -4 - \lambda & 1 \\ -3 & -\lambda \end{vmatrix} = (\lambda + 3)(\lambda + 1) = 0$$

has two roots  $\lambda_1 = -1$ ,  $\lambda_2 = -3$  (both are real, unequal and negative), the singular point  $(0, 0)$  is an improper node. Moreover, according to Theorem 5.2.5, it is stable. Figure 5.3.13 shows the orbits of some solutions. ■

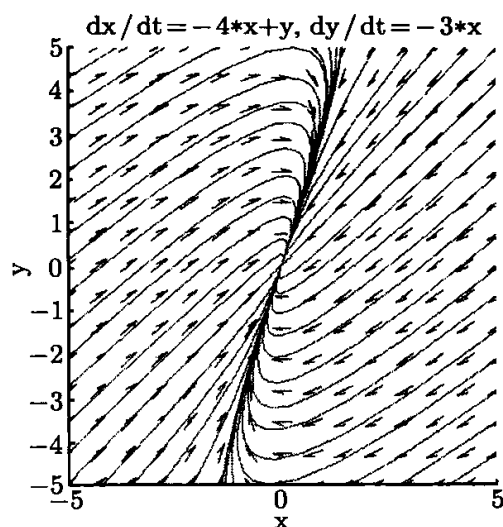


Figure 5.3.13

**Example 5.3.3** Determine the nature of singular point  $(0, 0)$  of the system of differential equations

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ -3 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

**Solution** Since the characteristic equation

$$\begin{vmatrix} 4 - \lambda & 1 \\ -3 & -\lambda \end{vmatrix} = (\lambda - 3)(\lambda - 1) = 0$$

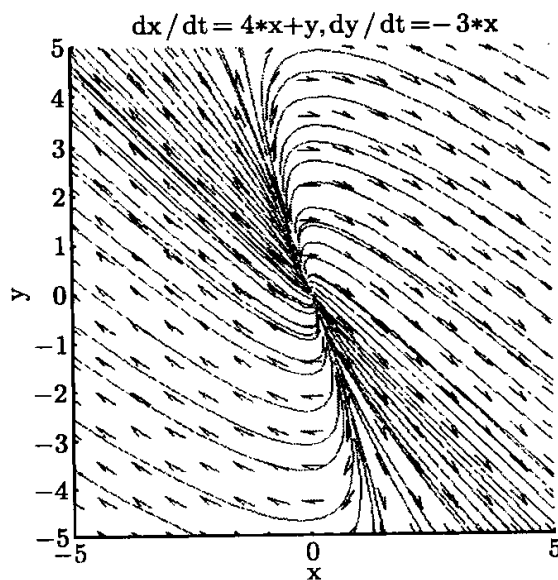


Figure 5.3.14

has two roots  $\lambda_1 = 1$ ,  $\lambda_2 = 3$  (both are real, unequal and positive), the singular point  $(0, 0)$  is an improper node. Moreover, according to Theorem 5.2.5, it is unstable. Figure 5.3.14 shows the orbits of some solutions. ■

### Exercise 5.3

Determine the nature of the zero solution of each of following systems of differential equations.

$$1. \dot{x} = \begin{pmatrix} 5 & -3 \\ 3 & -5 \end{pmatrix} x. \quad 2. \dot{x} = \begin{pmatrix} 1 & 5 \\ 5 & 1 \end{pmatrix} x.$$

$$3. \dot{x} = \frac{1}{8} \begin{pmatrix} 7 & 3 \\ -3 & 17 \end{pmatrix} x. \quad 4. \dot{x} = \frac{1}{8} \begin{pmatrix} -7 & -3 \\ 3 & -17 \end{pmatrix} x.$$

$$5. \dot{x} = \frac{1}{4} \begin{pmatrix} -9 & 15 \\ -15 & 9 \end{pmatrix} x. \quad 6. \dot{x} = \frac{1}{4} \begin{pmatrix} -10 & 15 \\ -15 & 8 \end{pmatrix} x.$$

Draw the phase portraits of following systems of differential equations.

$$7. \dot{x} = y, \dot{y} = -x.$$

$$8. \dot{x} = y(1 + x + y), \dot{y} = -x.$$

$$9. \dot{x} = y^3, \dot{y} = x^3.$$

$$10. \dot{x} = ax - bxy, \dot{y} = -cy + dxy \quad (a, b, c, d \text{ are positive}).$$

$$11. \dot{x} = 2xy, \dot{y} = x^2 - y^2.$$

Prove that all solutions of the following second-order equations are periodic.

$$12. \frac{d^2 z}{dt^2} + z^3 = 0. \quad 13. \frac{d^2 z}{dt^2} + z + z^5 = 0.$$

$$14. \frac{d^2 z}{dt^2} + e^{z^2} = 1. \quad 15. \frac{d^2 z}{dt^2} + \frac{z}{1 + z^2} = 0.$$

## 5.4 Applications

### Module 1 Volterra's Principle<sup>[41]</sup>

In the mid-1920's, the Italian biologist Umberto D'Ancona *came across*<sup>[42]</sup>, *in the course of his research on*<sup>[43]</sup> the struggle for existence between competing species, some data on *percentages-of-total-catch*<sup>[44]</sup> of *selachians*<sup>[45]</sup> (*sharks*<sup>[46]</sup>, *skate*<sup>[47]</sup>, *rays*<sup>[48]</sup>, etc.) that were brought into different Mediterranean ports in *the years*

[41] Volterra 原理

[42] 偶然发现

[43] 在研究 ... 的过程中

[44] 占总捕获量的百分比

[45] 软骨鱼

[46] 鲨鱼

[47] 鳐鱼

[48] 魷

[49] 第一次世界大战期间

[50] 困惑

[51] 推断

[52] 降低的捕获水平

[53] 一点也站不住脚

[54] 食用鱼

[55] 捕食对象、被捕食者

[56] 对 ... 更有利

[57] 捕食者

[58] 找到 ... 寻求帮助

[59] 被捕食者种群

[60] 捕食者种群

[61] 食用鱼种群密度不是很大

[62] 单位时间捕食者与被捕食者相遇的次数

that spanned World War I<sup>[49]</sup>. The data for the port of Fiume, Italy, during the years 1914—1923 is as Table 5.4.1.

Table 5.4.1

| 1914  | 1915  | 1916  | 1917  | 1918  | 1919  | 1920  | 1921  | 1922  | 1923  |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 11.9% | 21.4% | 22.1% | 21.2% | 36.4% | 27.3% | 16.0% | 15.9% | 14.8% | 10.7% |

What *puzzled*<sup>[50]</sup> D'Ancona was the very large increase in the percentage of selachians during the period of the war. At first, he *reasoned*<sup>[51]</sup> that the increase in the percentage of selachians was due to the greatly *reduced level of fishing*<sup>[52]</sup> during this period. However, this explanation *does not hold any water*<sup>[53]</sup> since *food fish*<sup>[54]</sup>, the selachian's *prey*<sup>[55]</sup>, were also more abundant during this period. D'Ancona's theory only shows that more selachians can be found when the level of fishing is reduced; it does not explain why a reduced level of fishing *is more beneficial to*<sup>[56]</sup> the *predators*<sup>[57]</sup> than to their prey.

After exhausting all possible biological explanations of this phenomenon, D'Ancona *turned to* his colleague, the famous Italian mathematician Vito Volterra *in the hope that*<sup>[58]</sup> Volterra would formulate a mathematical model of the growth of the selachians and their prey, the food fish, and that this model would provide the answer to D'Ancona's question. Volterra began his analysis of this problem by separating all the fish into the *prey population*<sup>[59]</sup>  $x(t)$  and the *predator population*<sup>[60]</sup>  $y(t)$ . Then he reasoned that the food fish do not compete very intensively among themselves for their food supply since the food supply is plentiful and *the fish population is not very dense*<sup>[61]</sup>. Hence, in the absence of the selachians, the food fish would grow according to the Malthusian law of population growth  $\dot{x} = ax$ , for some positive constant  $a$ . Next, reasoned Volterra, *the number of contacts per unit time between predators and prey*<sup>[62]</sup> is  $bxy$ , for some positive constant  $b$ . Similarly, Volterra concluded that the predators have a natural rate of decrease  $cy$  proportional to their present number  $y$ , and that they also increase at a rate  $dxy$  proportional to their present number  $y$  and their food supply  $x$ . Thus

$$\frac{dx}{dt} = ax - bxy, \quad \frac{dy}{dt} = -cy + dxy. \quad (5.4.1)$$

[63] 以年为周期的  
捕食者的平均  
比例

The system (5.4.1) governs the interaction of the selachians and food fish in the absence of fishing.

Now we introduce the effect of fishing to our model, and then show why a reduced level of fishing is more beneficial to the selachians than to the food fish.

Observe first that system (5.4.1) has two equilibrium solutions  $x(t) \equiv 0, y(t) \equiv 0$  (which is of no interest to us) and  $x(t) \equiv c/d, y(t) \equiv a/b$ . Then, we note that this system has the family of solutions

$$x(t) = x_0 e^{at}, y(t) = 0 \text{ and } x(t) = 0, y(t) = y_0 e^{-ct},$$

this implies that every solution  $x(t), y(t)$  of system (5.4.1) which starts in the first quadrant  $x > 0, y > 0$  at time  $t = t_0$  will remain there for all future time  $t \geq t_0$ . And in this region, the orbits of system (5.4.1) are closed, see Figure 5.4.1.

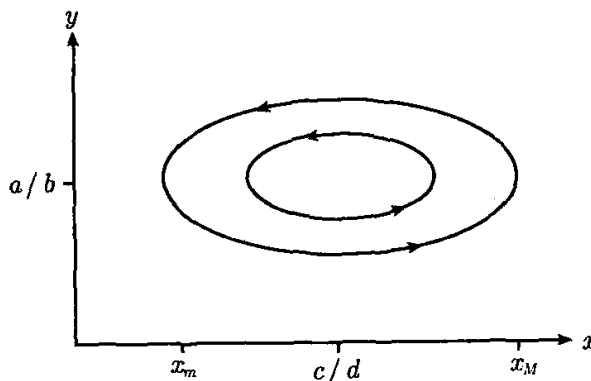


Figure 5.4.1 Orbits of the system (5.4.1) for  $x, y$  positive

D'Ancona's data are really *the averages over each one-year period of the proportion of predators*<sup>[63]</sup>. Thus, in order to compare these data with the predictions of system (5.4.1), we must compute the "average values" of  $x(t)$  and  $y(t)$ , for any solution  $x(t), y(t)$  of system (5.4.1).

**Lemma 5.4.1** Let  $x(t), y(t)$  be a periodic solution of system

[64] 捕获强度

[65] 真实的状态

(5.4.1), with period  $T > 0$ . Define the average values of  $x(t)$  and  $y(t)$  as

$$\bar{x} = \frac{1}{T} \int_0^T x(t) dt, \quad \bar{y} = \frac{1}{T} \int_0^T y(t) dt.$$

Then  $\bar{x} = c/d$  and  $\bar{y} = a/b$ .

**Proof** Dividing both sides of the first equation of system (5.4.1) by  $x$  gives

$$\frac{x'}{x} = a - by,$$

then

$$\frac{1}{T} \int_0^T \frac{x'(t)}{x(t)} dt = \frac{1}{T} \int_0^T [a - by(t)] dt.$$

Now,

$$\int_0^T \frac{x'(t)}{x(t)} dt = \ln x(T) - \ln x(0) = 0$$

since  $x(T) = x(0)$ . Consequently,

$$\frac{1}{T} \int_0^T by(t) dt = \frac{1}{T} \int_0^T a dt = a,$$

so that  $\bar{y} = a/b$ . Similarly, dividing both sides of the second equation of system (5.4.1) by  $Ty(t)$  and integrating from 0 to  $T$ , we obtain that  $\bar{x} = c/d$ . ■

Lemma 5.4.1 is remarkable since we can find these average values even though we cannot compute  $x(t)$  and  $y(t)$  exactly.

We are now ready to include the effects of fishing in our model. Observe that fishing decreases the population of food fish at a rate  $\varepsilon x(t)$  and decreases the population of selachians at a rate  $\varepsilon y(t)$ . The constant  $\varepsilon$  reflects *the intensity of fishing*<sup>[64]</sup>; i.e., the number of boats at sea, and the number of nets in the water. Thus *the true state of affairs*<sup>[65]</sup> is described by the modified system of differential equations.

$$\begin{aligned} \frac{dx}{dt} &= ax - bxy - \varepsilon x = (a - \varepsilon)x - bxy, \\ \frac{dy}{dt} &= -cy + dxy - \varepsilon y = -(c + \varepsilon)y + dxy. \end{aligned} \tag{5.4.2}$$

This system is exactly the same as system (5.4.1) (for  $a - \varepsilon > 0$ ) with  $a$  replaced by  $a - \varepsilon$ , and  $c$  replaced by  $c + \varepsilon$ . Hence the average values of  $x(t)$  and  $y(t)$  are now

$$\bar{x} = \frac{c + \varepsilon}{d}, \quad \bar{y} = \frac{a - \varepsilon}{b}. \quad (5.4.3)$$

Consequently, a moderate amount of fishing<sup>[66]</sup> ( $\varepsilon < a$ ) actually increases the number of food fish, on the average, and decreases the number of selachians. Conversely, a reduced level of fishing increases the number of selachians and decreases the number of food fish on the average. This remarkable result, which is known as Volterra's principle, explains the data of D'Ancona and completely solves our problem.

Volterra's principle has spectacular applications to insecticide treatments<sup>[67]</sup> which destroy both insect predators and their insect prey. It implies that the application of insecticides will actually increase the population of those insects which are kept in control by other predatory insects. A remarkable confirmation<sup>[68]</sup> comes from the cottony cushion<sup>[69]</sup> scale insect<sup>[70]</sup>, which, when accidentally introduced from Australia in 1868, threatened to destroy the American citrus industry<sup>[71]</sup>. Thereupon, its natural Australian predator<sup>[72]</sup>, a ladybird beetle<sup>[73]</sup> was introduced, and the beetles reduced the scale insects to a low level. When DDT was discovered to kill scale insects, it was applied by the orchardists<sup>[74]</sup> in the hope of further reducing the scale insects. However, in agreement with Volterra's principle<sup>[75]</sup>, the effect was an increase of the scale insect!

There exist several predator-prey interactions in nature which cannot be modeled by any system of ordinary differential equations. These situations occur when the prey are provided with a refuge<sup>[76]</sup> that is inaccessible to the predators<sup>[77]</sup>. In these situations, it is impossible to make any definitive statements about<sup>[78]</sup> the future number of predators and prey since we cannot predict how many prey will be stupid enough to leave their refuge. In other words, this process is now random<sup>[79]</sup>, rather than deterministic<sup>[80]</sup>, and therefore cannot be modeled by a system of ordi-

- [66] 适度的捕获
- [67] 杀虫剂
- [68] 一个非常著名的实例
- [69] 棉垫一样柔软的
- [70] 介壳虫
- [71] 柑橘业
- [72] 澳大利亚天敌
- [73] 瓢虫
- [74] 果农们
- [75] 与 Volterra 原理所预测的完全一样
- [76] 庇护所
- [77] 让捕食者无法接近
- [78] 对 ... 不能给出明确的说法
- [79] 随机的
- [80] 确定的

- [81] 草履虫
- [82] 毛虫
- [83] 试管
- [84] 关于战争的数学理论
- [85] 冲突
- [86] 爆发
- [87] 战争潜力
- [88] 武装力量、武器装备
- [89] 备战状况
- [90] 不满情绪
- [91] 制约影响
- [92] 联盟
- [93] 奥匈帝国
- [94] 第一次世界大战前夕
- [95] 关于... 长期以来一直争论不休

nary differential equations. This was verified directly in a famous experiment by Gause. He placed five *Paramecium*<sup>[81]</sup> and three *Didinium*<sup>[82]</sup> in each of 30 identical *test tubes*<sup>[83]</sup>, and provided the *Paramecium* with a refuge from the *Didinium*. Two days later, he found the predators dead in four tubes, and a mixed population containing from two to 38 *Paramecium* in the remaining 26 tubes.

## Module 2 Mathematical Theories of War<sup>[84]</sup>

In this article we construct a mathematical model which describes a *conflict*<sup>[85]</sup> between two nations. It is not an attempt to make scientific statements about foreign politics or to predict the date at which the next war will *breakout*<sup>[86]</sup>. This, of course, is clearly impossible. Our model is based on the work of Lewis Fry Richardson.

Let  $x = x(t)$  denote the *war potential*<sup>[87]</sup>, or *armaments*<sup>[88]</sup>, of the first nation, which we will call Jedesland, and let  $y(t)$  denote the war potential of the second nation, which we will call Andersland. The rate of change of  $x(t)$  depends, obviously, on the *war readiness*<sup>[89]</sup>  $y(t)$  of Andersland and the *grievances*<sup>[90]</sup> that Jedesland feels towards Andersland. In the most simplistic model we represent these terms by  $ky$  and  $g$  respectively, where  $k$  and  $g$  are positive constants. These two terms cause  $x$  to increase. On the other hand, the cost of armaments has a *restraining effect*<sup>[91]</sup> on  $dx/dt$ . We represent this term by  $-\alpha x$ , where  $\alpha$  is a positive constant. A similar analysis holds for  $dy/dt$ . Consequently,  $x = x(t)$ ,  $y = y(t)$  is a solution of the linear system of differential equations

$$\frac{dx}{dt} = ky - \alpha x + g, \quad \frac{dy}{dt} = lx - \beta y + h. \quad (5.4.4)$$

**Remark 5.4.1** Model (5.4.4) can also represent the relation between two *alliances*<sup>[92]</sup>. For example, Adersland and Jedesland can represent the alliances of France with Russian, and Germany with *Austria-Hungary*<sup>[93]</sup> during the years immediately prior to *World War I*<sup>[94]</sup>.

There has been a *constant debate*<sup>[95]</sup> on the cause of war. Sir

Edward Grey, the British Foreign Secretary<sup>[96]</sup> during World War I, writes: “The increase of armaments that is intended in each nation to produce consciousness of strength, and a sense of security, does not produce these effects<sup>[97]</sup>. On the contrary, it produces a consciousness of the strength of other nations and a sense of fear. The enormous growth of armaments in Europe, the sense of insecurity and fear caused by them – it was these that made war inevitable. This is the real and final account of the origin of the Great War<sup>[98]</sup>.” On the other hand, L.S. Amery, a member of Britain’s parliament during the 1930’s vehemently disagrees<sup>[99]</sup>. When the opinion of Sir Edward Grey was quoted in the House of Commons, Amery replied: “With all due respect to the memory of an eminent statesman<sup>[100]</sup>, I believe that statement to be entirely mistaken. The armaments were only the symptoms of the conflict of ambitions and ideals, of those nationalist forces which created the War<sup>[101]</sup>. The war brought about because Serbia, Italy and Rumania passionately desired the incorporation in their states of territories which at that time belonged to the Austrian Empire and which the Austrian government was not prepared to abandon without a struggle<sup>[102]</sup>. France was prepared, if the opportunity ever came, to make an effort to recover Alsace-Lorraine. It was in those facts, in those insoluble conflicts of ambitions<sup>[103]</sup>, and not in the armaments themselves, that the cause of the War lay<sup>[104]</sup>.”

The system of equations (5.4.4) takes both conflicting theories into account<sup>[105]</sup>. Sir Edward Grey would take  $g$  and  $h$  small compared to  $k$  and  $l$ , while Mr. Amery would take  $k$  and  $l$  small compared to  $g$  and  $h$ .

The system of equations (5.4.4) has several important implications. Suppose that  $g$  and  $h$  are both zero. Then,  $x(t) \equiv 0$ ,  $y(t) \equiv 0$  is an equilibrium solution of system (5.4.4). That is, if  $x$ ,  $y$ ,  $g$  and  $h$  are all made zero simultaneously, then  $x(t)$  and  $y(t)$  will always remain zero. This ideal condition is *permanent peace*<sup>[106]</sup> achieved by *disarmament*<sup>[107]</sup> and satisfaction. It has existed since 1817 on the border between Canada and the United States, and since 1905 on the border between Norway and Sweden.

[96] 英国外交大臣

[97] 军备增加并没有产生各国所预期的强国意识和安全感这样的效果

[98] 这场伟大的战争（指第一次世界大战）

[99] 强烈反对

[100] 真心地缅怀这位杰出的政治家

[101] 军备仅仅是表象，野心和理想冲突，以及民族主义者的力量才是引发战争的根源

[102] 战争爆发是因为塞尔维亚、意大利和俄国强烈地渴望将当时归属于奥地利帝国的领土合并，而奥地利政府又不准备和平地放弃这些土地

[103] 那些不能化解的野心冲突

[104] 为战争打下了基础

[105] 方程组 (5.4.4) 把这两种相冲突的理论都考虑进去了

[106] 永久和平

[107] 裁军

- [108] 不能令人满意的共同裁军  
 [109] 重整军备  
 [110] 单方面裁军  
 [111] 凡尔赛条约  
 [112] 军备竞赛  
 [113] 防务预算  
 [114] 合作促进相互合作

These equations further imply that *mutual disarmament without satisfaction*<sup>[108]</sup> is not permanent. Assume that  $x$  and  $y$  vanish simultaneously at some time  $t = t_0$ . At this time,  $dx/dt = g$  and  $dy/dt = h$ . Thus,  $x$  and  $y$  will not remain zero if  $g$  and  $h$  are positive. Instead, both nations will *rearm*<sup>[109]</sup>.

*Unilateral disarmament*<sup>[110]</sup> corresponds to set  $y = 0$  at a certain instant of time. At this time,  $dy/dt = lx + h$ . This implies that  $y$  will not remain zero if either  $h$  or  $x$  is positive. Thus, unilateral disarmament is never permanent. This accords with the historical fact that Germany, whose army was reduced by the *Treaty of Versailles*<sup>[111]</sup> to 100,000 men, a level far below that of several of her neighbors, insisting on rearming during the years 1933—1936.

A *race in armaments*<sup>[112]</sup> occurs when the “defense” terms predominate in system (5.4.4). In this case,

$$\frac{dx}{dt} = ky, \quad \frac{dy}{dt} = lx. \quad (5.4.5)$$

Every solution of system (5.4.5) is of the form

$$x(t) = Ae^{\sqrt{kl}t} + Be^{-\sqrt{kl}t}, \quad y(t) = \sqrt{\frac{l}{k}} \left[ Ae^{\sqrt{kl}t} - Be^{-\sqrt{kl}t} \right].$$

Therefore, both  $x(t)$  and  $y(t)$  approach infinity as  $t$  approaches infinity if  $A$  is positive. Thus infinity can be interpreted as war.

We find now that the system (5.4.4) is not quite correct since it does not take into effect the cooperation, or trade, between Andersland and Jedesland. As we see today, mutual cooperation between nations tends to decrease their fears and suspicions. We correct our model by changing the meaning of  $x(t)$  and  $y(t)$ ; we let the variables  $x(t)$  and  $y(t)$  stand for “threats” minus “cooperation.” Specifically, we set  $x = U - U_0$  and  $y = V - V_0$ , where  $U$  is the *defense budget*<sup>[113]</sup> of Jedesland,  $V$  is the defense budget of Andersland,  $U_0$  is the amount of goods exported by Jedesland to Andersland and  $V_0$  is the amount of goods exported by Andersland to Jedesland. Observe that *cooperation evokes reciprocal cooperation*<sup>[114]</sup>, just as armaments provoke more armaments. In

addition, *nations have a tendency to reduce cooperation on account of the expense which it involves*<sup>[115]</sup>. Thus, the system (5.4.4) still describes this more general state of affairs.

The system (5.4.4) has only one equilibrium solution

$$x \equiv x_0 = \frac{kh + \beta g}{\alpha\beta - kl}, \quad y \equiv y_0 = \frac{lg + \varepsilon h}{\alpha\beta - kl} \quad (5.4.6)$$

if  $\alpha\beta - kl \neq 0$ . We are interested in determining whether this equilibrium solution is stable or unstable. To this end, we write Eq.(5.4.4) in the form

$$\dot{\mathbf{w}} = \mathbf{A}\mathbf{w} + \mathbf{f},$$

where

$$\mathbf{w}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}, \quad \mathbf{f} = \begin{pmatrix} g \\ h \end{pmatrix} \quad \text{and} \quad \mathbf{A} = \begin{pmatrix} -\alpha & k \\ l & -\beta \end{pmatrix}.$$

The equilibrium solution is

$$\mathbf{w} \equiv \mathbf{w}_0 = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix},$$

where  $\mathbf{A}\mathbf{w}_0 + \mathbf{f} = \mathbf{0}$ . Setting  $\mathbf{z} = \mathbf{w} - \mathbf{w}_0$  we obtain that

$$\dot{\mathbf{z}} = \dot{\mathbf{w}} = \mathbf{A}\mathbf{w} + \mathbf{f} = \mathbf{A}(\mathbf{z} + \mathbf{w}_0) + \mathbf{f} = \mathbf{A}\mathbf{z} + \mathbf{A}\mathbf{w}_0 + \mathbf{f} = \mathbf{A}\mathbf{z}.$$

Clearly, the equilibrium solution  $\mathbf{w}(t) \equiv \mathbf{w}_0$  of  $\dot{\mathbf{w}} = \mathbf{A}\mathbf{w} + \mathbf{f}$  is stable if, and only if,  $\mathbf{z} \equiv \mathbf{0}$  is a stable solution of  $\dot{\mathbf{z}} = \mathbf{A}\mathbf{z}$ . To determine the stability of  $\mathbf{z} \equiv \mathbf{0}$ , we compute

$$p(\lambda) = \det \begin{pmatrix} -\alpha - \lambda & k \\ l & -\beta - \lambda \end{pmatrix} = \lambda^2 + (\alpha + \beta)\lambda + \alpha\beta - kl.$$

The roots of  $p(\lambda)$  are

$$\begin{aligned} \lambda &= \frac{-(\alpha + \beta) \pm [(\alpha + \beta)^2 - 4(\alpha\beta - kl)]^{1/2}}{2} \\ &= \frac{-(\alpha + \beta) \pm [(\alpha - \beta)^2 + 4kl]^{1/2}}{2}. \end{aligned}$$

Notice that both roots are real and unequal to zero. Moreover, both roots are negative if  $\alpha\beta - kl > 0$ , and one root is positive if

[115] 出于(时间、精力、金钱等)消耗方面的考虑各国又倾向于减少合作

- [116] 时间的倒数  $\alpha\beta - kl < 0$ . Thus,  $z(t) \equiv 0$ , and consequently the equilibrium  
 [117] 张弛时间 solution  $x(t) \equiv x_0, y(t) \equiv y_0$  is stable if  $\alpha\beta - kl > 0$ , and unstable  
 [118] 按 2.718 的 if  $\alpha\beta - kl < 0$ .  
 比例

Let us now tackle the difficult problem of estimating the coefficients  $\alpha, \beta, k, l, g$  and  $h$ . There is no way, obviously, of measuring  $g$  and  $h$ . However, it is possible to obtain reasonable estimates for  $\alpha, \beta, k$  and  $l$ . Observe that the units of these coefficients are *reciprocal times*<sup>[116]</sup>. Physicists and engineers would call  $\alpha^{-1}$  and  $\beta^{-1}$  *relaxation times*<sup>[117]</sup>, for  $y$  and  $g$  were identically zero, then

$$x(t) = e^{-\alpha(t-t_0)}x(t_0).$$

This implies that

$$x(t_0 + \alpha^{-1}) = x(t_0)/e.$$

Hence,  $\alpha^{-1}$  is the time required for Jedsland's armaments to be reduced *in the ratio* 2.718<sup>[118]</sup> if that nation has no grievances and no other nation has any armaments. Richardson estimates  $\alpha^{-1}$  to be the lifetime of Jedsland's parliament. Thus,  $\alpha = 0.2$  for Great Britain since the lifetime of Britain's parliament is five years.

To estimate  $k$  and  $l$  we take a hypothetical case in which  $g = 0$  and  $y = y_1$ , so that

$$\frac{dx}{dt} = ky_1 - \alpha x.$$

When  $x = 0$ ,  $1/k = y_1/(dx/dt)$ . Thus,  $1/k$  is the time required for Jedsland to catch up to Andersland and it provides that (i) Andersland's armaments remain constant, (ii) there are no grievances, and (iii) the cost of armaments doesn't slow Jedsland down. Consider now the German rearmament during 1933—1936. Germany started with nearly zero armaments and caught up with her neighbors in about three years. Assuming that the slowing effect of  $\alpha$  nearly balanced the Germans' very strong grievances  $g$ , we take  $k = 0.3 \text{ (year)}^{-1}$  for Germany. Further, we observe that  $k$  is obviously proportional to the amount of industry that a nation has. Thus,  $k = 0.15$  for a nation which has only half the

industrial capacity of Germany, and  $k = 0.9$  for a nation which has three times the industrial capacity of Germany.

Let us now check our model against the European arms race during 1909–1914. France was allied with Russia, and Germany was allied with Austria-Hungary. Neither Italy nor Britain was in a definite alliance with either party. Thus, let Jedsland represent the alliance of France with Russia, and let Adersland represent the alliance of Germany with Austria-Hungary. Since these two alliances were roughly equal in size we take  $k = l$ , and since each alliance was roughly three times the size of Germany, we take  $k = l = 0.9$ . We also assume that  $\alpha = \beta = 0.2$ . Then,

$$\frac{dx}{dt} = -\alpha x + ky + g, \quad \frac{dy}{dt} = kx - \alpha y + h. \quad (5.4.7)$$

Eq.(5.4.7) has a unique singular point

$$x_0 = \frac{kh + \alpha g}{\alpha^2 - k^2}, \quad y_0 = \frac{kg + \alpha h}{\alpha^2 - k^2}.$$

This point is unstable since

$$\alpha\beta - kl = \alpha^2 - k^2 = 0.04 - 0.81 = -0.77.$$

This, of course, is in agreement with the historical fact that these two alliances went to war with each other.

**Remark 5.4.2** The model we have constructed is very crude since it assumes that the grievances  $g$  and  $h$  are not even continuous functions of time since they jump instantaneously by large amounts. (It's safe to assume, though, that  $g$  and  $h$  are relatively constant over long periods of time.) In spite of this, the system (5.4.7) still provides a very accurate description of the arms race preceding World War I.

Now we add the two equations of system (5.4.7) together to obtain that

$$\frac{d}{dt}(x + y) = (k - \alpha)(x + y) + g + h. \quad (5.4.8)$$

Recall that  $x = U - U_0$  and  $y = V - V_0$ , where  $U$  and  $V$  are the defense budgets of the two alliances, and  $U_0$  and  $V_0$  are the

[119] 有着像机器一样的可预见性

amount of goods exported from each alliance to the other. Hence,

$$\frac{d}{dt}(U + V) = (k - \alpha) \left\{ U_0 + V_0 - \frac{g + h}{k - \alpha} - \frac{1}{k - \alpha} \frac{d}{dt}(U_0 + V_0) \right\}. \quad (5.4.9)$$

The defense budgets for the two alliances are set out in Table 5.4.2.

**Table 5.4.2 Defense budgets expressed in millions of sterling**

|                      | 1909  | 1910  | 1911  | 1912  | 1913  |
|----------------------|-------|-------|-------|-------|-------|
| France               | 48.6  | 50.9  | 57.1  | 63.2  | 74.7  |
| Russia               | 66.7  | 68.5  | 70.7  | 81.8  | 92.0  |
| Germany              | 63.1  | 62.0  | 62.5  | 68.2  | 95.4  |
| Austria-Hungary      | 20.8  | 23.4  | 24.6  | 25.5  | 26.9  |
| Total $U + V$        | 199.2 | 204.8 | 204.9 | 238.7 | 289.0 |
| $\Delta(U + V)$      |       | 5.6   | 10.1  | 23.8  | 50.3  |
| $U + V$ at same date |       | 202.0 | 209.8 | 226.8 | 263.8 |

From Table 5.4.2 we can see that how close the four points (represent the actual measurements), denoted by •, are to the straight line

$$\Delta(U + V) = 0.73(U + V - 194). \quad (5.4.10)$$

Thus, foreign politics does indeed have a *machine-like predictability*<sup>[119]</sup>. Eq.(5.4.9) and Eq.(5.4.10) imply that

$$g + h = (k - \alpha)(U_0 + V_0) - \Delta(U_0 + V_0) - 194$$

and  $k - \alpha = 0.73$ . This is in excellent agreement with Richardson's estimates of 0.9 for  $k$  and 0.2 for  $\alpha$ . Finally, observe from Eq.(5.4.10) that the total defense budgets of the two alliances will increase if  $U + V$  is greater than 194 million, and will decrease otherwise. In actual fact, the defense expenditures of the two alliances amounted to only 171.8 million. Thus began an arms race which led eventually to World War I, see Figure 5.4.2.

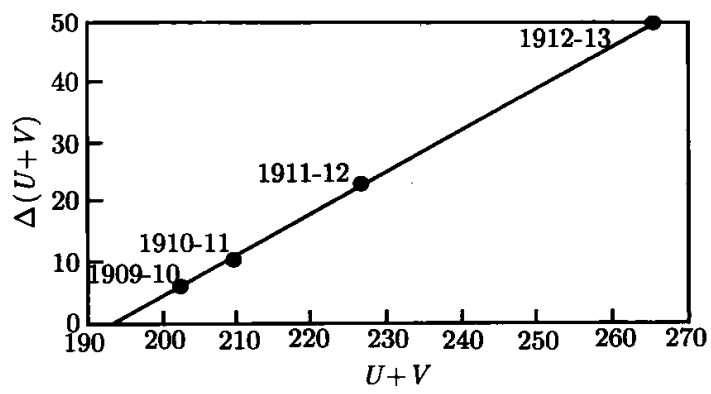


Figure 5.4.2

# Answers to Selected Exercises

## Chapter 1

### Exercise 1.2

1.  $y = ce^{-5t}$     2.  $y = ce^{5t}$     3.  $y = ce^{-t^3}$     4.  $y = ce^{t^3/3}$   
5.  $y = c/t^2$     6.  $y = ce^{-2/t}$     7.  $y = ce^{\cos t}$     8.  $y = ce^{\sin t}$   
9.  $y = ce^{3t} - 2$     10.  $y = \frac{c}{t^4} + \frac{1}{9}t^5$   
11.  $y = ce^{-t} + \frac{1}{2}\sin t - \frac{1}{2}\cos t$     12.  $y = 2 + c(1 + t^2)^{-3/2}$   
13.  $y = \frac{4t^2 + 40t + c}{10 + 2t} \quad (t > -5)$     14.  $y = \left( \frac{1}{ce^t - 2t - 1} \right)^{\frac{1}{3}}, y = 0$   
15.  $y = ce^{-3/t} - \frac{1}{3}$     16.  $y = \frac{e^{-t}}{c - t}$   
17.  $y = \left( \frac{1}{-\frac{1}{2} + ce^{3t^2}} \right)^{\frac{1}{3}}, y = 0$     18.  $y = ce^{-t^3/3} + 1$   
19.  $y = \frac{1}{ce^t - \sin t}, y = 0$     20.  $y = (6 + ce^{-t^2/4})^2$   
21.  $y = c_1 e^t - \frac{1}{2}t^2 - t + c_2$     22.  $y = c_1 \ln |t| + c_2$   
23.  $y = \frac{1}{4}(-t^{-2} + t^2)$     24.  $y = 5e^{-3(t^2 - \pi^2)}$   
25.  $y = \frac{3}{2}e^{(\cos t) - 1}$     26.  $y = \exp \left( - \int_0^t \sqrt{1 + s^2} e^{-s} ds \right)$   
27.  $y = \frac{4t^2 + 40t + 1304}{10 + 2t} \quad (t > -5)$     28.  $y = \frac{1}{2}(e^{\pi - t} + \sin t - \cos t)$   
29.  $y = 1 - 5e^{t^2/2}$     30.  $y = e^{-t^2/2} \int_{3/2}^t (1 + s)e^{s^2/2} ds$

### Exercise 1.3

1.  $y = \frac{-4}{t^4 + c}$     2.  $y = \pm \sqrt{\frac{2}{3}t^3 + 4t + c}$     3.  $y = -\ln |c - e^{t+3}|$   
4.  $y = \arcsin(c \sin t)$     5.  $y = 3 \tan(3t^2 + c)$     6.  $y = 1 +$

- $\tan(t+c)$  7.  $y = \pm(c+2t^2)^{1/4}$  8.  $2t^2 + 6t + 2y^3 + 3y^2 = c$   
 9.  $y = \frac{1}{c_1} \sinh(c_1 \pm c_2 t)$ ,  $|y'| > 1$ ;  $y = \frac{1}{c_1} \sin(c_1 t + c_2)$ ,  $|y'| < 1$   
 10.  $c_1 y^2 - 1 = (c_1 t + c_2)^2$  11.  $t \sin t + \cos t + 1 = y^6 - y$   
 12.  $y = \tan(t + \pi/4)$  13.  $y = (2e^{-4 \sin^2 t/2} - 1)^{1/2}$   
 14.  $y = (2e^{-\frac{1}{3}t^3} - 1)^{1/2}$ ,  $t \leq \sqrt[3]{3 \ln 2}$   
 15.  $y = \left[ 9 + 2 \ln \left( \frac{1+t^2}{5} \right) \right]^{1/2}$  16.  $y = \sqrt{2t - t^2}$   
 18.  $y = \pm \sqrt{ct^3 + t^2}$   
 19.  $y = c^2 e^{-2\sqrt{t/y}}$ ,  $t > 0$ ;  $y = -c^2 e^{2\sqrt{t/y}}$ ,  $t < 0$   
 20.  $y = \frac{t^2 - 1}{2}$

### Exercise 1.5

1. Yes 2. No 3. Yes 4. No 5.  $e^t + 3ty + \sin t = c$   
 6.  $\frac{1}{2}t^2 + t \sin y - y^2 = c$  7.  $y = \frac{-1 \pm \sqrt{1+ct}}{2t}$   
 8.  $2t + e^{ty} - y^2 = c$  9.  $t^3 y + 4t^2 y^2 + 4y^3 = 28$   
 10.  $t^4 e^{t+y} + t^2 + y^2 = 1$  11.  $y = -\frac{25}{t^2 + 1}$   
 12.  $y = \frac{-1 - \sqrt{1+8t}}{2t}$   
 13.  $\mu(t) = e^t$ ;  $\frac{y^2}{2} e^t + 2ye^{2t} + (ye^t + e^{2t}) \frac{dy}{dt} = 0$   
 14.  $\mu(t) = e^{-t^2}$ ;  $(-2tye^{-t^2} + te^{-t^2})dt + e^{-t^2} dy = 0$   
 15.  $\mu(y) = \frac{1}{y}$ ;  $ydt + tdy = 0$   
 16.  $\mu(t) = -\frac{1}{(ty)^2}$ ;  $\frac{ty-1}{t^2 y} dt - \frac{1}{ty} dy = 0$   
 17.  $a = -2$ ;  $y = \pm \left[ \frac{t(2t-1)}{2(1+ct)} \right]^{1/2}$  18.  $a = 1$ ;  $y^2 t^3 + e^{t+y} = c$

### Exercise 1.6

1.  $y = 3 \sum \frac{(2t^2)^n}{n!}$   
 3.  $y_1 = 1 + 2(t-1)$ ,  $y_2 = 1 + 2(t-1) + 3(t-1)^2 + 4(t-1)^3 + 2(t-1)^4$   
 4.  $y_1 = e^t - 1$ ,  $y_2 = t - e^t + \frac{1 + e^{2t}}{2}$ ,  
 $y_3 = -\frac{107}{48} + \frac{t}{4} + \frac{t^2}{2} + \frac{t^3}{3} + 2(1-t)e^t + \frac{(1+t)e^{2t}}{2} - \frac{e^{3t}}{3} + \frac{e^{4t}}{16}$

## Chapter 2

### Exercise 2.1

1. (a)  $W[e^t, e^{-t}] = -2$ ;      (b)  $W[\sin 3t, \cos 3t] = -3$ ;  
     (c)  $W[\sin^2 t, 1 - \cos 2t] = 0$ ;    (d)  $W[t, t \ln t] = t$   
 2. Yes    3.  $W[e^t, te^t] = e^{2t}$ , Yes    4.  $W[e^{-t}, 5e^{-t}] \equiv 0$ , No  
 10.  $y = (c_1 + c_2 t)e^{t^2}$     11.  $y = c_1 t + c_2(t^2 - 1)$   
 12.  $y = c_1(t + 1) + c_2 e^{2t}$

### Exercise 2.2

1.  $y = c_1 e^{-4t} + c_2 e^{-t}$     2.  $y = c_1 e^{-t} + c_2 e^{2t}$     3.  $y = c_1 + c_2 e^{7t}$   
 4.  $y = c_1 e^{-3t} + c_2 e^{-7t}$     5.  $y = c_1 e^{4t} + c_2 t e^{4t}$     6.  $y = c_1 + c_2 t$   
 7.  $y = c_1 e^{-0.1t} + c_2 t e^{-0.1t}$     8.  $y = e^{-2t}(c_1 \cos t + c_2 \sin t)$   
 9.  $y = e^{-t/2}(c_1 \cos t + c_2 \sin t)$   
 10.  $y = e^{3t/2} \left( c_1 \cos \frac{\sqrt{7}}{2} t + c_2 \sin \frac{\sqrt{7}}{2} t \right)$   
 11.  $y = c_1 \cos 2t + c_2 \sin 2t$     12.  $y = e^{3t}(c_1 \cos 4t + c_2 \sin 4t)$   
 13.  $y = \left( \frac{1}{2} + \frac{2}{\sqrt{6}} \right) e^{(-2+\sqrt{6})t} + \left( \frac{1}{2} - \frac{2}{\sqrt{6}} \right) e^{-(2+\sqrt{6})t}$   
 14.  $y = c_5 e^t + 2t e^{3t}$     15.  $y = e^{-t/3}[3 \cos(t/\sqrt{3}) + 5\sqrt{3} \sin(t/\sqrt{3})]$   
 16.  $y = (1 + 5t)e^{-2t}$     17.  $y = e^{-t}(c_1 \cos \sqrt{3}t + c_2 \sin \sqrt{3}t)$   
 18.  $y = e^{3t}(3 \cos 4t - 2 \sin 4t)$     19.  $V \geq -3$

### Exercise 2.3

3.  $y(t) = \left( -\frac{1}{4} + \frac{t}{2} \right) e^{3t}$     4.  $y(t) = t e^t \ln |t|$     5.  $y(t) = \frac{1}{4} e^{3t}$   
 6.  $y(t) = \frac{1}{6} \cos^2 2t + \frac{1}{12} \sin^2 2t$   
 7.  $y = \cos t + 2 \sin t - \cos t \ln(\sec t + \tan t)$   
 8.  $y = \int \sqrt{s+1} [e^{2(t-s)} - e^{(t-s)}] ds$   
 9.  $y = \frac{1}{13}(2 \cos t - 3 \sin t)e^{-t} - e^{-t} + \frac{24}{13}e^{-t/3}$     10.  $y(t) = -2t + t^2$   
 11.  $y(t) = -2t^2 + 2t - 3$   
 12.  $y(t) = t^2 e^{2t} \left( \frac{1}{1 \cdot 2} + \frac{t}{2 \cdot 3} + \cdots + \frac{t^{27}}{28 \cdot 29} \right)$   
 13.  $y(t) = -\frac{t}{4} \cos 2t$     14.  $y(t) = \frac{t}{4} \sin 2t$

15.  $y(t) = \frac{56}{663} \sin \frac{t}{2} - \frac{20}{663} \cos \frac{t}{2}$
16.  $y(t) = \frac{e^t}{125} [(15t - 4) \cos t + (20t - 22) \sin t]$
17.  $y(t) = t(e^{2t} - e^t)$
18.  $y(t) = \frac{-1}{50}(\cos t + 7 \sin t) + \left(\frac{t}{2} - \frac{1}{5}\right) \frac{te^t}{5}$

## Chapter 3

### Exercise 3.1

1. (a)  $x'_1 = x_2, x'_2 = x_3, x'_3 = -4x_3 - x_2^3 + 2x_1$   
 (b)  $x'_1 = x_2, x'_2 = x_3, x'_3 = x_4, x'_4 = x_5, x'_5 = te^t - x_2 - \cos x_1$   
 (c)  $x'_1 = x_2, x'_2 = x_1^2; \quad x_1(0) = 1, x_2(0) = 3/2$   
 (d)  $x'_1 = x_2, x'_2 = x_3, x'_3 = e^t - x_2^2 - 3x_1$
2. (a)  $\mathbf{x} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix} \mathbf{x}, \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$   
 (b)  $\mathbf{x} = \begin{pmatrix} 3 & -1 & 1 \\ 2 & 0 & 1 \\ 1 & -1 & 2 \end{pmatrix} \mathbf{x}, \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$   
 (c)  $\mathbf{x} = \begin{pmatrix} 5 & 5 \\ -1 & 7 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(3) = \begin{pmatrix} 0 \\ 6 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$   
 (d)  $\mathbf{x} = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(-1) = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$
4. Yes    5. No

### Exercise 3.2

1.  $\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ -3 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{5t}$
2.  $\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 3 \\ 2 \end{pmatrix} e^{4t}$

$$3. \mathbf{x}(t) = c_1 \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} e^{3t} + c_3 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} e^{-2t}$$

$$4. \mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} e^{-t} + c_3 \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} e^{8t}$$

$$5. \mathbf{x}(t) = \left[ c_1 \begin{pmatrix} \cos 3t \\ \sin 3t \end{pmatrix} + c_2 \begin{pmatrix} -\sin 3t \\ \cos 3t \end{pmatrix} \right] e^{4t}$$

$$6. \mathbf{x}(t) = \left[ c_1 \begin{pmatrix} 2 \\ 1 - \sin t \end{pmatrix} + c_2 \begin{pmatrix} 2 \sin t \\ \cos t + \sin t \end{pmatrix} \right] e^{-2t}$$

$$7. \mathbf{x}(t) = \left[ c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ -\sin t \\ \cos t \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ \cos t \\ \sin t \end{pmatrix} \right] e^t$$

$$8. \mathbf{x}(t) = \left[ c_1 \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ \cos 2t \\ \sin 2t \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ \sin 2t \\ -\cos 2t \end{pmatrix} \right] e^t$$

$$9. \mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^t + c_2 \begin{pmatrix} t \\ 1 \\ 0 \end{pmatrix} e^t + c_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} e^{2t}$$

$$10. \mathbf{x}(t) = c_1 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} e^{-2t} + \left[ c_2 \begin{pmatrix} -t \\ 1 \\ 0 \end{pmatrix} e^t + c_3 \begin{pmatrix} -t \\ 1 \\ 0 \end{pmatrix} \right] e^{-t}$$

$$11. \mathbf{x}(t) = \frac{1}{2} \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{7t} + \frac{1}{2} \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-5t}$$

$$12. \mathbf{x}(t) = \frac{7}{4} \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + \frac{1}{4} \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}$$

$$13. \mathbf{x}(t) = \begin{pmatrix} 1 \\ \cos t - \sin t \\ \cos t + \sin t \end{pmatrix} e^t \quad 14. \mathbf{x}(t) = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} e^{2t}$$

$$15. \mathbf{x}(t) = \begin{pmatrix} 1 + 5t - \frac{1}{2}t^2 \\ 2 - t \\ 1 \end{pmatrix} e^{2t}$$

$$16. \mathbf{x}(t) = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} e^{-2t} + \begin{pmatrix} -\sqrt{2} \sin \sqrt{2}t - \sqrt{2} \cos \sqrt{2}t \\ \cos \sqrt{2}t - \sqrt{2} \sin \sqrt{2}t \\ -3 \cos \sqrt{2}t \end{pmatrix}$$

$$17. \mathbf{x}(t) = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^t \quad 18. \mathbf{x}(t) = \begin{pmatrix} 1-t \\ t \\ t \end{pmatrix} e^{2t}$$

### Exercise 3.3

$$1. \mathbf{x}(t) = \begin{pmatrix} e^t & e^{-t} \\ -e^t & e^{-t} \end{pmatrix} \quad 2. \mathbf{x}(t) = \begin{pmatrix} -e^t & e^{3t} & -e^{-3t} \\ 4e^t & 2e^{3t} & e^{-3t} \\ e^t & e^{3t} & e^{-3t} \end{pmatrix}$$

$$3. \mathbf{x}(t) = \begin{pmatrix} e^{-2t} & e^{2t} & -e^{-4t} \\ e^{-2t} & -e^{2t} & e^{-4t} \\ e^{-2t} & e^{2t} & e^{-4t} \end{pmatrix}$$

$$4. e^{At} = \begin{pmatrix} e^{2t} & 2e^{5t} - 2e^{3t} & 4e^{5t} - (4+3t)e^{3t} \\ 0 & e^{5t} & 2e^{5t} - 2e^{3t} \\ 0 & 0 & e^{3t} \end{pmatrix}$$

$$5. e^{At} = \begin{pmatrix} e^t & -\frac{1}{2}e^t + \frac{1}{2}e^{3t} & -\frac{1}{2}e^{3t} + \frac{1}{2}e^{5t} \\ 0 & e^{3t} & -e^{3t} + e^{5t} \\ 0 & 0 & e^{5t} \end{pmatrix}$$

$$6. e^{At} = \begin{pmatrix} (t+1)e^{-t} & (t+1)e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -te^{-t} & -te^{-t} + e^{-2t} & e^{-2t} - e^{-t} \\ te^{-t} & te^{-t} & e^{-t} \end{pmatrix}$$

7. Yes    8. Yes    9. No

### Exercise 3.4

$$1. \mathbf{x} = \begin{pmatrix} \frac{31}{30}e^{-4t} + \frac{1}{6}e^{2t} - \frac{1}{5}e^t \\ -\frac{62}{15}e^{-4t} + \frac{1}{3}e^{2t} - \frac{1}{5}e^t \end{pmatrix}$$

$$2. \mathbf{x} = 2e^t \begin{pmatrix} t \cos t + 3t \sin t + \sin t \\ -t \sin t \end{pmatrix}$$

$$3. \mathbf{x} = \begin{pmatrix} [(-3c_1 + c_2)t + c_1]e^{3t} + \frac{1}{9}t + \frac{2}{27} \\ [(-9c_1 + 3c_2)t + c_2]e^{3t} + \frac{1}{9} \end{pmatrix}$$

$$4. \mathbf{x} = \begin{pmatrix} 3e^{3t} - 2e^{2t} - te^{2t} \\ e^{2t} \\ 3e^{5t} - 2e^{2t} \end{pmatrix}$$

$$5. \mathbf{x} = \begin{pmatrix} -2 - 6t + 2\cos t + 6\sin t \\ -6 + 6\cos t - 2\sin t \\ 3 - 2\cos t + 2\sin t \end{pmatrix}$$

## Chapter 4

### Exercise 4.1

$$\begin{array}{llll} 1. \frac{c}{s} & 2. \frac{\sqrt{3}}{s} & 3. \frac{1}{s^2} & 4. -\frac{3}{s^2} \end{array} \quad 5. \frac{5}{s^2 + 25} \quad 6. \frac{s}{s^2 + 4}$$

$$7. \frac{2}{s^3} \quad 8. \frac{1}{s+2} \quad 9. \frac{2e^{-4s}}{s} - \frac{1}{s}$$

$$10. \frac{1 - e^{-s}}{s} + \frac{e^{-(s-1)} - e^{-4(s-1)}}{s-1}$$

### Exercise 4.2

$$\begin{array}{ll} 1. \frac{1}{2} \left( \frac{1}{s} + \frac{s}{s^2 + 4a^2} \right) & 2. \frac{1}{2} \left( \frac{a+b}{s^2 + (a+b)^2} + \frac{a-b}{s^2 + (a-b)^2} \right) \\ 3. \frac{2s^3 - 6a^2s}{(s^2 + a^2)^3} & 4. \frac{2s}{(s^2 + 1)^2} \quad 5. \frac{4}{s^3} - \frac{3}{s^2} + \frac{4}{s} \\ 6. \frac{1}{s^5} + \frac{3}{s^2} + \frac{2}{s^2 + 4} + \frac{5}{s} & 7. \frac{1}{(s-4)^2} \quad 8. \frac{1}{s} + \frac{n!}{(s-a)^{n+1}} \\ 9. \frac{5}{(s+2)^2 + 25} & 10. \frac{s^2 - 7}{(s^2 + 7)^2} \quad 11. \frac{(s+1)^2 - 4}{[(s+1)^2 + 4]^2} \\ 12. \frac{2}{(s^2 - 1)^2} & 13. \frac{2(3s^2 + 12s + 13)}{s^2[(s+3)^2 + 4]^2} \quad 14. \frac{\pi}{2} - \arctan \frac{s}{3} \\ 15. \frac{1}{2} \ln \frac{s+1}{s-1} & 16. \arccot \frac{s+3}{2} \quad 17. \frac{1}{s} \arccot \frac{s+3}{2} \\ 18. \frac{\pi}{2(s-4)^2} - \frac{1}{(s-4)^2} \arctan \frac{s}{3} + \frac{3}{(s-4)(s^2 + 9)} \\ 19. \frac{1+bs}{s^2} - \frac{b}{s(1-e^{-bs})} & 20. \frac{1}{s} \tanh \frac{s}{2} \end{array}$$

### Exercise 4.3

$$\begin{array}{llll} 1. \frac{t^3}{6} & 2. e^{8t} & 3. e^t \cos 2t + \frac{1}{2} e^t \sin 2t & 4. \frac{1}{\sqrt{8}} e^t \sin \sqrt{8}t \\ 5. e^{(3/2)t} \cosh \frac{\sqrt{57}}{2}t + \frac{3}{\sqrt{57}} e^{(3/2)t} \sinh \frac{\sqrt{57}}{2}t \end{array}$$

6.  $e^{2t} \cos 3t + \frac{2}{3}e^{2t} \sin 3t$     7.  $\frac{5}{2}t \sin t$     8.  $\cosh 3t + \frac{1}{3} \sinh 3t$   
 9.  $e^{-2t} \cos 2t + e^{-2t} \sin 2t$     10.  $e^{(3/2)t} \cos \frac{\sqrt{7}}{2}t + \sqrt{7}e^{(3/2)t} \sin \frac{\sqrt{7}}{2}t$   
 11.  $\frac{5}{3}e^{2t} - \frac{2}{3}e^{-t}$     12.  $\frac{1}{2}e^{-t} - \frac{1}{2} \cos t + \frac{1}{2} \sin t$

#### Exercise 4.4

1.  $y(t) = e^{-2t}$     2.  $y(t) = te^{5t}$   
 3.  $y(t) = \frac{3}{2}e^{-t} - \frac{1}{2} \cos t + \frac{1}{2} \sin t$     4.  $y(t) = 2 \cos 2t + \sin 2t$   
 5.  $y(t) = e^{(3/2)t} \cos \frac{\sqrt{7}}{2}t + \sqrt{7}e^{(3/2)t} \sin \frac{\sqrt{7}}{2}t$   
 6.  $y(t) = 2e^{2t} + 2e^{-t} - 2t^2 + 2t - 3$     7.  $y(t) = \frac{1}{3!}t^3e^t$   
 8.  $y(t) = e^{-2t} \left( \frac{69}{65} \cos 2t + \frac{131}{130} \sin 2t \right) + \frac{7}{65} \sin t - \frac{4}{65} \cos t$   
 9.  $y(t) = \begin{cases} 2(1 - \cos t), & 0 \leq t \leq 3 \\ 3(t - \sin t) - 7(1 - \cos t), & t > 3 \end{cases}$  or  
 $y(t) = 2(1 - \cos t) + 3u_3(t)[(t - 3) - \sin(t - 3)]$   
 10.  $y(t) = 3 \cos 2t - \sin 2t + \frac{1}{4}(1 - \cos 2t) - \frac{1}{4}[1 - \cos 2(t - 4)]u_4(t)$   
 11.  $y(t) = 3 \cos t - \sin t + \frac{1}{2}t \sin t + \frac{1}{2}u_{\pi/2}(t) \left[ \left( t - \frac{1}{2}\pi \right) \sin t - \cos t \right]$   
 12.  $y(t) = \left\{ \frac{1}{49}(7t - 1) + \left( \cos \frac{\sqrt{27}}{2}t - \frac{13}{\sqrt{27}} \sin \frac{\sqrt{27}}{2}t \right) e^{-t/2} - u_2(t)(7t - 1) \right.$   
 $\left. - u_2(t) \left[ 13 \cos \frac{\sqrt{27}}{2}(t - 2) + \sqrt{27} \sin \frac{\sqrt{27}}{2}(t - 2) \right] e^{-(t-2)/2} \right\}$   
 13.  $y(t) = te^t + u_1(t)[2 + t + (2t - 5)e^{t-1}] - u_2(t)[1 + t + (2t - 7)e^{t-2}]$   
 14.  $y(t) = 3te^{-t} + \frac{1}{2}t^2e^{-t} + 3u_3(t)(t - 3)e^{-(t-3)}$   
 15.  $y(t) = \frac{1}{2}(\sin t - \cos t) - u_{\pi}(t) \sin t$   
 16.  $y(t) = (1 - t)e^{2t}3u_1(t)(t - 1)e^{2(t-1)} + u_2(t)(t - 2)e^{2(t-2)}$

#### Exercise 4.5

1.  $x(t) = e^{-t} \begin{pmatrix} 3 \\ 2 \end{pmatrix} + e^{4t} \begin{pmatrix} -3 \\ 3 \end{pmatrix}$

$$2. \mathbf{x}(t) = e^{-t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} + e^{2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$3. \mathbf{x}(t) = e^{-t} \begin{pmatrix} 6 \\ 2 \end{pmatrix} + e^t \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$4. \mathbf{x}(t) = \begin{pmatrix} -1/4 \\ t \end{pmatrix} + \begin{pmatrix} 7/8 \\ -7/4 \end{pmatrix} e^{2t} + \begin{pmatrix} 3/8 \\ 3/4 \end{pmatrix} e^{-2t}$$

$$5. \mathbf{x}(t) = \begin{pmatrix} 2 \\ 11/8 \end{pmatrix} e^{3t} + \begin{pmatrix} 1/2 \\ -1/4 \end{pmatrix} e^t - \begin{pmatrix} 1/2 \\ 1/8 \end{pmatrix} e^{-t}$$

$$6. \mathbf{x}(t) = -\frac{4}{3} \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-t} + \frac{1}{6} \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t} + \frac{1}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ -t \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t$$

$$7. \mathbf{x}(t) = e^t \begin{pmatrix} 1-t-t^2 \\ 1-\frac{3}{2}t^2 \end{pmatrix} \quad 8. \mathbf{x}(t) = 2e^t \begin{pmatrix} t \cos t + 3t \sin t + \sin t \\ -t \sin t \end{pmatrix}$$

$$9. \mathbf{x}(t) = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} e^t + \begin{pmatrix} 1 \\ \sin t \\ \cos t \end{pmatrix} \quad 10. \mathbf{x}(t) = \begin{pmatrix} 1-t \\ 1 \\ 1 \end{pmatrix} e^{2t}$$

$$11. \mathbf{x}(t) = \begin{pmatrix} 2 \\ -2 \\ -1 \end{pmatrix} e^t$$

## Chapter 5

### Exercise 5.1

$$1. (-1, -1), (1, 1) \quad 2. (-1, 1)$$

$$3. x = 0, y \in \mathbf{R}; \quad x \in \mathbf{R}, y = 1; \quad x \in \mathbf{R}, y = -1$$

$$4. (-1, 1), (1, -1) \quad 5. (0, 0), (0, 8), (7, 0), (4, 6)$$

$$6. x = y = z = 0; \quad x = \frac{c}{d}, y = \frac{a}{b}, z = -\left(\frac{c^2}{d^2} + \frac{a^2}{b^2}\right)$$

$$7. x = 0, y = -2, z = n\pi$$

### Exercise 5.2

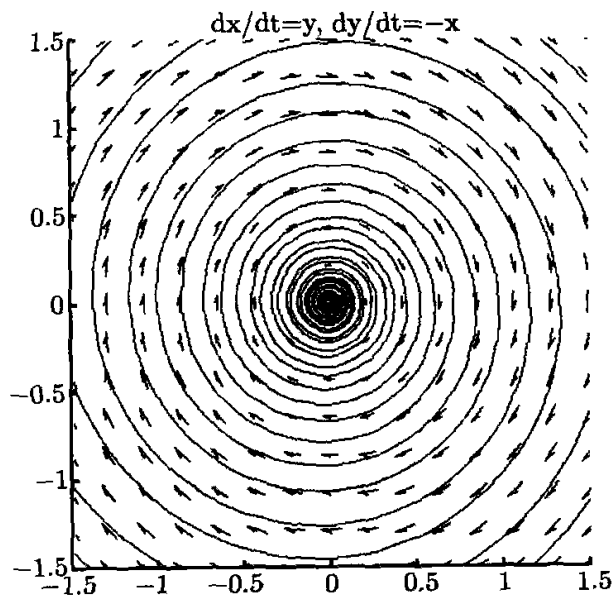
$$1. \text{Unstable} \quad 2. \text{Unstable} \quad 3. \text{Unstable} \quad 4. \text{Stable}$$

5. Stable    6. stable    7. Asymptotically stable  
 8. Asymptotically stable    9. Asymptotically stable  
 10. Asymptotically stable

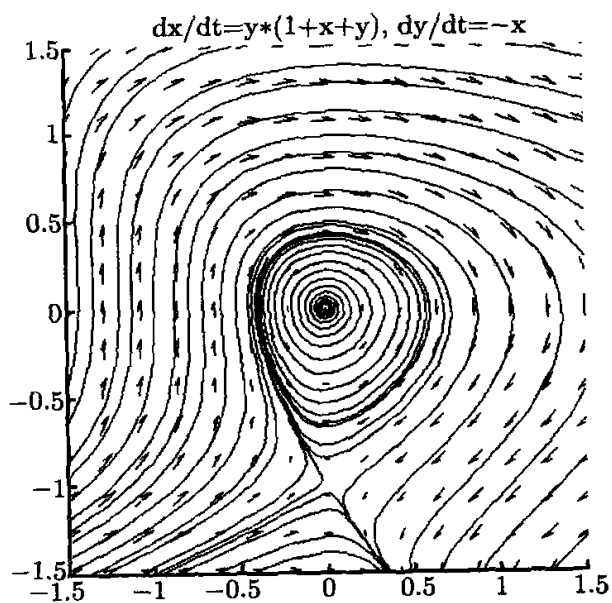
**Exercise 5.3**

1. Saddle point    2. Saddle point    3. Improper nodal  
 4. Improper nodal    5. Center    6. Spiral sink

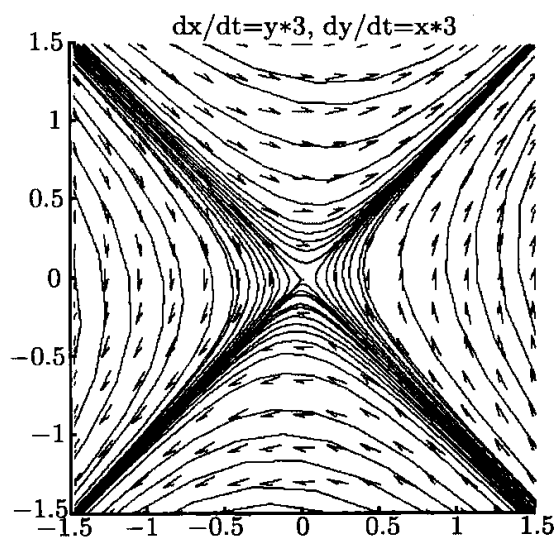
7.



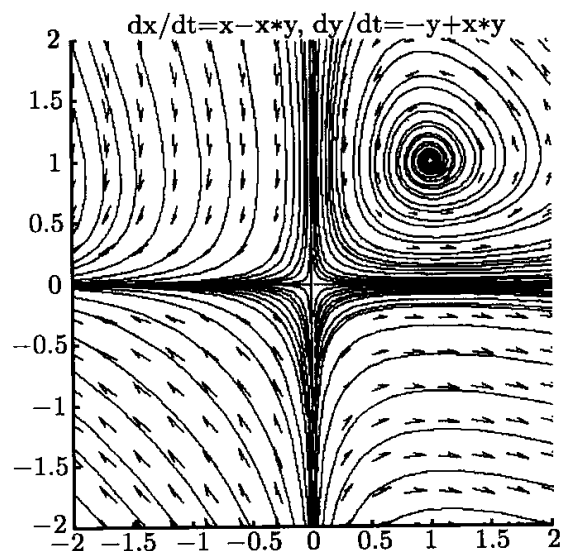
8.



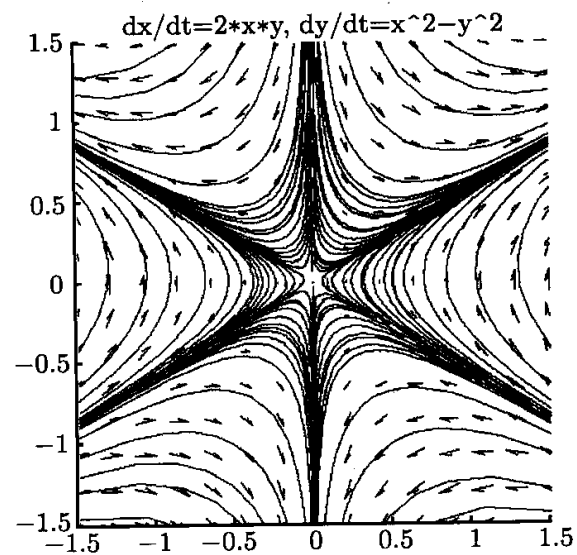
9.



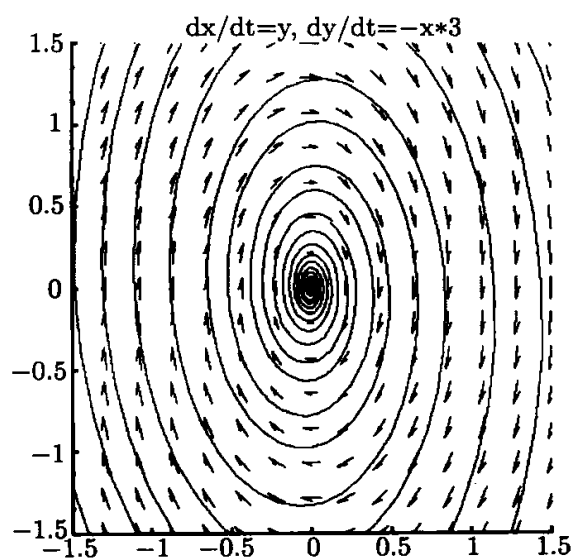
10.



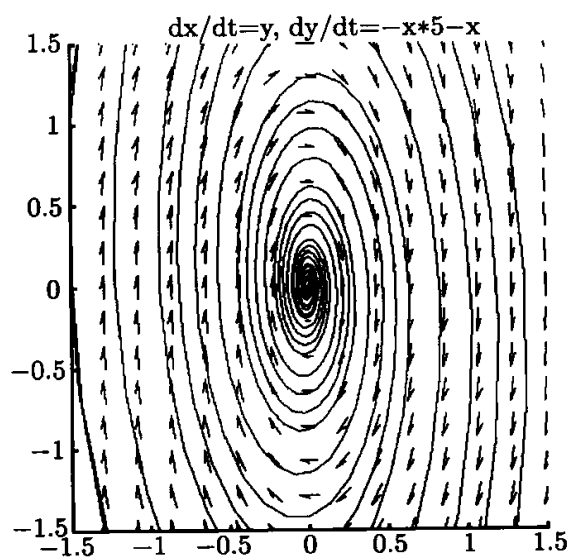
11.



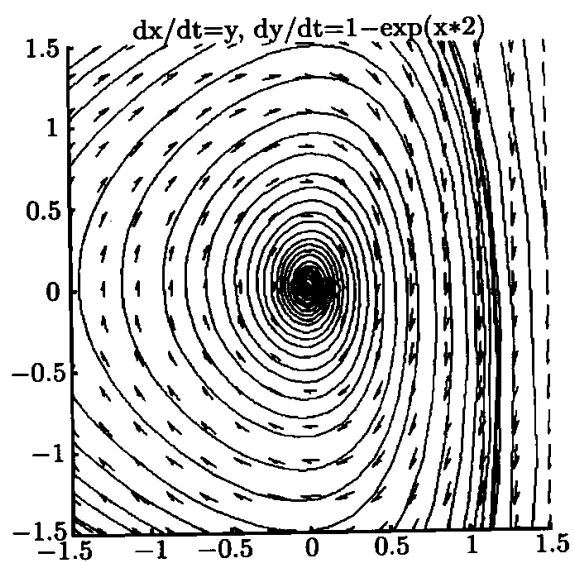
12.

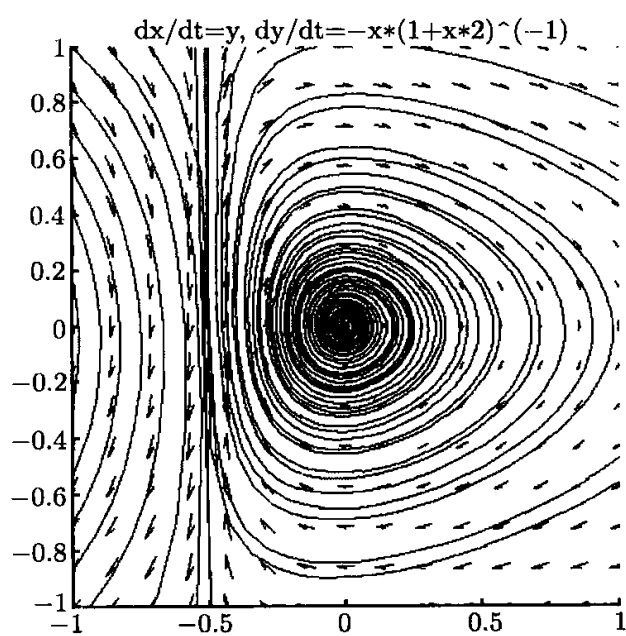


13.



14.



**15.**

## References

- 蔡燧林. 2007. 常微分方程. 2 版. 武汉: 武汉大学出版社
- 林武忠等. 2003. 常微分方程. 北京: 科学出版社
- 时宝等. 2007. 常微分方程基础及其应用. 北京: 科学出版社
- 叶宗泽等. 1985. 常微分方程组与运动稳定性理论. 天津: 天津大学出版社
- 张元林. 2008. 积分变换. 4 版. 北京: 高等教育出版社
- 周义仓等. 2004. 常微分方程及其应用——方法、理论、建模、计算机. 北京: 科学出版社
- 庄万. 2003. 常微分方程习题解. 济南: 山东科学技术出版社
- Arnol'd V I. 2003. Ordinary Differential Equations. New York: Springer-Verlag, 北京: 世界图书出版公司
- Braun M. 1998. Differential Equations and Their Applications. 4th ed. Springer-Verlag, 北京: 世界图书出版公司
- Brinkmann P, Laugesen R. 2004. User manual for Iode (graphical user interface), Urbana-Champaign, U.S.A.
- Bronson R. 2000. 题解精萃——微分方程 Differential Equations. 北京: 高等教育出版社, 麦格劳-希尔国际出版公司
- Henry C, et al. 2006. 常微分方程基础 (英文版). 原版第 5 版. 北京: 机械工业出版社
- Zill D G. 2004. A First Course in Differential Equations with Modeling Applications. 7th ed. 北京: 世界图书出版公司

## 附录 软件包 Iode 简介

### 1. Iode 简介

Iode 是美国伊利诺伊大学数学系的 Peter Brinkmann 和 Richard Laugesen 等开发的用于求解常微分方程的软件包. 其功能主要包括常微分方程 (包括二阶线性常微分方程) 的求解、方向场、相图、傅里叶级数以及热和波动方程的求解. Iode 是一个免费软件, 用户可以直接在 [www.math.uiuc.edu/iode](http://www.math.uiuc.edu/iode) 上下载该软件的压缩包. 解压后无须安装, Iode 可以直接运行在 Matlab 或 Octave 平台下 (两种语言大部分兼容), 类似于一个工具箱, 下面的介绍主要针对 Matlab 平台. 对于 Matlab 6.0 及其以后的版本, Iode 具有图形用户界面. 启动 Matlab, 将 Iode 所在文件夹设为当前目录, 在命令窗口中输入 Iode 即可启动 Iode, 其主菜单界面见图 1. 单击任一功能即可启动该模块, 也可以同时启动多个模块.

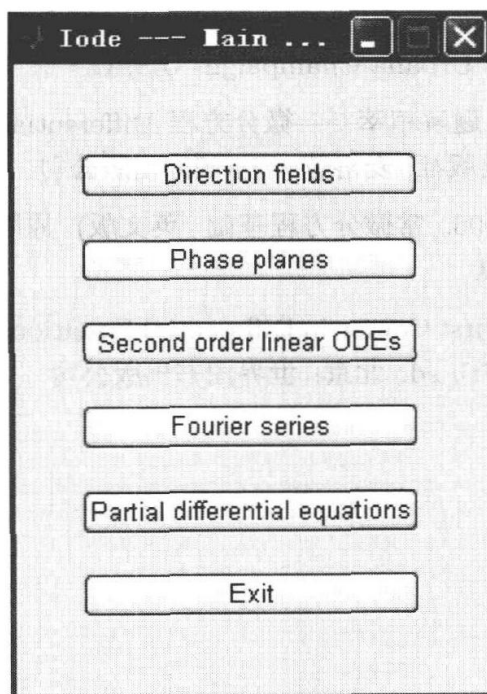


图 1 Iode 的主界面

每个模块具有相同的菜单来处理结果. 模块窗口的菜单栏含有三个菜单: File, Equation (or Function) 和 Options. File 用于与外界的数据交换, 如打开或保存文件. 打印; Equation 用于输入要求解的方程和相关参数的设定; Options 用于图形结果显示的设置. 下面以相平面 (Phase Plane) 为例, 来介绍模块的功能.

## 2. 相图绘制

相平面功能用于两个方程的自治系统:

$$\begin{cases} \frac{dx}{dt} = f(x, y), \\ \frac{dy}{dt} = g(x, y). \end{cases} \quad (1)$$

该模块可以画出与该自治系统相关的相图, 并计算和绘出系统在相平面上的数值解. 该模块启动后的界面见图 2.

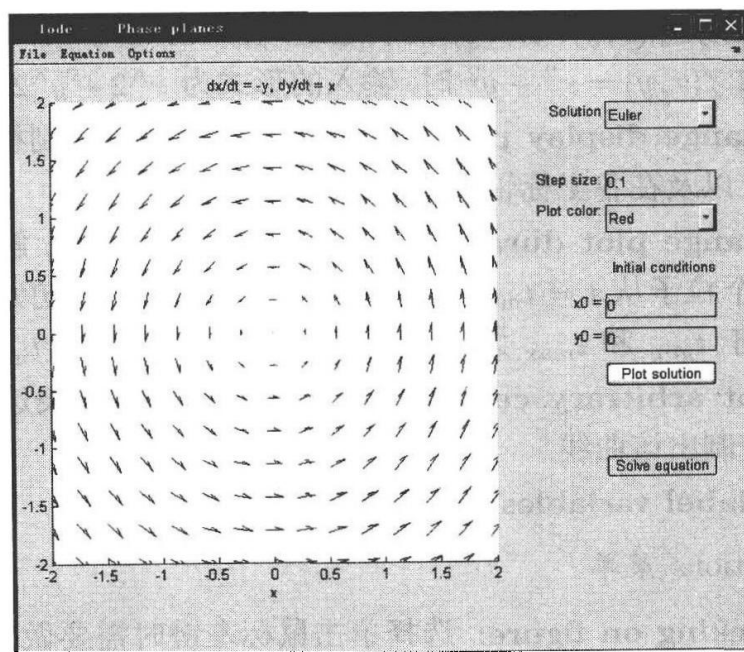


图 2 Phase planes 的主界面

### 2.1 图形界面的操作

**Solution method:** 选择求解的数值方法, 包括欧拉法、龙格-库塔法, 以及其他用户自定义的方法、精确计算 (需安装 Matlab 符号计算工具箱).

**Step size:** 数值求解步长, 一般是正实数.

**Plot color:** 画图的颜色.

**Initial conditions:** 数值求解的初值条件, 可以在对话框中输入初值, 也可以在需要绘制解曲线的位置用鼠标左键单击输入.

**Plot solution:** 画出解曲线.

## 2.2 File 菜单

**Open:** 打开一个 mat 文件, 载入保存的窗口设置.

**Save:** 保存当前窗口的设置, 文件扩展名为 mat.

**Print:** 打印结果, 既可以输出到打印机, 也可以打印为文件 (ps 或 pdf 格式, 需相关软件支持), 默认设置打印整个窗口, 选择 option 菜单中 “Suppress printing of user interface controls”, 即可只打印图形.

## 2.3 Equation 菜单

**Enter differential equation:** 输入需要求解的自治系统右侧两个函数  $f(x, y)$ ,  $g(x, y)$  的表达式, 输入时应用 Matlab 语言形式, 如  $f(x, y) = x^2 + y^2$  时, 输入的形式为  $x^2 + y^2$ .

**Change display parameters:** 设置求解和绘制相图的参数范围, 以及在各坐标轴上绘制曲线的段数.

**Change plot duration:** 设置绘图的持续时间, 解曲线可看作一个粒子从  $t = t_{\min}$  到  $t = t_{\max}$  的运动轨迹, 初始条件的  $t_0$  需落于  $t_{\min}$  和  $t_{\max}$  之间, 默认设置  $t_0 = t_{\min} = 0$ ,  $t_{\max} = 2\pi$ .

**Plot arbitrary curve:** 输入二维曲线的参数表达及参数范围, 绘制出该曲线.

**Relabel variables:** 重新标记参数.

## 2.4 Options 菜单

**Clicking on figure:** 选择单击鼠标左键时需要做什么, 有四个选项.

(1) plots solution through click point (this is the default action), 单击鼠标左键绘制以该点为初值的解曲线.

(2) zooms in on click point, 以该点为中心将参数范围缩小.

(3) zooms out on click point, 以该点为中心将参数范围扩大.

(4) recenters figure on click point, 使相图以该点为中心.

**Change zoom factors:** 改变缩放因子, 默认缩放因子为 2.

**Clear plot:** 清除所有曲线.

**Refresh plot:** 刷新.

**Show coordinate maps:** 显示坐标  $x$  相对于  $t, y$  相对于  $t$  的曲线, 以及  $x, y$  相对于  $t$  的曲线.

**Enter caption:** 在相图的下方输入标题.

### 3. 举例

作为例子, 用 Iode 的 Phase Plane 功能画出下面自治系统的相图:

$$\begin{cases} dx/dt = -4x + y, \\ dy/dt = -3x. \end{cases} \quad (2)$$

启动 Matlab, 将 iode\_dos 设为当前目录, 在命令窗口中输入 iode 启动 iode, 选择 Phase planes, 即可启动画相平面功能. 在 Phase planes 界面下, 选择 Equation-Enter differential equation,  $f(x, y)$  输入  $-4 * x + y$ ,  $g(x, y)$  输入  $-3 * x$ , 参数范围设置为  $x, y \in [-2, 2]$ ,  $x$  轴,  $y$  轴上各画 30 段. 画出相图见图 3, 选择初值  $x_0 = 2, y_0 = 0.5$ , 求解步长为 0.1, 用欧拉法求解的解曲线如图 3 中的实线.

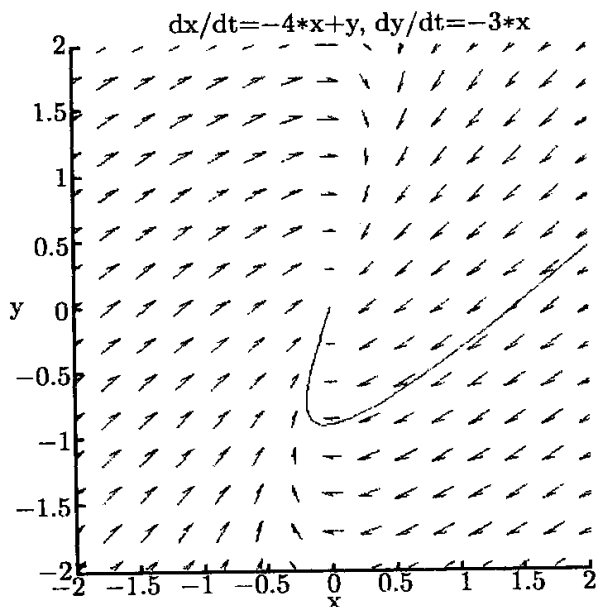


图 3 自治系统的相平面和一条解曲线

这里主要介绍了 Iode 的相平面功能, 关于 Iode 的更多细节可以参考文献 (Brinkmann et al., 2004).

Images have been losslessly embedded. Information about the original file can be found in PDF attachments. Some stats (more in the PDF attachments):

```
{
  "filename": "MTI3Njk5NjNf5bi45b6u5YiG5pa556iL5Y+K5YW25bqU55So55CG6K665LiO5qih5Z6LLnppcA==",
  "filename_decoded":
"12769963_\u5e38\u5fae\u5206\u65b9\u7a0b\u53ca\u5176\u5e94\u7528\u7406\u8bba\u4e0e\u6a21\u578b.zip",
  "filesize": 11183433,
  "md5": "3805ee67b9e408c211db18f916facf27",
  "header_md5": "17426d5d34613bbde58fd7641a7636ac",
  "sha1": "b4d283c4cec100a6469b2c07b81799fecaaeab94",
  "sha256": "91882a7766074c528b22750f506f4b20426ae08187c84840c2127f852d16e5f8",
  "crc32": 613061837,
  "zip_password": "",
  "uncompressed_size": 11485840,
  "pdg_dir_name": "12769963_\u2502\u00fa\u256c\u00f3\u2556\u2553\u2556\u255c\u2502\u2560\u255d\u2591\u255e\u03a3\u2559\u00aa\u2559\u251c\u2514\u03c6\u252c\u2588\u2559\u03b4\u2500\u00fa\u2568\u2550",
  "pdg_main_pages_found": 213,
  "pdg_main_pages_max": 213,
  "total_pages": 221,
  "total_pixels": 1143510992,
  "pdf_generation_missing_pages": false
}
```